HY-TTC 500

System Manual

Programmable ECU for Sensor-Actuator Management

Product Version 01.04

Original Instructions

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Hardware Description

1 Introduction

HY-TTC 500 is a family of programmable electronic control units for sensor/actuator management. Lots of configurable I/Os allow the use of HY-TTC 500 with different sensor and actuator types. Being part of a complete and compatible product family, the control unit is specifically designed for vehicles and machines that operate in rough environments and at extreme operating temperatures. The robust die-cast aluminum housing of HY-TTC 500 provides protection against electromagnetic disturbances and mechanical stress. A 180 MHz TI TMS570-integrated microprocessor provides the necessary processing power. To provide individual features that are scalable to system integrator needs, the HY-TTC 500 family is available in several variants, with different assembly options. See Section 1.5 on page 4 for variants overview.

1.1 Inputs and Outputs

All inputs and outputs of each HY-TTC 500 variant are protected against electrical surges and short circuits. In addition, internal safety measures allow the detection of open load, overload and short circuit conditions at the outputs. Proportional hydraulic components can be directly connected to the current-controlled PWM outputs. The HY-TTC 500 family is designed to support a variety of analog and digital sensor types. Many software-configurable input options can be selected to adapt to different sensor types. A group of individually configurable analog inputs with a voltage ranging from 0. . . 5V to 0. . . 32V are provided. Those analog inputs can be set to different voltage ranges by software in order to achieve the best analog accuracy and resolution. The analog inputs can also be configured as a current input or for resistive measurements.

1.2 Communication Interfaces

On the fully equipped variant HY-TTC 580, 7 x CAN (according to CAN 2.0B), 1x RS-232 and 1x LIN interface are available for serial communication. Additionally a 10 Mbit/s Ethernet interface for application download and debug purpose is provided.

1.3 Safety and Certification

The HY-TTC 500 family is designed to comply with the international standards IEC 61508 [15] and ISO 13849 [24].

1.4 Programming Options

The unit may be programmed in C or CODESYS. CODESYS is one of the most common IEC 61131-3 programming systems [17] running under Microsoft Windows. CODESYS supports several editors, including the **Instruction List Editor**, **Sequential Function Chart Editor**, and **Function Block Diagram Editor**. CODESYS produces native machine code for the main processor of HY-TTC 500.

1.5 HY-TTC 500 Variants

The following HY-TTC 500 variants are described in this System Manual:

- **HY-TTC 580**
- **HY-TTC 540**
- **HY-TTC 520 (customer-specific variant only)**
- **HY-TTC 510**

The main difference of the listed HY-TTC 500 variants is the amount of available I/Os, whereas the HY-TTC 580 is the powerfulest variant with the maximum number of I/Os. See Chapter 3 on page 29 which main- and alternative functions are available on which variant.

Unless other specified in Chapter 4 on page 63, the functionality of the available main- and alternative functions do not differ between each variants.

Feature	HY-TTC 580	HY-TTC 540	HY-TTC 520	HY-TTC 510
CPU				
32-bit TI TMS570	x	x	x	x
Int. FLASH	3 MB	3 MB	3 MB	3 MB
Int. RAM	256 kB	256 kB	256 kB	256 kB
Memory				
Ext. FLASH	8 MB	-	-	-
Ext. RAM	2 MB	2 MB	2 MB	2 MB
Ext. EEPROM	64 kB	64 kB	64 kB	64 kB
Interface				
CAN	7	4	4	3
CAN bus termination	4	4	4	3
Ethernet	1	-	-	-
LIN	1	-	-	1
RS232	1	-	-	-
Real time clock	1	-	-	-
Outputs				
High-Side PWM with CM	36	28	18	16
High-Side digital	8	8	8	8
High-Side digital, PVG, VOUT	8	-	-	8
Low-Side digital	8	8	8	8
Inputs				
Analog input 3 modes (V)(I)(R)	8	8	8	8
Analog input 2 modes (V)(I)	16	16	16	16
Analog input (V)	-	8	-	-
Timer input	12	20	20	20
Terminal 15	1	1	1	1
Wake-up	1	1	1	1
Sensor supply				
+5V/500mA	2	2	2	2
+5-10V/2.5W	1	1	1	1

Feature	HY-TTC 580	HY-TTC 540	HY-TTC 520	HY-TTC 510
Safety Switch				
Number of secondary shut-off path	3	2	2	2
Software				
C Programming	x	x	x	x
CODESYS® Safety SIL 2, CANopen® Safety Master	x	x	-	x
CODESYS® V3, CANopen® Master	x	x	-	x

Table 1.1: Variants overview

1.5.1 HY-TTC 580 Variant

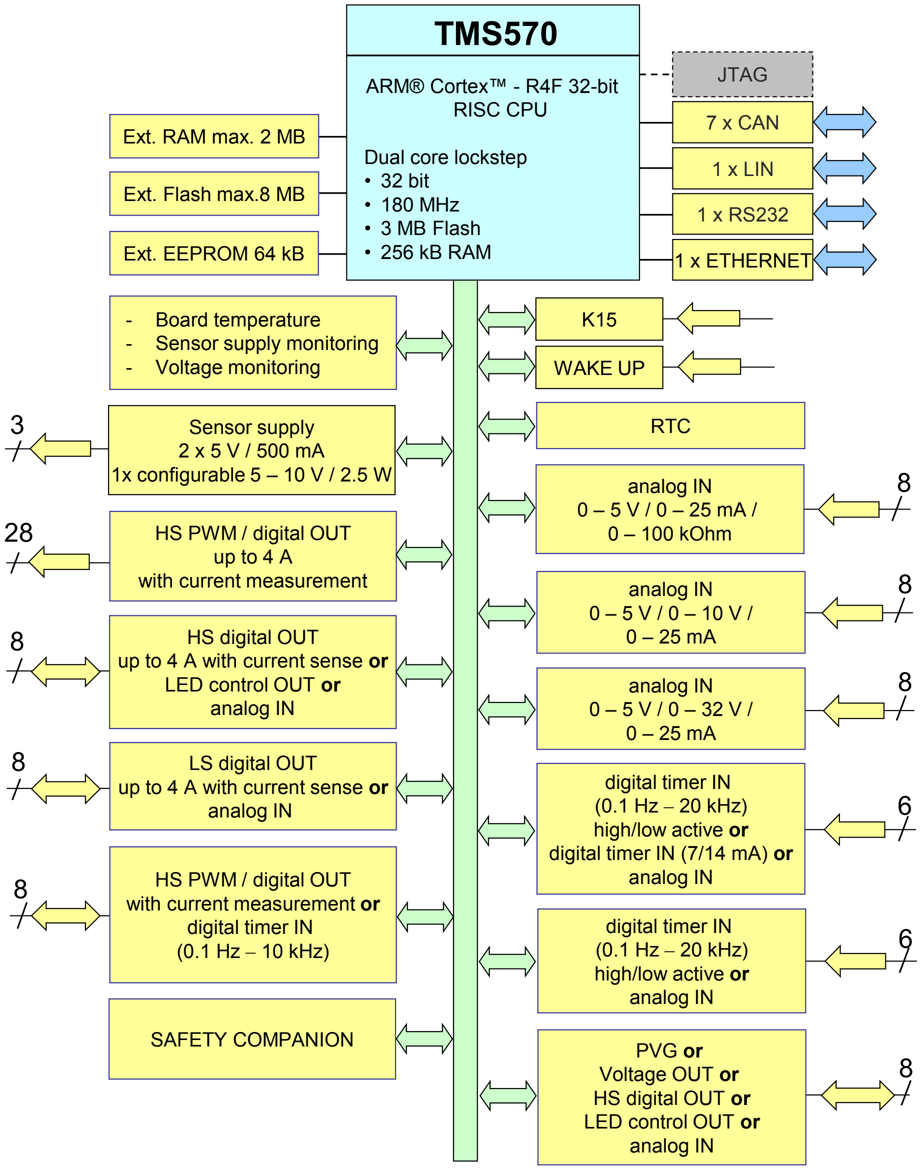

Figure 1.1: HY-TTC 580 Variant

System CPU

- TMS570LS3137 CPU running at 180 MHz, 3 MB internal Flash, 256 kB internal RAM, 64 kB configuration Flash
- Safety companion
- 12 bit ADC with 5 V reference voltage
- Real Time Clock (RTC)

Memory

External EEPROM	64 kB More than 1,000,000 write cycles. More than 40-year data retention.
External Flash	8 MB More than 100,000 write/erase cycles per block. More than 20-year data retention.
Internal Flash	3 MB Program Flash. More than 1000 write/erase cycles. More than 15-year data retention.
Configuration Flash	64 kB More than 100,000 write/erase cycles. More than 15-year data retention.
External SRAM	2 MB
Internal SRAM	256 kB

Communication Interfaces

- 7 x CAN (50 to 1000 kbit/s)
- 1 x LIN (up to 20 kBd)
- 1 x RS232 (up to 115 kBd)
- 1 x Ethernet (10 Mbit/s)

Power Supply

- Supply voltage: 8 to 32 V
- Separate supply pins for CPU subsystem and I/O subsystem
- Load dump protection
- Low current consumption: 0.4 A at 12 V
- 2 x 5 V / 500 mA sensor supply
- 1 x 5 to 10 V / 2.5 W sensor supply, voltage selected by software
- Board temperature, sensor supply and battery monitoring

Inputs

8 x analog input 3 modes	Voltage measurement: 0 to 5 V Current measurement: 0 to 25 mA Resistor measurement: 0 to 100 kΩ
16 x analog input 2 modes	Voltage measurement: 8 x 0 to 5 V/ 0 to 10 V Voltage measurement: 8 x 0 to 5 V/ 0 to 32 V Current measurement: 16 x 0 to 25 mA
6 x timer input	Frequency and pulse width measurement Input pair as encoder Voltage measurement 0 to 32 V
6 x timer input	Frequency and pulse width measurement Input pair as encoder Digital (7/14 mA) current loop speed sensor Voltage measurement 0 to 32 V

All modes are configurable in software. The analog inputs provide 12-bit resolution. Each voltage input can be configured as a digital input with an adjustable threshold.

Outputs

36 x PWM-controlled HS Outputs	
	PWM mode (50 Hz to 1 kHz) Nominal current 4 A Digital output mode with current feedback
when used as an input	Digital input 8 PWM-controlled HS outputs can be alternatively used as frequency or pulse width measurement input
8 x digital HS outputs	Digital output mode Nominal current 4 A with voltage feedback
when used as an input	Voltage measurement 0 to 32 V Digital input
8 x digital LS outputs	Digital output mode Nominal current 4 A
when used as an input	Voltage measurement 0 to 32 V Digital input
8 x HS outputs	Digital output mode PVG output Voltage output (VOUT)
when used as an input	Voltage measurement 0 to 32 V Digital input

Specifications

Dimensions	See [35]
Weight	See [35]
Operating ambient temperature	-40 ℃ to +85 ℃
Storage temperature	-40 ℃ to +85 ℃
Housing	IP67- and IP6k9k-rated die-cast aluminum housing and 154-pin connector
	Pressure equalization with water barrier
Operating altitude	0 to 4000 m

1.5.2 HY-TTC 540 Variant

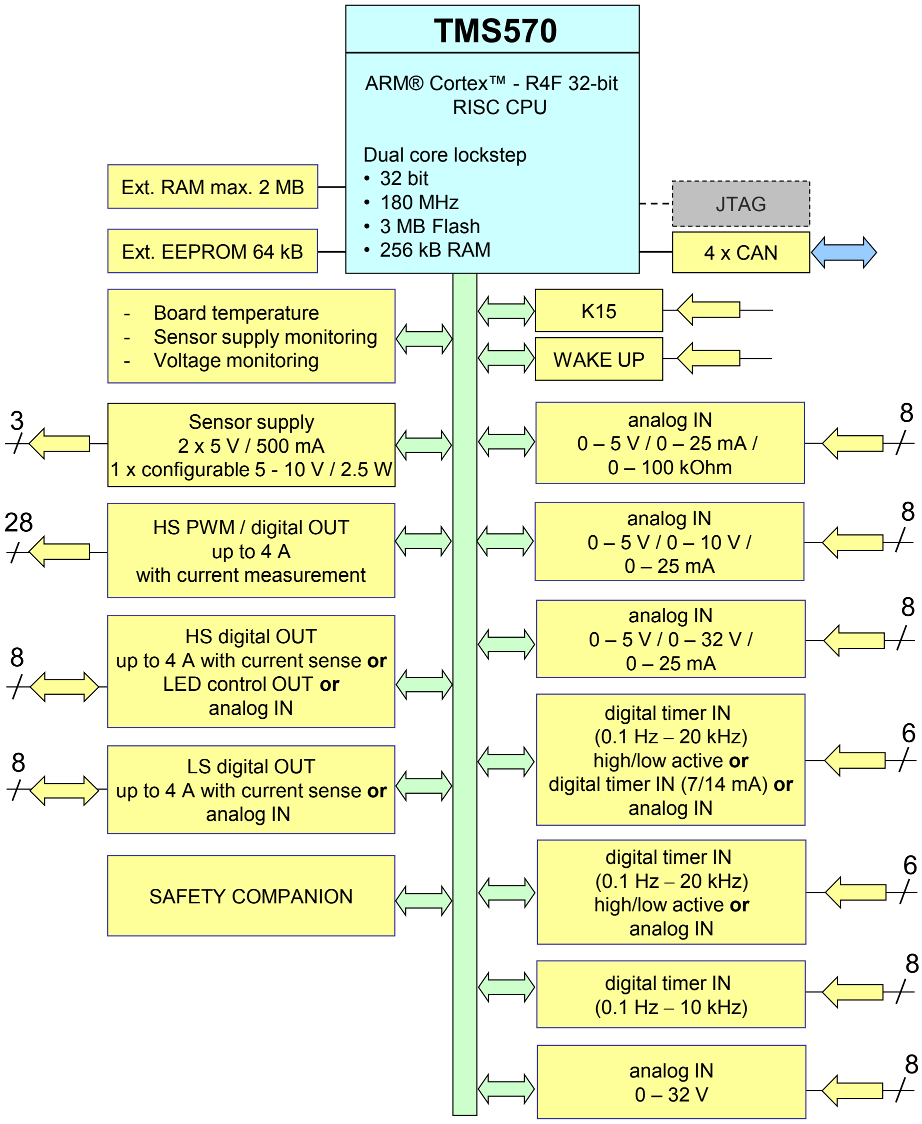

Figure 1.2: HY-TTC 540 Variant

System CPU

- TMS570LS3137 CPU running at 180 MHz, 3 MB internal Flash, 256 kB internal RAM, 64 kB configuration Flash
- Safety companion
- 12 bit ADC with 5 V reference voltage

Memory

External EEPROM	64 kB More than 1,000,000 write cycles. More than 40-year data retention.
Internal Flash	3 MB Program Flash. More than 1000 write/erase cycles. More than 15-year data retention.
Configuration Flash	64 kB More than 100,000 write/erase cycles. More than 15-year data retention.
External SRAM	2 MB
Internal SRAM	256 kB

Communication Interfaces

- 4 x CAN (50 to 1000 kbit/s)

Power Supply

- Supply voltage: 8 to 32 V
- Separate supply pins for CPU subsystem and I/O subsystem
- Load dump protection
- Low current consumption: 0.4 A at 12 V
- 2 x 5 V / 500 mA sensor supply
- 1 x 5 to 10 V / 2.5 W sensor supply, voltage selected by software
- Board temperature, sensor supply and battery monitoring

Inputs

8 x analog input 3 modes	Voltage measurement: 0 to 5 V Current measurement: 0 to 25 mA Resistor measurement: 0 to 100 kΩ
16 x analog input 2 modes	Voltage measurement: 8 x 0 to 5 V/ 0 to 10 V Voltage measurement: 8 x 0 to 5 V/ 0 to 32 V Current measurement: 16 x 0 to 25 mA
6 x timer input	Frequency and pulse width measurement Input pair as encoder Voltage measurement 0 to 32 V
6 x timer input	Frequency and pulse width measurement Input pair as encoder Digital (7/14 mA) current loop speed sensor Voltage measurement 0 to 32 V
8 x timer input	Frequency and pulse width measurement
8 x analog input	0 to 32 V

All modes are configurable in software. The analog inputs provide 12-bit resolution. Each voltage input can be configured as a digital input with an adjustable threshold.

Outputs

28 x PWM-controlled HS Outputs when used as an input	 PWM mode (50 Hz to 1 kHz) Nominal current 4 A Digital output mode with current feedback Digital input
8 x digital HS outputs when used as an input	Digital output mode Nominal current 4 A with voltage feedback Voltage measurement 0 to 32 V Digital input
8 x digital LS outputs when used as an input	Digital output mode Nominal current 4 A Voltage measurement 0 to 32 V Digital input

Specifications

Dimensions	See [35]
Weight	See [35]
Operating ambient temperature	-40 ℃ to +85 ℃
Storage temperature	-40 ℃ to +85 ℃
Housing	IP67- and IP6k9k-rated die-cast aluminum housing and 154-pin connector
	Pressure equalization with water barrier
Operating altitude	0 to 4000 m

1.5.3 HY-TTC 520 Variant (Customer-specific variant only)

This is an customer-specific variant only.

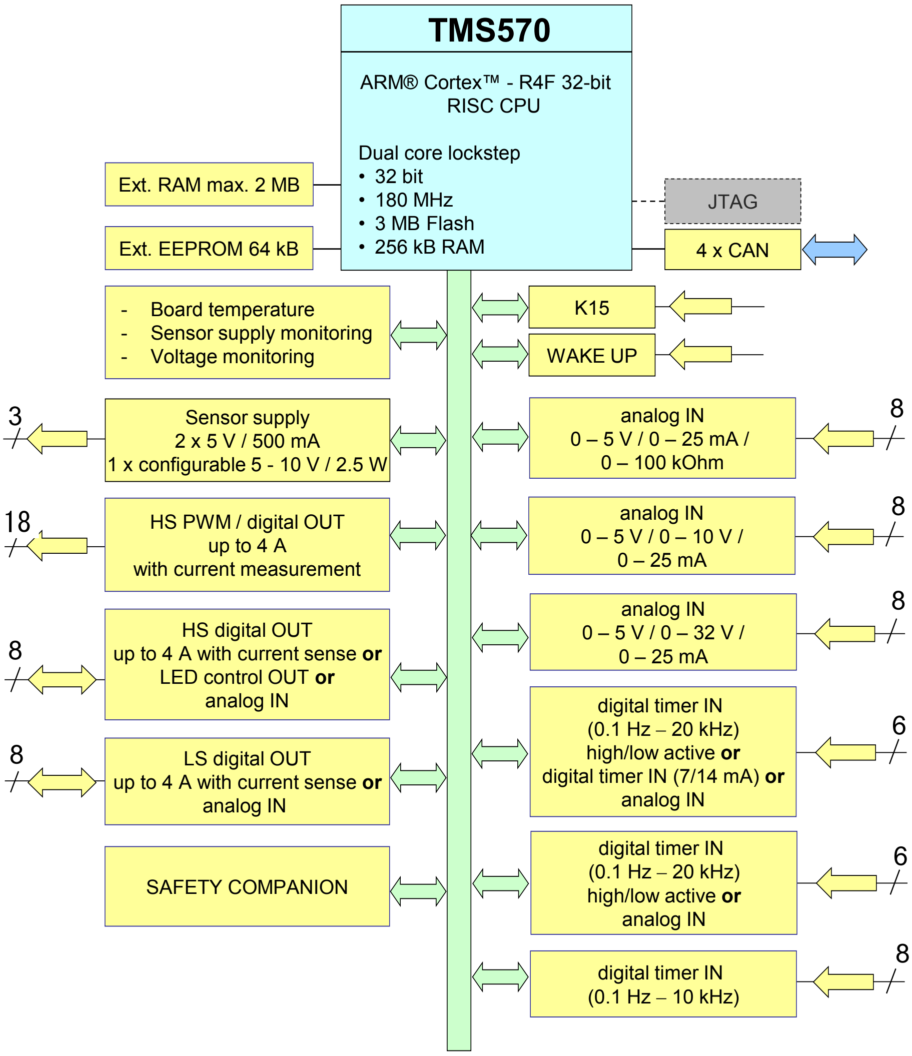

Figure 1.3: HY-TTC 520 Variant

System CPU

- TMS570LS3137 CPU running at 180 MHz, 3 MB internal Flash, 256 kB internal RAM, 64 kB configuration Flash
- Safety companion
- 12 bit ADC with 5 V reference voltage

Memory

External EEPROM	64 kB More than 1,000,000 write cycles. More than 40-year data retention.
Internal Flash	3 MB Program Flash. More than 1000 write/erase cycles. More than 15-year data retention.
Configuration Flash	64 kB More than 100,000 write/erase cycles. More than 15-year data retention.
External SRAM	2 MB
Internal SRAM	256 kB

Communication Interfaces

- 4 x CAN (50 to 1000 kbit/s)

Power Supply

- Supply voltage: 8 to 32 V
- Separate supply pins for CPU subsystem and I/O subsystem
- Load dump protection
- Low current consumption: 0.4 A at 12 V
- 2 x 5 V / 500 mA sensor supply
- 1 x 5 to 10 V / 2.5 W sensor supply, voltage selected by software
- Board temperature, sensor supply and battery monitoring

Inputs

8 x analog input 3 modes	Voltage measurement: 0 to 5 V Current measurement: 0 to 25 mA Resistor measurement: 0 to 100 kΩ
16 x analog input 2 modes	Voltage measurement: 8 x 0 to 5 V/ 0 to 10 V Voltage measurement: 8 x 0 to 5 V/ 0 to 32 V Current measurement: 16 x 0 to 25 mA
8 x timer input	Frequency and pulse width measurement
6 x timer input	Frequency and pulse width measurement Input pair as encoder Voltage measurement 0 to 32 V
6 x timer input	Frequency and pulse width measurement Input pair as encoder Digital (7/14 mA) current loop speed sensor Voltage measurement 0 to 32 V

All modes are configurable in software. The analog inputs provide 12-bit resolution. Each voltage input can be configured as a digital input with an adjustable threshold.

Outputs

18 x PWM-controlled HS Outputs	
	PWM mode (50 Hz to 1 kHz) Nominal current 4 A Digital output mode with current feedback
when used as an input	Digital input
8 x digital HS outputs	Digital output mode Nominal current 4 A with voltage feedback
when used as an input	Voltage measurement 0 to 32 V Digital input
8 x digital LS outputs	Digital output mode Nominal current 4 A
when used as an input	Voltage measurement 0 to 32 V Digital input

Specifications

Dimensions	See [35]
Weight	See [35]
Operating ambient temperature	-40 ℃ to +85 ℃
Storage temperature	-40 ℃ to +85 ℃
Housing	IP67- and IP6k9k-rated die-cast aluminum housing and 154-pin connector
	Pressure equalization with water barrier
Operating altitude	0 to 4000 m

1.5.4 HY-TTC 510 Variant

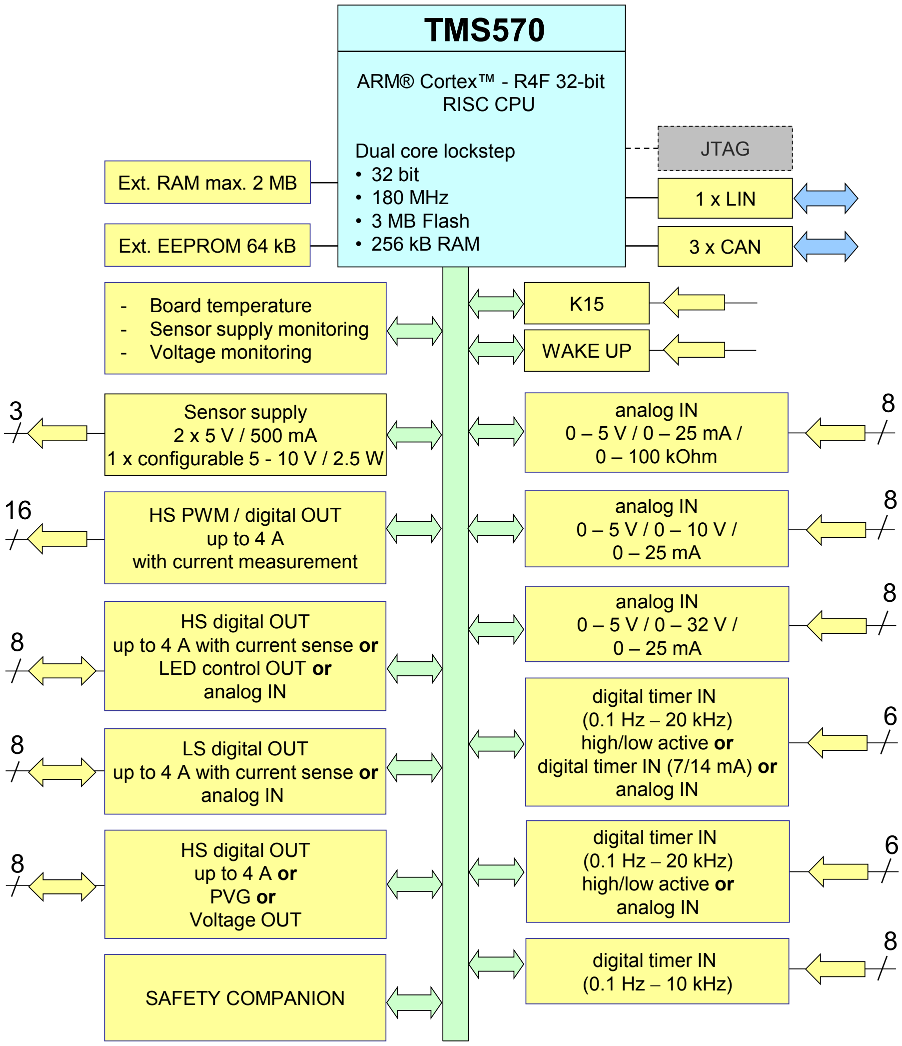

Figure 1.4: HY-TTC 510 Variant

System CPU

- TMS570LS3137 CPU running at 180 MHz, 3 MB internal Flash, 256 kB internal RAM, 64 kB configuration Flash
- Safety companion
- 12 bit ADC with 5 V reference voltage

Memory

External EEPROM	64 kB More than 1,000,000 write cycles. More than 40-year data retention.
Internal Flash	3 MB Program Flash. More than 1000 write/erase cycles. More than 15-year data retention.
Configuration Flash	64 kB More than 100,000 write/erase cycles. More than 15-year data retention.
External SRAM	2 MB
Internal SRAM	256 kB

Communication Interfaces

- 3 x CAN (50 to 1000 kbit/s)
- 1 x LIN (up to 20 kBd)

Power Supply

- Supply voltage: 8 to 32 V
- Separate supply pins for CPU subsystem and I/O subsystem
- Load dump protection
- Low current consumption: 0.4 A at 12 V
- 2 x 5 V / 500 mA sensor supply
- 1 x 5 to 10 V / 2.5 W sensor supply, voltage selected by software
- Board temperature, sensor supply and battery monitoring

Inputs

8 x analog input 3 modes	Voltage measurement: 0 to 5 V Current measurement: 0 to 25 mA Resistor measurement: 0 to 100 kΩ
16 x analog input 2 modes	Voltage measurement: 8 x 0 to 5 V/ 0 to 10 V Voltage measurement: 8 x 0 to 5 V/ 0 to 32 V Current measurement: 16 x 0 to 25 mA
8 x timer input	Frequency and pulse width measurement
6 x timer input	Frequency and pulse width measurement Input pair as encoder Voltage measurement 0 to 32 V
6 x timer input	Frequency and pulse width measurement Input pair as encoder Digital (7/14 mA) current loop speed sensor Voltage measurement 0 to 32 V

All modes are configurable in software. The analog inputs provide 12-bit resolution. Each voltage input can be configured as a digital input with an adjustable threshold.

Outputs

16 x PWM-controlled HS Outputs	
	PWM mode (50 Hz to 1 kHz)
	Nominal current 4 A
	Digital output mode with current feedback
when used as an input	Digital input
8 x digital HS outputs	Digital output mode
	Nominal current 4 A with voltage feedback
when used as an input	Voltage measurement 0 to 32 V
	Digital input
8 x digital LS outputs	Digital output mode
	Nominal current 4 A
when used as an input	Voltage measurement 0 to 32 V
	Digital input
8 x HS outputs	Digital output mode
	PVG output
	Voltage output (VOUT)
when used as an input	Voltage measurement 0 to 32 V
	Digital input

Specifications

Dimensions	See [35]
Weight	See [35]
Operating ambient temperature	-40 ℃ to +85 ℃
Storage temperature	-40 ℃ to +85 ℃
Housing	IP67- and IP6k9k-rated die-cast aluminum housing and 154-pin connector
	Pressure equalization with water barrier
Operating altitude	0 to 4000 m

1.6 Standards and Guidelines

The HY-TTC 500 was developed to comply with several international standards and guidelines. This section lists the relevant standards and guidelines and the applied limits and severity levels.

Environmental Criteria – ISO 16750 Code:

CODE: ISO 16750 (B[1] F[2]), L, G, D, Z, (IP6k7; IP6k9k)

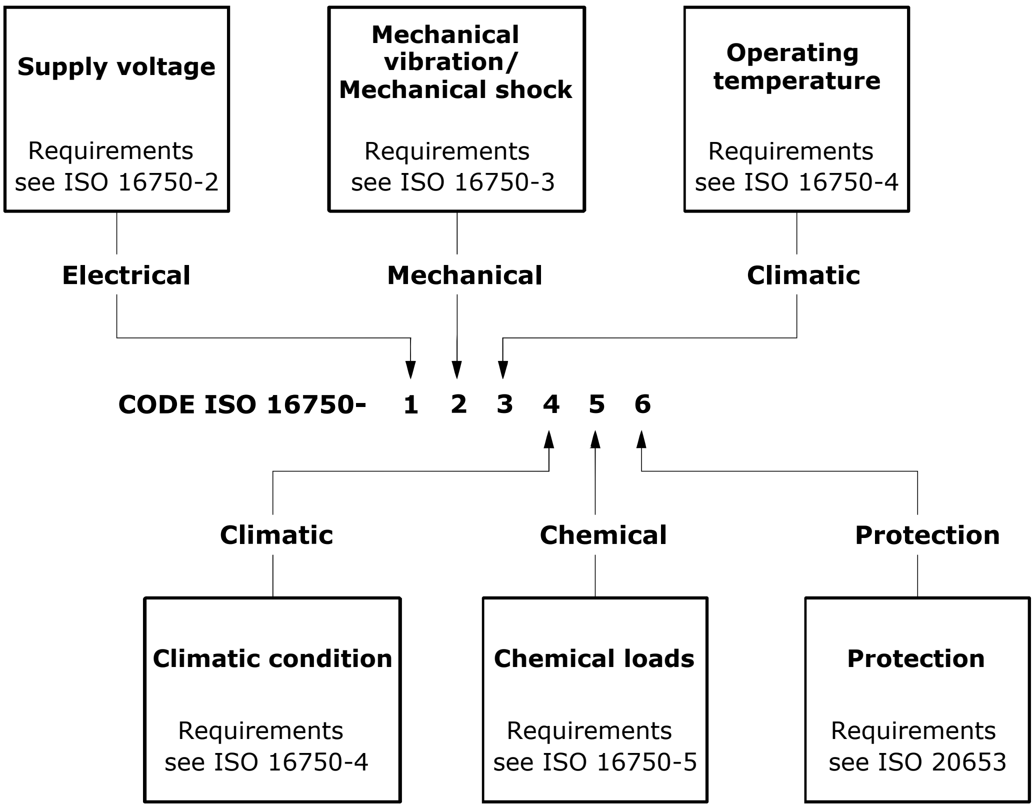

Figure 1.5: ISO 16750-1:2006 [20], Figure 1 – Code allocation

[1] According to ISO 16750-2:2012 [29], Table 3, *Starting profile values for systems with 12 V nominal voltage*, functional status C at level II test-pulse can be achieved.

[2] According to ISO 16750-2:2012 [29], Table 4, *Starting profile values for systems with 24 V nominal voltage*, functional status A at level II and functional status B at level III test-pulse can be achieved.

1.6.1 Electrical Capability

ISO 16750-2:2012 [29]	
ISO 7637-2:2011 [28]	Electrical transient conduction along supply lines. Test pluse: 1: -600 V, 1 ms 2a: +50 V, 50 µs 2b: +20 V, 200 ms 3a: -200 V, 0.1 µs 3b: +200 V, 0.1 µs 4: 12 V system, -6V drop (6 V remaining voltage) 24 V system, -18 V drop (6 V remaining voltage) 5a: +174 V, 2 Ω, 350 ms
ISO 7637-3:2007 [21]	Electrical transient transmission along signal lines. Tested for 24 V parameters, severity I

1.6.2 Mechanical Capability

ISO 16750-3:2012 [30]	Free fall tests, 1 m high, 6 falls per side Random vibration, broad-band 3 axes, 32 h per axis 57.9 m/s^2 -10 Hz to 2 kHz, temperature profile superimposed Shock, half-sine 3 axes, 60 shocks 500 m/s^2, 6 ms

1.6.3 Climatic Capability

ISO 16750-4:2012 [26]	Humid Heat Cyclic, DIN EN 60068-2-30:2006-06 [5], DIN EN 60068-2-38:2009 [7] Damp Heat, DIN EN 60068-2-78:2014-02[16] Salt spray, DIN EN 60068-2-11:2000-02 [4], DIN EN 60068-2-38:1996-10 [7]

1.6.4 Chemical Capability

The list of applied chemical agents for tests according to **IEC 16750-5:2010**[27] is given in Table 1.2 on the current page.

ID	Chemical Agent	Application Method
AA	Diesel fuel	I, III, IV, V
AB	"Bio" diesel	I, III, IV, V
AC	Petrol/gasoline unleaded	I, III, IV, V
AE	Methanol	II, III, IV, V, VI
BA	Engine oil	II, III, IV, V
BB	Differential oil	II, III, IV, V, VI
BC	Transmission fluid	II, III, IV, V, VI
BD	Hydraulic fluid	II, III, IV, V
CA	Battery fluid	III, V
CB	Brake fluid	II, III, V
CC	Antifreeze fluid	I, III, IV, V, VI
CE	Cavity prodection	II, III
CF	Protective lacquer	I, II
CG	Protective lacquer remover	I, III, IV, V
DA	Windscreen washer fluid	II, III, IV, V
DB	Vehicle washing chemicals	I, II, III, IV, V
DC	Interior cleaner	I, III
DD	Glass cleaner	I, III
DE	Wheel cleaner	I, II, III, IV
DF	Cold cleaning agent	I, II, III, IV, V, VI
DK	Denatured alcohol	I, II, III, IV, V
ED	Refreshment containing caffeine and sugar	III, IV
YYA	Gasoline with 15% methanol	I, III, IV, V
YYB	FAM test fuel	I, III, IV, V

Table 1.2: List of chemical agents

1.6.5 Ingress Protection Capability

ISO 20653:2013 [25] IP6k7 and IP6k9k

1.6.6 ESD and EMC Capability for Road Vehicles

UNECE 10.4	Uniform provisions concerning the approval of vehicles with regard to EMC
DIN EN 13309:2010-12 (E) [6]	Construction machinery – Electromagnetic compatibility of machines with internal power supply
ISO 14982:1998 [18]	Agricultural and forestry machinery – Electromagnetic compatibility – Test methods and acceptance criteria
ISO 11452-2:2004 [19]	100 V/m, 20 MHz to 3 GHz
CISPR 25 ED. 3.0 B:2008 [1]	Conducted emissions, Class 3
ISO 10605:2008 [22]	ESD powered and unpowered ±6 kV contact discharge ±8 kV air discharge

1.6.7 ESD and EMC Capability for Industrial Applications

IEC 61000-6-2:2005 [12]	Immunity for industrial environments. Conformance to surge immunity is only given if signal line wire length is less than 30 m.
IEC 61000-6-4:2007 [13]	Radiated emission for industry

1.6.8 Functional Safety

ISO 13849:2008-12 [24]	Safety of machinery – Safety-related parts of control systems Performance level d. (PL d)
IEC 61508:2010 [15]	Functional safety of electrical/electronic/programmable electronic safety-related systems (E/E/PE, or E/E/PES), Safety Integrity Level 2 (SIL 2)

1.7 Instructions for Safe Operation

For safe operation of the HY-TTC 500 family ECUs, the following rules have to be obeyed:

1.7.1 General

- Carefully read the instructions and specifications listed in this document before operating the ECU
- The ECU has to be operated by using the type of connectors specified in section section 3.2 on page 31
- It is not allowed to use any other connector or cable harness than one of the specified ones
- It is not allowed to operate the ECU in an environment that violates the specified operational range
- The ECU has to be operated by skilled personnel only
- When operating the ECU in an environment close to humans, it has to be considered that the ECU contain power electronics and therefore the housing can have high temperatures
- It is not allowed to open a sealed ECU
- It is not allowed to operate an unsealed ECU outside the laboratory
- It is not allowed to operate a prototype ECU in a production environment (no matter if it is sealed or unsealed)
- Only skilled and trained personnel is allowed to operate a prototype ECU (no matter if it is sealed or unsealed)
- The ECU does not require maintenance activities by the user/system integrator. The only maintenance activity allowed for the user is exchanging the ECU (for example after it has reached its specified lifetime)

1.7.2 Checks to be done before commissioning the ECU

- Check the supply voltage before connecting the ECU
- Check the ECU connector and the cable harness to be free of defects
- Check the correct dimensioning of the wires in the cable harness
- Be sure that the ECU is mounted in a way that humans are not directly exposed to it and physical contact is avoided
- Be sure to choose a mounting location for the ECU that eliminates the possibility of operation temperatures greater than the maximum temperature allowed for the ECU
- The power supply of the ECU has to be secured with a fuse. The fuse trip current has to match to the maximum specified input current of the ECU and the cable harness. See Section 3.3 on page 37 for more information

2 Mounting and Label

2.1 Mounting Requirements

Any requirements about the housing and mounting is defined in the Mounting Requirements Document (MRD) [34]. Furthermore, product dimensions and tolerances are defined in the Product Drawing [35].

2.2 Label Information

Any information about the label and its content is defined in the Product Drawing [35].

3 Pinning and Connector

3.1 Connector

Figure 3.1 on this page shows the main connector of HY-TTC 500 ECU, which has 154 pins and divided into two segments.

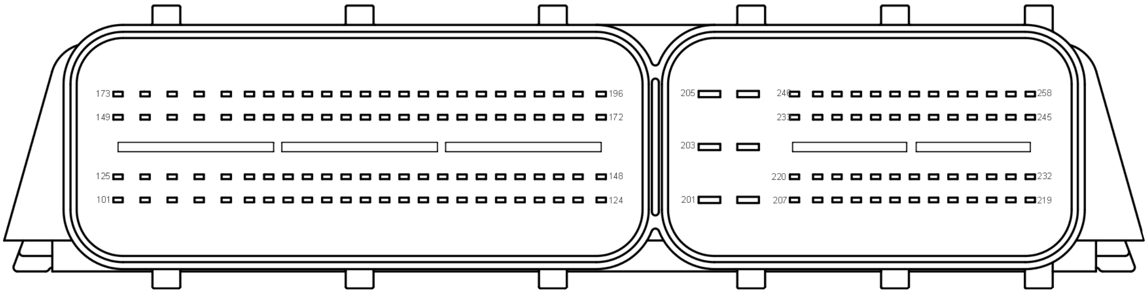

Figure 3.1: Main connector

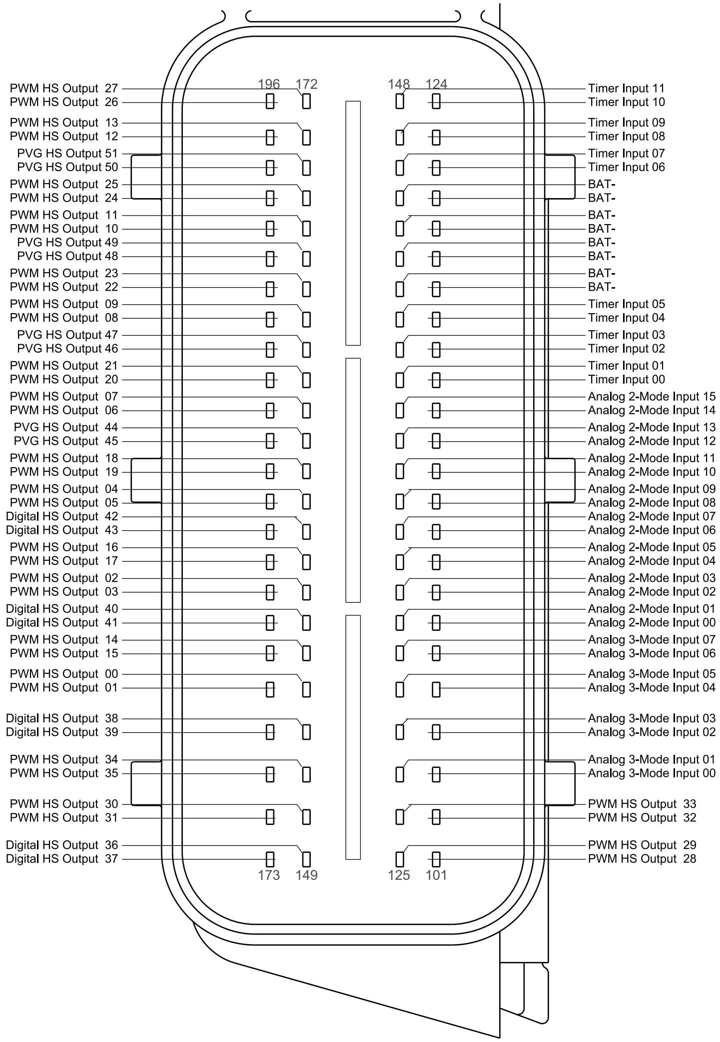

Figure 3.2: Main connector pinout (left-hand segment)

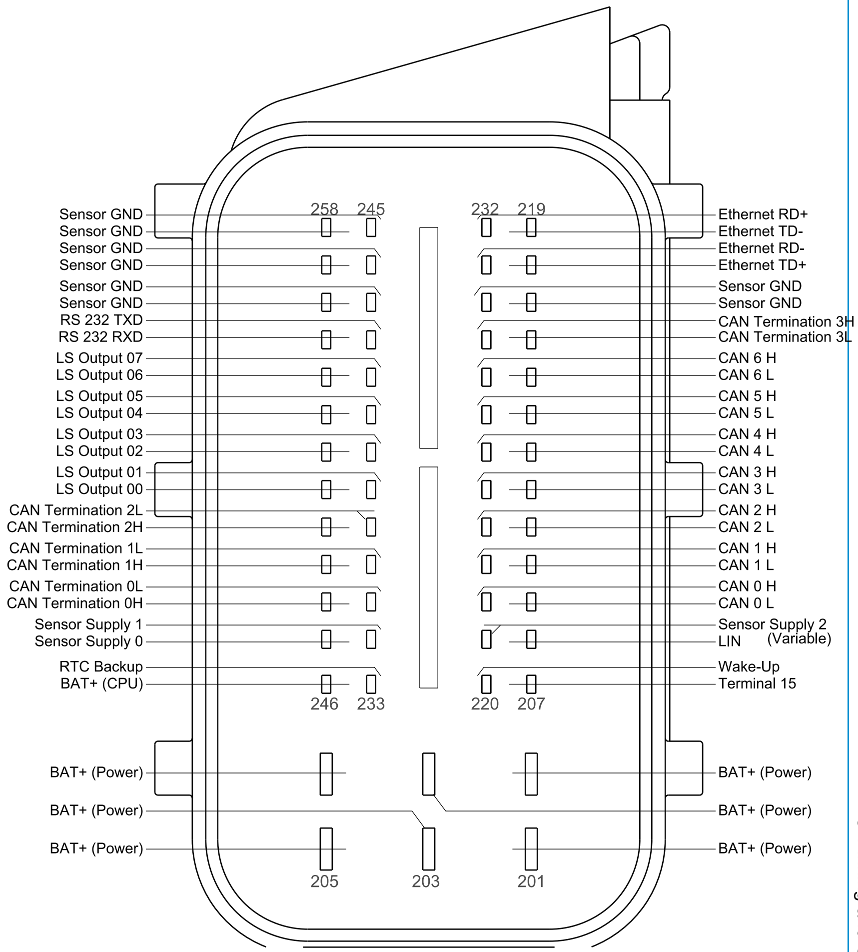

Figure 3.3: Main connector pinout (right-hand segment)

3.2 Mating Connector

The listed part numbers can be ordered from HERTH+BUSS, Kostal or Bosch with a minimum quantity of, for example, 100 pieces.

For lower quantities, TTControl GmbH provides complete kits with Bosch connectors, crimp contacts and sealings.

TTControl Order Numbers	Description
10619	Connector kit for HY-TTC 500, 58-positions
10620	Connector kit for HY-TTC 500, 96-positions

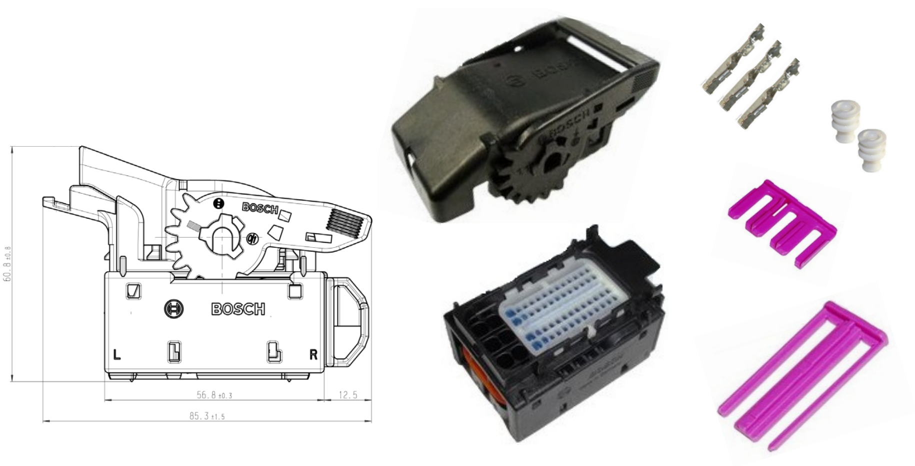

Figure 3.4: 58 terminal plug housing set

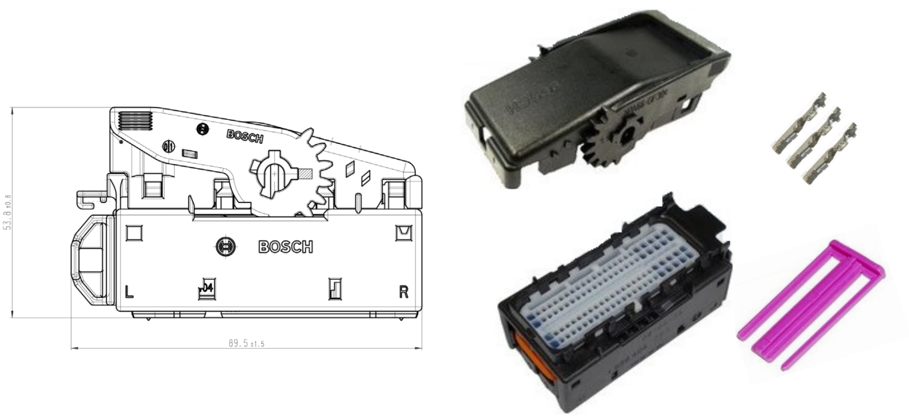

Figure 3.5: 96 terminal plug housing set

3.2.1 Kostal Mating Connector

Please follow the Process Specification below for the right handling of the Kostal mating connector:

Process Specification DOC00105005,
Receptacle Housing 58-way and 96-way for control units

3.2.1.1 Mating Connector 96-positions

KOSTAL Part Number	HERTH+BUSS Part Number	Description	Quantity
9409601	50390395	Plug housing	1
10400794061	50390397	Holder	1
10400794071	50390398	Secondary lock	1
22400794081	50390399	Cover cap (cable exit on the left side)	1
22400172011	50390567	Cover cap (cable exit on the upper right side)	1
10800794051	50282066 (discontinued)	Sealing-/protection plug [1]	1
10204225	50282099	Sealing-/protection plug [1]	1

Table 3.1: KOSTAL / HERTH+BUSS part numbers

[1] In order to achieve the specified IP rating each single wire must be populated either with a single-, blind- or a total protection-plug.

3.2.1.2 Mating Connector 58-positions

KOSTAL Part Number	HERTH+BUSS Part Number	Description	Quantity
9405801	50390396	Plug housing	1
10400758951	50390400	Holder	1
10400758991	50390401	Secondary lock	1
22400794001	50390402	Cover cap (cable exit on the right side)	1
22400172001	50390449	Cover cap (cable exit on the upper left side)	1
10800758941	50282065	Sealing-/protection plug [1]	1
-	50282062	High power pin single sealing-/protection plug [1]	6
10800472631	50282030	High power pin single blind-sealing-/protection plug [1]	-

Table 3.2: KOSTAL / HERTH+BUSS part numbers

3.2.1.3 Crimp Contacts

For I/O pins (148) KOSTAL Part Number	HERTH+BUSS Part Number	Description
22140734080 32140734080	50253445[2] 50253445088[3]	KKS MLK 1.2 m, crimp contact tinned, wire size 0.75 mm^2 to 1 mm^2
22140734070 32140734070	50253443 50253443088	KKS MLK 1.2 m, crimp contact tinned, wire size 0.5 mm^2 to 0.75 mm^2
22140734060 32140734060	50253441 50253441088	KKS MLK 1.2 m, crimp contact tinned, wire size 0.35 mm^2 to 0.5 mm^2
22140734050 32140734050	50253439 50253439088	KKS MLK 1.2 m, crimp contact tinned, wire size 0.1 mm^2 to 0.25 mm^2
For High Power pins (6) Kostal Part Number	**HERTH+BUSS Part Number**	**Description**
22140499580 32140499580	50253230 50253230088	KKS SLK 2.8 ELA, crimp contact tinned, wire size 1mm^2 to 2.5mm^2
22140499570 32140499570	50253229 50253229088	KKS SLK 2.8 ELA, crimp contact tinned, wire size 0.5mm^2 to 0.75 mm^2

Table 3.3: KOSTAL / HERTH+BUSS part numbers

3.2.1.4 Tools

KOSTAL Part Number	HERTH+BUSS Part Number	Description
80411002	95942166	Crimping pliers without inserts
80411504	95942167	Crimping pliers insert set for MLK 1,2
80411631	95942169	Crimping pliers insert set for SLK 2,8
80495003	95945400	Unlocking tool for MLK 1,2
-	95945402	Unlocking tool for SLK 2,8

Table 3.4: KOSTAL / HERTH+BUSS part numbers

[2] Loose crimp contact order code, minimum 50 pieces per packing unit.
[3] Strip (reel) crimp contact order code, minimum 4000 pieces per packing unit.

3.2.2 Bosch Mating Connector

Please follow the assembly instruction and technical customer information below for the right handling of the Bosch mating connector:

Assembly Instruction 1 928 A00 48M

Technical Customer Information 1 928 A00 45T

3.2.2.1 Mating Connector 96-positions

Bosch Part Number	Description	Quantity
1 928 404 781	Plug housing	1
1 928 404 927	Plug housing, 90-degree angled (alt.)	-
1 928 404 773	Cover cap	1
1 928 404 762	Secondary lock	1

Table 3.5: Bosch part numbers

3.2.2.2 Mating Connector 58-positions

Bosch Part Number	Description	Quantity
1 928 404 916	Plug housing	1
1 928 404 780	Plug housing, 90-degree angled (alt.)	-
1 928 404 917	Cover cap	1
1 928 404 760	Secondary lock	1
1 928 404 761	Secondary lock	1
1 928 300 600	High power pin single sealing-/protection plug [1]	6
1 928 300 601	High power pin single blind-sealing-/protection plug [1]	-

Table 3.6: Bosch part numbers

[1] In order to achieve the specified IP rating each single wire must be populated either with a single protection-, blind protection- or a total protection-plug.

3.2.2.3 Crimp Contacts

For I/O pins (148) Bosch Part Number	Description
1 928 498 679 1 928 498 137 (alt.)	Matrix 1.2 m, crimp contact tinned, wire size 0.35 mm^2 to 0.5 mm^2
1 928 498 680 1 928 498 138 (alt.)	Matrix 1.2 m, crimp contact tinned, wire size 0.75 mm^2 to 1.0 mm^2
For High Power pins (6) Bosch Part Number	**Description**
1 928 498 057	BDK 2.8, crimp contact tinned, wire size 1.5mm^2 to 2.5mm^2

Table 3.7: Bosch crimp contacts part numbers

3.2.2.4 Tools

Bosch Part Number	Description
1 928 498 161	Crimping pliers / BDK / BSK 2,8 / 0,5 - 1 mm^2
1 928 498 162	Crimping pliers / BDK / BSK 2,8 / 1,5 - 2,5 mm^2
1 928 498 212	Crimping pliers / Matrix 1,2 / 0,35 - 0,5 mm^2
1 928 498 213	Crimping pliers / Matrix 1,2 / 0,75 - 1,0 mm^2
1 928 498 167	Unlocking tool for BDK / BSK 2,8
1 928 498 168	Unlocking tool spare pin for BDK / BSK 2,8
1 928 498 218	Unlocking tool for Matrix 1,2
1 928 498 219	Unlocking tool spare pin for Matrix 1,2

Table 3.8: Bosch Tools part numbers

3.3 Fuse

For cable harness protection, it is recommended to protect each power supply path by a dedicated fuse. Please take note that the selected fuse type has to match to the current capability of the cable harness.

Symbol	Parameter	Note	Min.	Max.	Unit
BAT+ CPU	Fuse trip current of Positive Power Supply of Internal Electronics		2		A
BAT+ Power	Fuse trip current of Positive Power Supply of Power Stages	1		60	A

Note 1 The maximum fuse trip current is dependent on the specific customer application. The maximum total load current must be considered, see Section 4.1.3 on page 65

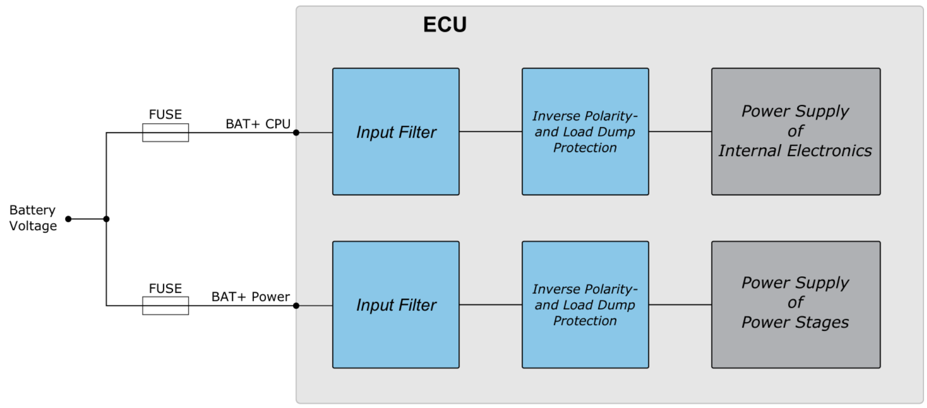

Figure 3.6: Battery Supply - Fuse

3.4 HY-TTC 500 Family Pinning

The following subsections describe the HY-TTC 500 family variant dependent main- and the respective alternative functions on connector pin-level.
Each hardware function in the table, e.g. Pin 101 - HS PWM Output, is referenced to its main- (*IO_PWM_28*) and alternative (*IO_DO_44, IO_PWD_12, IO_DI_28*) function. *IO_xx_yy* represents the software define for each function. For more information about API documentation, please refer to Chapter 8 on page 204.

Please take note that the main function shall be used preferably and the technical specifications must particularly be regarded when alternative functions are to be used. See Chapter 4 on page 63 for limits and restrictions.

3.4.1 HY-TTC 580 Variant

		Alternative Function													
Pin No.	Main Function	*HS Digital Output*	*Timer Input*	*PVG Output*	*VOUT Output*	*A/D Input (HS Output/PVG/VOUT)*	*Current Loop Input*	*A/D Input (Timer Input)*	*A/D Input (HS Digital Output)*	*A/D Input (LS Digital Output)*	*Analog Current Input 2M*	*Digital Input 2M*	*Analog Current Input 3M*	*Analog Resistance Input 3M*	*Digital Input 3M*
P101	HS PWM Output IO_PWM_28	IO_DO_44	IO_PWD_12 IO_DI_28												
P102	HS PWM Output IO_PWM_32	IO_DO_48	IO_PWD_16 IO_DI_32												
P103	Analog Voltage Input IO_ADC_00												IO_ADC_00	IO_ADC_00	IO_DI_48
P104	Analog Voltage Input IO_ADC_02												IO_ADC_02	IO_ADC_02	IO_DI_50
P105	Analog Voltage Input IO_ADC_04												IO_ADC_04	IO_ADC_04	IO_DI_52
P106	Analog Voltage Input IO_ADC_06												IO_ADC_06	IO_ADC_06	IO_DI_54
P107	Analog Voltage Input IO_ADC_08										IO_ADC_08	IO_DI_56			
P108	Analog Voltage Input IO_ADC_10										IO_ADC_10	IO_DI_58			
P109	Analog Voltage Input IO_ADC_12										IO_ADC_12	IO_DI_60			
P110	Analog Voltage Input IO_ADC_14										IO_ADC_14	IO_DI_62			
P111	Analog Voltage Input IO_ADC_16										IO_ADC_16	IO_DI_64			
P112	Analog Voltage Input IO_ADC_18										IO_ADC_18	IO_DI_66			
P113	Analog Voltage Input IO_ADC_20										IO_ADC_20	IO_DI_68			
P114	Analog Voltage Input IO_ADC_22										IO_ADC_22	IO_DI_70			
P115	Timer Input IO_PWD_00						IO_PWD_00	IO_ADC_24 IO_DI_36							
P116	Timer Input IO_PWD_02						IO_PWD_02	IO_ADC_26 IO_DI_38							
P117	Timer Input IO_PWD_04						IO_PWD_04	IO_ADC_28 IO_DI_40							
P118	BAT-														
P119	BAT-														
P120	BAT-														
P121	BAT-														
P122	Timer Input IO_PWD_06							IO_ADC_30 IO_DI_42							
P123	Timer Input IO_PWD_08							IO_ADC_32 IO_DI_44							
P124	Timer Input IO_PWD_10							IO_ADC_34 IO_DI_46							

Pin No.	Main Function	Alternative Function: HS Digital Output	Timer Input	PVG Output	VOUT Output	A/D Input (HS Output/PVG/VOUT)	Current Loop Input	A/D Input (Timer Input)	A/D Input (HS Digital Output)	A/D Input (LS Digital Output)	Analog Current Input 2M	Digital Input 2M	Analog Current Input 3M	Analog Resistance Input 3M	Digital Input 3M
P125	HS PWM Output IO_PWM_29	IO_DO_45	IO_DI_29 IO_PWD_13												
P126	HS PWM Output IO_PWM_33	IO_DO_49	IO_PWD_17 IO_DI_33												
P127	Analog Voltage Input IO_ADC_01												IO_ADC_01	IO_ADC_01	IO_DI_49
P128	Analog Voltage Input IO_ADC_03												IO_ADC_03	IO_ADC_03	IO_DI_51
P129	Analog Voltage Input IO_ADC_05												IO_ADC_05	IO_ADC_05	IO_DI_53
P130	Analog Voltage Input IO_ADC_07												IO_ADC_07	IO_ADC_07	IO_DI_55
P131	Analog Voltage Input IO_ADC_09										IO_ADC_09	IO_DI_57			
P132	Analog Voltage Input IO_ADC_11										IO_ADC_11	IO_DI_59			
P133	Analog Voltage Input IO_ADC_13										IO_ADC_13	IO_DI_61			
P134	Analog Voltage Input IO_ADC_15										IO_ADC_15	IO_DI_63			
P135	Analog Voltage Input IO_ADC_17										IO_ADC_17	IO_DI_65			
P136	Analog Voltage Input IO_ADC_19										IO_ADC_19	IO_DI_67			
P137	Analog Voltage Input IO_ADC_21										IO_ADC_21	IO_DI_69			
P138	Analog Voltage Input IO_ADC_23										IO_ADC_23	IO_DI_71			
P139	Timer Input IO_PWD_01						IO_PWD_01	IO_ADC_25 IO_DI_37							
P140	Timer Input IO_PWD_03						IO_PWD_03	IO_ADC_27 IO_DI_39							
P141	Timer Input IO_PWD_05						IO_PWD_05	IO_ADC_29 IO_DI_41							
P142	BAT -														
P143	BAT-														
P144	BAT-														
P145	BAT-														
P146	Timer Input IO_PWD_07							IO_ADC_31 IO_DI_43							
P147	Timer Input IO_PWD_09							IO_ADC_33 IO_DI_45							
P148	Timer Input IO_PWD_11							IO_ADC_35 IO_DI_47							
P149	HS Digital Output IO_DO_00								IO_ADC_36 IO_DI_72						
P150	HS PWM Output IO_PWM_30	IO_DO_46	IO_PWD_14 IO_DI_30												

		Alternative Function													
Pin No.	Main Function	*HS Digital Output*	*Timer Input*	*PVG Output*	*VOUT Output*	*A/D Input (HS Output/PVG/VOUT)*	*Current Loop Input*	*A/D Input (Timer Input)*	*A/D Input (HS Digital Output)*	*A/D Input (LS Digital Output)*	*Analog Current Input 2M*	*Digital Input 2M*	*Analog Current Input 3M*	*Analog Resistance Input 3M*	*Digital Input 3M*
P151	HS PWM Output IO_PWM_34	IO_DO_50	IO_PWD_18 IO_DI_34												
P152	HS Digital Output IO_DO_02								IO_ADC_38 IO_DI_74						
P153	HS PWM Output IO_PWM_00	IO_DO_16	IO_DI_00												
P154	HS PWM Output IO_PWM_14	IO_DO_30	IO_DI_14												
P155	HS Digital Output IO_DO_04								IO_ADC_40 IO_DI_76						
P156	HS PWM Output IO_PWM_02	IO_DO_18	IO_DI_02												
P157	HS PWM Output IO_PWM_16	IO_DO_32	IO_DI_16												
P158	HS Digital Output IO_DO_06								IO_ADC_42 IO_DI_78						
P159	HS PWM Output IO_PWM_04	IO_DO_20	IO_DI_04												
P160	HS PWM Output IO_PWM_18	IO_DO_34	IO_DI_18												
P161	HS Digital Output IO_DO_52			IO_PVG_00	IO_VOUT_00	IO_ADC_52 IO_DI_88									
P162	HS PWM Output IO_PWM_07	IO_DO_23	IO_DI_07												
P163	HS PWM Output IO_PWM_21	IO_DO_37	IO_DI_21												
P164	HS Digital Output IO_DO_55			IO_PVG_03	IO_VOUT_03	IO_ADC_55 IO_DI_91									
P165	HS PWM Output IO_PWM_09	IO_DO_25	IO_DI_09												
P166	HS PWM Output IO_PWM_23	IO_DO_39	IO_DI_23												
P167	HS Digital Output IO_DO_57			IO_PVG_05	IO_VOUT_05	IO_ADC_57 IO_DI_93									
P168	HS PWM Output IO_PWM_11	IO_DO_27	IO_DI_11												
P169	HS PWM Output IO_PWM_25	IO_DO_41	IO_DI_25												
P170	HS Digital Output IO_DO_59			IO_PVG_07	IO_VOUT_07	IO_ADC_59 IO_DI_95									
P171	HS PWM Output IO_PWM_13	IO_DO_29	IO_DI_13												
P172	HS PWM Output IO_PWM_27	IO_DO_43	IO_DI_27												
P173	HS Digital Output IO_DO_01								IO_ADC_37 IO_DI_73						
P174	HS PWM Output IO_PWM_31	IO_DO_47	IO_PWD_15 IO_DI_31												

		Alternative Function													
Pin No.	Main Function	HS Digital Output	Timer Input	PVG Output	VOUT Output	A/D Input (HS Output/PVG/VOUT)	Current Loop Input	A/D Input (Timer Input)	A/D Input (HS Digital Output)	A/D Input (LS Digital Output)	Analog Current Input 2M	Digital Input 2M	Analog Current Input 3M	Analog Resistance Input 3M	Digital Input 3M
P175	HS PWM Output IO_PWM_35	IO_DO_51	IO_PWD_19 IO_DI_35												
P176	HS Digital Output IO_DO_03								IO_ADC_39 IO_DI_75						
P177	HS PWM Output IO_PWM_01	IO_DO_17	IO_DI_01												
P178	HS PWM Output IO_PWM_15	IO_DO_31	IO_DI_15												
P179	HS Digital Output IO_DO_05								IO_ADC_41 IO_DI_77						
P180	HS PWM Output IO_PWM_03	IO_DO_19	IO_DI_03												
P181	HS PWM Output IO_PWM_17	IO_DO_33	IO_DI_17												
P182	HS Digital Output IO_DO_07								IO_ADC_43 IO_DI_79						
P183	HS PWM Output IO_PWM_05	IO_DO_21	IO_DI_05												
P184	HS PWM Output IO_PWM_19	IO_DO_35	IO_DI_19												
P185	HS Digital Output IO_DO_53			IO_PVG_01	IO_VOUT_01	IO_ADC_53 IO_DI_89									
P186	HS PWM Output IO_PWM_06	IO_DO_22	IO_DI_06												
P187	HS PWM Output IO_PWM_20	IO_DO_36	IO_DI_20												
P188	HS Digital Output IO_DO_54			IO_PVG_02	IO_VOUT_02	IO_ADC_54 IO_DI_90									
P189	HS PWM Output IO_PWM_08	IO_DO_24	IO_DI_08												
P190	HS PWM Output IO_PWM_22	IO_DO_38	IO_DI_22												
P191	HS Digital Output IO_DO_56			IO_PVG_04	IO_VOUT_04	IO_ADC_56 IO_DI_92									
P192	HS PWM Output IO_PWM_10	IO_DO_26	IO_DI_10												
P193	HS PWM Output IO_PWM_24	IO_DO_40	IO_DI_24												
P194	HS Digital Output IO_DO_58			IO_PVG_06	IO_VOUT_06	IO_ADC_58 IO_DI_94									
P195	HS PWM Output IO_PWM_12	IO_DO_28	IO_DI_12												
P196	HS PWM Output IO_PWM_26	IO_DO_42	IO_DI_26												
P201	BAT+ Power														
P202	BAT+ Power														
P203	BAT+ Power														
P204	BAT+ Power														

Pin No.	Main Function	Alternative Function													
		HS Digital Output	Timer Input	PVG Output	VOUT Output	A/D Input (HS Output/PVG/VOUT)	Current Loop Input	A/D Input (Timer Input)	A/D Input (HS Digital Output)	A/D Input (LS Digital Output)	Analog Current Input 2M	Digital Input 2M	Analog Current Input 3M	Analog Resistance Input 3M	Digital Input 3M
P205	BAT+ Power														
P206	BAT+ Power														
P207	Terminal 15 IO_ADC_K15														
P208	LIN IO_LIN														
P209	CAN 0 Low IO_CAN_CHANNEL_0														
P210	CAN 1 Low IO_CAN_CHANNEL_1														
P211	CAN 2 Low IO_CAN_CHANNEL_2														
P212	CAN 3 Low IO_CAN_CHANNEL_3														
P213	CAN 4 Low IO_CAN_CHANNEL_4														
P214	CAN 5 Low IO_CAN_CHANNEL_5														
P215	CAN 6 Low IO_CAN_CHANNEL_6														
P216	CAN Termination 3L														
P217	Sensor GND														
P218	Ethernet TD+ IO_Download														
P219	Ethernet TD- IO_Download														
P220	Wake-Up IO_ADC_WAKE_UP														
P221	Sensor Supply Var. IO_ADC_SENSOR_SUPPLY_2														
P222	CAN 0 High IO_CAN_CHANNEL_0														
P223	CAN 1 High IO_CAN_CHANNEL_1														
P224	CAN 2 High IO_CAN_CHANNEL_2														
P225	CAN 3 High IO_CAN_CHANNEL_3														
P226	CAN 4 High IO_CAN_CHANNEL_4														
P227	CAN 5 High IO_CAN_CHANNEL_5														
P228	CAN 6 High IO_CAN_CHANNEL_6														
P229	CAN Termination 3H														
P230	Sensor GND														
P231	Ethernet RD- IO_Download														

Pin No.	Main Function	Alternative Function: HS Digital Output	Timer Input	PVG Output	VOUT Output	A/D Input (HS Output/PVG/VOUT)	Current Loop Input	A/D Input (Timer Input)	A/D Input (HS Digital Output)	A/D Input (LS Digital Output)	Analog Current Input 2M	Digital Input 2M	Analog Current Input 3M	Analog Resistance Input 3M	Digital Input 3M
P232	Ethernet RD+ IO_Download														
P233	RTC_BACKUP_BAT														
P234	Sensor Supply 5 V IO_ADC_SENSOR_SUPPLY_1														
P235	CAN Termination 0L														
P236	CAN Termination 1L														
P237	CAN Termination 2L														
P238	LS Digital Output IO_DO_09									IO_ADC_45 IO_DI_81					
P239	LS Digital Output IO_DO_11									IO_ADC_47 IO_DI_83					
P240	LS Digital Output IO_DO_13									IO_ADC_49 IO_DI_85					
P241	LS Digital Output IO_DO_15									IO_ADC_51 IO_DI_87					
P242	RS232 TXD IO_UART														
P243	Sensor GND														
P244	Sensor GND														
P245	Sensor GND														
P246	BAT+ CPU IO_ADC_UBAT														
P247	Sensor Supply 5 V IO_ADC_SENSOR_SUPPLY_0														
P248	CAN Termination 0H														
P249	CAN Termination 1H														
P250	CAN Termination 2H														
P251	LS Digital Output IO_DO_08									IO_ADC_44 IO_DI_80					
P252	LS Digital Output IO_DO_10									IO_ADC_46 IO_DI_82					
P253	LS Digital Output IO_DO_12									IO_ADC_48 IO_DI_84					
P254	LS Digital Output IO_DO_14									IO_ADC_50 IO_DI_86					
P255	RS232 RXD IO_UART														
P256	Sensor GND														
P257	Sensor GND														
P258	Sensor GND														

Table 3.9: Pinning of HY-TTC 580

3.4.2 HY-TTC 540 Variant

Pin No.	Main Function	Alternative Function: HS Digital Output	Timer Input	PVG Output	VOUT Output	A/D Input (HS Output/PVG/VOUT)	Current Loop Input	A/D Input (Timer Input)	A/D Input (HS Digital Output)	A/D Input (LS Digital Output)	Analog Current Input 2M	Digital Input 2M	Analog Current Input 3M	Analog Resistance Input 3M	Digital Input 3M
P101	HS PWM Output		IO_PWD_12 IO_DI_28												
P102	HS PWM Output		IO_PWD_16 IO_DI_32												
P103	Analog Voltage Input IO_ADC_00												IO_ADC_00	IO_ADC_00	IO_DI_48
P104	Analog Voltage Input IO_ADC_02												IO_ADC_02	IO_ADC_02	IO_DI_50
P105	Analog Voltage Input IO_ADC_04												IO_ADC_04	IO_ADC_04	IO_DI_52
P106	Analog Voltage Input IO_ADC_06												IO_ADC_06	IO_ADC_06	IO_DI_54
P107	Analog Voltage Input IO_ADC_08										IO_ADC_08	IO_DI_56			
P108	Analog Voltage Input IO_ADC_10										IO_ADC_10	IO_DI_58			
P109	Analog Voltage Input IO_ADC_12										IO_ADC_12	IO_DI_60			
P110	Analog Voltage Input IO_ADC_14										IO_ADC_14	IO_DI_62			
P111	Analog Voltage Input IO_ADC_16										IO_ADC_16	IO_DI_64			
P112	Analog Voltage Input IO_ADC_18										IO_ADC_18	IO_DI_66			
P113	Analog Voltage Input IO_ADC_20										IO_ADC_20	IO_DI_68			
P114	Analog Voltage Input IO_ADC_22										IO_ADC_22	IO_DI_70			
P115	Timer Input IO_PWD_00						IO_PWD_00	IO_ADC_24 IO_DI_36							
P116	Timer Input IO_PWD_02						IO_PWD_02	IO_ADC_26 IO_DI_38							
P117	Timer Input IO_PWD_04						IO_PWD_04	IO_ADC_28 IO_DI_40							
P118	BAT-														
P119	BAT-														
P120	BAT-														
P121	BAT-														
P122	Timer Input IO_PWD_06							IO_ADC_30 IO_DI_42							
P123	Timer Input IO_PWD_08							IO_ADC_32 IO_DI_44							
P124	Timer Input IO_PWD_10							IO_ADC_34 IO_DI_46							

Pin No.	Main Function	Alternative Function													
		HS Digital Output	*Timer Input*	*PVG Output*	*VOUT Output*	*A/D Input (HS Output/PVG/VOUT)*	*Current Loop Input*	*A/D Input (Timer Input)*	*A/D Input (HS Digital Output)*	*A/D Input (LS Digital Output)*	*Analog Current Input 2M*	*Digital Input 2M*	*Analog Current Input 3M*	*Analog Resistance Input 3M*	*Digital Input 3M*
P125	HS PWM Output		IO_DI_29 IO_PWD_13												
P126	HS PWM Output		IO_PWD_17 IO_DI_33												
P127	Analog Voltage Input IO_ADC_01												IO_ADC_01	IO_ADC_01	IO_DI_49
P128	Analog Voltage Input IO_ADC_03												IO_ADC_03	IO_ADC_03	IO_DI_51
P129	Analog Voltage Input IO_ADC_05												IO_ADC_05	IO_ADC_05	IO_DI_53
P130	Analog Voltage Input IO_ADC_07												IO_ADC_07	IO_ADC_07	IO_DI_55
P131	Analog Voltage Input IO_ADC_09										IO_ADC_09	IO_DI_57			
P132	Analog Voltage Input IO_ADC_11										IO_ADC_11	IO_DI_59			
P133	Analog Voltage Input IO_ADC_13										IO_ADC_13	IO_DI_61			
P134	Analog Voltage Input IO_ADC_15										IO_ADC_15	IO_DI_63			
P135	Analog Voltage Input IO_ADC_17										IO_ADC_17	IO_DI_65			
P136	Analog Voltage Input IO_ADC_19										IO_ADC_19	IO_DI_67			
P137	Analog Voltage Input IO_ADC_21										IO_ADC_21	IO_DI_69			
P138	Analog Voltage Input IO_ADC_23										IO_ADC_23	IO_DI_71			
P139	Timer Input IO_PWD_01						IO_PWD_01	IO_ADC_25 IO_DI_37							
P140	Timer Input IO_PWD_03						IO_PWD_03	IO_ADC_27 IO_DI_39							
P141	Timer Input IO_PWD_05						IO_PWD_05	IO_ADC_29 IO_DI_41							
P142	BAT -														
P143	BAT-														
P144	BAT-														
P145	BAT-														
P146	Timer Input IO_PWD_07							IO_ADC_31 IO_DI_43							
P147	Timer Input IO_PWD_09							IO_ADC_33 IO_DI_45							
P148	Timer Input IO_PWD_11							IO_ADC_35 IO_DI_47							
P149	HS Digital Output IO_DO_00								IO_ADC_36 IO_DI_72						
P150	HS PWM Output		IO_PWD_14 IO_DI_30												

Pin No.	Main Function	Alternative Function													
		HS Digital Output	*Timer Input*	*PVG Output*	*VOUT Output*	*A/D Input (HS Output/PVG/VOUT)*	*Current Loop Input*	*A/D Input (Timer Input)*	*A/D Input (HS Digital Output)*	*A/D Input (LS Digital Output)*	*Analog Current Input 2M*	*Digital Input 2M*	*Analog Current Input 3M*	*Analog Resistance Input 3M*	*Digital Input 3M*
P151	HS PWM Output		IO_PWD_18 IO_DI_34												
P152	HS Digital Output IO_DO_02								IO_ADC_38 IO_DI_74						
P153	HS PWM Output IO_PWM_00	IO_DO_16	IO_DI_00												
P154	HS PWM Output IO_PWM_14	IO_DO_30	IO_DI_14												
P155	HS Digital Output IO_DO_04								IO_ADC_40 IO_DI_76						
P156	HS PWM Output IO_PWM_02	IO_DO_18	IO_DI_02												
P157	HS PWM Output IO_PWM_16	IO_DO_32	IO_DI_16												
P158	HS Digital Output IO_DO_06								IO_ADC_42 IO_DI_78						
P159	HS PWM Output IO_PWM_04	IO_DO_20	IO_DI_04												
P160	HS PWM Output IO_PWM_18	IO_DO_34	IO_DI_18												
P161	HS Digital Output					IO_ADC_52 IO_DI_88									
P162	HS PWM Output IO_PWM_07	IO_DO_23	IO_DI_07												
P163	HS PWM Output IO_PWM_21	IO_DO_37	IO_DI_21												
P164	HS Digital Output					IO_ADC_55 IO_DI_91									
P165	HS PWM Output IO_PWM_09	IO_DO_25	IO_DI_09												
P166	HS PWM Output IO_PWM_23	IO_DO_39	IO_DI_23												
P167	HS Digital Output					IO_ADC_57 IO_DI_93									
P168	HS PWM Output IO_PWM_11	IO_DO_27	IO_DI_11												
P169	HS PWM Output IO_PWM_25	IO_DO_41	IO_DI_25												
P170	HS Digital Output					IO_ADC_59 IO_DI_95									
P171	HS PWM Output IO_PWM_13	IO_DO_29	IO_DI_13												
P172	HS PWM Output IO_PWM_27	IO_DO_43	IO_DI_27												
P173	HS Digital Output IO_DO_01								IO_ADC_37 IO_DI_73						
P174	HS PWM Output		IO_PWD_15 IO_DI_31												

Pin No.	Main Function	Alternative Function: HS Digital Output	Timer Input	PVG Output	VOUT Output	A/D Input (HS Output/PVG/VOUT)	Current Loop Input	A/D Input (Timer Input)	A/D Input (HS Digital Output)	A/D Input (LS Digital Output)	Analog Current Input 2M	Digital Input 2M	Analog Current Input 3M	Analog Resistance Input 3M	Digital Input 3M
P175	HS PWM Output		IO_PWD_19 IO_DI_35												
P176	HS Digital Output IO_DO_03								IO_ADC_39 IO_DI_75						
P177	HS PWM Output IO_PWM_01	IO_DO_17	IO_DI_01												
P178	HS PWM Output IO_PWM_15	IO_DO_31	IO_DI_15												
P179	HS Digital Output IO_DO_05								IO_ADC_41 IO_DI_77						
P180	HS PWM Output IO_PWM_03	IO_DO_19	IO_DI_03												
P181	HS PWM Output IO_PWM_17	IO_DO_33	IO_DI_17												
P182	HS Digital Output IO_DO_07								IO_ADC_43 IO_DI_79						
P183	HS PWM Output IO_PWM_05	IO_DO_21	IO_DI_05												
P184	HS PWM Output IO_PWM_19	IO_DO_35	IO_DI_19												
P185	HS Digital Output					IO_ADC_53 IO_DI_89									
P186	HS PWM Output IO_PWM_06	IO_DO_22	IO_DI_06												
P187	HS PWM Output IO_PWM_20	IO_DO_36	IO_DI_20												
P188	HS Digital Output					IO_ADC_54 IO_DI_90									
P189	HS PWM Output IO_PWM_08	IO_DO_24	IO_DI_08												
P190	HS PWM Output IO_PWM_22	IO_DO_38	IO_DI_22												
P191	HS Digital Output					IO_ADC_56 IO_DI_92									
P192	HS PWM Output IO_PWM_10	IO_DO_26	IO_DI_10												
P193	HS PWM Output IO_PWM_24	IO_DO_40	IO_DI_24												
P194	HS Digital Output					IO_ADC_58 IO_DI_94									
P195	HS PWM Output IO_PWM_12	IO_DO_28	IO_DI_12												
P196	HS PWM Output IO_PWM_26	IO_DO_42	IO_DI_26												
P201	BAT+ Power														
P202	BAT+ Power														
P203	BAT+ Power														
P204	BAT+ Power														

Pin No.	Main Function	Alternative Function													
		HS Digital Output	Timer Input	PVG Output	VOUT Output	A/D Input (HS Output/PVG/VOUT)	Current Loop Input	A/D Input (Timer Input)	A/D Input (HS Digital Output)	A/D Input (LS Digital Output)	Analog Current Input 2M	Digital Input 2M	Analog Current Input 3M	Analog Resistance Input 3M	Digital Input 3M
P205	BAT+ Power														
P206	BAT+ Power														
P207	Terminal 15 IO_ADC_K15														
P208															
P209	CAN 0 Low IO_CAN_CHANNEL_0														
P210	CAN 1 Low IO_CAN_CHANNEL_1														
P211	CAN 2 Low IO_CAN_CHANNEL_2														
P212	CAN 3 Low IO_CAN_CHANNEL_3														
P213															
P214															
P215															
P216	CAN Termination 3L														
P217	Sensor GND														
P218															
P219															
P220	Wake-Up IO_ADC_WAKE_UP														
P221	Sensor Supply Var. IO_ADC_SENSOR_SUPPLY_2														
P222	CAN 0 High IO_CAN_CHANNEL_0														
P223	CAN 1 High IO_CAN_CHANNEL_1														
P224	CAN 2 High IO_CAN_CHANNEL_2														
P225	CAN 3 High IO_CAN_CHANNEL_3														
P226															
P227															
P228															
P229	CAN Termination 3H														
P230	Sensor GND														
P231															
P232															
P233															
P234	Sensor Supply 5 V IO_ADC_SENSOR_SUPPLY_1														
P235	CAN Termination 0L														
P236	CAN Termination 1L														
P237	CAN Termination 2L														
P238	LS Digital Output IO_DO_09									IO_ADC_45 IO_DI_81					

		Alternative Function													
Pin No.	Main Function	*HS Digital Output*	*Timer Input*	*PVG Output*	*VOUT Output*	*A/D Input (HS Output/PVG/VOUT)*	*Current Loop Input*	*A/D Input (Timer Input)*	*A/D Input (HS Digital Output)*	*A/D Input (LS Digital Output)*	*Analog Current Input 2M*	*Digital Input 2M*	*Analog Current Input 3M*	*Analog Resistance Input 3M*	*Digital Input 3M*
P239	LS Digital Output IO_DO_11									IO_ADC_47 IO_DI_83					
P240	LS Digital Output IO_DO_13									IO_ADC_49 IO_DI_85					
P241	LS Digital Output IO_DO_15									IO_ADC_51 IO_DI_87					
P242															
P243	Sensor GND														
P244	Sensor GND														
P245	Sensor GND														
P246	BAT+ CPU IO_ADC_UBAT														
P247	Sensor Supply 5 V IO_ADC_SENSOR_SUPPLY_0														
P248	CAN Termination 0H														
P249	CAN Termination 1H														
P250	CAN Termination 2H														
P251	LS Digital Output IO_DO_08									IO_ADC_44 IO_DI_80					
P252	LS Digital Output IO_DO_10									IO_ADC_46 IO_DI_82					
P253	LS Digital Output IO_DO_12									IO_ADC_48 IO_DI_84					
P254	LS Digital Output IO_DO_14									IO_ADC_50 IO_DI_86					
P255															
P256	Sensor GND														
P257	Sensor GND														
P258	Sensor GND														

Table 3.10: Pinning of HY-TTC 540

3.4.3 HY-TTC 520 Variant (Customer-specific variant only)

Pin No.	Main Function	Alternative Function													
		HS Digital Output	Timer Input	PVG Output	VOUT Output	A/D Input (HS Output/PVG/VOUT)	Current Loop Input	A/D Input (Timer Input)	A/D Input (HS Digital Output)	A/D Input (LS Digital Output)	Analog Current Input 2M	Digital Input 2M	Analog Current Input 3M	Analog Resistance Input 3M	Digital Input 3M
P101	HS PWM Output		IO_PWD_12 IO_DI_28												
P102	HS PWM Output		IO_PWD_16 IO_DI_32												
P103	Analog Voltage Input IO_ADC_00												IO_ADC_00	IO_ADC_00	IO_DI_48
P104	Analog Voltage Input IO_ADC_02												IO_ADC_02	IO_ADC_02	IO_DI_50
P105	Analog Voltage Input IO_ADC_04												IO_ADC_04	IO_ADC_04	IO_DI_52
P106	Analog Voltage Input IO_ADC_06												IO_ADC_06	IO_ADC_06	IO_DI_54
P107	Analog Voltage Input IO_ADC_08										IO_ADC_08	IO_DI_56			
P108	Analog Voltage Input IO_ADC_10										IO_ADC_10	IO_DI_58			
P109	Analog Voltage Input IO_ADC_12										IO_ADC_12	IO_DI_60			
P110	Analog Voltage Input IO_ADC_14										IO_ADC_14	IO_DI_62			
P111	Analog Voltage Input IO_ADC_16										IO_ADC_16	IO_DI_64			
P112	Analog Voltage Input IO_ADC_18										IO_ADC_18	IO_DI_66			
P113	Analog Voltage Input IO_ADC_20										IO_ADC_20	IO_DI_68			
P114	Analog Voltage Input IO_ADC_22										IO_ADC_22	IO_DI_70			
P115	Timer Input IO_PWD_00						IO_PWD_00	IO_ADC_24 IO_DI_36							
P116	Timer Input IO_PWD_02						IO_PWD_02	IO_ADC_26 IO_DI_38							
P117	Timer Input IO_PWD_04						IO_PWD_04	IO_ADC_28 IO_DI_40							
P118	BAT-														
P119	BAT-														
P120	BAT-														
P121	BAT-														
P122	Timer Input IO_PWD_06							IO_ADC_30 IO_DI_42							
P123	Timer Input IO_PWD_08							IO_ADC_32 IO_DI_44							
P124	Timer Input IO_PWD_10							IO_ADC_34 IO_DI_46							

		Alternative Function													
Pin No.	Main Function	HS Digital Output	Timer Input	PVG Output	VOUT Output	A/D Input (HS Output/PVG/VOUT)	Current Loop Input	A/D Input (Timer Input)	A/D Input (HS Digital Output)	A/D Input (LS Digital Output)	Analog Current Input 2M	Digital Input 2M	Analog Current Input 3M	Analog Resistance Input 3M	Digital Input 3M
P125	HS PWM Output		IO_DI_29 IO_PWD_13												
P126	HS PWM Output		IO_PWD_17 IO_DI_33												
P127	Analog Voltage Input IO_ADC_01												IO_ADC_01	IO_ADC_01	IO_DI_49
P128	Analog Voltage Input IO_ADC_03												IO_ADC_03	IO_ADC_03	IO_DI_51
P129	Analog Voltage Input IO_ADC_05												IO_ADC_05	IO_ADC_05	IO_DI_53
P130	Analog Voltage Input IO_ADC_07												IO_ADC_07	IO_ADC_07	IO_DI_55
P131	Analog Voltage Input IO_ADC_09										IO_ADC_09	IO_DI_57			
P132	Analog Voltage Input IO_ADC_11										IO_ADC_11	IO_DI_59			
P133	Analog Voltage Input IO_ADC_13										IO_ADC_13	IO_DI_61			
P134	Analog Voltage Input IO_ADC_15										IO_ADC_15	IO_DI_63			
P135	Analog Voltage Input IO_ADC_17										IO_ADC_17	IO_DI_65			
P136	Analog Voltage Input IO_ADC_19										IO_ADC_19	IO_DI_67			
P137	Analog Voltage Input IO_ADC_21										IO_ADC_21	IO_DI_69			
P138	Analog Voltage Input IO_ADC_23										IO_ADC_23	IO_DI_71			
P139	Timer Input IO_PWD_01						IO_PWD_01	IO_ADC_25 IO_DI_37							
P140	Timer Input IO_PWD_03						IO_PWD_03	IO_ADC_27 IO_DI_39							
P141	Timer Input IO_PWD_05						IO_PWD_05	IO_ADC_29 IO_DI_41							
P142	BAT -														
P143	BAT-														
P144	BAT-														
P145	BAT-														
P146	Timer Input IO_PWD_07							IO_ADC_31 IO_DI_43							
P147	Timer Input IO_PWD_09							IO_ADC_33 IO_DI_45							
P148	Timer Input IO_PWD_11							IO_ADC_35 IO_DI_47							
P149	HS Digital Output IO_DO_00								IO_ADC_36 IO_DI_72						
P150	HS PWM Output		IO_PWD_14 IO_DI_30												

Pin No.	Main Function	Alternative Function													
		HS Digital Output	Timer Input	PVG Output	VOUT Output	A/D Input (HS Output/PVG/VOUT)	Current Loop Input	A/D Input (Timer Input)	A/D Input (HS Digital Output)	A/D Input (LS Digital Output)	Analog Current Input 2M	Digital Input 2M	Analog Current Input 3M	Analog Resistance Input 3M	Digital Input 3M
P151	HS PWM Output		IO_PWD_18 IO_DI_34												
P152	HS Digital Output IO_DO_02								IO_ADC_38 IO_DI_74						
P153	HS PWM Output IO_PWM_00	IO_DO_16	IO_DI_00												
P154	HS PWM Output IO_PWM_14	IO_DO_30	IO_DI_14												
P155	HS Digital Output IO_DO_04								IO_ADC_40 IO_DI_76						
P156	HS PWM Output IO_PWM_02	IO_DO_18	IO_DI_02												
P157	HS PWM Output IO_PWM_16	IO_DO_32	IO_DI_16												
P158	HS Digital Output IO_DO_06								IO_ADC_42 IO_DI_78						
P159	HS PWM Output IO_PWM_04	IO_DO_20	IO_DI_04												
P160	HS PWM Output IO_PWM_18	IO_DO_34	IO_DI_18												
P161															
P162	HS PWM Output IO_PWM_07	IO_DO_23	IO_DI_07												
P163	HS PWM Output IO_PWM_21	IO_DO_37	IO_DI_21												
P164															
P165	HS PWM Output IO_PWM_09	IO_DO_25	IO_DI_09												
P166															
P167															
P168															
P169															
P170															
P171															
P172															
P173	HS Digital Output IO_DO_01								IO_ADC_37 IO_DI_73						
P174	HS PWM Output		IO_PWD_15 IO_DI_31												
P175	HS PWM Output		IO_PWD_19 IO_DI_35												
P176	HS Digital Output IO_DO_03								IO_ADC_39 IO_DI_75						
P177	HS PWM Output IO_PWM_01	IO_DO_17	IO_DI_01												
P178	HS PWM Output IO_PWM_15	IO_DO_31	IO_DI_15												

		Alternative Function													
Pin No.	Main Function	HS Digital Output	Timer Input	PVG Output	VOUT Output	A/D Input (HS Output/PVG/VOUT)	Current Loop Input	A/D Input (Timer Input)	A/D Input (HS Digital Output)	A/D Input (LS Digital Output)	Analog Current Input 2M	Digital Input 2M	Analog Current Input 3M	Analog Resistance Input 3M	Digital Input 3M
P179	HS Digital Output IO_DO_05								IO_ADC_41 IO_DI_77						
P180	HS PWM Output IO_PWM_03	IO_DO_19	IO_DI_03												
P181	HS PWM Output IO_PWM_17	IO_DO_33	IO_DI_17												
P182	HS Digital Output IO_DO_07								IO_ADC_43 IO_DI_79						
P183	HS PWM Output IO_PWM_05	IO_DO_21	IO_DI_05												
P184	HS PWM Output IO_PWM_19	IO_DO_35	IO_DI_19												
P185															
P186	HS PWM Output IO_PWM_06	IO_DO_22	IO_DI_06												
P187	HS PWM Output IO_PWM_20	IO_DO_36	IO_DI_20												
P188															
P189	HS PWM Output IO_PWM_08	IO_DO_24	IO_DI_08												
P190															
P191															
P192															
P193															
P194															
P195															
P196															
P201	BAT+ Power														
P202	BAT+ Power														
P203	BAT+ Power														
P204	BAT+ Power														
P205	BAT+ Power														
P206	BAT+ Power														
P207	Terminal 15 IO_ADC_K15														
P208															
P209	CAN 0 Low IO_CAN_CHANNEL_0														
P210	CAN 1 Low IO_CAN_CHANNEL_1														
P211	CAN 2 Low IO_CAN_CHANNEL_2														
P212	CAN 3 Low IO_CAN_CHANNEL_3														
P213															
P214															
P215															

Pin No.	Main Function	HS Digital Output	Timer Input	PVG Output	VOUT Output	A/D Input (HS Output/PVG/VOUT)	Current Loop Input	A/D Input (Timer Input)	A/D Input (HS Digital Output)	A/D Input (LS Digital Output)	Analog Current Input 2M	Digital Input 2M	Analog Current Input 3M	Analog Resistance Input 3M	Digital Input 3M
		Alternative Function													
P216	CAN Termination 3L														
P217	Sensor GND														
P218															
P219															
P220	Wake-Up IO_ADC_WAKE_UP														
P221	Sensor Supply Var. IO_ADC_SENSOR_SUPPLY_2														
P222	CAN 0 High IO_CAN_CHANNEL_0														
P223	CAN 1 High IO_CAN_CHANNEL_1														
P224	CAN 2 High IO_CAN_CHANNEL_2														
P225	CAN 3 High IO_CAN_CHANNEL_3														
P226															
P227															
P228															
P229	CAN Termination 3H														
P230	Sensor GND														
P231															
P232															
P233															
P234	Sensor Supply 5 V IO_ADC_SENSOR_SUPPLY_1														
P235	CAN Termination 0L														
P236	CAN Termination 1L														
P237	CAN Termination 2L														
P238	LS Digital Output IO_DO_09									IO_ADC_45 IO_DI_81					
P239	LS Digital Output IO_DO_11									IO_ADC_47 IO_DI_83					
P240	LS Digital Output IO_DO_13									IO_ADC_49 IO_DI_85					
P241	LS Digital Output IO_DO_15									IO_ADC_51 IO_DI_87					
P242															
P243	Sensor GND														
P244	Sensor GND														
P245	Sensor GND														
P246	BAT+ CPU IO_ADC_UBAT														
P247	Sensor Supply 5 V IO_ADC_SENSOR_SUPPLY_0														
P248	CAN Termination 0H														
P249	CAN Termination 1H														

Pin No.	Main Function	HS Digital Output	Timer Input	PVG Output	VOUT Output	A/D Input (HS Output/PVG/VOUT)	Current Loop Input	A/D Input (Timer Input)	A/D Input (HS Digital Output)	A/D Input (LS Digital Output)	Analog Current Input 2M	Digital Input 2M	Analog Current Input 3M	Analog Resistance Input 3M	Digital Input 3M
		Alternative Function													
P250	CAN Termination 2H														
P251	LS Digital Output IO_DO_08									IO_ADC_44 IO_DI_80					
P252	LS Digital Output IO_DO_10									IO_ADC_46 IO_DI_82					
P253	LS Digital Output IO_DO_12									IO_ADC_48 IO_DI_84					
P254	LS Digital Output IO_DO_14									IO_ADC_50 IO_DI_86					
P255															
P256	Sensor GND														
P257	Sensor GND														
P258	Sensor GND														

Table 3.11: Pinning of HY-TTC 520

3.4.4 HY-TTC 510 Variant

		Alternative Function													
Pin No.	Main Function	*HS Digital Output*	*Timer Input*	*PVG Output*	*VOUT Output*	*A/D Input (HS Output/PVG/VOUT)*	*Current Loop Input*	*A/D Input (Timer Input)*	*A/D Input (HS Digital Output)*	*A/D Input (LS Digital Output)*	*Analog Current Input 2M*	*Digital Input 2M*	*Analog Current Input 3M*	*Analog Resistance Input 3M*	*Digital Input 3M*
P101	HS PWM Output		IO_PWD_12 IO_DI_28												
P102	HS PWM Output		IO_PWD_16 IO_DI_32												
P103	Analog Voltage Input IO_ADC_00												IO_ADC_00	IO_ADC_00	IO_DI_48
P104	Analog Voltage Input IO_ADC_02												IO_ADC_02	IO_ADC_02	IO_DI_50
P105	Analog Voltage Input IO_ADC_04												IO_ADC_04	IO_ADC_04	IO_DI_52
P106	Analog Voltage Input IO_ADC_06												IO_ADC_06	IO_ADC_06	IO_DI_54
P107	Analog Voltage Input IO_ADC_08										IO_ADC_08	IO_DI_56			
P108	Analog Voltage Input IO_ADC_10										IO_ADC_10	IO_DI_58			
P109	Analog Voltage Input IO_ADC_12										IO_ADC_12	IO_DI_60			
P110	Analog Voltage Input IO_ADC_14										IO_ADC_14	IO_DI_62			
P111	Analog Voltage Input IO_ADC_16										IO_ADC_16	IO_DI_64			
P112	Analog Voltage Input IO_ADC_18										IO_ADC_18	IO_DI_66			
P113	Analog Voltage Input IO_ADC_20										IO_ADC_20	IO_DI_68			
P114	Analog Voltage Input IO_ADC_22										IO_ADC_22	IO_DI_70			
P115	Timer Input IO_PWD_00						IO_PWD_00	IO_ADC_24 IO_DI_36							
P116	Timer Input IO_PWD_02						IO_PWD_02	IO_ADC_26 IO_DI_38							
P117	Timer Input IO_PWD_04						IO_PWD_04	IO_ADC_28 IO_DI_40							
P118	BAT-														
P119	BAT-														
P120	BAT-														
P121	BAT-														
P122	Timer Input IO_PWD_06							IO_ADC_30 IO_DI_42							
P123	Timer Input IO_PWD_08							IO_ADC_32 IO_DI_44							
P124	Timer Input IO_PWD_10							IO_ADC_34 IO_DI_46							

Pin No.	Main Function	Alternative Function: HS Digital Output	Timer Input	PVG Output	VOUT Output	A/D Input (HS Output/PVG/VOUT)	Current Loop Input	A/D Input (Timer Input)	A/D Input (HS Digital Output)	A/D Input (LS Digital Output)	Analog Current Input 2M	Digital Input 2M	Analog Current Input 3M	Analog Resistance Input 3M	Digital Input 3M
P125	HS PWM Output		IO_DI_29 IO_PWD_13												
P126	HS PWM Output		IO_PWD_17 IO_DI_33												
P127	Analog Voltage Input IO_ADC_01												IO_ADC_01	IO_ADC_01	IO_DI_49
P128	Analog Voltage Input IO_ADC_03												IO_ADC_03	IO_ADC_03	IO_DI_51
P129	Analog Voltage Input IO_ADC_05												IO_ADC_05	IO_ADC_05	IO_DI_53
P130	Analog Voltage Input IO_ADC_07												IO_ADC_07	IO_ADC_07	IO_DI_55
P131	Analog Voltage Input IO_ADC_09										IO_ADC_09	IO_DI_57			
P132	Analog Voltage Input IO_ADC_11										IO_ADC_11	IO_DI_59			
P133	Analog Voltage Input IO_ADC_13										IO_ADC_13	IO_DI_61			
P134	Analog Voltage Input IO_ADC_15										IO_ADC_15	IO_DI_63			
P135	Analog Voltage Input IO_ADC_17										IO_ADC_17	IO_DI_65			
P136	Analog Voltage Input IO_ADC_19										IO_ADC_19	IO_DI_67			
P137	Analog Voltage Input IO_ADC_21										IO_ADC_21	IO_DI_69			
P138	Analog Voltage Input IO_ADC_23										IO_ADC_23	IO_DI_71			
P139	Timer Input IO_PWD_01						IO_PWD_01	IO_ADC_25 IO_DI_37							
P140	Timer Input IO_PWD_03						IO_PWD_03	IO_ADC_27 IO_DI_39							
P141	Timer Input IO_PWD_05						IO_PWD_05	IO_ADC_29 IO_DI_41							
P142	BAT -														
P143	BAT-														
P144	BAT-														
P145	BAT-														
P146	Timer Input IO_PWD_07							IO_ADC_31 IO_DI_43							
P147	Timer Input IO_PWD_09							IO_ADC_33 IO_DI_45							
P148	Timer Input IO_PWD_11							IO_ADC_35 IO_DI_47							
P149	HS Digital Output IO_DO_00								IO_ADC_36 IO_DI_72						
P150	HS PWM Output		IO_PWD_14 IO_DI_30												

		Alternative Function													
Pin No.	Main Function	HS Digital Output	Timer Input	PVG Output	VOUT Output	A/D Input (HS Output/PVG/VOUT)	Current Loop Input	A/D Input (Timer Input)	A/D Input (HS Digital Output)	A/D Input (LS Digital Output)	Analog Current Input 2M	Digital Input 2M	Analog Current Input 3M	Analog Resistance Input 3M	Digital Input 3M
P151	HS PWM Output		IO_PWD_18 IO_DI_34												
P152	HS Digital Output IO_DO_02								IO_ADC_38 IO_DI_74						
P153	HS PWM Output IO_PWM_00	IO_DO_16	IO_DI_00												
P154	HS PWM Output IO_PWM_14	IO_DO_30	IO_DI_14												
P155	HS Digital Output IO_DO_04								IO_ADC_40 IO_DI_76						
P156	HS PWM Output IO_PWM_02	IO_DO_18	IO_DI_02												
P157	HS PWM Output IO_PWM_16	IO_DO_32	IO_DI_16												
P158	HS Digital Output IO_DO_06								IO_ADC_42 IO_DI_78						
P159	HS PWM Output IO_PWM_04	IO_DO_20	IO_DI_04												
P160	HS PWM Output IO_PWM_18	IO_DO_34	IO_DI_18												
P161	HS Digital Output IO_DO_52			IO_PVG_00	IO_VOUT_00	IO_ADC_52 IO_DI_88									
P162	HS PWM Output IO_PWM_07	IO_DO_23	IO_DI_07												
P163	HS PWM Output IO_PWM_21	IO_DO_37	IO_DI_21												
P164	HS Digital Output IO_DO_55			IO_PVG_03	IO_VOUT_03	IO_ADC_55 IO_DI_91									
P165															
P166															
P167	HS Digital Output IO_DO_57			IO_PVG_05	IO_VOUT_05	IO_ADC_57 IO_DI_93									
P168															
P169															
P170	HS Digital Output IO_DO_59			IO_PVG_07	IO_VOUT_07	IO_ADC_59 IO_DI_95									
P171															
P172															
P173	HS Digital Output IO_DO_01								IO_ADC_37 IO_DI_73						
P174	HS PWM Output		IO_PWD_15 IO_DI_31												
P175	HS PWM Output		IO_PWD_19 IO_DI_35												
P176	HS Digital Output IO_DO_03								IO_ADC_39 IO_DI_75						
P177	HS PWM Output IO_PWM_01	IO_DO_17	IO_DI_01												

Pin No.	Main Function	Alternative Function													
		HS Digital Output	*Timer Input*	*PVG Output*	*VOUT Output*	*A/D Input (HS Output/PVG/VOUT)*	*Current Loop Input*	*A/D Input (Timer Input)*	*A/D Input (HS Digital Output)*	*A/D Input (LS Digital Output)*	*Analog Current Input 2M*	*Digital Input 2M*	*Analog Current Input 3M*	*Analog Resistance Input 3M*	*Digital Input 3M*
P178	HS PWM Output IO_PWM_15	IO_DO_31	IO_DI_15												
P179	HS Digital Output IO_DO_05								IO_ADC_41 IO_DI_77						
P180	HS PWM Output IO_PWM_03	IO_DO_19	IO_DI_03												
P181	HS PWM Output IO_PWM_17	IO_DO_33	IO_DI_17												
P182	HS Digital Output IO_DO_07								IO_ADC_43 IO_DI_79						
P183	HS PWM Output IO_PWM_05	IO_DO_21	IO_DI_05												
P184	HS PWM Output IO_PWM_19	IO_DO_35	IO_DI_19												
P185	HS Digital Output IO_DO_53			IO_PVG_01	IO_VOUT_01	IO_ADC_53 IO_DI_89									
P186	HS PWM Output IO_PWM_06	IO_DO_22	IO_DI_06												
P187	HS PWM Output IO_PWM_20	IO_DO_36	IO_DI_20												
P188	HS Digital Output IO_DO_54			IO_PVG_02	IO_VOUT_02	IO_ADC_54 IO_DI_90									
P189															
P190															
P191	HS Digital Output IO_DO_56			IO_PVG_04	IO_VOUT_04	IO_ADC_56 IO_DI_92									
P192															
P193															
P194	HS Digital Output IO_DO_58			IO_PVG_06	IO_VOUT_06	IO_ADC_58 IO_DI_94									
P195															
P196															
P201	BAT+ Power														
P202	BAT+ Power														
P203	BAT+ Power														
P204	BAT+ Power														
P205	BAT+ Power														
P206	BAT+ Power														
P207	Terminal 15 IO_ADC_K15														
P208	LIN IO_LIN														
P209	CAN 0 Low IO_CAN_CHANNEL_0														
P210	CAN 1 Low IO_CAN_CHANNEL_1														
P211	CAN 2 Low IO_CAN_CHANNEL_2														

Pin No.	Main Function	Alternative Function: HS Digital Output	Timer Input	PVG Output	VOUT Output	A/D Input (HS Output/PVG/VOUT)	Current Loop Input	A/D Input (Timer Input)	A/D Input (HS Digital Output)	A/D Input (LS Digital Output)	Analog Current Input 2M	Digital Input 2M	Analog Current Input 3M	Analog Resistance Input 3M	Digital Input 3M
P212															
P213															
P214															
P215															
P216															
P217	Sensor GND														
P218															
P219															
P220	Wake-Up IO_ADC_WAKE_UP														
P221	Sensor Supply Var. IO_ADC_SENSOR_SUPPLY_2														
P222	CAN 0 High IO_CAN_CHANNEL_0														
P223	CAN 1 High IO_CAN_CHANNEL_1														
P224	CAN 2 High IO_CAN_CHANNEL_2														
P225															
P226															
P227															
P228															
P229															
P230	Sensor GND														
P231															
P232															
P233															
P234	Sensor Supply 5 V IO_ADC_SENSOR_SUPPLY_1														
P235	CAN Termination 0L														
P236	CAN Termination 1L														
P237	CAN Termination 2L														
P238	LS Digital Output IO_DO_09									IO_ADC_45 IO_DI_81					
P239	LS Digital Output IO_DO_11									IO_ADC_47 IO_DI_83					
P240	LS Digital Output IO_DO_13									IO_ADC_49 IO_DI_85					
P241	LS Digital Output IO_DO_15									IO_ADC_51 IO_DI_87					
P242															
P243	Sensor GND														
P244	Sensor GND														
P245	Sensor GND														
P246	BAT+ CPU IO_ADC_UBAT														

Pin No.	Main Function	HS Digital Output	Timer Input	PVG Output	VOUT Output	A/D Input (HS Output/PVG/VOUT)	Current Loop Input	A/D Input (Timer Input)	A/D Input (HS Digital Output)	A/D Input (LS Digital Output)	Analog Current Input 2M	Digital Input 2M	Analog Current Input 3M	Analog Resistance Input 3M	Digital Input 3M
		Alternative Function													
P247	Sensor Supply 5 V IO_ADC_SENSOR_SUPPLY_0														
P248	CAN Termination 0H														
P249	CAN Termination 1H														
P250	CAN Termination 2H														
P251	LS Digital Output IO_DO_08									IO_ADC_44 IO_DI_80					
P252	LS Digital Output IO_DO_10									IO_ADC_46 IO_DI_82					
P253	LS Digital Output IO_DO_12									IO_ADC_48 IO_DI_84					
P254	LS Digital Output IO_DO_14									IO_ADC_50 IO_DI_86					
P255															
P256	Sensor GND														
P257	Sensor GND														
P258	Sensor GND														

Table 3.12: Pinning of HY-TTC 510

4 Specification of Inputs and Outputs

4.1 Positive Power Supply of Power Stages (BAT+ Power)

4.1.1 Pinout

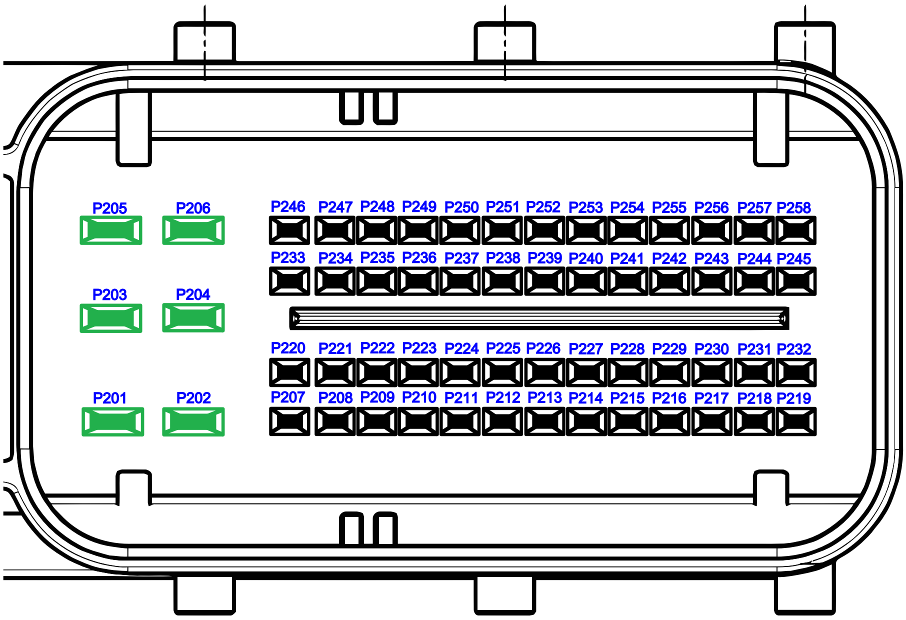

Figure 4.1: Pinout of BAT+ Power

Pin No.	Function
P201	Battery (+) Supply of Power Stages / BAT+ Power
P202	Battery (+) Supply of Power Stages / BAT+ Power
P203	Battery (+) Supply of Power Stages / BAT+ Power
P204	Battery (+) Supply of Power Stages / BAT+ Power
P205	Battery (+) Supply of Power Stages / BAT+ Power
P206	Battery (+) Supply of Power Stages / BAT+ Power

4.1.2 Functional Description

Supply pins for power stage supply.

The nominal supply voltage for full operation is between 6 and 32 V, including supply voltage ranges for 12 and 24 V battery operation. In this voltage range, all the I/Os work, as described in the system manual. BAT+ Power pins are equipped with inverse polarity protection.

TTControl GmbH recommends to use these pins in parallel and the maximum possible wire size (FLRY type) in case of maximum load current to reduce voltage drop and prevent overheating of the crimp contact.

4.1.3 Maximum Ratings

T_{ECU} = -40. . . +125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
$BAT+_{max}$	Permanent non-destructive supply voltage	1	-32	32	V
$BAT+_{lim}$	Peak non-destructive supply clamping voltage <1 ms	1, 2	-40	45	V
$I_{in\text{-}lim}$	Peak non-destructive supply clamping current <1 ms	1, 2	-10	+100	A
T_d	Load dump protection time according to ISO 7637-2 [28], Pulse 5, Level IV (superimposed 174 V, R_i = 2 Ω)	1		350	ms
$I_{in\text{-}max}$	Permanent battery supply current (all 6 pins in parallel with symmetrical wire connection))	3		60	A
$I_{in\text{-}max}$	Permanent battery supply current per pin	3		10	A
$I_{in\text{-}total}$	Total load current, 12 V and 24 V battery operation	4		45	A
$I_{in\text{-}total}$	Total load current, 12 V and 24 V battery operation	4,5		60	A

Note 1 The control unit is protected by a transient suppressor, specified by clamp voltage, current and duration of voltage transient

Note 2 1 ms pulse width, non-repetitive. The pulse width is defined as the point at which the peak current decreases to 50 % of the maximum value.

Note 3 This battery supply current is related to the total load current of all high-side power-stages. In worst case, all outputs are in non-PWM mode or with maximum duty cycle operated, the battery current equals the total load current. With typical PWM-operation the battery supply current is significant lower than the total load current. For more details please see Section 6.2.2 on page 169.

Note 4 HY-TTC 500 variant-dependent, see Section 6.2.2 on page 169

Note 5 T_{ECU} = -40. . . +85 °C

4.1.4 Characteristics

T_{ECU} = -40. . . +125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
C_{in}	Pin input capacitance			250	µF
BAT+	Supply voltage for full operation		6	32	V

4.2 Positive Power Supply of Internal Electronics (BAT+ CPU)

4.2.1 Pinout

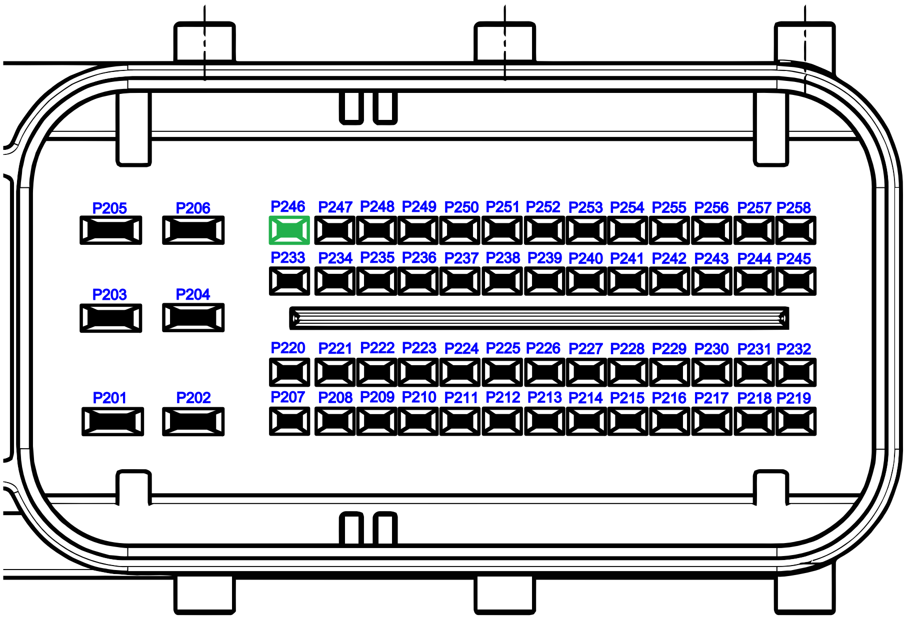

Figure 4.2: Pinout of BAT+ CPU

Pin No.	Function	SW-define
P246	Battery supply of internal electronics	IO_ADC_UBAT

4.2.2 Functional Description

Supply pin for positive power supply of internal electronics, sensor supply and PVG/V_{out} output stages.

As the output voltage of the PVG/V_{out} outputs is defined as a percentage value in relation to the battery voltage, the voltage drop on the wire to this pin has a direct influence on the accuracy of the PVG output voltage.

BAT+ CPU pin is equipped with inverse polarity protection.

TTControl GmbH recommends to use the maximum possible wire size (FLRY type) in case of maximum load current to reduce voltage drop and prevent overheating of the crimp contact.

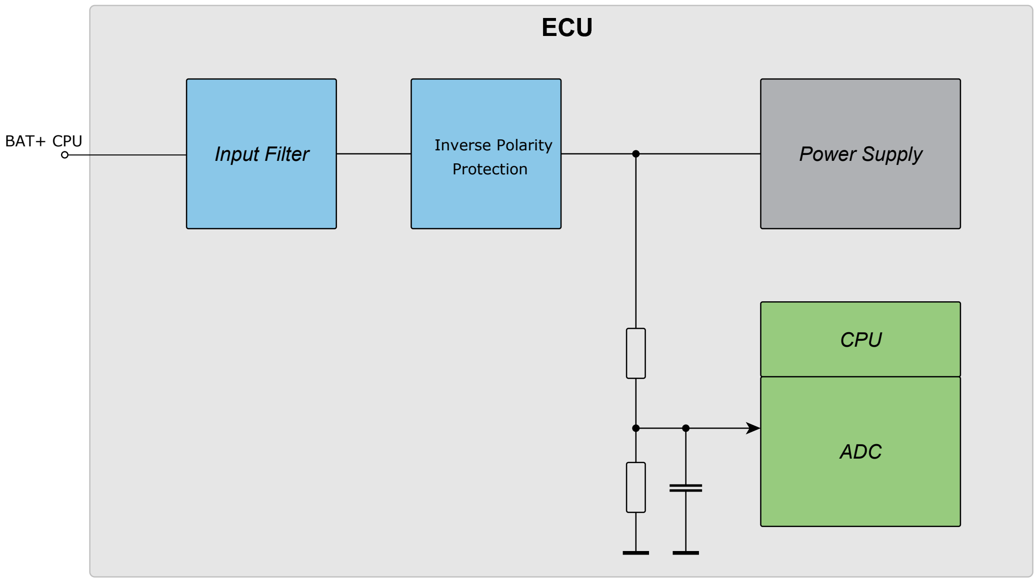

Figure 4.3: Supply pin for the internal ECU logic

4.2.3 Maximum Ratings

T_{ECU} = -40. . . +125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
$U_{in\text{-}max}$	Permanent non-destructive supply voltage	1	-32	32	V
$U_{in\text{-}lim}$	Peak non-destructive supply clamping voltage <1 ms	1,2	-40	45	V
$I_{in\text{-}lim}$	Peak non-destructive supply clamping current <1 ms	1,2	-10	+100	A
T_d	Load dump protection time according to ISO 7637-2 [28], Pulse 5, Level IV (superimposed 174 V, $R_i = 2\ \Omega$)	1		350	ms
$I_{in\text{-}max}$	Permanent input current	3		3	A

Note 1 The control unit is protected by a transient suppressor, specified by clamp voltage, current and duration of voltage transient.

Note 2 1 ms pulse width, non-repetitive. The pulse width is defined as the point at which the peak current decreases to 50 % of the maximum value.

Note 3 Current internally self-limited

4.2.4 Characteristics

T_{ECU} = -40. . . +125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
C_{in}	Pin input capacitance			250	µF
BAT+	Supply voltage for start-up	1	8	32	V
BAT+	Supply voltage for full operation	2	6	32	V
$I_{in\text{-}idle}$	Supply current of unit without load (8 V)			600	mA
$I_{in\text{-}idle}$	Supply current of unit without load (12 V)			400	mA
$I_{in\text{-}idle}$	Supply current of unit without load (24 V)			200	mA
$I_{in\text{-}STBY}$	Standby supply current (Terminal 15 and Wake-up off)			1	mA
$I_{in\text{-}STBY}$	Standby supply current (Terminal 15 and Wake-up off)	3		<0.5	mA

Note 1 8 V is the initial voltage for start-up at the beginning of the drive cycle

Note 2 See Section 4.2.5 on the facing page

Note 3 T_{ECU} = -40. . . +85 °C

4.2.5 Low-Voltage Operation

The HY-TTC 500 core system is designed for full operation after start-up between 6 V and 32 V, including supply voltage ranges for 12 V and 24 V battery operation and cold-start cranking according to ISO 16750-2 [29]. The initial minimum supply voltage at the beginning of the drive cycle is 8 V. After start-up, the CPU will remain operational down to 6 V, e.g. during cold-start cranking.

The minimum supply voltage during cold-start cranking is defined by ISO 16750-2:2012 (see Table 3, *Starting profile values for systems with 12 V nominal voltage, U_N*, and Table 4, *Values for systems with 24 V nominal voltage, U_N*). The HY-TTC 500 core system complies with ISO 16750-2:2012, level I, II (functional status C), III and IV for 12-V systems and level I, II (functional status A) and III (functional status B) for 24-V systems, see Table 4.1 on page 71 and Table 4.2 on page 71.

Restrictions during cold-start cranking apply also for sensor supplies. For more information, see Section 4.7 on page 82 and Section 4.8 on page 85.

HY-TTC 500 ISO 16750 code specification see Section 1.6 on page 23.

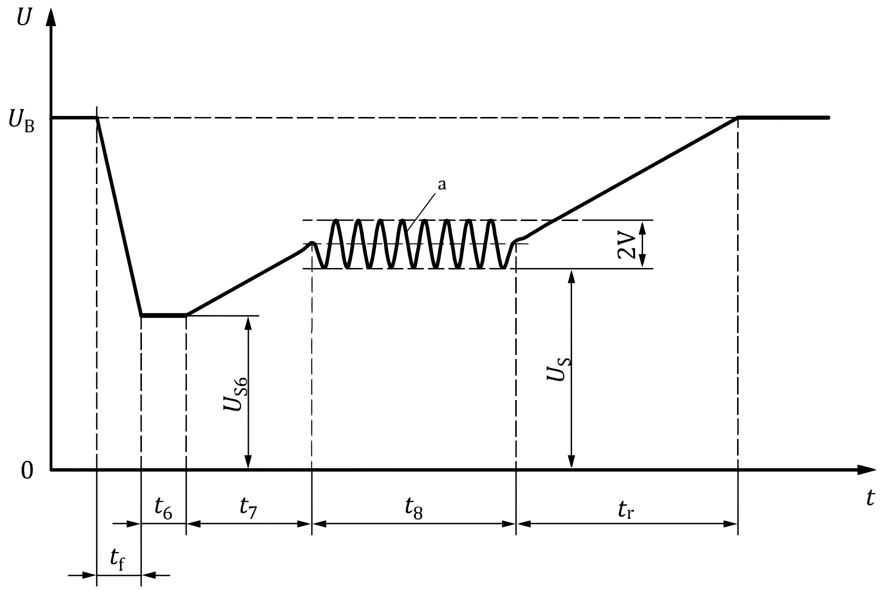

Figure 4.4: ISO 16750-2, Figure 7 – Starting profile

Key

t	time
U	test voltage
t_f	falling slope
t_r	rising slope
t_6, t_7, t_8	duration parameters (in accordance with Table 3 and Table 4 of ISO 16750-2)
U_B	supply voltage for generator not in operation (see ISO 16750-1 [20])
U_S	supply voltage
U_{S6}	supply voltage at t_6
a	f = 2 Hz

4.2.5.1 HY-TTC 500 ISO 16750 functional status

ISO 16750-2:2012 starting profile - functional status for 12 V system nominal voltage

	Functional Status		
	A	B	C
Level I	X		
Level II			X
Level III			X
Level IV		X	

Table 4.1: ISO 16750 functional status for 12 V systems

ISO 16750-2:2012 starting profile - functional status for 24 V system nominal voltage

	Functional Status		
	A	B	C
Level I	X		
Level II	X		
Level III		X [1]	

Table 4.2: ISO 16750 functional status for 24 V systems

[1] After start-up

4.2.6 Voltage Monitoring

The battery voltage on pin BAT+ CPU is connected to an ADC input. Battery voltage measurement can be used for diagnostic purposes.

T_{ECU} = -40. . . +125 ℃

Symbol	Parameter	Note	Min.	Max.	Unit
τ_{in}	First order low pass filter		1.5	2.5	ms
V_{nom}	Nominal battery supply range	1	0	33	V
$V_{tol\text{-}0}$	Zero reading error		-80	+80	mV
$V_{tol\text{-}0}$	Zero reading error	2	-67	+67	mV
$V_{tol\text{-}p}$	Proportional error		-1.2	+1.2	%
$V_{tol\text{-}p}$	Proportional error	2	-1	+1	%
LSB	Nominal value of 1 LSB		-	13.4	mV

Note 1 The nominal battery supply range is only a value to calculate the actual voltage.
Note 2 T_{ECU} = -40. . . +85 ℃

4.3 Negative Power Supply (BAT-)

4.3.1 Pinout

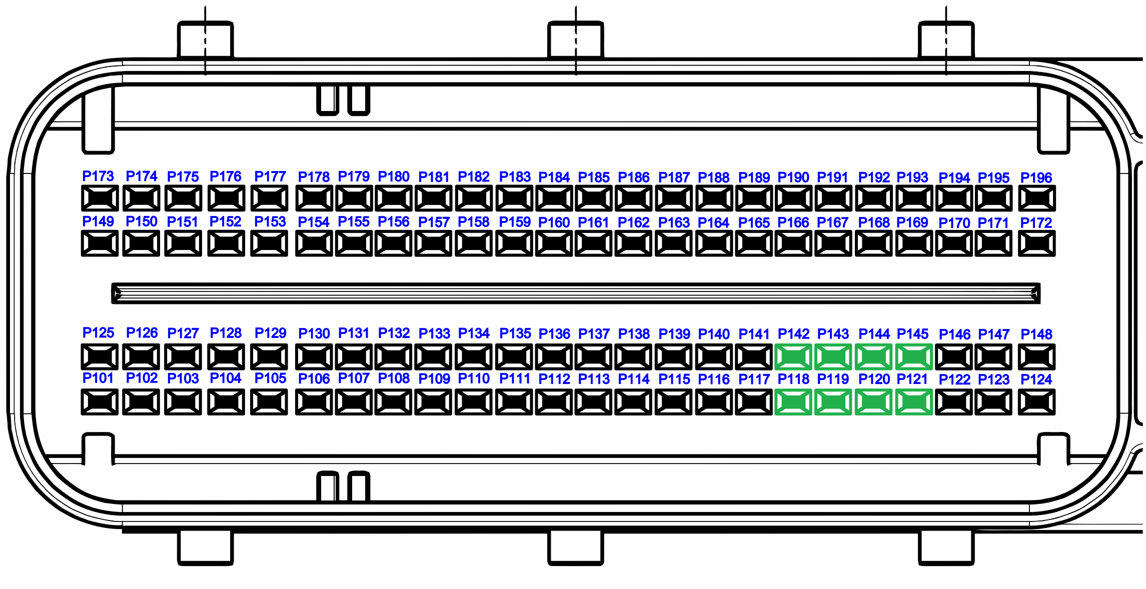

Figure 4.5: Pinout of BAT-

Pin No.	Function
P118	Negative Power Supply (BAT-)
P119	Negative Power Supply (BAT-)
P120	Negative Power Supply (BAT-)
P121	Negative Power Supply (BAT-)
P142	Negative Power Supply (BAT-)
P143	Negative Power Supply (BAT-)
P144	Negative Power Supply (BAT-)
P145	Negative Power Supply (BAT-)

4.3.2 Functional Description

Supply pins for BAT- connection.

The HY-TTC 500 housing is not directly connected to BAT-, for more information see Section 6.4 on page 171.

TTControl GmbH recommends to use these pins in parallel and the maximum possible wire size (FLRY type) in case of maximum load current to reduce voltage drop and prevent overheating of the crimp contact.

4.3.3 Maximum Ratings

T_{ECU} = -40. . . +125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
$I_{out-max}$	Permanent current per pin			4	A
$I_{out-max}$	Permanent current all pins			28	A

4.4 Sensor GND

4.4.1 Pinout

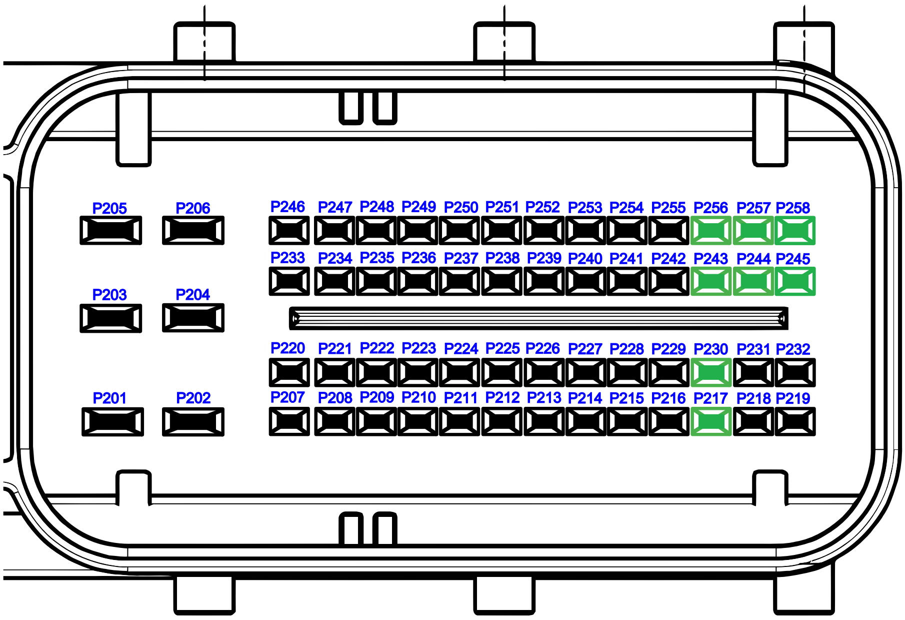

Figure 4.6: Pinout of sensor GND

Pin No.	Function
P217	Sensor GND
P230	Sensor GND
P243	Sensor GND
P244	Sensor GND
P245	Sensor GND
P256	Sensor GND
P257	Sensor GND
P258	Sensor GND

4.4.2 Functional Description

Supply pins for analog sensor GND connection.

These pins can also be used as GND connection for digital sensors.

4.4.3 Maximum Ratings

T_{ECU} = -40. . . +125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
$I_{out\text{-}max}$	Permanent current per pin	1		1	A

Note 1 It is recommended to use all sensor ground pins simultaneously to ensure load distribution and minimize voltage drop on the external wiring.

4.5 Terminal 15

4.5.1 Pinout

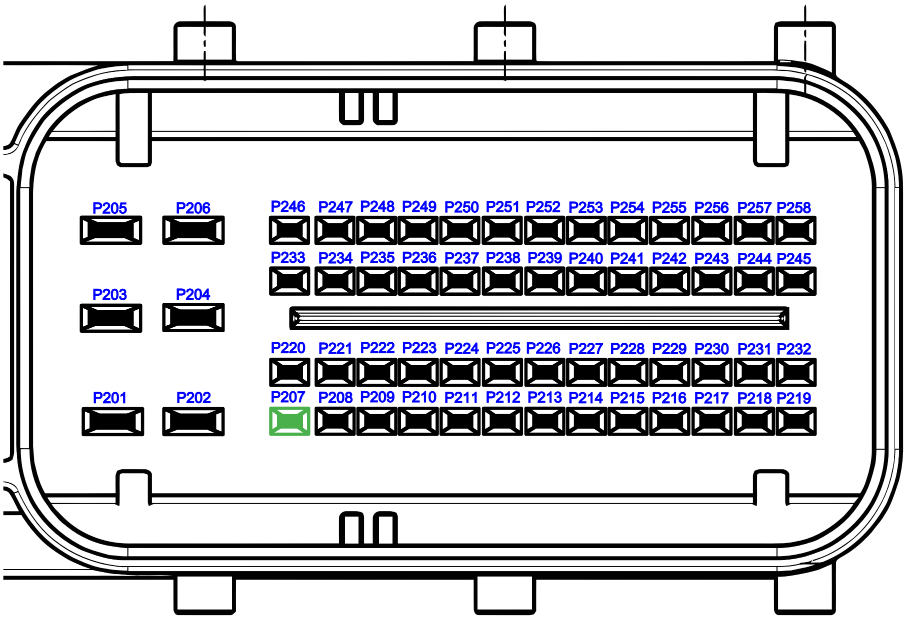

Figure 4.7: Pinout of terminal 15

Pin No.	Function	SW-define
P207	Terminal 15 Input	IO_ADC_K15

4.5.2 Functional Description

This is the power control input for permanently supplied systems. When switched to positive supply, this input gives the command to power up the ECU, regardless of the `Wake-up` pin status. When switched off, the ECU can activate its keep-alive functionality (if keep-alive functionality is enabled by SW) and is finally switched off by software after a user-defined period of time.

For systems with a main power switch (not permanently supplied) this pin must be tied to BAT+ CPU. This input is monitored by the CPU via an ADC input.

See also section *POWER - Driver for ECU power functions* of the API documentation [38].

4.5.3 Maximum Ratings

T_{ECU} = -40. . . +125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
V_{in}	Permanent (DC) input voltage		-33	+33	V
V_{in}	Transient peak input voltage 500 ms		-50	+50	V
V_{in}	Transient peak input voltage 1ms		-100	+100	V

4.5.4 Characteristics

T_{ECU} = -40. . . +125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
C_{in}	Pin input capacitance		8	12	nF
R_{pd}	Pull-down resistor		6.5	11.5	kΩ
I_{in}	Input current	1	2	2.5	mA
I_{in}	Input current	2	4	4.5	mA
V_{il}	Input voltage for low level		0	6	V
V_{ih}	Input voltage for high level	3	8	32	V
τ_{in}	Input low pass filter		180	280	µs

Note 1 at 16 V input voltage
Note 2 at 32 V input voltage
Note 3 8 V is the initial voltage for start-up at the beginning of the drive cycle

4.6 Wake-Up

4.6.1 Pinout

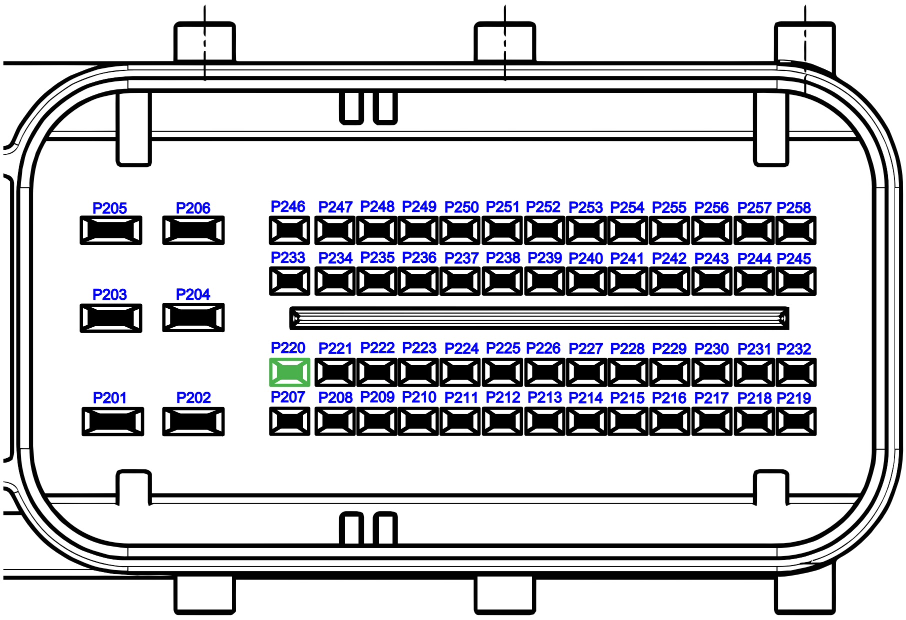

Figure 4.8: Pinout of wake-up

Pin No.	Function	SW-define
P220	Wake-up input	IO_ADC_WAKE_UP

4.6.2 Functional Description

This is the `Wake-up` input for permanently supplied systems. When switched to positive supply (rising edge triggered), this input gives the command to power up the ECU, regardless of the `Terminal 15` pin status. When switched off, the ECU can activate its keep-alive functionality (provided that keep-alive functionality is enabled by SW) and finally switches off by software after a user-defined period of time. The application software can command the ECU to switch off even if the `Wake-up` pin is high, but only if `Terminal 15` is off. This input is monitored by the CPU via an ADC input.

Use case pre-boot sequence

For example, it is possible to start the boot sequence of the ECU by opening the vehicle door by connecting the `Wake-up` pin with the vehicle door contact. When the driver enters the vehicle and turns the ignition key on (`Terminal 15` to high), the ECU boot process is already finished. With this feature the ECU is ready for operation without any delay.

If the `Wake-up` pin is in a continuous high state (e.g. vehicle door is left open) and `Terminal 15` pin is not switched on, the application software shall power down the ECU after an application dependent timeout.

After forcing the ECU shutdown via application software, it is necessary to externally toggle the `Wake-up` pin or activate `Terminal 15` pin to restart the ECU.

See also section *POWER - Driver for ECU power functions* of the API documentation [38].

4.6.3 Maximum Ratings

T_{ECU} = -40. . . +125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
V_{in}	Permanent (DC) input voltage		-33	33	V
V_{in}	Transient peak input voltage 500 ms		-50	50	V
V_{in}	Transient peak input voltage 1 ms		-100	100	V

4.6.4 Characteristics

T_{ECU} = -40. . . +125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
C_{in}	Pin input capacitance		8	12	nF
R_{pd}	Pull-down resistor		6.5	11.5	kΩ
I_{in}	Input current	1	2	2.5	mA
I_{in}	Input current	2	4	4.5	mA
V_{il}	Input voltage for low level		0	6	V
V_{ih}	Input voltage for high level		8	32	V
τ_{in}	Input low pass filter		180	280	µs

Note 1 at 16 V input voltage
Note 2 at 32 V input voltage
Note 3 8 V is the initial voltage for start-up at the beginning of the drive cycle

4.7 Sensor Supplies 5 V

4.7.1 Pinout

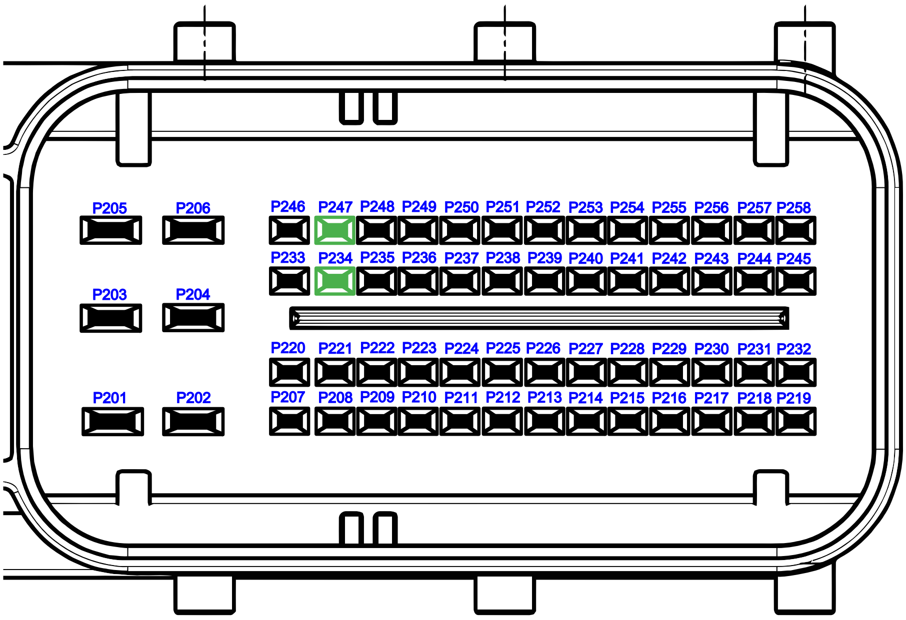

Figure 4.9: Pinout of sensor supply 5 V

Pin No.	Function	SW-define
P247	Sensor Supply 0	IO_ADC_SENSOR_SUPPLY_0
P234	Sensor Supply 1	IO_ADC_SENSOR_SUPPLY_1

4.7.2 Functional Description

Two independent sensor supplies 5 V for 3-wire sensors (e.g. potentiometers, pressure sensors etc.).

For fully redundant sensors with 2 sensor-supply connections, both supplies must be connected to different sensor supplies.

If the input voltage on the BAT+ CPU pin is lower than typically 6 V (at 5 mA sensor supply load current), the sensor supply output voltage will be out of specification. One example of such low input voltage situations may be cold-start cranking in 12/24 V systems where the supply voltage can drop below 6 V. If the sensor supply output voltage drops below 4.7 V, the application software will be informed about this error situation after glitch filtering.

TTControl GmbH recommends to use the maximum possible wire size (FLRY type) in case of maximum load current to reduce voltage drop and prevent overheating of the crimp contact.

See also section *POWER - Driver for ECU power functions* of the API documentation [38].

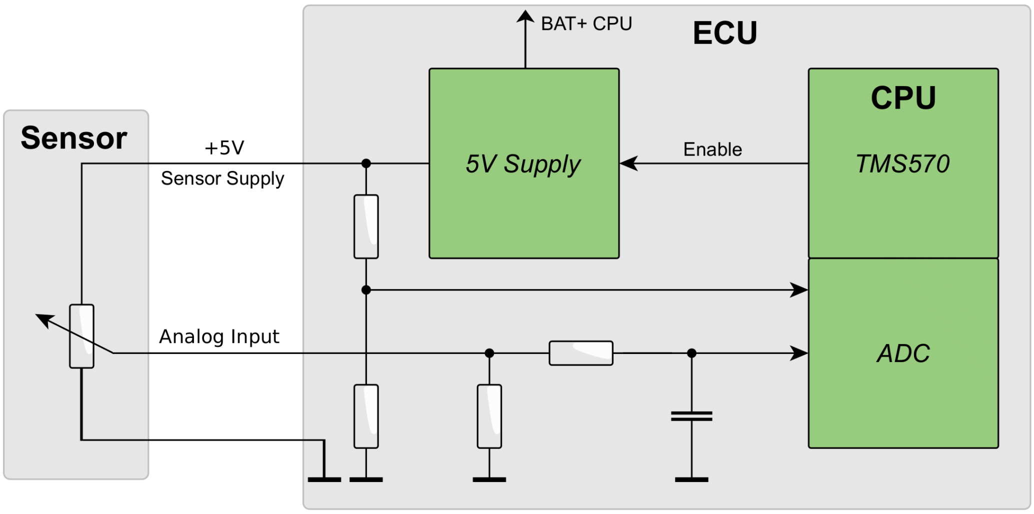

Figure 4.10: Sensor supply 5 V

4.7.3 Maximum Ratings

T_{ECU} = -40. . . +125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
V_{in}	Output voltage under overload conditions (e.g. short circuit to supply voltage)		-1	+33	V

4.7.4 Characteristics

T_{ECU} = -40. . . +125 ℃

Symbol	Parameter	Note	Min.	Max.	Unit
C_{out}	Pin output capacitance		0.8	1.2	µF
V_{out}	Output voltage, at I_{load}		4.85	5.15	V
I_{load}	Load current		0	500	mA

4.8 Sensor Supply Variable

4.8.1 Pinout

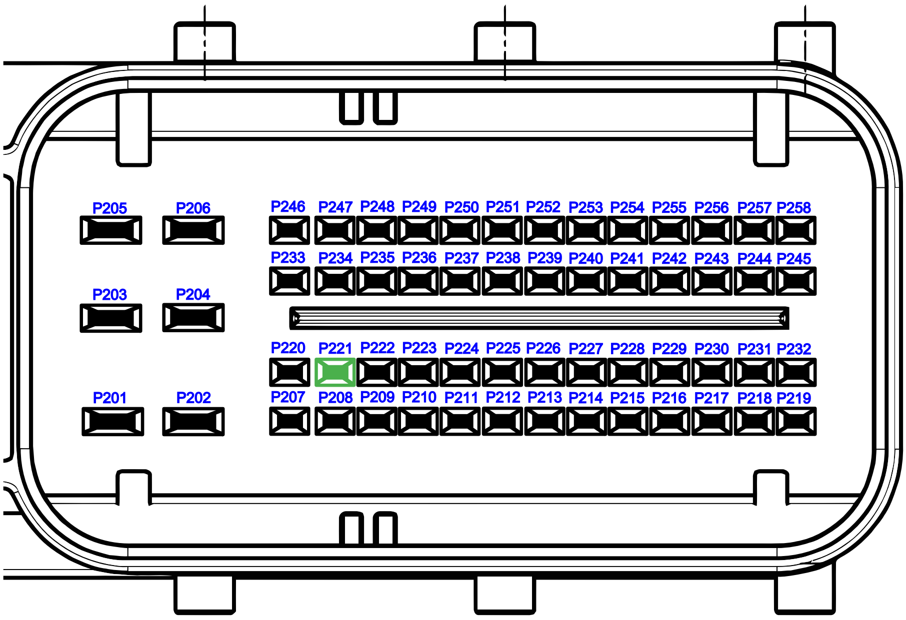

Figure 4.11: Pinout of sensor supply variable

Pin No.	Function	SW-define
P221	Sensor Supply 2	IO_ADC_SENSOR_SUPPLY_2

4.8.2 Functional Description

One independent sensor supply with variable output voltage, configurable in 1 V steps, is provided for 3-wire sensors (e.g. potentiometers, pressure sensors etc.).

As already described in Section 4.7 on page 82, 5 V sensor supply, Functional Description, the BAT+ CPU pin voltage must be at least 1 V higher than the required sensor supply output voltage VSSUP. If the BAT+ CPU pin voltage is lower than VSSUP -1 V, the sensor supply output voltage will be out of specification.

TTControl GmbH recommends to use the maximum possible wire size (FLRY type) in case of maximum load current to reduce voltage drop and prevent overheating of the crimp contact.

See also section *POWER - Driver for ECU power functions* of the API documentation [38].

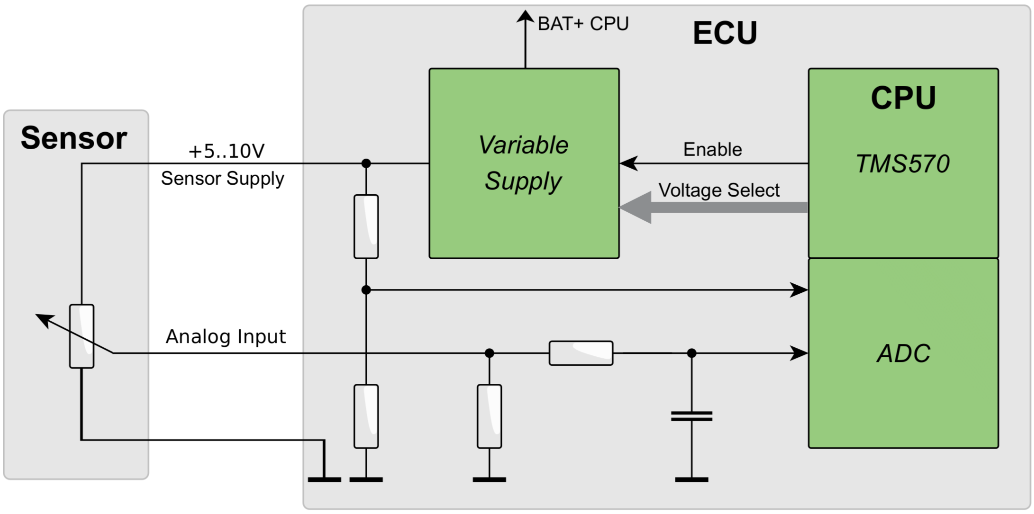

Figure 4.12: Sensor supply variable

4.8.3 Maximum Ratings

T_{ECU} = -40. . . +125 ℃

Symbol	Parameter	Note	Min.	Max.	Unit
V_{in}	Output voltage under overload conditions (e.g. short circuit to supply voltages)		-1	33	V

4.8.4 Characteristics

T_{ECU} = -40. . . +125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
C_{out}	Pin output capacitance		0.8	1.2	µF
V_{out}	Output voltage, 5-V setting		4.85	5.15	V
V_{out}	Output voltage, 6-V setting		5.80	6.20	V
V_{out}	Output voltage, 7-V setting		6.75	7.25	V
V_{out}	Output voltage, 8-V setting		7.70	8.30	V
V_{out}	Output voltage, 9-V setting		8.65	9.35	V
V_{out}	Output voltage, 10-V setting		9.60	10.40	V
P_{load}	Maximum output power, 5-V. . . 10-V setting			2.5	W

4.9 Analog Input 3 Modes

4.9.1 Pinout

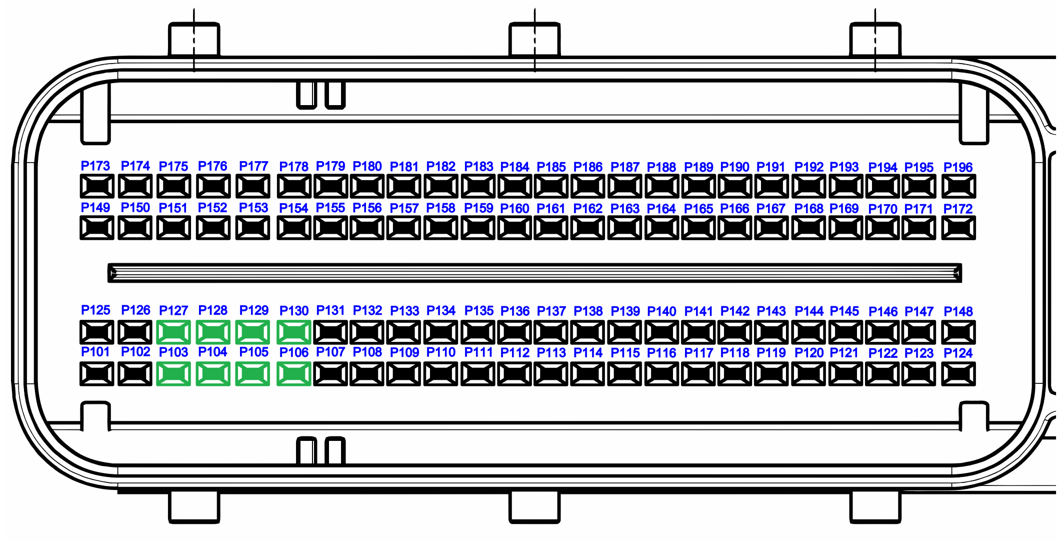

Figure 4.13: Pinout of analog input 3 mode

Pin No.	Function 1	Function 2	Function 3	SW-define
P103	Analog 0...5 V Input	Analog 0...25 mA Input	Analog 0...100 kΩ Input	IO_ADC_00
P127	Analog 0...5 V Input	Analog 0...25 mA Input	Analog 0...100 kΩ Input	IO_ADC_01
P104	Analog 0...5 V Input	Analog 0...25 mA Input	Analog 0...100 kΩ Input	IO_ADC_02
P128	Analog 0...5 V Input	Analog 0...25 mA Input	Analog 0...100 kΩ Input	IO_ADC_03
P105	Analog 0...5 V Input	Analog 0...25 mA Input	Analog 0...100 kΩ Input	IO_ADC_04
P129	Analog 0...5 V Input	Analog 0...25 mA Input	Analog 0...100 kΩ Input	IO_ADC_05
P106	Analog 0...5 V Input	Analog 0...25 mA Input	Analog 0...100 kΩ Input	IO_ADC_06
P130	Analog 0...5 V Input	Analog 0...25 mA Input	Analog 0...100 kΩ Input	IO_ADC_07

4.9.2 Functional Description

8 multipurpose analog inputs with 12-bit resolution.

The inputs can be configured to 3 different operation modes individually by software:

- Analog Voltage Input: 8 x 0 to 5 V, ratiometric or with absolute reference
- Analog Current Input: 8 x 0 to 25 mA, analog current loop sensors
- Analog Resistance Input: 8 x 0 to 100 kΩ

All inputs are short-circuit protected, independent of application software (included in low-level driver software). Each input is provided with a first-order low pass filter with 3 ms time constant, allowing 2 ms sample rate.

See also section *ADC - Analog to Digital Converter driver* of the API documentation [38].

4.9.3 Maximum Ratings

T_{ECU} = -40. . . +125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
V_{in}	Input voltage under overload conditions	1	-1	+33	V

Note 1 Due to thermal reasons only one of the 8 inputs may be shorted to 33 V at the same time. A connection to any supply voltage higher than 5 V is not allowed for normal operation.

4.9.4 Analog Voltage Input

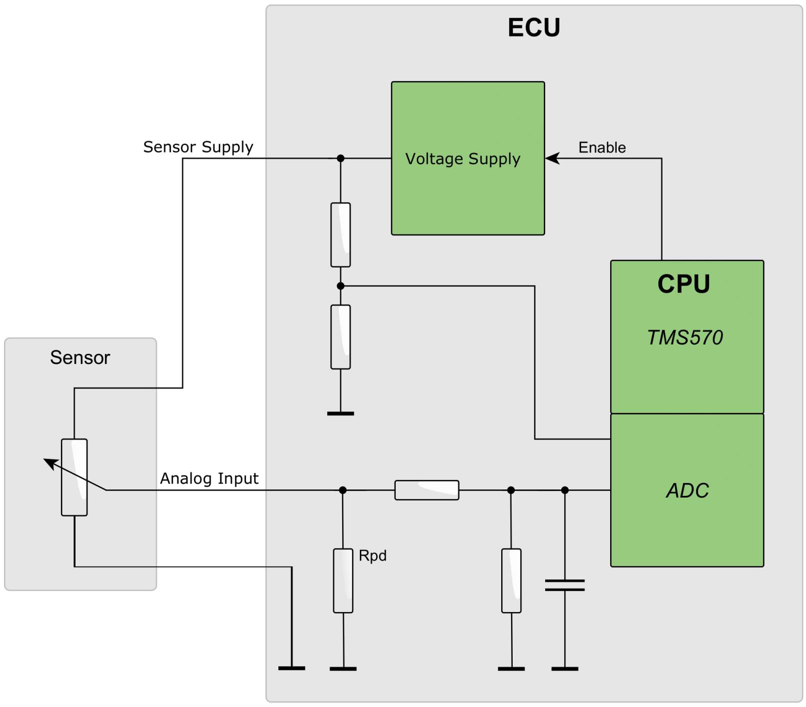

Figure 4.14: Analog voltage input (ratiometric)

Absolute vs. ratiometric voltage measurement:

Many sensor types are available in absolute or ratiometric measurement variant.

- **Absolute**: The sensor output voltage is a fixed value and directly corresponds to a physical value. For example, 2.5 V corresponds to 1 bar. Any tolerance in the reference voltage of the sensor and the ECU generates additional measurement inaccuracy.
- **Ratiometric**: The sensor output voltage is a fixed percentage of the sensor supply, the ratio corresponds to a physical value. For example, 50 % corresponds to 1 bar (or 2.5 V if the sensor supply is exactly 5.00 V). Any tolerance in the reference voltage of the sensor and the ECU is completely compensated and will not generate additional measurement inaccuracy.

Due to the described behavior, it is generally recommended to use ratiometric sensors.

Absolute or ratiometric function selection is done by software for each input individually.

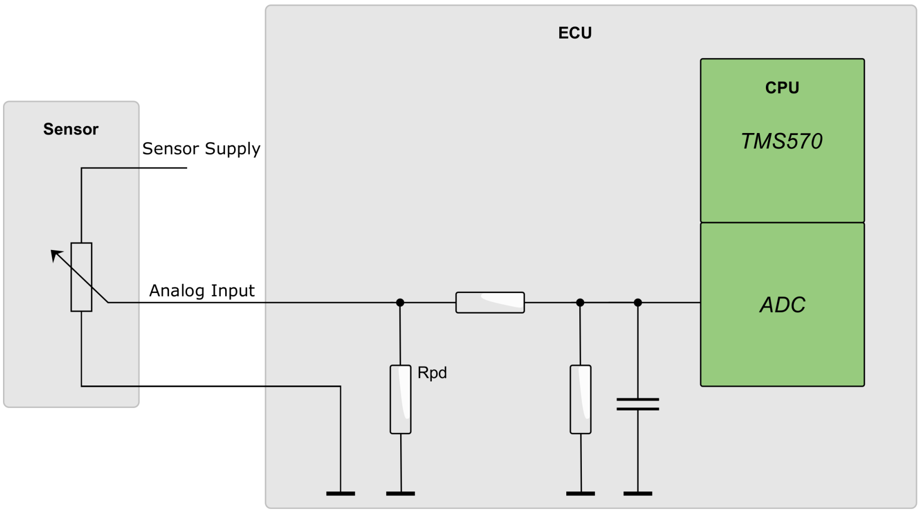

Figure 4.15: Analog voltage input

4.9.4.1 Characteristics of 5 V Input (Ratiometric)

T_{ECU} = -40. . . +125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
C_{in}	Pin input capacitance		8	12	nF
R_{pd}	Pull-down resistor		98	107	kΩ
τ_{in}	Input low pass filter		2.2	3.8	ms
V_{nom}	Nominal input voltage range		0	5	V
V_{In}	Input voltage range	1	0.2	4.8	V
V_{tol-0}	Zero reading error	4,5	-15	+15	mV
V_{tol-0}	Zero reading error	2,4,5	-10	+10	mV
V_{tol-p}	Proportional error	4,5	-0.2	+0.2	%
V_{tol-p}	Proportional error	2,4,5	-0.2	+0.2	%
LSB	Nominal value of 1 LSB		-	1.2207	mV

4.9.4.2 Characteristics of 5 V Input (Absolute)

T_{ECU} = -40. . . +125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
C_{in}	Pin input capacitance		8	12	nF
R_{pd}	Pull-down resistor		98	107	kΩ
τ_{in}	Input low pass filter		2.2	3.8	ms
V_{nom}	Nominal input voltage range		0	5	V
V_{in}	Input voltage range	1	0.2	4.8	V
V_{tol-0}	Zero reading error	3,5	-15	+15	mV
V_{tol-0}	Zero reading error	2,3,5	-10	+10	mV
V_{tol-p}	Proportional error	3,5	-0.8	+0.8	%
V_{tol-p}	Proportional error	2,3,5	-0.6	+0.6	%
LSB	Nominal value of 1 LSB		-	1.2207	mV

4.9.4.3 Characteristics of 5 V Digital Input

T_{ECU} = -40. . . +125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
C_{in}	Pin input capacitance		8	12	nF
R_{pu}	Pull-up resistor		4.8	5	kΩ
V_{pu}	Pull-up voltage		4.9	5.1	V
τ_{in}	Input low pass filter		2.2	3.8	ms
V_{il}	Input voltage for low level		0	●	V
V_{ih}	Input voltage for high level		●	5	V
LSB	Nominal value of 1 LSB		-	1.2207	mV

Note 1 For full accuracy

Note 2 T_{ECU} = -40. . . +85 °C

Note 3 Absolute measurement. This includes the conversion error of the HY-TTC 500 only. For the calculation of the total measurement error, it is necessary to sum the error of HY-TTC 500 and the absolute sensor error (measurement tolerance plus tolerance of external sensor reference).

Note 4 Ratiometric mode. This includes the conversion error of the HY-TTC 500 and the sensor supply error. For the calculation of the total measurement error, the error of HY-TTC 500 and the error of the ratiometric sensor (measurement tolerance) must be added.

Note ● Configuration by application software

Note 5 The total measurement error is the sum of zero reading error and the proportional error.

4.9.5 Analog Current Input

Analog input for 0...25mA sensor measurement.

Due to the wider measurement range of the input compared to the output range of popular sensors with 4...20 mA, short to GND, short to BAT+ and cable defects can be easily detected.

In case of an overload situation, the pin is switched to a high impedance state. The protection mechanism tries re-enabling the output 10 times per drive cycle.

During power down (`Terminal 15` off), the ECU does not disconnect the current sensor input. It is not recommended to supply the sensors permanently in order to prevent battery discharge.

TTControl GmbH recommends one of the following 2 options:

1. **Option 1**: Use a digital output for supplying the sensor. When the device is switched off, the ECU can perform an application-controlled shutdown, e. g., in order to operate a cooling fan to cool down an engine until the temperature is low enough or to store data in the non-volatile memory of the ECU. If the application controlled shut-down is finished, the ECU switches off and consumes less than 1 mA of battery current (including sensors).
2. **Option 2**: `Terminal 15` is used to supply the current loop sensor directly. Note that `Terminal 15` is often used to switch relays or other inductive loads directly. This may cause transients in excess of ±50 V, for which the sensor must be specified.

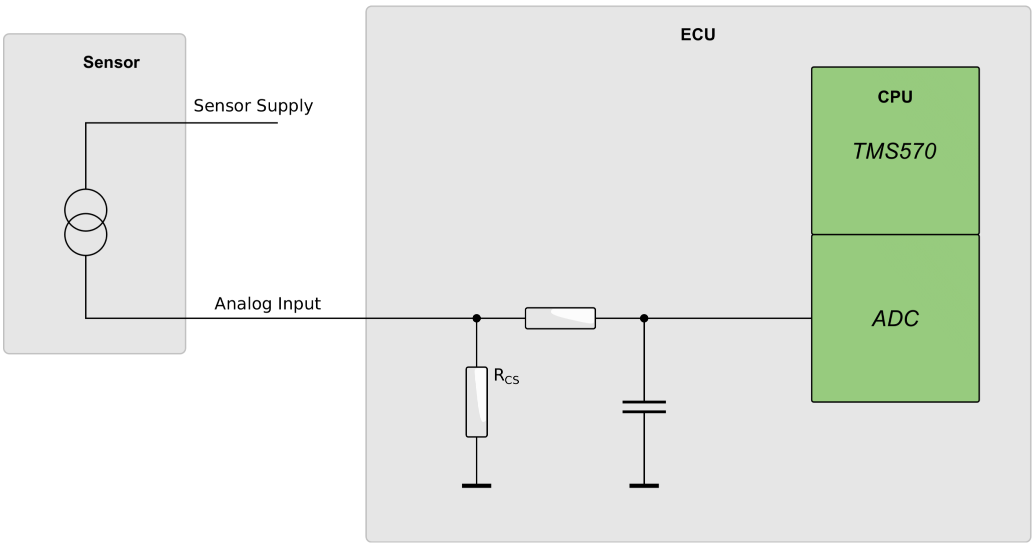

Figure 4.16: Analog current input

4.9.5.1 Characteristics of Analog Current Input

T_{ECU} = -40. . . +125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
C_{in}	Pin input capacitance		8	12	nF
R_{CS}	Current sense resistor	1	177	185	Ω
τ_{in}	Input low pass filter		2.2	3.8	ms
I_{in}	Input current range		0	25	mA
LSB	Nominal value of 1 LSB		-	6.78	µA
$I_{tol\text{-}0}$	Zero reading error	3	-70	+70	µA
$I_{tol\text{-}0}$	Zero reading error	2,3	-40	+40	µA
$I_{tol\text{-}p}$	Proportional error	3	-2.5	+2.5	%
$I_{tol\text{-}p}$	Proportional error	2,3	-2.0	+2.0	%

Note 1 This is the load resistor value for the current loop sensor.
Note 2 T_{ECU} = -40. . . +85 °C
Note 3 The total measurement error is the sum of zero reading error and the proportional error.

4.9.6 Analog Resistance Input

Input for 0. . . 100kΩ resistance sensor measurement.

Resistive sensors are for example NTC or PTC resistors for temperature measurement.

The actual resistor value of the sensor is computed from the measured input voltage together with the known reference resistor value. Be aware that this measurement setup has the highest accuracy and resolution if the sensors resistance is in the magnitude of the reference resistors value.

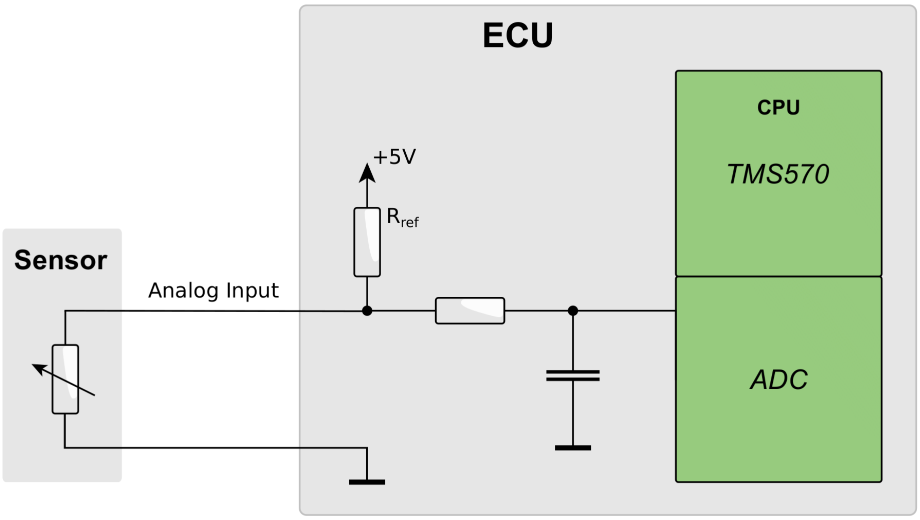

Figure 4.17: Analog resistance input

The resistance mode may also be used as digital input with switches connected to ground, see Figure 4.19 on the following page. The use of switches to BAT+ is not allowed.

To enhance the diagnostic coverage, use switches of type Namur. With a Namur-type switch sensor, short to ground, short to BAT+ and cable defects can be easily detected.

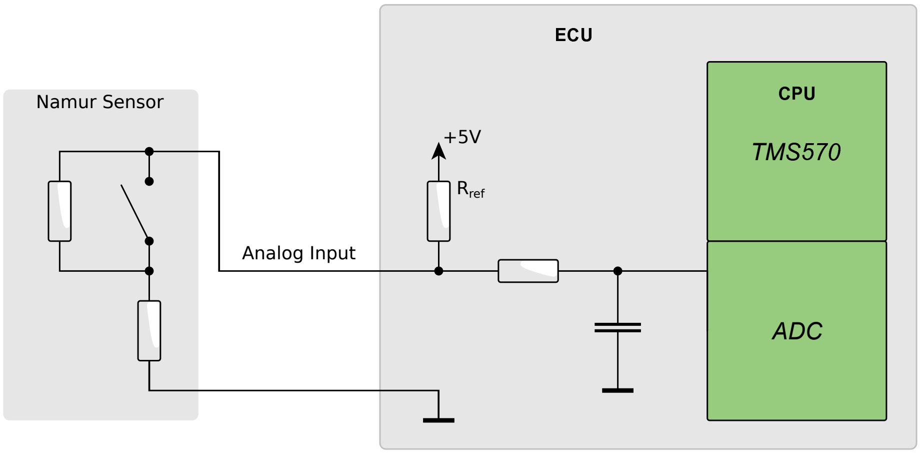

Figure 4.18: Namur type sensor (only for switches to ground)

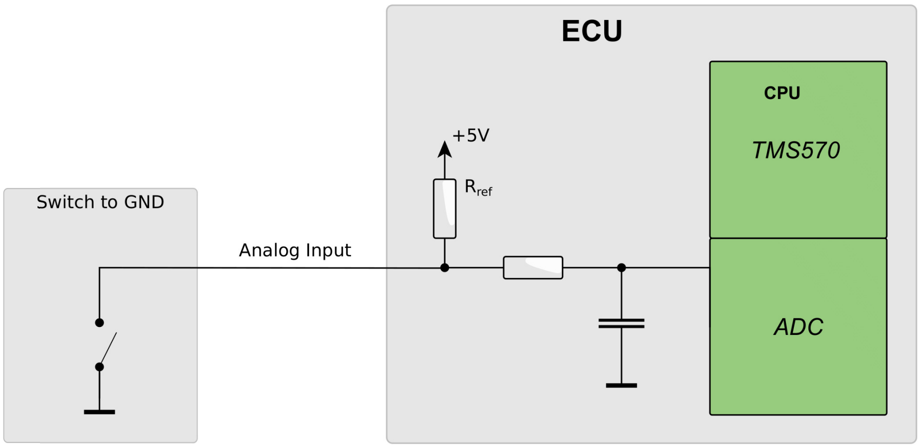

Figure 4.19: Switch input (only for switches to ground)

4.9.6.1 Characteristics of Analog Resistance Input

Resistance sensors are, e. g., PTC resistors for temperature measurement.

T_{ECU} = -40. . . +125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
C_{in}	Pin input capacitance		8	12	nF
R_{ref}	Reference resistor		4831	4929	Ω
τ_{in}	Input low pass filter		2.2	3.8	ms
R_{ext_range}	Resistance measurement range		0	100	kΩ

T_{ECU} = -40. . . +85 °C

Symbol	Parameter	Note	Min.	Max.	Unit
$R_{tol\text{-}m}$	Measurement tolerance for 0. . . 99 Ω		-5	+5	Ω
$R_{tol\text{-}m}$	Measurement tolerance for 100 Ω		-5	+5	%
$R_{tol\text{-}m}$	Measurement tolerance for 200 Ω		-4	+4	%
$R_{tol\text{-}m}$	Measurement tolerance for 500 Ω		-2.5	+2.5	%
$R_{tol\text{-}m}$	Measurement tolerance for 1 k. . . 20 kΩ		-2	+2	%
$R_{tol\text{-}m}$	Measurement tolerance for 50 kΩ		-3	+3	%
$R_{tol\text{-}m}$	Measurement tolerance for 100 kΩ		-4	+4	%

T_{ECU} = +85. . . +125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
$R_{tol\text{-}m}$	Measurement tolerance for 0. . . 99 Ω		-10	+10	Ω
$R_{tol\text{-}m}$	Measurement tolerance for 100 Ω		-10	+10	%
$R_{tol\text{-}m}$	Measurement tolerance for 200 Ω		-6	+6	%
$R_{tol\text{-}m}$	Measurement tolerance for 500 Ω. . . 20 kΩ		-3	+3	%
$R_{tol\text{-}m}$	Measurement tolerance for 50 kΩ		-4	+4	%
$R_{tol\text{-}m}$	Measurement tolerance for 100 kΩ		-5	+5	%

The resistance measurement tolerance is given at specific sensor resistance value. Any value in between needs to be linear interpolated.

4.10 Analog Input 2 Modes

4.10.1 Pinout

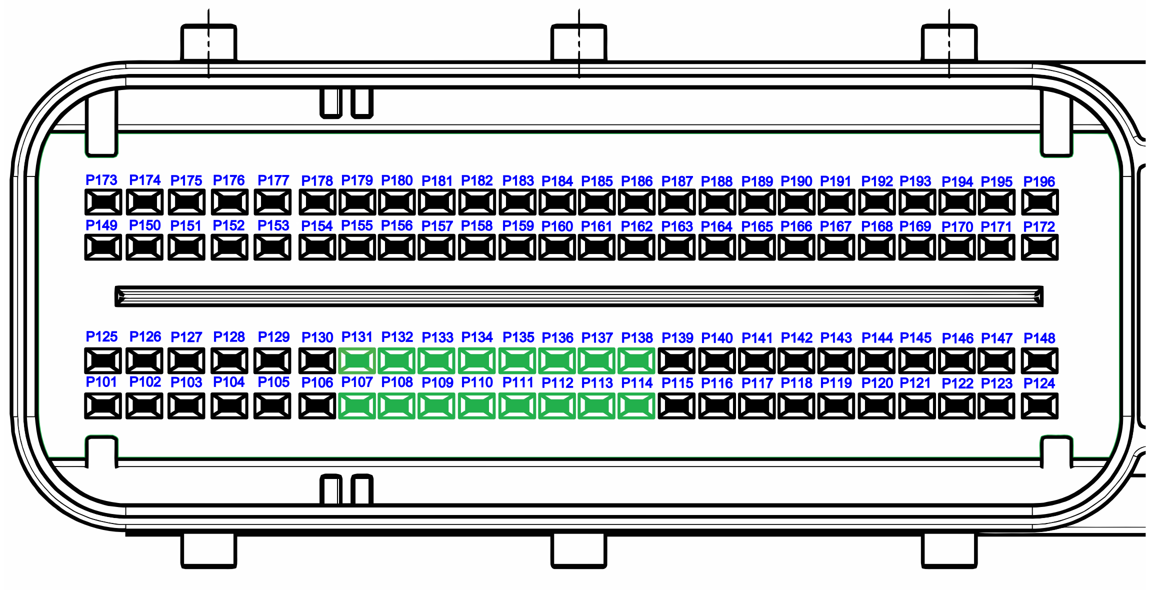

Figure 4.20: Pinout of analog input 2 mode

Pin No.	Function 1	Function 2	SW-define
P107	Analog 0...5 V, 0...10 V Input	Analog 0...25 mA Input	IO_ADC_08
P131	Analog 0...5 V, 0...10 V Input	Analog 0...25 mA Input	IO_ADC_09
P108	Analog 0...5 V, 0...10 V Input	Analog 0...25 mA Input	IO_ADC_10
P132	Analog 0...5 V, 0...10 V Input	Analog 0...25 mA Input	IO_ADC_11
P109	Analog 0...5 V, 0...10 V Input	Analog 0...25 mA Input	IO_ADC_12
P133	Analog 0...5 V, 0...10 V Input	Analog 0...25 mA Input	IO_ADC_13
P110	Analog 0...5 V, 0...10 V Input	Analog 0...25 mA Input	IO_ADC_14
P134	Analog 0...5 V, 0...10 V Input	Analog 0...25 mA Input	IO_ADC_15
P111	Analog 0...5 V, 0...32 V Input	Analog 0...25 mA Input	IO_ADC_16
P135	Analog 0...5 V, 0...32 V Input	Analog 0...25 mA Input	IO_ADC_17
P112	Analog 0...5 V, 0...32 V Input	Analog 0...25 mA Input	IO_ADC_18
P136	Analog 0...5 V, 0...32 V Input	Analog 0...25 mA Input	IO_ADC_19
P113	Analog 0...5 V, 0...32 V Input	Analog 0...25 mA Input	IO_ADC_20
P137	Analog 0...5 V, 0...32 V Input	Analog 0...25 mA Input	IO_ADC_21
P114	Analog 0...5 V, 0...32 V Input	Analog 0...25 mA Input	IO_ADC_22
P138	Analog 0...5 V, 0...32 V Input	Analog 0...25 mA Input	IO_ADC_23

4.10.2 Functional Description

16 multipurpose analog inputs with 12-bit resolution.

The inputs can be configured to 3 different operation modes individually by software:

- Analog Voltage Input: 8 x 0 to 5 V/ 0 to 10 V, ratiometric or with absolute reference.
- Analog Voltage Input: 8 x 0 to 5 V/ 0 to 32 V, ratiometric or with absolute reference.
- Analog Current Input: 16 x 0 to 25 mA, analog current loop sensors.

All inputs are short-circuit protected, independent of application software (included in low-level driver software). Each input is provided with a first-order low pass filter with 3 ms time constant, and converted with 2 ms sample rate.

See also section *ADC - Analog to Digital Converter driver* of the API documentation [38].

4.10.3 Maximum Ratings

T_{ECU} = -40. . . +125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
V_{in}	Input voltage	1	-1	+33	V

Note 1 Due to thermal reasons only one of 8 inputs (except 32 V setting) may be shorted to 33 V at the same time. A connection to any supply voltage higher than the configured voltage setting (5 V, 10 V or 32 V) is not allowed for normal operation.

4.10.4 Analog Voltage Input

See Section 4.9.4 on page 90 for more information.

4.10.4.1 Characteristics of 5 V Input (Ratiometric)

Ratiometric mode: This includes the conversion error of the HY-TTC 500 and the sensor supply error. For the calculation of the total measurement error, the error of HY-TTC 500 and the error of the ratiometric sensor (measurement tolerance) must be added.

T_{ECU} = -40. . . +125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
C_{in}	Pin input capacitance		8	12	nF
R_{pd}	Pull-down resistor		98	107	kΩ
τ_{in}	Input low pass filter		2.2	3.8	ms
V_{nom}	Nominal input voltage range		0	5	V
V_{in}	Input voltage range	1	0.2	4.8	V
$V_{tol\text{-}0}$	Zero reading error	4,5	-15	+15	mV
$V_{tol\text{-}0}$	Zero reading error	2,4,5	-10	+10	mV
$V_{tol\text{-}p}$	Proportional error	4,5	-0.2	+0.2	%
$V_{tol\text{-}p}$	Proportional error	2,4,5	-0.2	+0.2	%
LSB	Nominal value of 1 LSB		-	1.2207	mV

4.10.4.2 Characteristics of 5 V Input (Absolute)

Absolute measurement: This includes the conversion error of the HY-TTC 500 only. For the calculation of the total measurement error the error of the HY-TTC 500 and the error of the absolute sensor (measurement tolerance plus tolerance of external sensor reference) must be added.

T_{ECU} = -40. . . +125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
C_{in}	Pin input capacitance		8	12	nF
R_{pd}	Pull-down resistor		98	107	kΩ
τ_{in}	Input low pass filter		2.2	3.8	ms
V_{nom}	Nominal input voltage range		0	5	V
V_{in}	Input voltage range	1	0.2	4.8	V
$V_{tol\text{-}0}$	Zero reading error	3,5	-15	+15	mV
$V_{tol\text{-}0}$	Zero reading error	2,3,5	-10	+10	mV
$V_{tol\text{-}p}$	Proportional error	3,5	-0.8	+0.8	%
$V_{tol\text{-}p}$	Proportional error	2,3,5	-0.6	+0.6	%
LSB	Nominal value of 1 LSB		-	1.2207	mV

4.10.4.3 Characteristics of 10 V Input (Absolute)

T_{ECU} = -40. . . +125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
C_{in}	Pin input capacitance		8	12	nF
R_{pd}	Pull-down resistor		19	22.5	kΩ
τ_{in}	Input low pass filter		2.2	3.8	ms
V_{nom}	Nominal input voltage range		0	10	V
V_{in}	Input voltage range	1	0.2	10	V
V_{tol-0}	Zero reading error	3,5	-18	+18	mV
V_{tol-0}	Zero reading error	2,3,5	-13	+13	mV
V_{tol-p}	Proportional error	3,5	-1.8	+1.8	%
V_{tol-p}	Proportional error	2,3,5	-1.6	+1.6	%
LSB	Nominal value of 1 LSB		-	2.57	mV

4.10.4.4 Characteristics of 32 V Input (Absolute)

T_{ECU} = -40. . . +125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
C_{in}	Pin input capacitance		8	12	nF
R_{pd}	Pull-down resistor		14.7	15.3	kΩ
τ_{in}	Input low pass filter		2.2	3.8	ms
V_{nom}	Nominal input voltage range		0	32	V
V_{in}	Input voltage range	1	0.2	32	V
V_{tol-0}	Zero reading error	5	-50	+50	mV
V_{tol-0}	Zero reading error	2,5	-45	+45	mV
V_{tol-p}	Proportional error	3,5	-4	+4	%
V_{tol-p}	Proportional error	2,5	-3	+3	%
LSB	Nominal value of 1 LSB		-	8.59	mV

4.10.4.5 Characteristics of 32 V Digital Input

T_{ECU} = -40...+125 ℃

Symbol	Parameter	Note	Min.	Max.	Unit
C_{in}	Pin input capacitance		8	12	nF
R_{pd}	Pull-down resistor		14.7	15.3	kΩ
τ_{in}	Input low pass filter		2.2	3.8	ms
V_{il}	Input voltage for low level		0	●	V
V_{ih}	Input voltage for high level		●	32	V
LSB	Nominal value of 1 LSB		-	8.59	mV

Note 1 For full accuracy

Note 2 T_{ECU} = -40...+85 ℃

Note 3 Absolute measurement. This includes the conversion error of the HY-TTC 500 only. For the calculation of the total measurement error, it is necessary to sum the error of HY-TTC 500 and the absolute sensor error (measurement tolerance plus tolerance of external sensor reference).

Note 4 Ratiometric mode. This includes the conversion error of the HY-TTC 500 and the sensor supply error. For the calculation of the total measurement error, the error of HY-TTC 500 and the error of the ratiometric sensor (measurement tolerance) must be added.

Note 5 The total measurement error is the sum of zero reading error and the proportional error.

Note ● Configuration by application software

4.10.5 Analog Current Input

See Section 4.9.5 on page 93 for more information.

4.10.5.1 Characteristics of Analog Current Input

T_{ECU} = -40...+125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
C_{in}	Pin input capacitance		8	12	nF
R_{CS}	Current sense resistor	1	177	185	Ω
τ_{in}	Input low pass filter		2.2	3.8	ms
I_{in}	Input current range		0	25	mA
LSB	Nominal value of 1 LSB		-	6.78	µA
$I_{tol\text{-}0}$	Zero reading error	3	-70	+70	µA
$I_{tol\text{-}0}$	Zero reading error	2,3	-40	+40	µA
$I_{tol\text{-}p}$	Proportional error	3	-2.5	+2.5	%
$I_{tol\text{-}p}$	Proportional error	2,3	-2.0	+2.0	%

Note 1 This is the load resistor value for the current loop sensor.
Note 2 T_{ECU} = -40...+85 °C
Note 3 The total measurement error is the sum of zero reading error and the proportional error.

4.11 Timer Inputs

4.11.1 Pinout

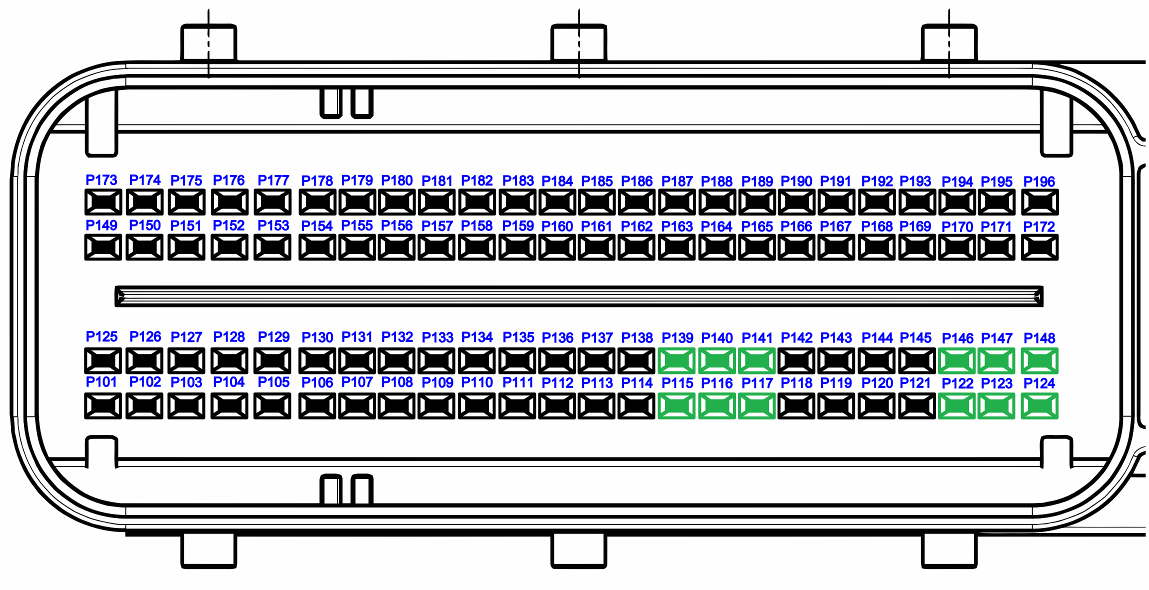

Figure 4.21: Pinout of timer input

Pin No.	Function	SW-define
P115	Timer Input	IO_PWD_00
P139	Timer Input	IO_PWD_01
P116	Timer Input	IO_PWD_02
P140	Timer Input	IO_PWD_03
P117	Timer Input	IO_PWD_04
P141	Timer Input	IO_PWD_05
P122	Timer Input	IO_PWD_06
P146	Timer Input	IO_PWD_07
P123	Timer Input	IO_PWD_08
P147	Timer Input	IO_PWD_09
P124	Timer Input	IO_PWD_10
P148	Timer Input	IO_PWD_11

4.11.2 Functional Description

12 digital inputs with timer function to process input signals such as frequency (rotational speed), pulse count and quadrature decoding (incremental length measurement), PWM etc.

6 out of 12 inputs can be alternatively used as digital (7/14 mA) current loop speed sensors.

The inputs can be individually configured by software with a pull-up/pull-down resistor to adapt them to different sensor types,
The timer input can be used as an analog voltage input as well. For diagnosis, it is possible to measure the analog voltage and frequency at the same channel at the same time.

See also sections *PWD - Pulse Width Decode and digital timer input driver* and *DIO - Driver for digital inputs and outputs* of the API documentation [38].

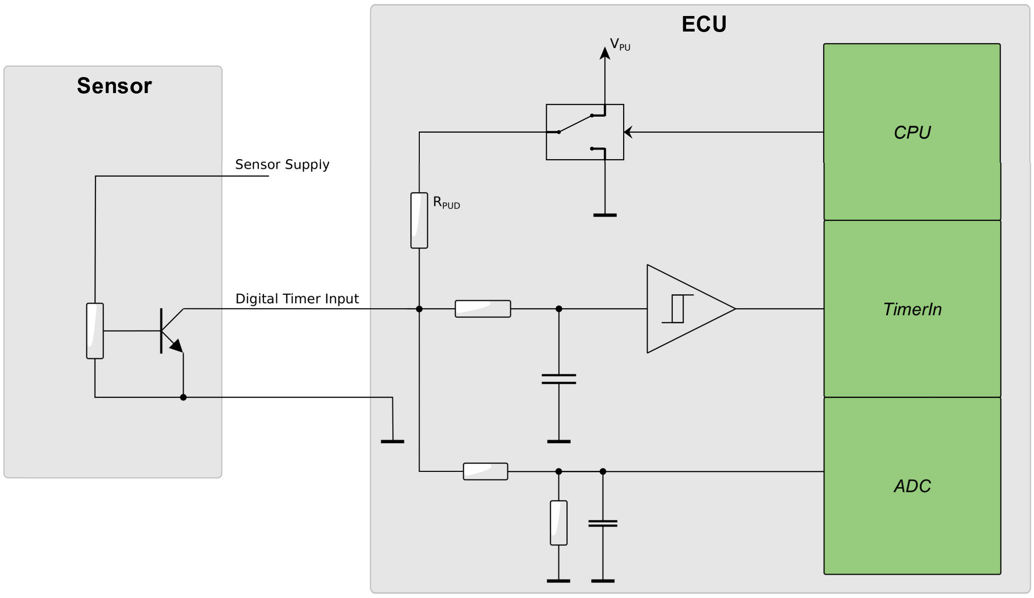

Figure 4.22: Digital input for frequency/timing measurement with NPN-type 2-pole sensor

4.11.3 Maximum Ratings

T_{ECU} = -40. . . +125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
V_{in}	Input voltage under overload conditions		-1	+33	V

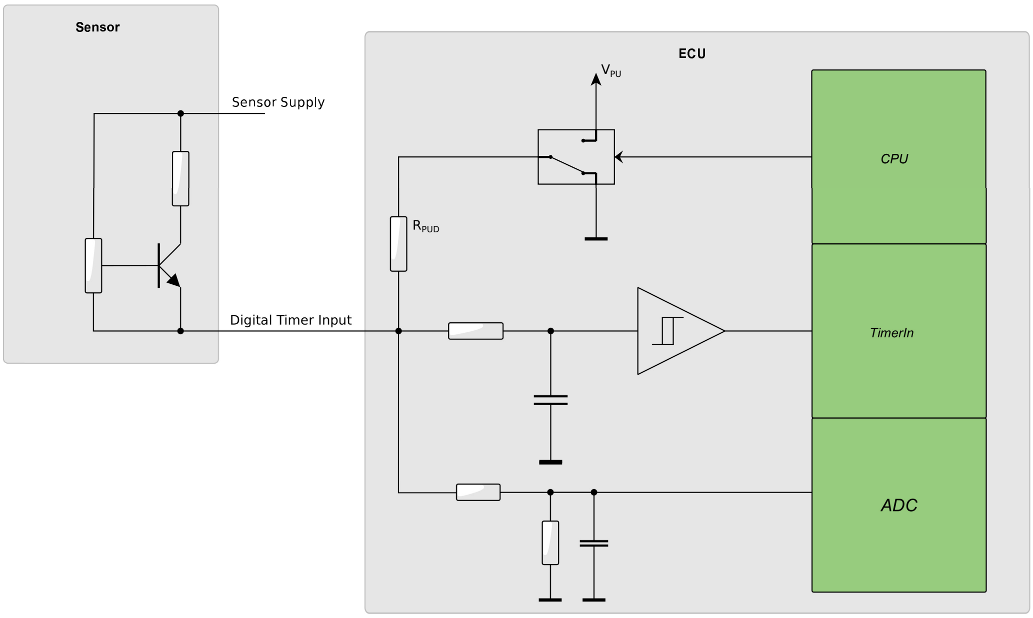

Figure 4.23: Digital input for frequency/timing measurement with PNP-type 2-pole sensor

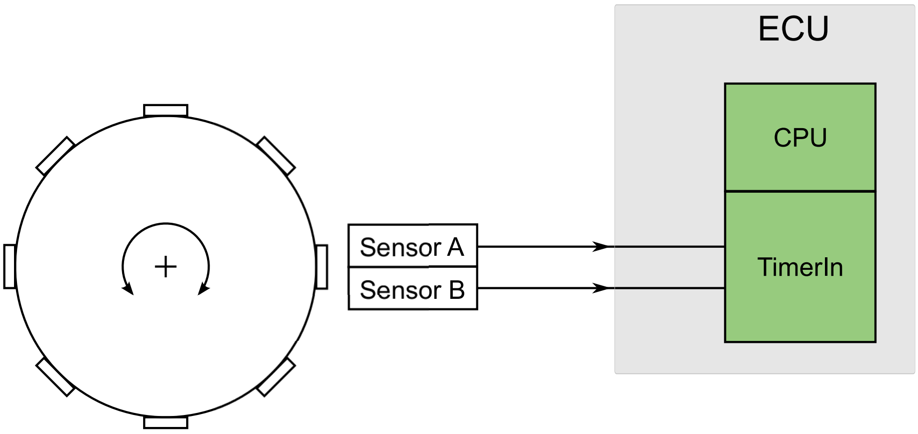

Figure 4.24: Digital input pair for encoder and direction

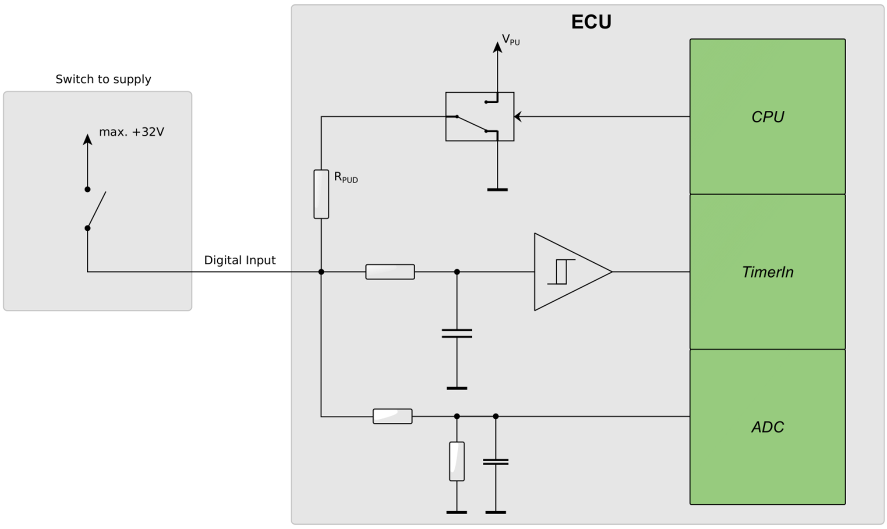

Figure 4.25: Digital input for switch connected to (battery) supply voltage

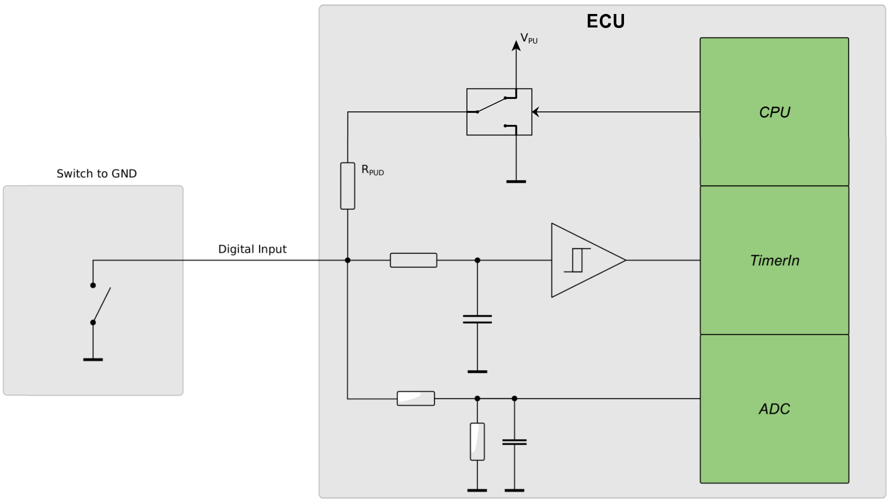

Figure 4.26: Digital input for switch connected to ground

4.11.4 Timer and Current Loop Inputs

4.11.4.1 Characteristics of Timer Input

T_{ECU} = -40. . . +125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
R_{pud}	Pull-up/pull-down resistor		7.5	10	kΩ
V_{pu}	Pull-up voltage (open load)	1	4.25	4.8	V
τ_{in}	Input low pass filter		1.4	3.4	µs
F_{max}	Maximum input frequency range			20	kHz
F_{min}	Minimum input frequency		0.1		Hz
τ_{min}	Minimum on/off time to be measured by timer unit		20		µs
V_{il}	Input voltage for low level		0	2	V
V_{ih}	Input voltage for high level		3	32	V
t_{res}	Timer resolution	2	2	2	µs
t_{res}	Timer resolution	3	0.5	0.5	µs

Note 1 This is the input voltage with pull-up setting, without the sensor connected.
Note 2 IO_PWD_00 - IO_PWD_05
Note 3 IO_PWD_06 - IO_PWD_11

4.11.4.2 Characteristics of Current Loop Input

The timer input IO_PWD_00 to IO_PWD_05 can be alternatively also used as digital (7/14 mA) current loop sensor inputs. See Figure 4.27 on the next page.

During power down (`Terminal 15` off), the ECU does not disconnect the timer and current loop sensor inputs. It is not recommended to supply the sensors permanently in order to prevent battery discharge.

TTControl GmbH recommends one of the following 2 options:

1. **Option 1**: Use a digital output for supplying the sensor. When the device is switched off, the ECU can perform an application-controlled shutdown, e. g., in order to operate a cooling fan to cool down an engine until the temperature is low enough or to store data in the non-volatile memory of the ECU. If the application controlled shut-down is finished, the ECU switches off and consumes less than 1 mA of battery current (including sensors).
2. **Option 2**: `Terminal 15` is used to supply the current loop sensor directly. Note that `Terminal 15` is often used to switch relays or other inductive loads directly. This may cause transients in excess of ±50 V, for which the sensor must be specified.

T_{ECU} = -40. . . +125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
R_{pdc}	Pull-down resistor (current loop config.)	1	88	93	Ω
R_{pud}	Pull-up/pull-down resistor		7.5	10	kΩ
V_{pu}	Pull-up voltage (open load)		4.25	4.8	V
τ_{in}	Input low pass filter		1.4	1.6	µs
F_{max}	Maximum input frequency range			20	kHz
F_{min}	Minimum input frequency		0.1		Hz
τ_{min}	Minimum on/off time to be measured by timer unit		20		µs
I_{il}	Input current for low level(current loop config.)		4	8.5	mA
I_{ih}	Input current for high level (current loop config.)		11	20	mA
$I_{il\ SRC}$	Input current (7/14 mA) sensor SRC too low	2	0	4	mA
$I_{ih\ SRC}$	Input current (7/14 mA) sensor SRC too high	3	20		mA
t_{res}	Timer resolution		2	2	µs

Note 1 With software setting for digital (7/14 mA) current loop sensor inputs (ABS-type sensors).

Note 2 Fault detection window for defect digital (7/14 mA) current loop sensor inputs with too low current.

Note 3 Fault detection window for defect digital (7/14 mA) current loop sensor inputs with too high current. If the current exceeds the maximum input current, then overload protection gets active.

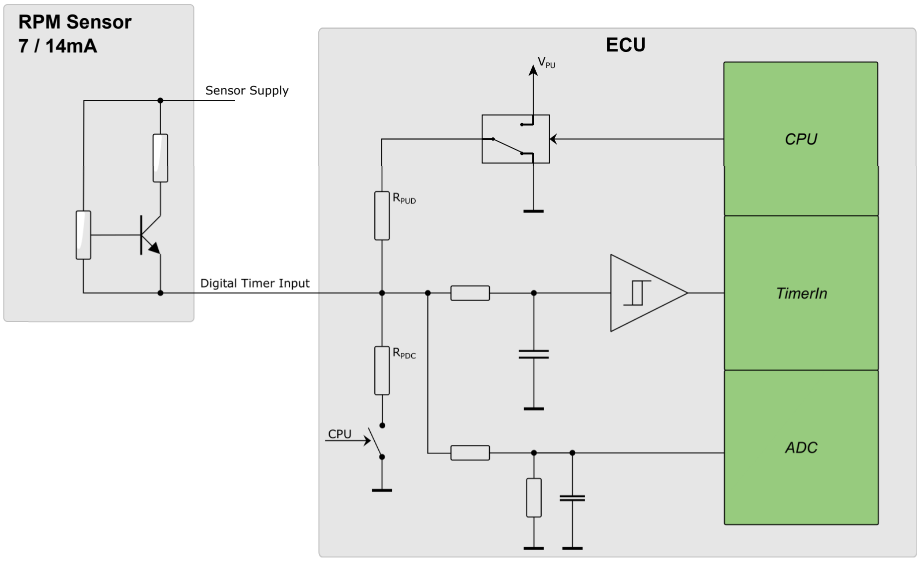

Figure 4.27: Digital input for frequency measurement with ABS-type 7/14 mA, 2 pole sensor

4.11.5 Analog and Digital Inputs

4.11.5.1 Characteristics of Analog Voltage Input

T_{ECU} = -40. . . +125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
	Resolution		12	12	bit
R_{pud}	Pull-up/pull-down resistor		7.5	10	kΩ
V_{pu}	Pull-up voltage (open load)	2	4.25	4.8	V
V_{in}	Input voltage range		0	32	V
τ_{in}	Input low pass filter (analog path)		8	12	ms
$V_{tol\text{-}0}$	Zero reading error	3	-55	+55	mV
$V_{tol\text{-}0}$	Zero reading error	1,3	-50	+50	mV
$V_{tol\text{-}p}$	Proportional error	3	-4	+4	%
$V_{tol\text{-}p}$	Proportional error	1,3	-3	+3	%
LSB	Nominal value of 1 LSB		-	8.00	mV

4.11.5.2 Characteristics of Digital Input

T_{ECU} = -40. . . +125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
R_{pud}	Pull-up/pull-down resistor		7.5	10	kΩ
V_{pu}	Pull-up voltage (open load)	2	4.25	4.8	V
V_{in}	Input voltage range		0	32	V
τ_{in}	Input low pass filter		2.8	3.2	µs
V_{il}	Input voltage for low level		0	2	V
V_{ih}	Input voltage for high level		3	32	V

Note 1 T_{ECU} = -40. . . +85 °C

Note 2 This is the input voltage with pull-up setting, without the sensor connected.

Note 3 The total measurement error is the sum of zero reading error and the proportional error.

4.12 High-Side PWM Outputs

4.12.1 Pinout

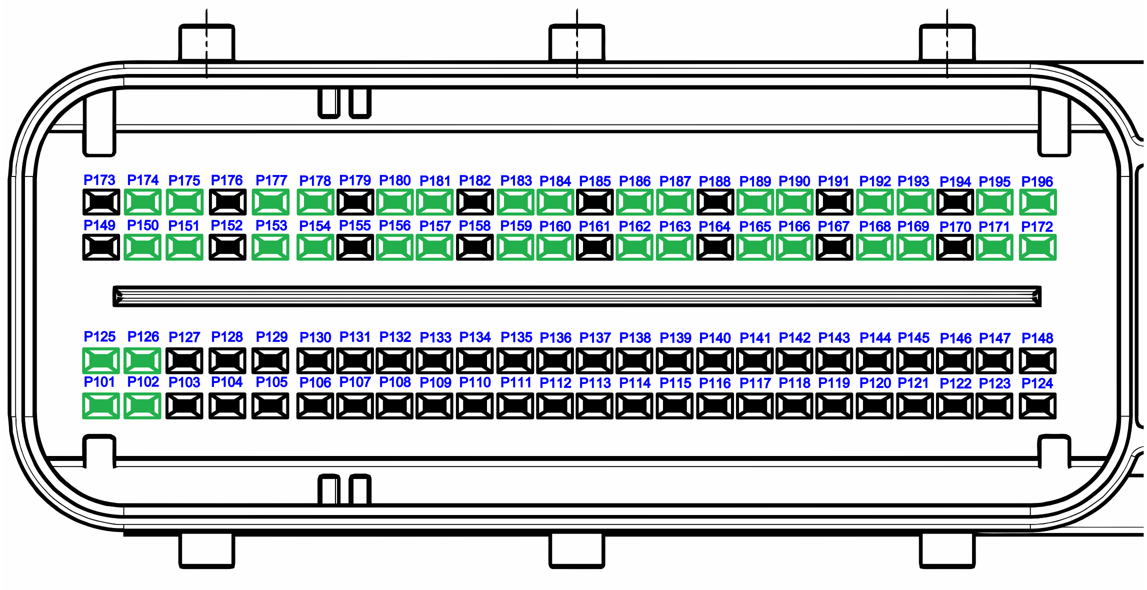

Figure 4.28: Pinout of high-side PWM outputs

The following table depicts the high-side PWM outputs together with their corresponding external and secondary shut-off group/paths and power stages.

Pin No.	Function	SW-define	Ext./Second. Shut-off	Power stage
P153	High-Side PWM Output	IO_PWM_00	A	a
P177	High-Side PWM Output	IO_PWM_01	A	a
P156	High-Side PWM Output	IO_PWM_02	A	b
P180	High-Side PWM Output	IO_PWM_03	A	b
P159	High-Side PWM Output	IO_PWM_04	A	c
P183	High-Side PWM Output	IO_PWM_05	A	c
P186	High-Side PWM Output	IO_PWM_06	A	d
P162	High-Side PWM Output	IO_PWM_07	A	d
P189	High-Side PWM Output	IO_PWM_08	A	e
P165	High-Side PWM Output	IO_PWM_09	A	e
P192	High-Side PWM Output	IO_PWM_10	A	f
P168	High-Side PWM Output	IO_PWM_11	A	f
P195	High-Side PWM Output	IO_PWM_12	A	g
P171	High-Side PWM Output	IO_PWM_13	A	g
P154	High-Side PWM Output	IO_PWM_14	B	h
P178	High-Side PWM Output	IO_PWM_15	B	h
P157	High-Side PWM Output	IO_PWM_16	B	i
P181	High-Side PWM Output	IO_PWM_17	B	i

Pin No.	Function	SW-define	Ext./Second. Shut-off	Power stage
P160	High-Side PWM Output	IO_PWM_18	B	j
P184	High-Side PWM Output	IO_PWM_19	B	j
P187	High-Side PWM Output	IO_PWM_20	B	k
P163	High-Side PWM Output	IO_PWM_21	B	k
P190	High-Side PWM Output	IO_PWM_22	B	l
P166	High-Side PWM Output	IO_PWM_23	B	l
P193	High-Side PWM Output	IO_PWM_24	B	m
P169	High-Side PWM Output	IO_PWM_25	B	m
P196	High-Side PWM Output	IO_PWM_26	B	n
P172	High-Side PWM Output	IO_PWM_27	B	n
P101	High-Side PWM Output	IO_PWM_28	C	o
P125	High-Side PWM Output	IO_PWM_29	C	o
P150	High-Side PWM Output	IO_PWM_30	C	p
P174	High-Side PWM Output	IO_PWM_31	C	p
P102	High-Side PWM Output	IO_PWM_32	C	q
P126	High-Side PWM Output	IO_PWM_33	C	q
P151	High-Side PWM Output	IO_PWM_34	C	r
P175	High-Side PWM Output	IO_PWM_35	C	r

4.12.2 Functional Description

Power output stages with freewheeling diodes for inductive loads with low-side connection.
The load current is controlled with PWM. For better accuracy and diagnostics, a current measurement/feedback loop is provided.

If an error is detected in a safety-critical system, the watchdog or the main CPU can disable the output stage (off state), triggered by the application software.

For diagnostic and safety reasons, the actual PWM output signal is looped back to a timer input, and the measured value is compared to the set value. For safety-critical applications, fast error detection is necessary. For this reason, a permanent PWM output is available, setting a minimum on/off time to 100/200 µs instead of 0 or 100 % duty cycle. This means, there is a reliable periodical state change of the output allowing permanent load monitoring that is independent of the operation point. So, even when the load is switched off, a short on the load can be detected.

TTControl GmbH recommends to use the maximum possible wire size (FLRY type) in case of maximum load current to reduce voltage drop and prevent overheating of the crimp contact.

See also section *PWM - Pulse Width Modulation driver* of the API documentation [38].

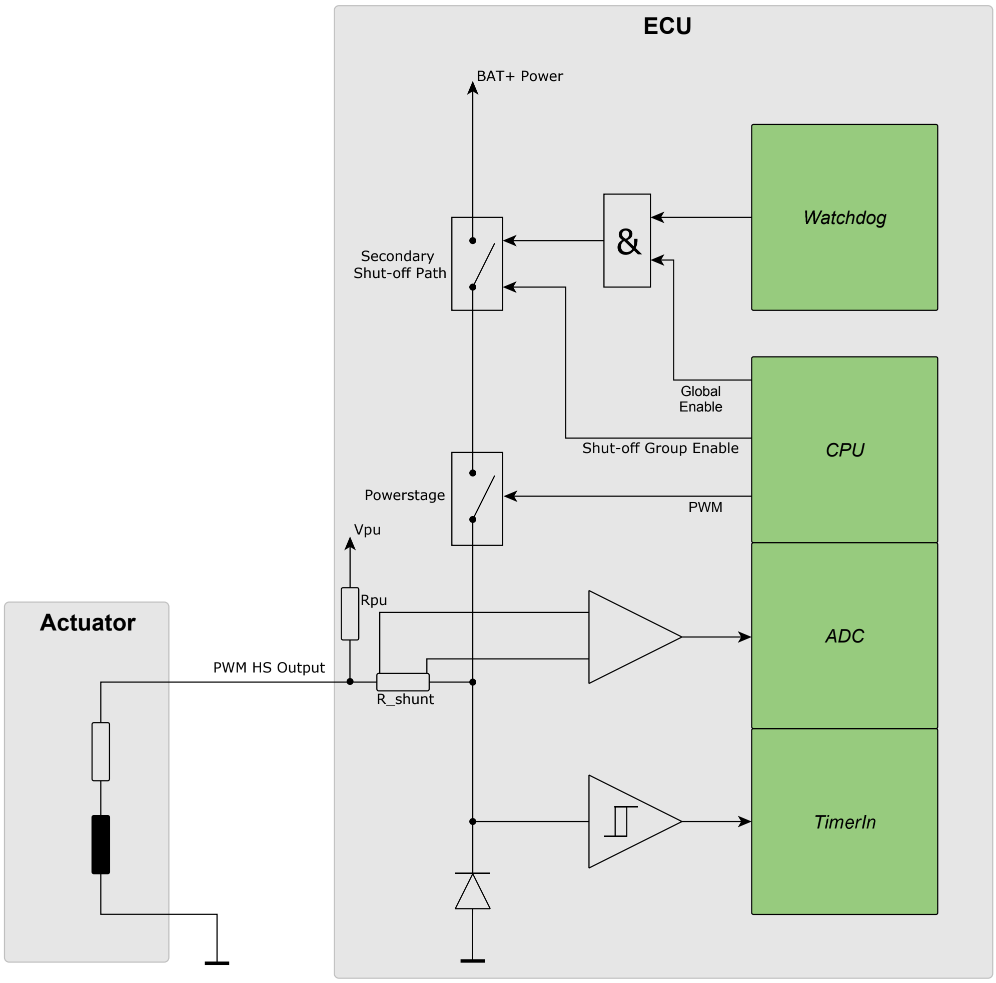

Figure 4.29: High-side output stage with PWM functionality

4.12.2.1 Power Stage Pairing

If outputs shall be used in parallel, always combine two channels from the same double-channel power stage and use the digital output mode. See Section 4.12.1 on page 112 for using the right power stage outputs in parallel.

Due to thermal limits, the resulting total load current of this output pair has to be de-rated by a factor of 0.85 (e.g. combining two 3 A outputs would result in a total load current of 3 A x 2 x 0.85 = 5.1 A). The application software has to make sure that both outputs are switched on at the same point in time, otherwise the over-current protection may trip.

For balanced current distribution through each of the pin pairs, the cable routing shall be symmetrical if pin-pairs or multiple pins shall be wired parallel to support higher load currents.

4.12.2.2 Mutual Exclusive Mode

The HY-TTC 500 uses double-channel high-side power stages. For load leveling it is a benefit if loads, which are switched on mutually exclusive (which means either load A, or load B is on, but not A and B at the same time), are connected to the same double-channel power stage. This reduces the thermal stress of the components. The power stage pairing is given in Section 4.12.1 on page 112.

For HS PWM output stage operating in 400 - 1000 Hz mode, the current limit is increased to 1 A if used in mutual exclusive mode, see Section 4.12.4.1 on the facing page.

4.12.3 Maximum Ratings

T_{ECU} = -40. . . +125 ℃

Symbol	Parameter	Note	Min.	Max.	Unit
V_{in}	Input voltage under overload conditions		-0.5	BAT+ Power +0.5	V

4.12.4 High-Side PWM Outputs

4.12.4.1 Characteristics of High-Side PWM Output

T_{ECU} = -40. . . +125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
C_{out}	Pin input capacitance		15	25	nF
R_{pu}	Pull-up resistor		4	5	kΩ
V_{pu}	Pull-up voltage (open load)		4.2	4.7	V
f_{PWM}	PWM frequency	1	50	1000	Hz
$T_{min\text{-}on}$	Minimum on time	2	100		µs
$T_{min\text{-}off}$	Minimum off time	2	200		µs
R_{on}	On-resistance			150	mΩ
I_{max}	Maximum load current (f = 50. . . 200 Hz)			3	A
I_{max}	Maximum load current (f = 50. . . 200 Hz)	3		4	A
I_{max}	Maximum load current (f = 200. . . 400 Hz)			1.2	A
I_{max}	Maximum load current (f = 400. . . 1000 Hz)			0.1	A
I_{max}	Maximum load current (f = 400. . . 1000 Hz)	4		1	A
I_{peak}	Peak load current limit	5		5.5	A
R_{load_min}	Minimum coil resistance (12 V)	6	2		Ω
R_{load_min}	Minimum coil resistance (24 V)	6	4		Ω
R_{load_max}	Maximum load resistance	7		1	kΩ

Note 1 Not all values for PWM frequency are possible, see section *PWM - Pulse Width Modulation driver* of the API documentation [38] for supported frequency values

Note 2 Instead of 0 % or 100 % output, minimum on/off time is inserted automatically when the output is configured to be safety critical. This is necessary for optimal load diagnostics.

Note 3 T_{ECU} = -40. . . +85 °C

Note 4 HS power stage used in mutual exclusive mode

Note 5 For t < 5 ms

Note 6 Additionally to observing the maximum load current limit, there is also a minimum load resistance limit, depending on the battery supply voltage.

Note 7 Exceeding this value will trigger open load detection.

4.12.4.2 Diagnostic Functions

Load monitoring is the detection of overloads, external short circuits of the load output to positive or negative supply (BAT+/BAT-) or any other power output and the detection of load loss.

The diagnostic functions are different between PWM and digital operation:

- **PWM-operated high-side output (duty cycle 0 % < X < 100 %)**:
  Under normal load conditions, the feedback signals to the timer unit and the ADC follow the corresponding PWM output. In case of a disconnected load (open load), the output is pulled to 5 V by an internal resistor. If there is a short circuit to ground, the feedback signals are constantly low. A short circuit to BAT+ implicates that the feedback signals are pulled to 5 V, which also results in a constantly high level. By merging the measurement results from the

timer and the ADC unit, it is possible to differentiate the diagnostic functions, as shown in the table below.

Output Signal	Status Signal			
	Normal	Open Load	Short to GND	Short to U_{BAT}
0 % < X < 100 %	●	●	●	●

● = detected

- **Digitally operated high-side output (true duty cycle of 0 or 100 %, without min. and max. pulses)**:
  When the power stage is switched off, the monitoring interface will read back low level if the load is properly connected or if there is a short circuit to ground. In case of open load or short circuit to BAT+ the monitoring interface will read back high level.
  When the power stage is switched on, a high level will be read back in case of normal operation. In case of excessive overload or short circuit to ground, the output switches off in order to protect the output stage. In this case, the monitoring interface will read back a low level. The possible diagnostic functions of the digital operation are shown in the table below.

Output Signal	Status Signal			
	Normal	Open Load	Short to GND	Short to U_{BAT}
On	●	×	●	×
Off	●	●	×	●

● = detected
× = not detected

4.12.4.3 Characteristics of Current Measurement

T_{ECU} = -40. . . +125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
$I_{tol\text{-}p}$	Proportional error	1,2	-2.5	+2.5	%
$I_{tol\text{-}p}$	Proportional error	4,2	-2.0	+2.0	%
$I_{tol\text{-}0}$	Zero reading error	1,2	-40	+40	mA
$I_{tol\text{-}0}$	Zero reading error	1,2,4	-35	+35	mA
LSB	Nominal value of 1 LSB		-	1	mA
f_{g_LP}	Cut off frequency of 3rd order low pass filter	3	30	50	Hz

Note 1 The measured value is clipped in software if below zero. So at some devices a small output current is necessary to get ADC-values greater than zero.

Note 2 The total error (I_{tol}) is the sum of proportional error and zero reading error:
$I_{tol} = \pm\left(I_{tol-p} \cdot I_{load} + I_{tol-0}\right)$

Note 3 An active low pass filter (3rd order) is provided to reduce current ripple from the ADC input. Further digital filtering is applied to eliminate the current ripple completely and provide a stable measurement value for the application.

Note 4 T_{ECU} = -40. . . +85 °C

4.12.5 High-Side Digital Outputs

When pulse width modulation is not necessary, the output can be configured as a simple digital output. This output mode is only allowed for non-safety critical applications, otherwise the high-side PWM mode need to be used.

4.12.5.1 Characteristics of High-Side Digital Output

T_{ECU} = -40. . . +125 ℃

Symbol	Parameter	Note	Min.	Max.	Unit
C_{out}	Pin input capacitance		15	25	nF
R_{pu}	Pull-up resistor		4	5	kΩ
V_{pu}	Pull-up voltage (open load)		4.2	4.7	V
R_{on}	On-resistance			150	mΩ
I_{max}	Maximum load current			3	A
I_{max}	Maximum load current	1		4	A
I_{peak}	Peak load current limit	2		5.5	A
R_{load_min}	Minimum coil resistance (12 V)	3	2		Ω
R_{load_min}	Minimum coil resistance (24 V)	3	4		Ω
R_{load_max}	Maximum load resistance	4		1	kΩ

Note 1 T_{ECU} = -40. . . +85 ℃
Note 2 For t < 5 ms
Note 3 Additionally to observing the maximum load current limit, there is also a minimum load resistance limit, depending on the battery supply voltage.
Note 4 Exceeding this value will trigger open load detection.

See Section 4.12.4.3 on the preceding page for characteristics of current measurement.

4.12.6 Digital and Frequency Inputs

If a high-side output is not needed on IO_PWM_00 - IO_PWM_35, the loop-back path of these output stages can be alternatively used as a digital or frequency input.

External switches which are directly switching to battery voltage must not be used with alternative inputs.

See Section 6.6 on page 194 for more information.

4.12.6.1 Characteristics of Digital Input

The alternative digital input function (use case: external switch connected to GND) is available for high-side PWM channels IO_PWM_00 - IO_PWM_35.

T_{ECU} = -40. . . +125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
C_{in}	Pin input capacitance		15	25	nF
τ_{in}	Input low pass filter		1.4	3.4	µs
R_{pu}	Pull-up resistor		4	5	kΩ
V_{pu}	Pull-up voltage		4.25	4.8	V
V_{il}	Input voltage for low level		0	2	V
V_{ih}	Input voltage for high level	1	3	32	V
V_{in}	Input voltage range	1	-0.5	BAT+ Power +0.5	V

Note 1 The input voltage may go up to 32 V but must never exceed battery supply voltage.

4.12.6.2 Characteristics of Timer Input

The alternative timer input function is only available for high-side PWM channels IO_PWM_28 - IO_PWM_35.

T_{ECU} = -40. . . +125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
C_{in}	Pin input capacitance		15	25	nF
τ_{in}	Input low pass filter		1.4	3.4	µs
R_{pu}	Pull-up resistor		4	5	kΩ
V_{pu}	Pull-up voltage		4.25	4.8	V
f_{max}	Maximum input frequency	1		2	kHz
f_{max}	Maximum input frequency	2		10	kHz
f_{min}	Minimum input frequency	3	0.1		Hz
t_{res}	Timer resolution		1	1	µs
t_{min}	Minimum on/off time to be measured by timer input		20		µs
V_{il}	Input voltage for low level		0	2	V
V_{ih}	Input voltage for high level	4	3	32	V
V_{in}	Input voltage range	4	-0.5	BAT+ Power +0.5	V

Note 1 With open collector sensor output.

Note 2 With push-pull sensor output stage.

Note 3 Due to the dynamic range of the timer, there is a minimum frequency when a timer overflow occurs. Even at a lower frequency the output value will be read as 0 Hz.

Note 4 The input voltage may go up to 32 V but must never exceed battery supply voltage.

4.12.7 Secondary Shut-off Paths

The available PWM output stages are allocated to three independent secondary shut-off paths (=Safety switches). See Section 4.12.1 on page 112, column *Ext./Second. Shut-Off* of pinout table. For safety-critical applications, these paths allow to selectively activate/deactivate a set of specific PWM outputs in case of a detected actuator failure. Thus, the ECU can be operated in a reduced (limp home) mode that allows the vehicle to be safely driven to the repair shop.

Figure 4.30 on this page shows the detailed distribution of the PWM output stages to the shut-off paths.

See Section 5.1.1 on page 164 for safety concept overview.

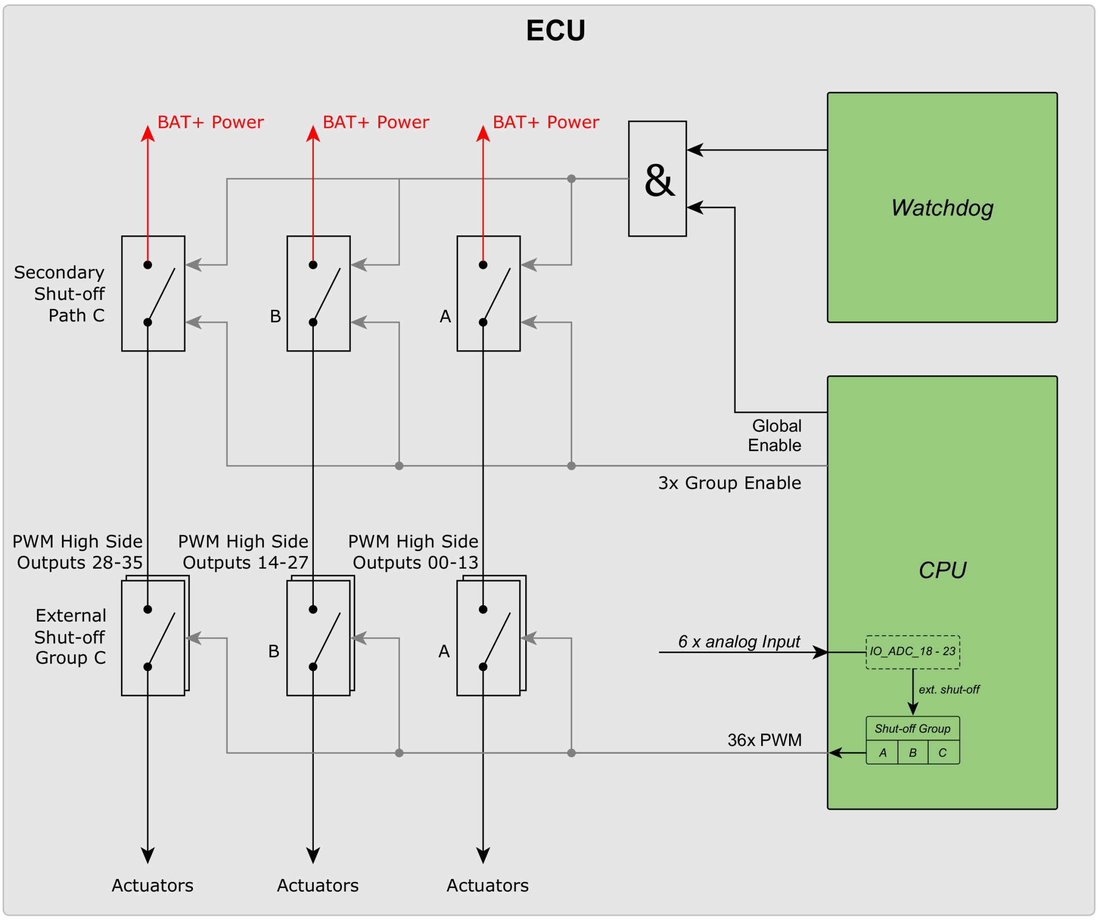

Figure 4.30: Distribution of PWM output stages to shut-off groups/paths

4.12.8 External Shut-off Groups

6 analog inputs can be configured acting as external activating or deactivating switch input for shut-off groups A, B and C, see Figure 4.30 on the facing page.

The external analog input pins can deactivate the shut-off groups independently of the watchdog functionality and the application software as well.

List of analog input pins to shut-off group mapping:

Pin No.	Function	SW-define	Shut-Off Group
P112	0...32 V Input	IO_ADC_18	A
P136	0...32 V Input	IO_ADC_19	A
P113	0...32 V Input	IO_ADC_20	B
P137	0...32 V Input	IO_ADC_21	B
P114	0...32 V Input	IO_ADC_22	C
P138	0...32 V Input	IO_ADC_23	C

Depending on the voltage level on the external pin(s), the software driver switches the corresponding shut-off group ON or OFF within the previously configured failure reaction time. 2 Analog Inputs have to be used in parallel for each shut-off group to guarantee SIL 2 Level.

4.12.8.1 Characteristics of External Shut-off

T_{ECU} = -40...+125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
C_{in}	Pin input capacitance		8	12	nF
R_{pd}	Pull-down resistor		14.7	15.3	kΩ
V_{in}	Input voltage range		0	32	V
V_{il}	Input voltage level to signal a low level input		0	4	V
V_{ih}	Input voltage level to signal a high level input		6	32	V
$V_{tol\text{-}0}$	Zero reading error	2	-50	+50	mV
$V_{tol\text{-}0}$	Zero reading error	1,2	-45	+45	mV
$V_{tol\text{-}p}$	Proportional error	2	-4	+4	%
$V_{tol\text{-}p}$	Proportional error	1,2	-3	+3	%
LSB	Nominal value of 1 LSB		-	8.59	mV

Note 1 T_{ECU} = -40...+85 °C

Note 2 The total measurement error is the sum of zero reading error and the proportional error.

4.12.9 Tertiary Shut-off Path

4.12.9.1 Functional description

Loads on safety-critical high-side PWM channels can be connected to the ground as shown in Figure 4.29 on page 115 or to a low-side digital output as shown in Figure 4.31 on this page.

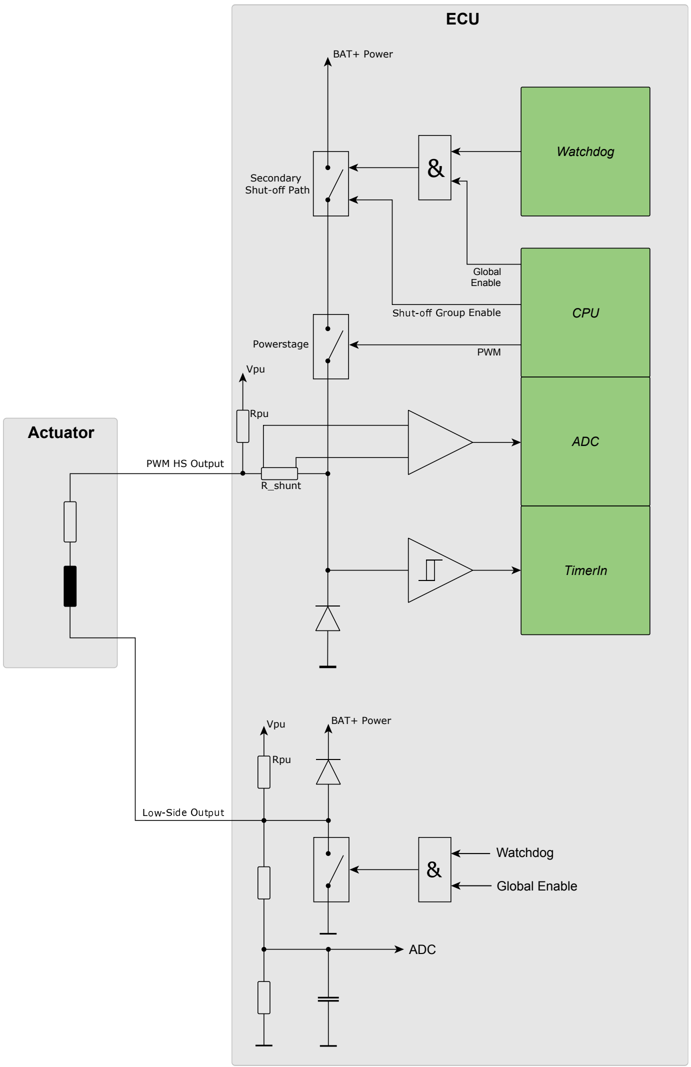

Figure 4.31: Tertiary Shut-off Path

With the tertiary shut-off configuration shown in 4.31 the HY-TTC 500 is able to switch off the high-side PWM channels even if the load has a short circuit to the positive battery supply.

Please refer to section *PWM - Pulse Width Modulation driver* of the API documentation [38] for more information how to configure such an output.

4.13 High-Side Digital Outputs

4.13.1 Pinout

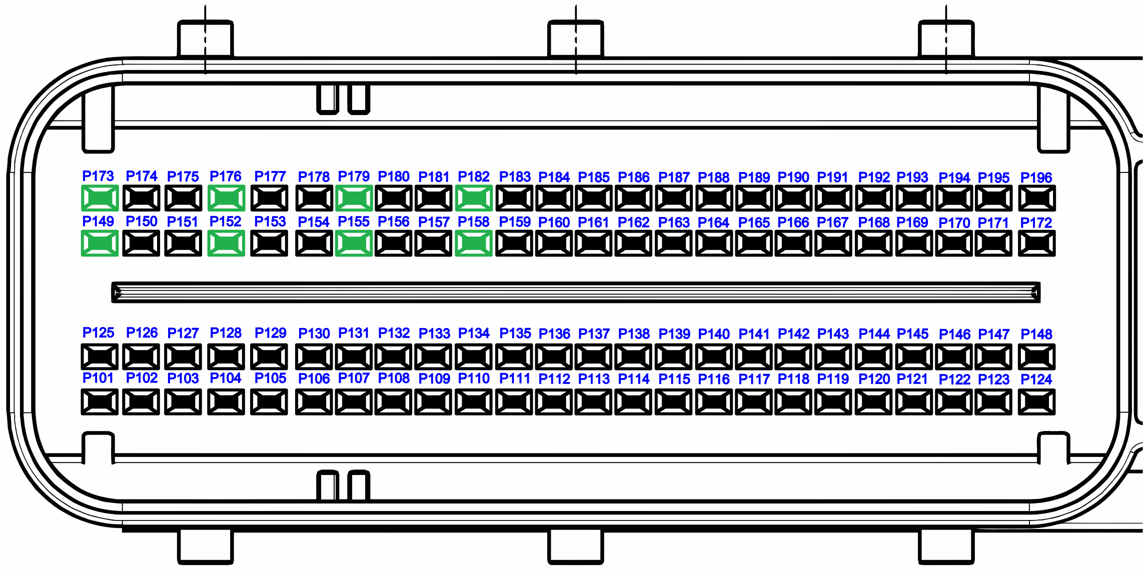

Figure 4.32: Pinout of high-side digital outputs

Pin No.	Function	SW-define	Power stage
P149	High-Side Digital Output	IO_DO_00	a
P173	High-Side Digital Output	IO_DO_01	a
P152	High-Side Digital Output	IO_DO_02	b
P176	High-Side Digital Output	IO_DO_03	b
P155	High-Side Digital Output	IO_DO_04	c
P179	High-Side Digital Output	IO_DO_05	c
P158	High-Side Digital Output	IO_DO_06	d
P182	High-Side Digital Output	IO_DO_07	d

4.13.2 Functional Description

Eight power output stages with freewheeling diodes for inductive and resistive loads with low-side connection.

Suitable loads are lamps, valves, relays etc.

If an error is detected in a safety-critical resource, the watchdog CPU or the main CPU will disable the output stage (off state).

For diagnostic reasons, the output signal is looped back to the CPU, and the value measured is compared with the value set. When the output is not used, the loop-back signal can be used as an analog input or as a digital input.

TTControl GmbH recommends to use the maximum possible wire size (FLRY type) in case of maximum load current to reduce voltage drop and prevent overheating of the crimp contact.

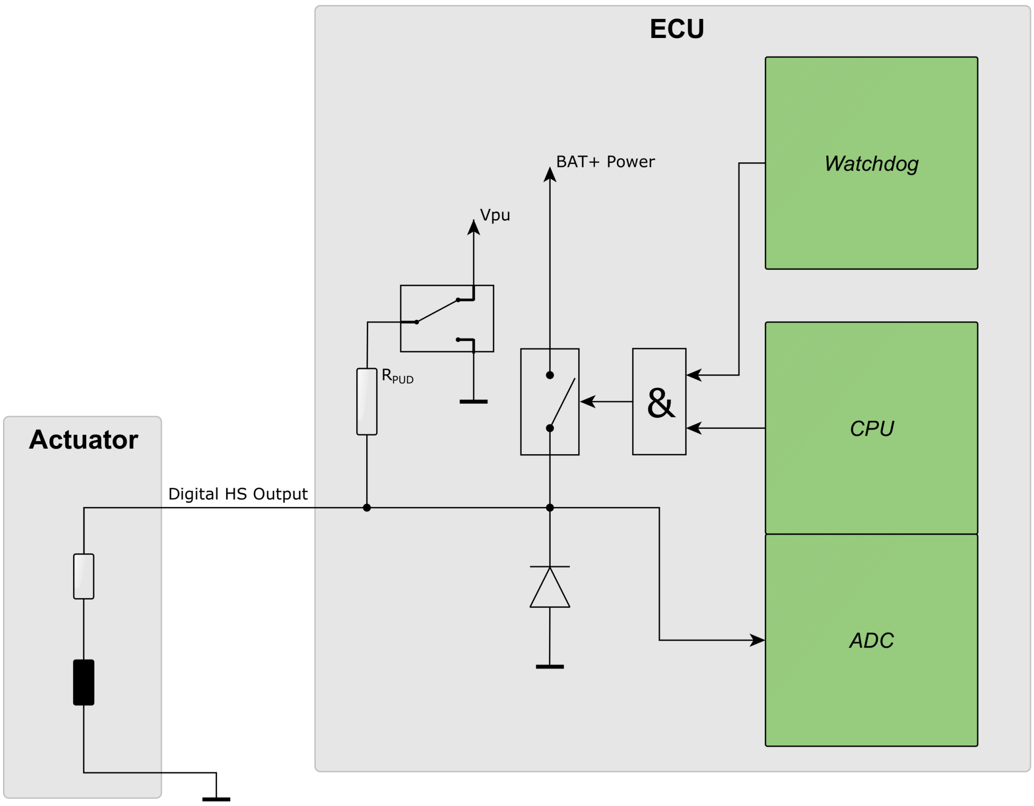

Figure 4.33: Digital high-side power stage

4.13.2.1 Power Stage Pairing

If outputs shall be used in parallel, always combine two channels from the same double-channel power stage and use the digital output mode. See Section 4.13.1 on page 125 for using the right power stage outputs in parallel.

Due to thermal limits, the resulting total load current of this output pair has to be de-rated by a factor of 0.85 (e.g. combining two 3 A outputs would result in a total load current of 3 A x 2 x 0.85 = 5.1 A). The application software has to make sure that both outputs are switched on at the same point in time, otherwise the over-current protection may trip.

For balanced current distribution through each of the pin pairs, the cable routing shall be symmetrical if pin-pairs or multiple pins shall be wired parallel to support higher load currents.

4.13.2.2 Mutual Exclusive Mode

The HY-TTC 500 uses double-channel high-side power stages. For load leveling it is a benefit if loads, which are switched on mutually exclusive (which means either load A, or load B is on, but not A and B at the same time), are connected to the same double-channel power stage. This reduces the thermal stress of the components. The power stage pairing is given in Section 4.13.1 on page 125.

4.13.3 Diagnostic Functions

Load monitoring is the detection of overloads, external short circuits of the load output to positive or negative supply (BAT+/BAT-) or any other power output and the detection of load loss.

Output Signal	Status Signal			
	Normal	Open Load	Short to GND	Short to U_{BAT}
On	●	×	●	×
Off	●	●	×	●

● = detected
× = not detected

4.13.4 Maximum Ratings

T_{ECU} = -40. . . +125 ℃

Symbol	Parameter	Note	Min.	Max.	Unit
V_{in}	Input/output voltage under overload conditions		-0.5	BAT+ Power +0.5	V

4.13.5 High-Side Digital Outputs

4.13.5.1 Characteristics of High-Side Output

T_{ECU} = -40. . . +125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
C_{in}	Pin input capacitance		8	12	nF
R_{pud}	Pull-up/pull-down resistor		7.5	10	kΩ
V_{pu}	Pull-up voltage (open load)		4	4.5	V
R_{on}	On-resistance			100	mΩ
I_{load}	Nominal load current		0	3	A
I_{load}	Nominal load current	1	0	4	A
I_{peak}	Peak load current	2	0	6	A
I_{tol-p}	Proportional error	3	-14	+14	%
I_{tol-p}	Proportional error	1,3	-10	+10	%
I_{tol-0}	Zero reading error	3	-150	+150	mA
I_{tol-0}	Zero reading error	1,3	-150	+150	mA

Note 1 T_{ECU} = -40. . . +85 °C

Note 2 Peak current for not more than 100 ms. Exceeding this value will trigger the overload protection and switch off the power stage. Steady state operation goes only up to 3 A/4 A depending on temperature.

Note 3 The total measurement error is the sum of zero reading error and the proportional error.

4.13.6 Analog and Digital Inputs

External switches which are directly switching to battery voltage must not be used with alternative inputs.

See Section 6.6 on page 194 for more information.

4.13.6.1 Characteristics of Analog Voltage Input

T_{ECU} = -40. . . +125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
C_{in}	Pin input capacitance		8	12	nF
τ_{in}	Input low pass filter (analog path)		1.5	2.5	ms
R_{pud}	Pull-up/pull-down	1	7.5	10	kΩ
V_{pu}	Pull-up voltage (open load)		4	4.5	V
	Resolution		12	12	bit
$V_{tol\text{-}0}$	Zero reading error	3	-55	+55	mV
$V_{tol\text{-}0}$	Zero reading error	1,3	-50	+50	mV
$V_{tol\text{-}p}$	Proportional error	3	-4	+4	%
$V_{tol\text{-}p}$	Proportional error	1,3	-3	+3	%
LSB	Nominal value of 1 LSB		-	13.4	mV
V_{in}	Input voltage measurement range	2	0	32	V
V_{in}	Input voltage range	2	-0.5	BAT+ Power +0.5	V

Note 1 T_{ECU} = -40. . . +85 °C
Note 2 The input voltage may go up to 32 V but must never exceed battery supply voltage.
Note 3 The total measurement error is the sum of zero reading error and the proportional error.

4.13.6.2 Characteristics of Digital Input

T_{ECU} = -40. . . +125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
C_{in}	Pin input capacitance		8	12	nF
τ_{in}	Input low pass filter		1.5	2.5	ms
R_{pud}	Pull-up/pull-down resistor		7.5	10	kΩ
V_{pu}	Pull-up voltage (open load)		4	4.5	V
	Resolution		12	12	bit
V_{il}	Input voltage for low level		0	●	V
V_{ih}	Input voltage for high level	1	●	32	V
V_{in}	Input voltage range	1	-0.5	BAT+ Power +0.5	V

Note ● Configuration by application software
Note 1 The input voltage may go up to 32 V but must never exceed battery supply voltage.

4.14 High-Side Digital/PVG/VOUT Outputs

4.14.1 Pinout

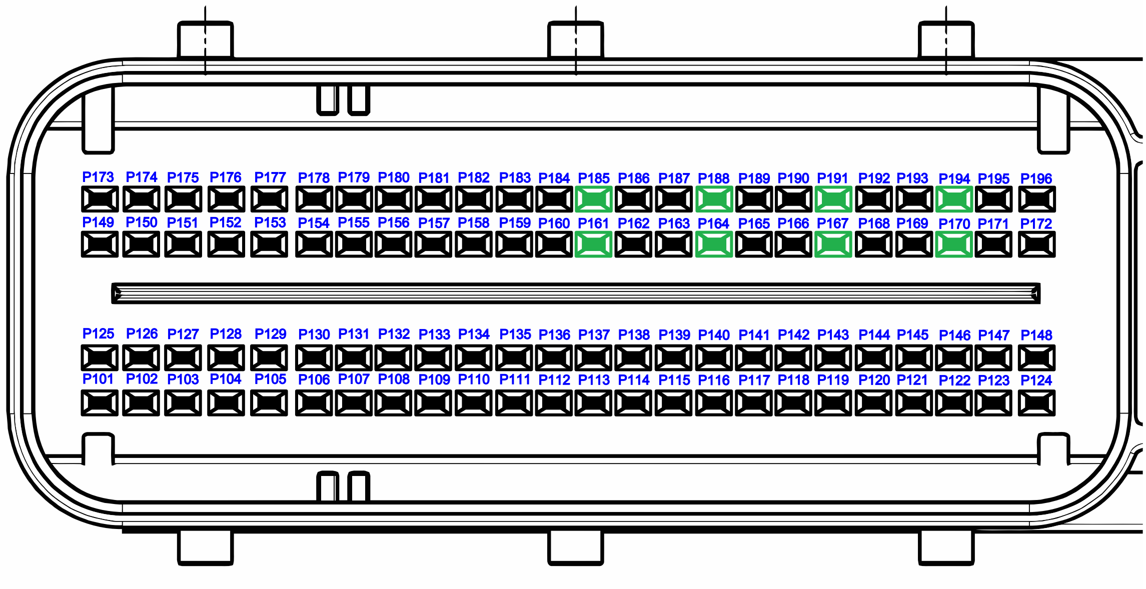

Figure 4.34: Pinout of High-side digital/PVG/VOUT outputs

Pin No.	Function	SW-define	Power stage
P161	High-Side Digital Output/PVG/V_{OUT}	IO_PVG_00	a
P185	High-Side Digital Output/PVG/V_{OUT}	IO_PVG_01	a
P188	High-Side Digital Output/PVG/V_{OUT}	IO_PVG_02	b
P164	High-Side Digital Output/PVG/V_{OUT}	IO_PVG_03	b
P191	High-Side Digital Output/PVG/V_{OUT}	IO_PVG_04	c
P167	High-Side Digital Output/PVG/V_{OUT}	IO_PVG_05	c
P194	High-Side Digital Output/PVG/V_{OUT}	IO_PVG_06	d
P170	High-Side Digital Output/PVG/V_{OUT}	IO_PVG_07	d

4.14.2 Functional Description

Output stages for interfacing to PVG type hydraulic valve groups (PVG mode), for low power analog voltage loads (V_{OUT} mode) or for high power inductive/resistive loads (HS mode).

All eight outputs can be configured independently of each other.

TTControl GmbH recommends to use the maximum possible wire size (FLRY type) in case of maximum load current to reduce voltage drop and prevent overheating of the crimp contact.

See also section *PVG - Proportional Valve output driver* of the API documentation [38].

4.14.2.1 Power Stage Pairing

If outputs shall be used in parallel, always combine two channels from the same double-channel power stage and use the digital output mode. See Section 4.14.1 on the preceding page for using the right power stage outputs in parallel.

Due to thermal limits, the resulting total load current of this output pair has to be de-rated by a factor of 0.85 (e.g. combining two 3 A outputs would result in a total load current of 3 A x 2 x 0.85 = 5.1 A). The application software has to make sure that both outputs are switched on at the same point in time, otherwise the over-current protection may trip.

For balanced current distribution through each of the pin pairs, the cable routing shall be symmetrical if pin-pairs or multiple pins shall be wired parallel to support higher load currents.

4.14.2.2 Mutual Exclusive Mode

The HY-TTC 500 uses double-channel high-side power stages. For load leveling it is a benefit if loads, which are switched on mutually exclusive (which means either load A, or load B is on, but not A and B at the same time), are connected to the same double-channel power stage. This reduces the thermal stress of the components. The power stage pairing is given in Section 4.14.1 on the previous page.

4.14.3 Maximum Ratings

T_{ECU} = -40. . . +125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
V_{in}	Input/output voltage under overload conditions		-0.5	BAT+ Power +0.5	V

4.14.4 High-Side Digital Outputs

Eight power output stages with freewheeling diodes for inductive and resistive loads with low-side connection.

Suitable loads are lamps, valves, relays etc.

If an error is detected in a safety-critical resource, the watchdog CPU or the main CPU can disable the output stage (off state).

For diagnostic reasons the output signal is looped back to the CPU, and the value measured is compared with the value set. When the output is not used, the loop-back signal can be used as an analog input or as a digital input.

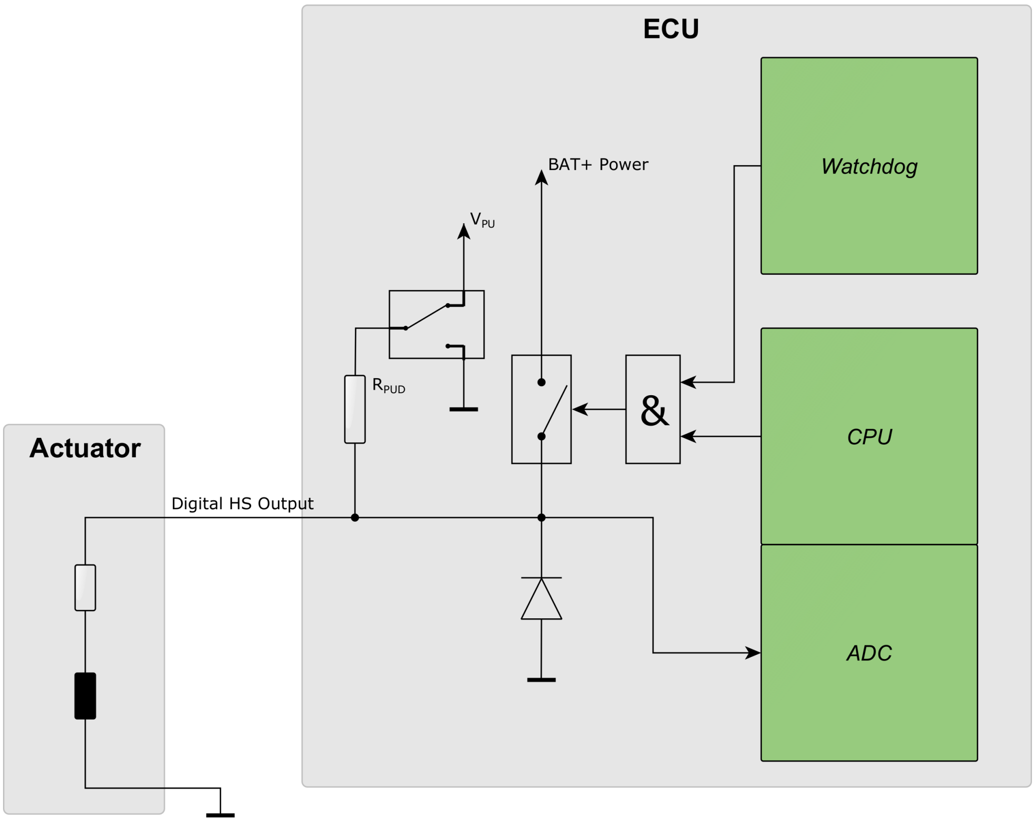

Figure 4.35: Digital high-side power stage

4.14.4.1 Diagnostic Functions

Load monitoring is the detection of overloads, external short circuits of the load output to positive or negative supply (BAT+/BAT-) or any other power output and detection of load loss.

Output Signal	Status Signal			
	Normal	Open Load	Short to GND	Short to U_{BAT}
On	●	×	●	×
Off	●	●	×	●

● = detected
× = not detected

4.14.4.2 Characteristics of High-Side Digital

T_{ECU} = -40. . . +125 ℃

Symbol	Parameter	Note	Min.	Max.	Unit
C_{in}	Pin input capacitance		8	12	nF
R_{pud}	Pull-up/pull-down resistor		2.5	2.6	kΩ
V_{pu}	Pull-up voltage (open load)		2.15	2.45	V
R_{on}	On-resistance			100	mΩ
I_{load}	Nominal load current		0	3	A
I_{max}	Maximum load current	1		4	A
I_{peak}	Peak load current	2	0	6	A
$I_{tol\text{-}p}$	Proportional error	3	-14	+14	%
$I_{tol\text{-}p}$	Proportional error	1,3	-10	+10	%
$I_{tol\text{-}0}$	Zero reading error	3	-150	+150	mA
$I_{tol\text{-}0}$	Zero reading error	1,3	-150	+150	mA

Note 1 T_{ECU} = -40. . . +85 ℃

Note 2 Peak current for maximal 100 ms. Exceeding this value will trigger overload protection and switch off the power stage.Steady state operation goes only up to 3 A/4 A depending on temperature.

Note 3 The total measurement error is the sum of zero reading error and the proportional error.

4.14.5 PVG Outputs

Proportional Valve Groups (PVG) are a group of hydraulic load-sensing valves with integrated electronics allowing advanced flow controllability, e.g., for load-independent flow control.

For diagnostic reasons in output mode, the output signal is looped back to the CPU, and the measured value is compared to the set value. If the difference between these two values is above a fixed limit, an overload is detected, and the output is disabled. The protection mechanism tries re-enabling the output 10 times per drive cycle. It is not allowed to use two outputs in parallel to increase driving strength.

The PVG output can be used to control PVG valves of the types PVEA, PVEH and PVES. These types of valves apply a low pass filter to the input signal and use the resulting DC voltage in relation to the valves supply voltage (BAT+) as a parameter for flow control.

The HY-TTC 500 uses the BAT+ CPU pin as a reference voltage input. The principle schematic is shown in Figure 4.36 on this page. The output is open loop controlled. The ADC input is for diagnostic purposes only and can be evaluated by the application software.

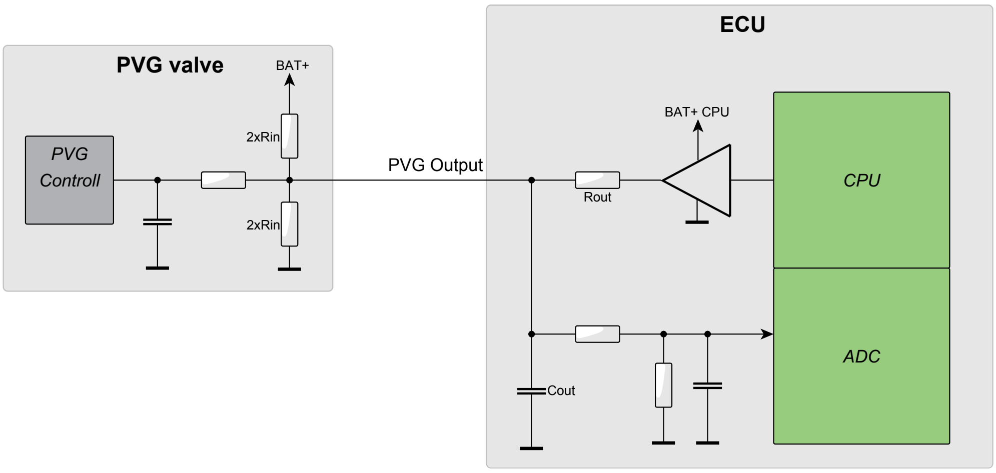

Figure 4.36: Output stage in PVG mode

4.14.5.1 Characteristics of PVG

T_{ECU} = -40. . . +125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
C_{out}	Pin capacitance		430	530	nF
R_{out}	Output resistance		2.5	2.6	kΩ
V_{nom}	Nominal voltage range (BAT+ = 8. . . 32 V, nominal load resistance)	1,2	0.1 · BAT+	0.9 · BAT+	V
$V_{tol\text{-}p}$	Proportional error (BAT+ = 8. . . 32 V, nominal load resistance)	1,3	-2	+2	%
$V_{tol\text{-}0}$	Zero reading error (BAT+ = 8. . . 32 V, nominal load resistance)	1,3	-100	+100	mV

Note 1 The PVG valves use a voltage divider with 2 · 24 k Ω between ground and BAT+. This is equivalent to 1 · 12 k Ω against 50 % of BAT+.

Note 2 The standard PVG valves are controllable between 25 % and 75 % of BAT+. This specification parameter is to show the HW-related control limits only.

Note 3 The total measurement error is the sum of zero reading error and the proportional error.

4.14.6 VOUT Outputs

In V_{OUT} mode, the outputs generate a DC voltage that can be used to connect to any high-impedance analog input. The load resistance of the receiving device defines the maximum possible output voltage.

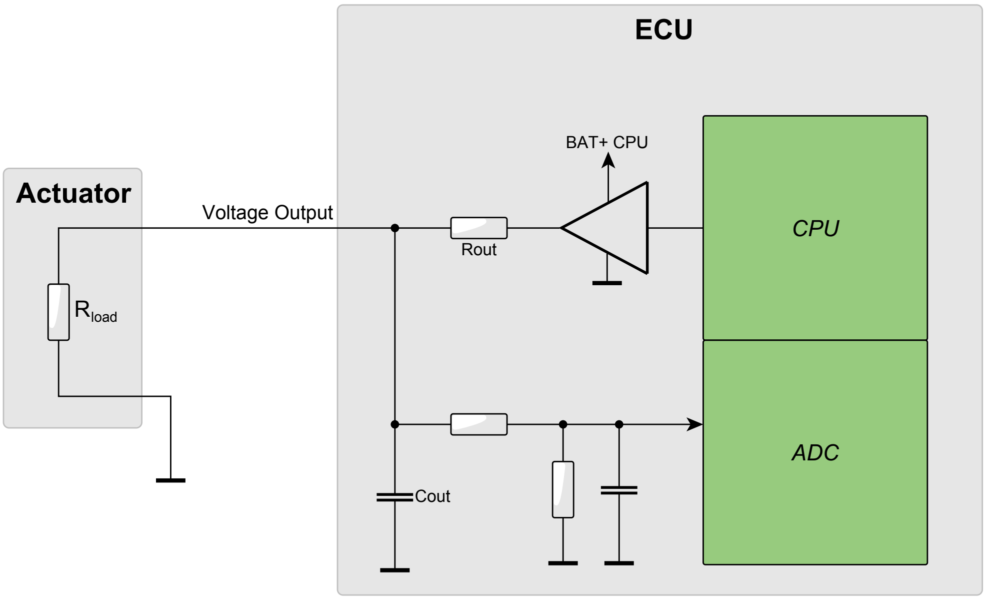

Figure 4.37: Output stage in V_{OUT} mode

In PVG mode, PVG valves have a well-defined input resistance, and the output signal settings can be calculated in advance by considering the characteristics of the output stage. In voltage mode, however, a PID controller must be applied to generate the desired output voltage. This results in a certain settling time, which depends on the parameter set of the PID controllers.

The V_{OUT} mode output can be used to control PVG valves of the type PVEU or other generic resistive loads.

For diagnostic reasons, the output signal is looped back to the CPU in output mode, and the measured value is compared to the set value. If the difference between these two values is above a fixed limit, an overload is detected and the output is disabled. The protection mechanism tries to re-enable the output 10 times per drive cycle. It is not allowed to use two outputs in parallel to increase driving strength.

4.14.6.1 Characteristics of VOUT

T_{ECU} = -40. . . +125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
C_{out}	Pin capacitance		430	530	nF
R_{out}	Output resistance		2.5	2.6	kΩ
I_{load}	Nominal load current		0	5	mA
V_{nom}	Nominal voltage range (open load), BAT+ = 8. . . 32 V	1	0.0 · BAT+	0.99 · BAT+	
V_{tol-p}	Proportional error (BAT+ = 8. . . 32 V)	3	-2	+2	%
V_{tol-p}	Proportional error (BAT+ = 8. . . 32 V)	2,3	-1.5	+1.5	%
V_{tol-0}	Zero error (BAT+ = 8. . . 32 V)	3	-300	+300	mV
V_{tol-0}	Zero error (BAT+ = 8. . . 32 V)	2,3	-200	+200	mV

Note 1 In the VOUT setting, the open load voltage is only open loop controlled. The load creates a voltage divider with the well-defined output resistance (Rout) of the VOUT stage. This effect must be considered in the application SW. To get a desired (loaded) output voltage the proper open load voltage must be calculated and set to a (higher) open load voltage level. For example, with a load RL = 10 kΩ to ground the open load voltage (Vset) must be set to 12.55 V ($Vset = Vout \frac{RL+2.55\ kohm}{RL}$) to get an output voltage of 10 V.

Note 2 T_{ECU} = -40. . . +85 °C

Note 3 The total error is the sum of zero error and the proportional error.

4.14.7 Analog and Digital Inputs

This input type is suitable for low impedance switches and sensors only. For standard analog senors please use Analog Input 2 Modes (Section 4.10 on page 98) and Analog Input 3 Modes (Section 4.9 on page 88).

External switches which are directly switching to battery voltage must not be used with alternative inputs.

See Section 6.6 on page 194 for more information.

4.14.7.1 Characteristics of Analog Voltage Input

T_{ECU} = -40. . . +125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
C_{in}	Pin input capacitance		384	576	nF
τ_{in}	Input low pass filter		8	12	ms
R_{pu}	Pull-up resistor		2.5	2.6	kΩ
V_{pu}	Pull-up voltage (open load)		4.7	5.1	V
	Resolution		12	12	bit
$V_{tol\text{-}0}$	Zero reading error	3	-55	+55	mV
$V_{tol\text{-}0}$	Zero reading error	1,3	-50	+50	mV
$V_{tol\text{-}p}$	Proportional error	3	-4	+4	%
$V_{tol\text{-}p}$	Proportional error	1,3	-3	+3	%
LSB	Nominal value of 1 LSB		-	8	mV
V_{in}	Input voltage measurement range	2	0	32	V
V_{in}	Input voltage range	2	-0.5	BAT+ Power +0.5	V

Note 1 T_{ECU} = -40. . . +85 °C

Note 2 The input voltage may go up to 32 V but must never exceed battery supply voltage.

Note 3 The total measurement error is the sum of zero reading error and the proportional error.

4.14.7.2 Characteristics of Digital Input

T_{ECU} = -40. . . +125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
C_{in}	Pin input capacitance		384	576	nF
τ_{in}	Input low pass filter		8	12	ms
R_{pu}	Pull-up resistor		2.5	2.6	kΩ
V_{pu}	Pull-up voltage (open load)		4.7	5.1	V
	Resolution		12	12	bit
V_{il}	Input voltage for low level		0	●	V
V_{ih}	Input voltage for high level	1	●	32	V
V_{in}	Input voltage range	1	-0.5	BAT+ Power +0.5	V

Note ● Configuration by application software

Note 1 The input voltage may go up to 32 V but must never exceed battery supply voltage.

4.15 Low-Side Digital Outputs

4.15.1 Pinout

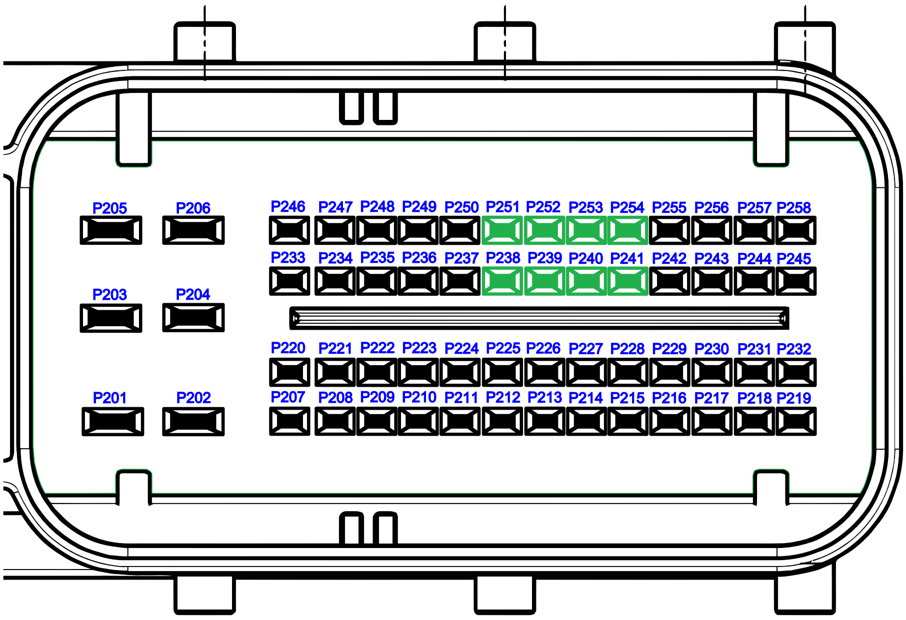

Figure 4.38: Pinout of low-side digital outputs

Pin No.	Function	SW-define
P251	Low-Side Digital Output	IO_DO_08
P238	Low-Side Digital Output	IO_DO_09
P252	Low-Side Digital Output	IO_DO_10
P239	Low-Side Digital Output	IO_DO_11
P253	Low-Side Digital Output	IO_DO_12
P240	Low-Side Digital Output	IO_DO_13
P254	Low-Side Digital Output	IO_DO_14
P241	Low-Side Digital Output	IO_DO_15

4.15.2 Functional Description

Eight low-side switches with freewheeling diodes to BAT+ Power for inductive and resistive loads.

It is important to use the same voltage (no switch between) for BAT+ of the load and the BAT+ Power path to not interrupt the internal freewheeling path for dissipating the energy in case of switching off inductive loads.

The current through the low-side switch is monitored and triggers the opening in case of over-current. Short-circuit and overload protection is included in low-level driver software. Before a tripped channel can be re-enabled, the overload situation has to be removed.

TTControl GmbH recommends to use the maximum possible wire size (FLRY type) in case of maximum load current to reduce voltage drop and prevent overheating of the crimp contact.

See also section *DIO - Driver for digital inputs and outputs* of the API documentation [38].

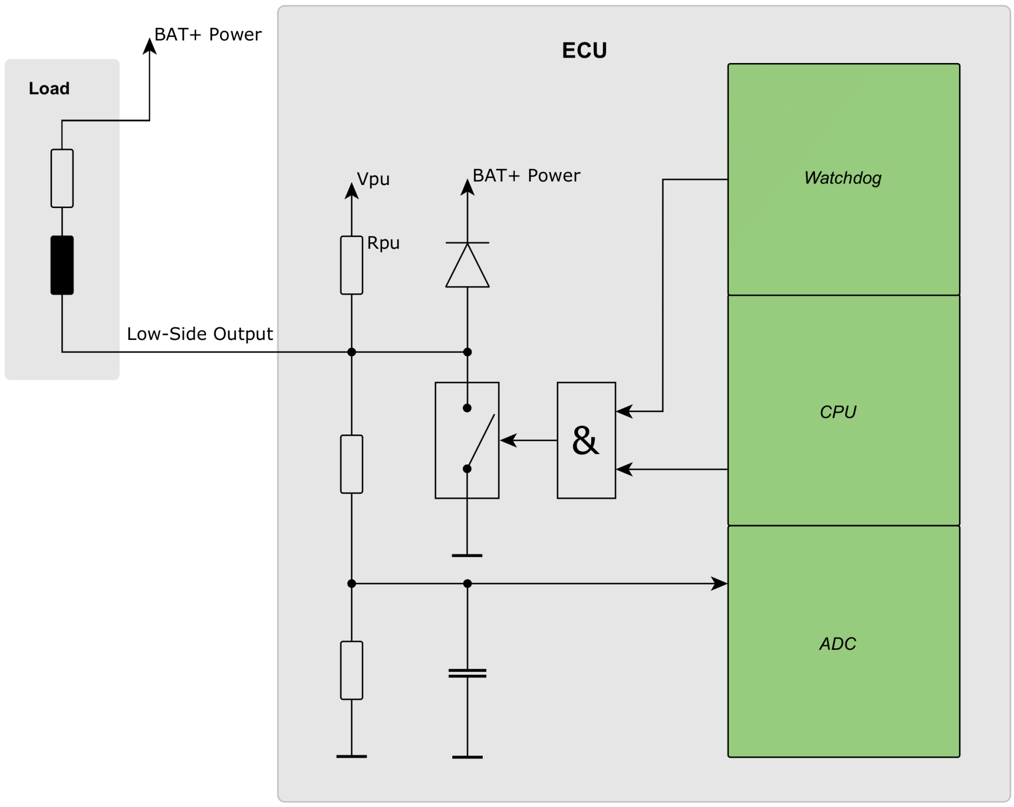

Figure 4.39: Low-side switch for resistive and inductive loads

4.15.2.1 Power Stage Pairing

If higher load current is needed, the output stages can be used in parallel.

Due to thermal limits, the resulting total load current of this output pair has to be de-rated by a factor of 0.85 (e.g. combining two 3 A outputs would result in a total load current of 3 A x 2 x 0.85 = 5.1 A). The application software has to make sure that both outputs are switched on at the same point in time, otherwise the over-current protection may trip.

For balanced current distribution through each of the pin pairs, the cable routing shall be symmetrical if pin-pairs or multiple pins shall be wired parallel to support higher load currents.

4.15.3 Maximum Ratings

T_{ECU} = -40. . . +125 ℃

Symbol	Parameter	Note	Min.	Max.	Unit
V_{in}	Input/output voltage under overload conditions		-0.5	BAT+ Power +0.5	V

4.15.4 Characteristics of Low-Side Digital Output

T_{ECU} = -40. . . +125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
C_{in}	Pin input capacitance		8	12	nF
R_{pu}	Pull-up resistor		8	12	kΩ
V_{pu}	Pull-up voltage		4	4.5	V
R_{on}	On-resistance			50	mΩ
I_{load}	Nominal load current		0	3	A
I_{max}	Maximum load current	1		4	A
I_{peak}	Peak load current	2	0	6	A
$I_{tol\text{-}0}$	Zero reading error	3	-400	+400	mA
$I_{tol\text{-}0}$	Zero reading error	1,3	-300	+300	mA
$I_{tol\text{-}p}$	Proportional error	3	-14	+14	%
$I_{tol\text{-}p}$	Proportional error	1,3	-10	+10	%

Note 1 T_{ECU} = -40. . . +85 °C.

Note 2 Peak current for not more than 100 ms. Exceeding this value will trigger overload protection and switch off the power stage. Steady state operation goes only up to 3 A/4 A depending on temperature

Note 3 The total measurement error is the sum of zero reading error and the proportional error.

4.15.5 Analog and Digital Inputs

External switches which are directly switching to battery voltage must not be used with alternative inputs.

See Section 6.6 on page 194 for more information.

4.15.5.1 Characteristics of Analog Voltage Input

T_{ECU} = -40. . . +125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
C_{in}	Pin input capacitance		8	12	nF
τ_{in}	Input low pass filter		1.5	2.5	ms
R_{pu}	Pull-up resistor		8	12	kΩ
V_{pu}	Pull-up voltage		4	4.5	V
	Resolution		12	12	bit
$V_{tol\text{-}0}$	Zero reading error	3	-55	+55	mV
$V_{tol\text{-}0}$	Zero reading error	1,3	-50	+50	mV
$V_{tol\text{-}p}$	Proportional error	3	-4	+4	%
$V_{tol\text{-}p}$	Proportional error	1,3	-3	+3	%
LSB	Nominal value of 1 LSB		-	13.4	mV
V_{in}	Input voltage measurement range	2	0	32	V
V_{in}	Input voltage range	2	-0.5	BAT+ Power +0.5	V

Note 1 T_{ECU} = -40. . . +85 °C
Note 2 The input voltage may go up to 32 V but must never exceed battery supply voltage.
Note 3 The total measurement error is the sum of zero reading error and the proportional error.

4.15.5.2 Characteristics of Digital Input

T_{ECU} = -40. . . +125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
C_{in}	Pin input capacitance		8	12	nF
τ_{in}	Input low pass filter		1.5	2.5	ms
R_{pu}	Pull-up resistor		8	12	kΩ
V_{pu}	Pull-up voltage		4	4.5	V
	Resolution		12	12	bit
V_{il}	Input voltage for low level		0	●	V
V_{ih}	Input voltage for high level	1	●	32	V
V_{in}	Input voltage range	1	-0.5	BAT+ Power +0.5	V

Note ● Configuration by application software
Note 1 The input voltage may go up to 32 V but must never exceed battery supply voltage.

4.16 LIN

4.16.1 Pinout

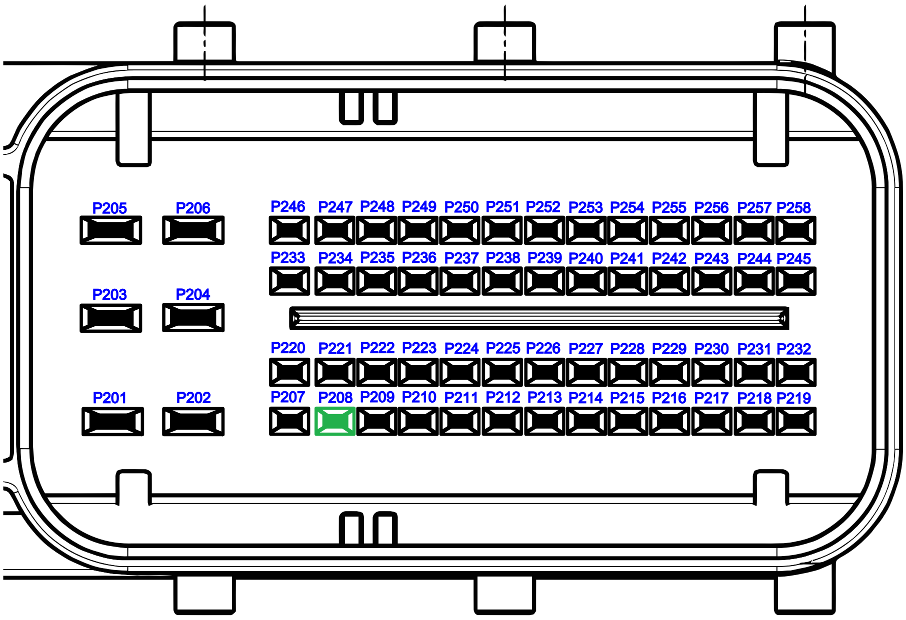

Figure 4.40: LIN pinout

Pin No.	Function	SW-define
P208	LIN	IO_LIN

4.16.2 Functional Description

LIN is configured in master mode.

LIN is a bidirectional half duplex serial bus for up to 10 nodes.

Note: A common ground (chassis) or a proper ground connection is necessary for LIN operation. If you connect to an external device (e.g., to a PC with a LIN interface), make sure not to violate the maximum voltage ratings when connecting to the LIN connection.

See also section *LIN - Local Interconnect Network driver* of the API documentation [38].

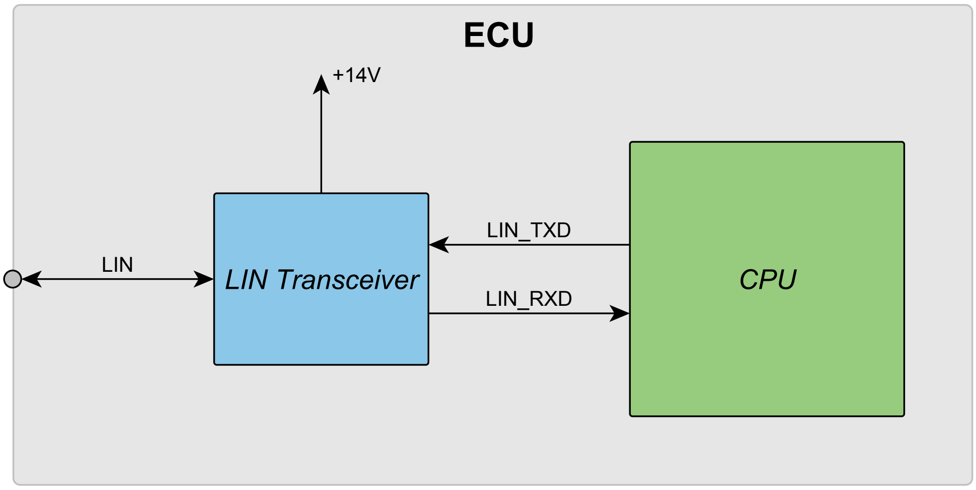

Figure 4.41: LIN interface

4.16.3 Maximum Ratings

T_{ECU} = -40. . . +125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
V_{LIN}	Bus voltage under overload conditions		-1	+33	V

4.16.4 Characteristics

T_{ECU} = -40. . . +125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
C_{out}	Pin output capacitance		0.8	1.2	nF
V_{BUSdom}	Receiver dominant state			$0.4 \cdot V_{Bat\_LIN}$	V
V_{BUSrec}	Receiver recessive state		$0.6 \cdot V_{Bat\_LIN}$	–	V
V_{OL}	Output low voltage			2.0	V
V_{Bat_LIN}	LIN supply voltage	1	13	15	V
R_{pu}	Pull-up resistor		0.95	1.05	kΩ
S_{Tr}	Baud rate			20	kBd

Note 1 For battery voltages higher than V_Bat_LIN +0.5 V

4.17 RS232

4.17.1 Pinout

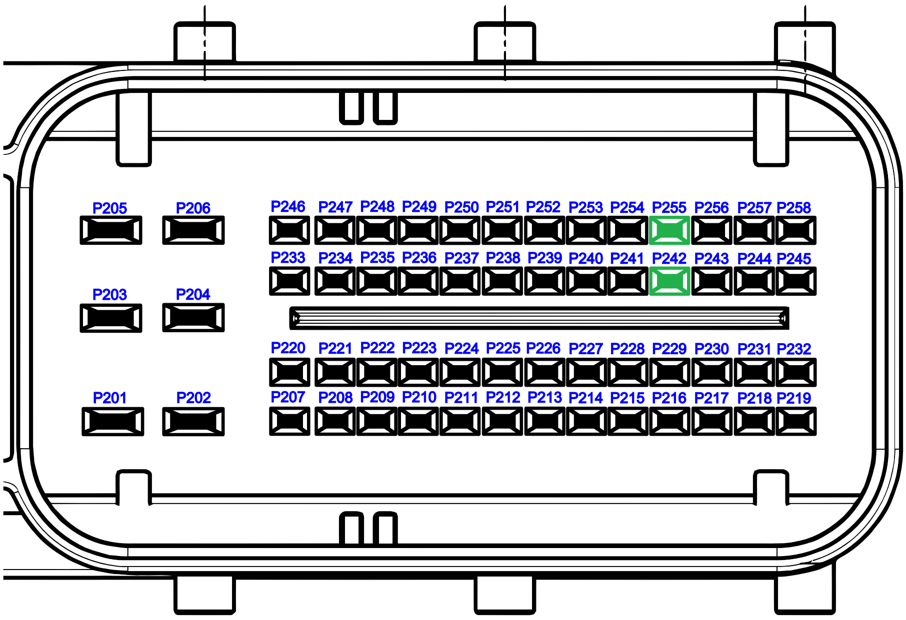

Figure 4.42: RS232 pinout

Pin No.	Function	SW-define
P242	RS232 Serial Interface Output(TX)	IO_UART
P255	RS232 Serial Interface Output(RX)	

4.17.2 Functional Description

RS232 is used as a full duplex serial interface.

Note: Handshake lines (RTS, CTS) are not available. A common ground (chassis) or a proper ground connection is necessary for RS232 operation. If you connect to an external device (e.g., to a PC with an RS232 interface), make sure not to violate the maximum ratings.

See also section *UART - Universal Asynchronous Receiver Transmitter driver* of the API documentation [38].

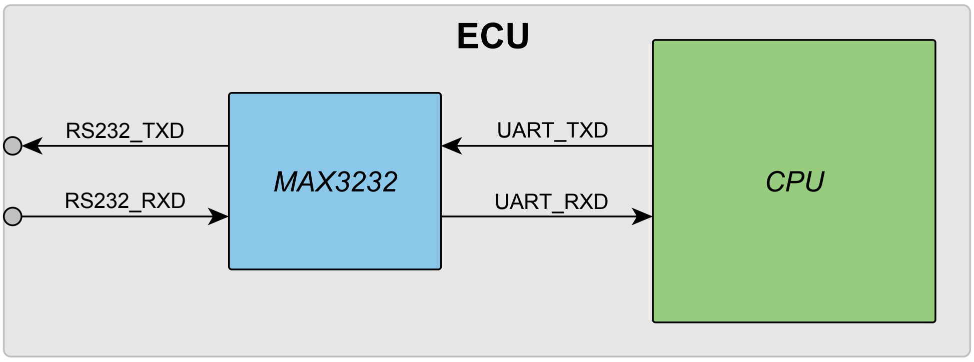

Figure 4.43: RS232 interface

4.17.3 Maximum Ratings

T_{ECU} = -40. . . +125 ℃

Symbol	Parameter	Note	Min.	Max.	Unit
V_{RS232x}	Bus voltage under overload conditions (e.g. short circuit to supply voltages)		-15	+33	V

4.17.4 Characteristics

T_{ECU} = -40. . . +125 ℃

Symbol	Parameter	Note	Min.	Max.	Unit
C_{out}	Pin output capacitance		100	150	pF
V_{il}	Input voltage for low level		-22	0.8	V
V_{ih}	Input voltage for high level		2.4	22	V
V_{ol}	Output voltage for low level		-5		V
V_{oh}	Output voltage for high level		+5		V
S_{Tr}	Baud rate			115	kBd

4.18 CAN

4.18.1 Pinout

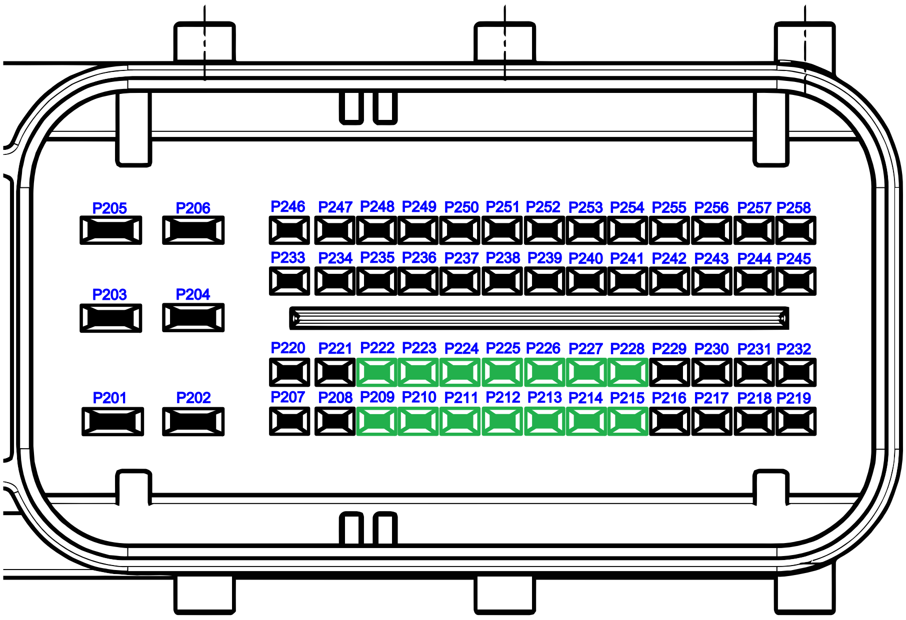

Figure 4.44: CAN pinout

Pin No.	Function	SW-define
P209	CAN Interface 0 – Low Line	IO_CAN_CHANNEL_0
P222	CAN Interface 0 – High Line	
P210	CAN Interface 1 – Low Line	IO_CAN_CHANNEL_1
P223	CAN Interface 1 – High Line	
P211	CAN Interface 2 – Low Line	IO_CAN_CHANNEL_2
P224	CAN Interface 2 – High Line	
P212	CAN Interface 3 – Low Line	IO_CAN_CHANNEL_3
P225	CAN Interface 3 – High Line	
P213	CAN Interface 4 – Low Line	IO_CAN_CHANNEL_4
P226	CAN Interface 4 – High Line	
P214	CAN Interface 5 – Low Line	IO_CAN_CHANNEL_5
P227	CAN Interface 5 – High Line	
P215	CAN Interface 6 – Low Line	IO_CAN_CHANNEL_6
P228	CAN Interface 6 – High Line	

4.18.2 Functional Description

CAN is a bidirectional twisted pair bus. Needs termination with 120 Ω in 2-control units, whereas the others remain unterminated.

Termination must be fit at the ends of the bus line to prevent wave reflection. Termination is necessary to enter the recessive state.

Note: A common ground (chassis) or a proper ground connection is necessary for CAN operation. In case of connecting with an external device (e.g. PC with CAN-interface for downloading software) please make sure that the maximum voltage ratings are not violated when connecting to or disconnecting from the CAN bus.

The CAN interface is fully ISO 11898-2/-5 compliant.

See also section *CAN - Controller Area Network driver* of the API documentation [38].

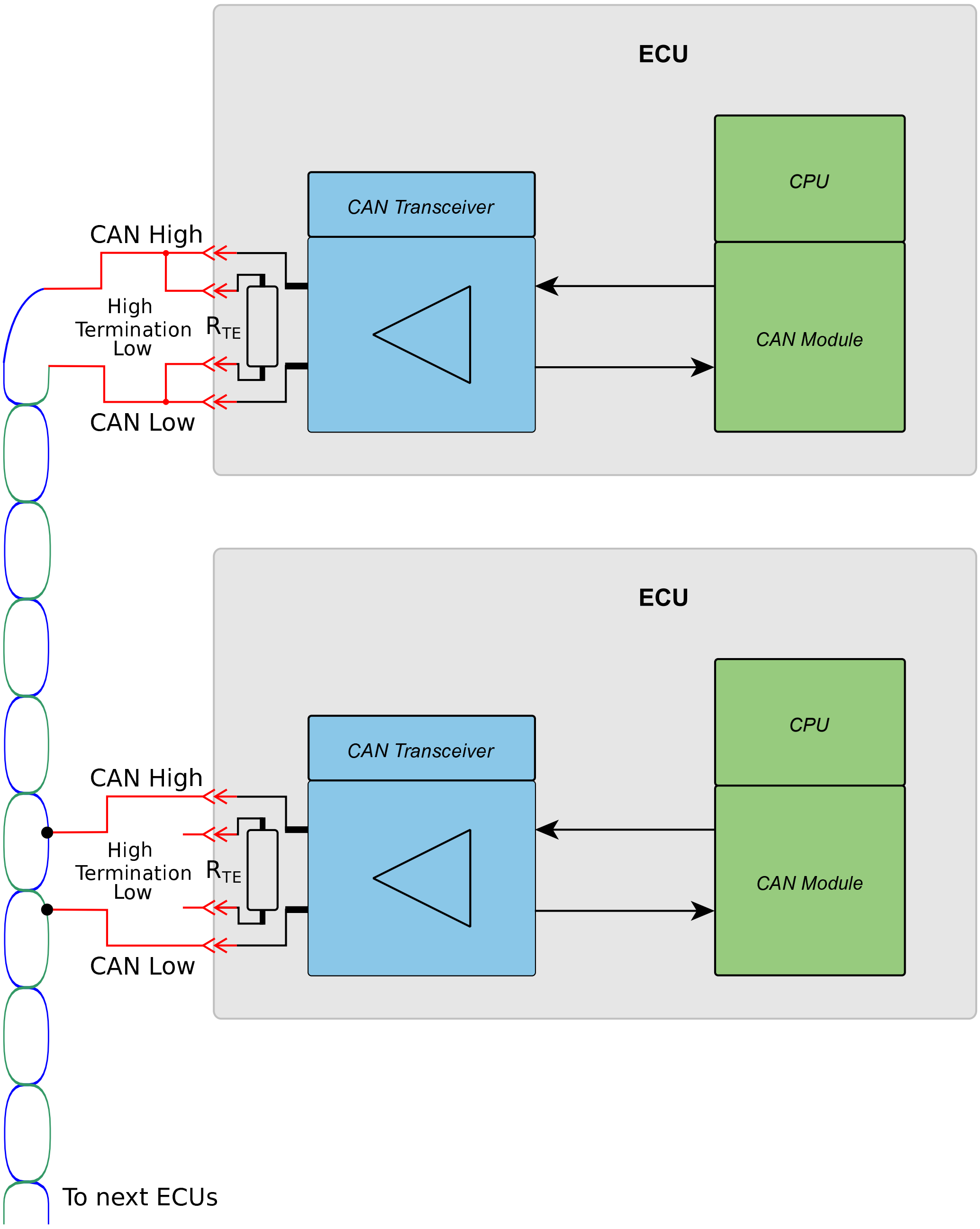

Figure 4.45: CAN interface

4.18.3 Maximum Ratings

T_{ECU} = -40...+125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
V_{CAN_H}, V_{CAN_L}	Bus voltage under overload conditions (e.g. short circuit to supply voltages)		-25	+33	V

4.18.4 Characteristics

T_{ECU} = -40...+125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
C_{in}	Pin output capacitance			100	pF
$V_{in\text{-}CMM}$	Input common mode range	1	-12	12	V
$V_{in\text{-}dif}$	Differential input threshold voltage (V_{CAN_H} - V_{CAN_L})		0.5	0.9	V
$V_{out\text{-}dif}$	Differential output voltage dominant state (V_{CAN_H} - V_{CAN_L})		1.5	3	V
$V_{out\text{-}dif}$	Differential output voltage recessive state (V_{CAN_H} - V_{CAN_L})		-0.1	+0.1	V
V_{CAN_L}, V_{CAN_H}	Common mode idle voltage (recessive state)		2	3	V
I_{CAN_CNL}	Output current limit		-40	-200	mA
I_{CAN_CNH}	Output current limit		40	200	mA
S_{Tr}	Bit rate	2		1000	kbit/s
S_{Tr}	Bit rate	3		500	kbit/s
S_{Tr}	Bit rate			250	kbit/s
S_{Tr}	Bit rate			125	kbit/s
S_{Tr}	Bit rate			100	kbit/s
S_{Tr}	Bit rate			50	kbit/s

Note 1 Due to high current in the wiring harness, the individual ground potential of the control units can differ up to several V. This difference will also appear as common mode voltage between a transmitting and a receiving control unit and not influence the differential bus signal, as long as it is within the common mode limits.

Note 2 Pay attention to the limitations of CAN. The arbitration process allows 1 Mbit/s operation only in small networks and reduced wire length. By way of example, a so-called “private CAN”, which is a short point-to-point connection (less than 10 m) between two nodes only, can be operated at 1 MBit/s.

Note 3 For typical network sizes and topologies (networks with stub wires) and more than two nodes, the practical limit is 500 kbit/s.

4.19 CAN Termination

4.19.1 Pinout

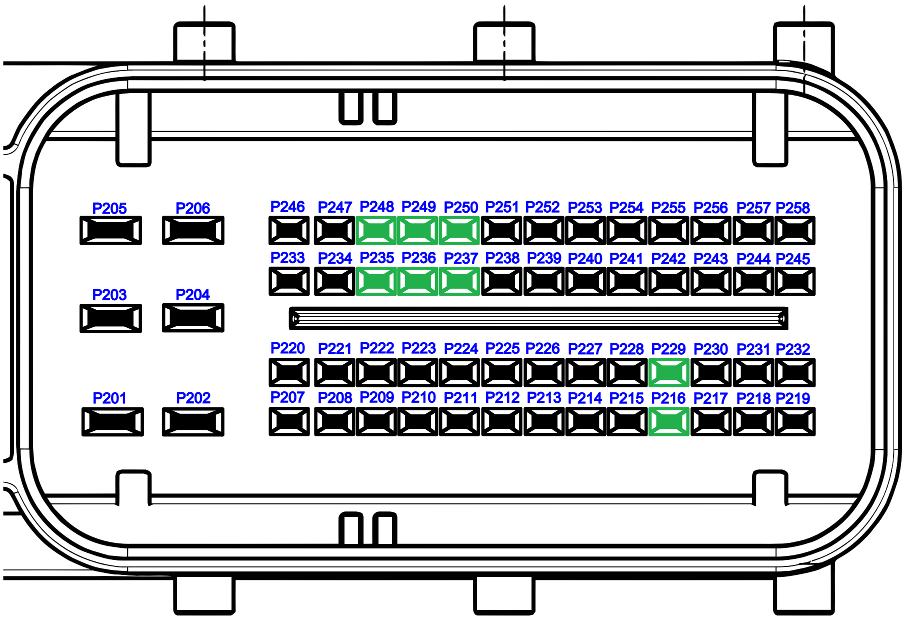

Figure 4.46: Pinout of CAN termination

Pin No.	Function
P235	120 ΩCAN Termination 0 Low
P248	120 ΩCAN Termination 0 High
P236	120 ΩCAN Termination 1 Low
P249	120 ΩCAN Termination 1 High
P237	120 ΩCAN Termination 2 Low
P250	120 ΩCAN Termination 2 High
P216	120 ΩCAN Termination 3 Low
P229	120 ΩCAN Termination 3 High

4.19.2 Functional Description

For easy termination of the CAN bus, there are four built in 120 Ω termination resistors on the HY-TTC 500 which are accessible via 2 connector pins each. They can be used for any CAN port.

The 120 Ω termination of a control unit is realized with two serial 60 Ω resistors (split termination). To get an impedance of 60 Ω on the whole bus, a termination resistor of 120 Ω is required in two control units.

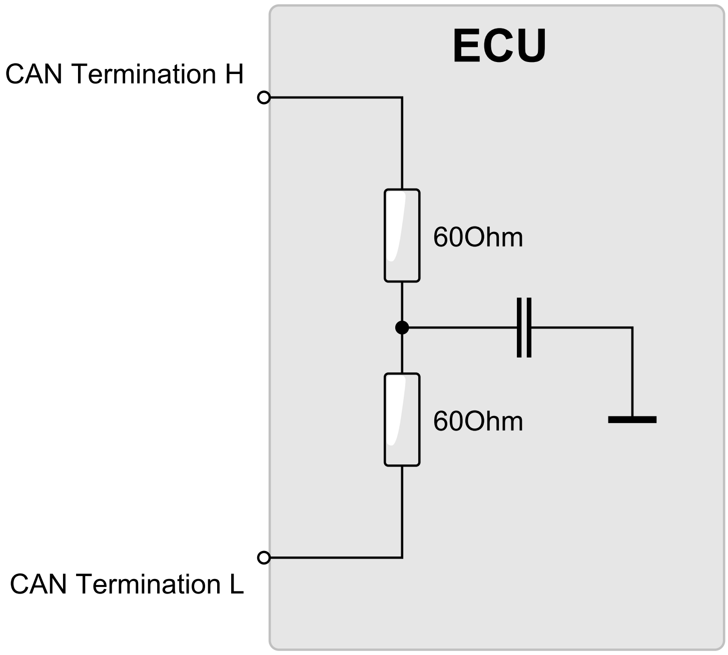

Figure 4.47: Optional CAN termination

4.20 Ethernet

4.20.1 Pinout

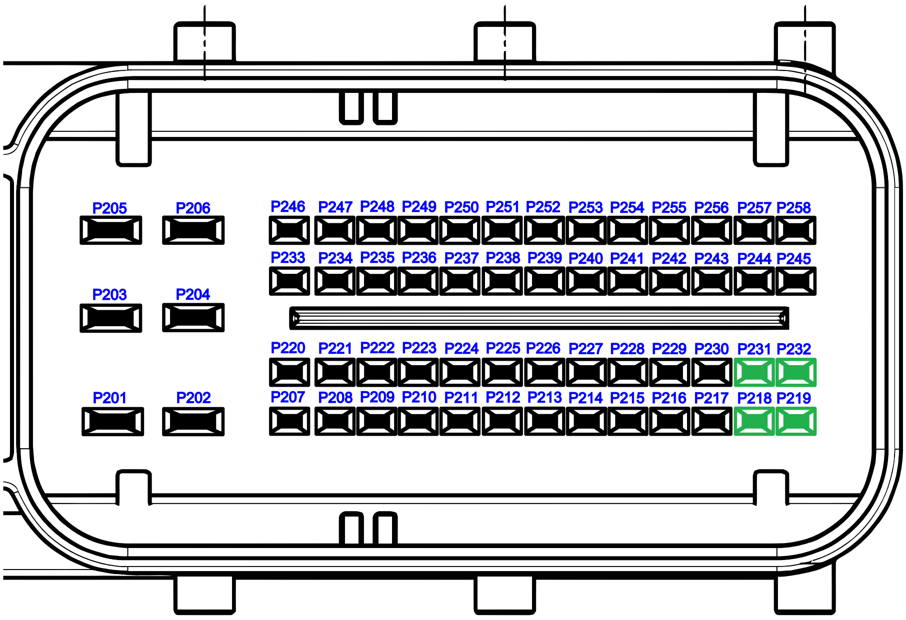

Figure 4.48: Ethernet pinout

Pin No.	Function	SW-define
P218	Ethernet Differential Transmit – TD+	IO_Download
P219	Ethernet Differential Transmit – TD-	
P231	Ethernet Differential Receive – RD-	
P232	Ethernet Differential Receive – RD+	

4.20.2 Functional Description

Ethernet interface for application download and debug purpose.

The 10 Mbit/s full duplex Ethernet port is designed for IEEE 802.3 compliance. However, there is no standard Ethernet connector provided, the Ethernet signals are located on the main connector.

Make sure to use appropriate cabling according to the Ethernet standard. Use at least an Ethernet CAT3 cable for a transmission speed of 10 Mbit/s. In a noisy environment, it is recommended to use shielded cables.

See also section *DOWNLOAD - Driver for Ethernet download* of the API documentation [38].

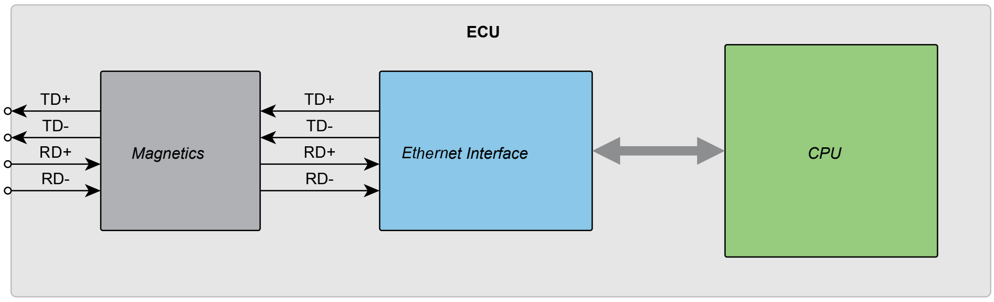

Figure 4.49: Ethernet interface

4.20.3 Maximum Ratings

T_{ECU} = −40...+125 ℃

Symbol	Parameter	Note	Min.	Max.	Unit
$V_{in\text{-}CMM}$	Input common mode voltage range 50/60 HZ AC, 60 s test duration		0	750	V_{RMS}

4.21 Real-Time Clock (RTC)

4.21.1 Pinout

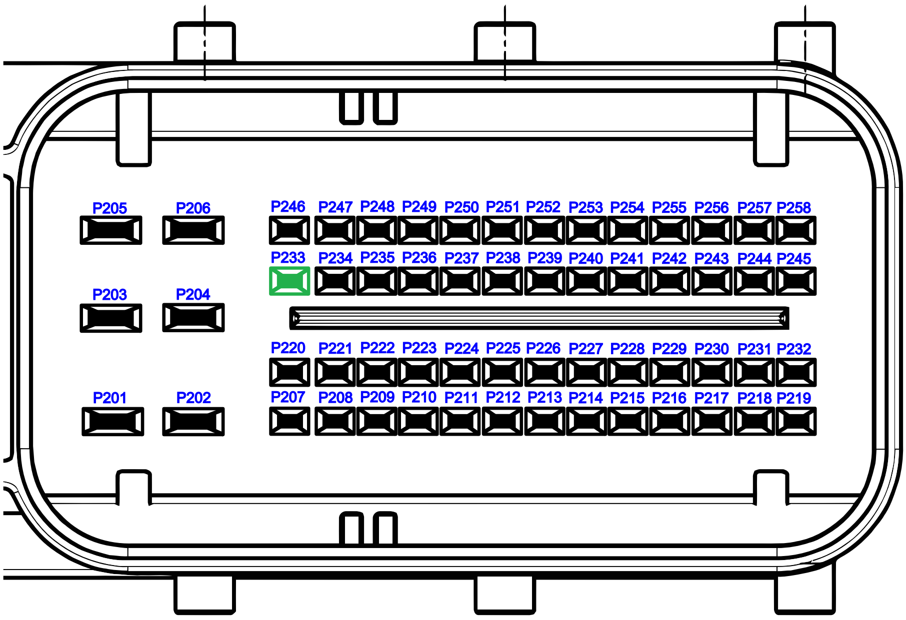

Figure 4.50: RTC pinout

Pin No.	Function	SW-define
P233	Real-Time Clock Backup Battery	IO_RTC

4.21.2 Functional Description

The HY-TTC 500 includes a real-time clock with a backup power system for exact system time- and date-keeping after every power cycle.

The real-time clock module is either supplied by the internal 5 V supply, vehicle battery, or by the external RTC battery pin.

When HY-TTC 500 is connected to the vehicle battery via the BAT+ CPU pin, the RTC is supplied by the vehicle battery independent of whether the device is operational or in power-off mode. When both the RTC backup battery and BAT+ CPU are disconnected, the real-time clock is supplied by an internal capacitor. The capacitor provides approximately 10 minutes of backup time when fully charged. Charging an empty capacitor takes at least 5 minutes.

See also section *RTC - Real Time Clock driver* of the API documentation [38].

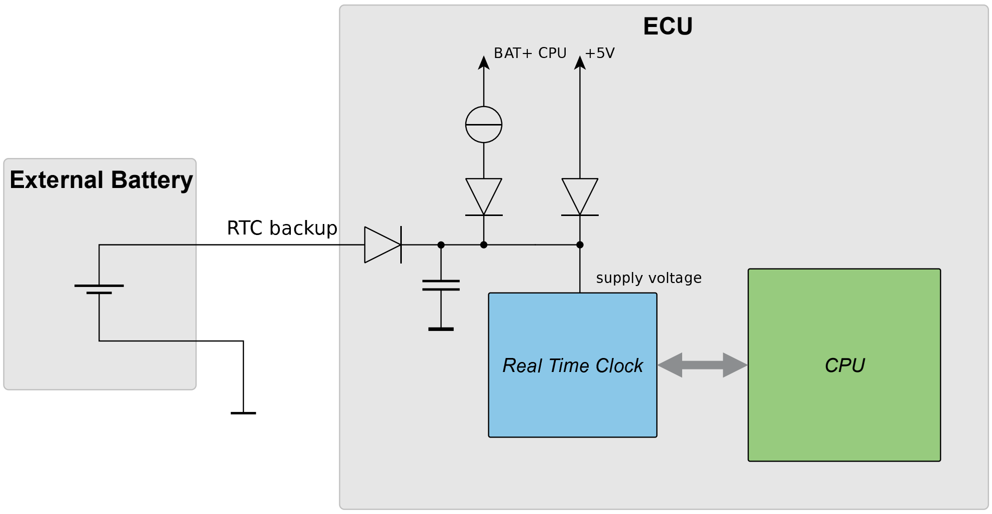

Figure 4.51: Voltage supply of real-time clock

4.21.3 Maximum Ratings

T_{ECU} = -40. . . +125 ℃

Symbol	Parameter	Note	Min.	Max.	Unit
V_{in}	Input voltage		-1	+33	V

4.21.4 Characteristics

T_{ECU} = -40. . . +125 ℃

Symbol	Parameter	Note	Min.	Max.	Unit
C_{in}	Pin input capacitance		8	12	nF
I_{in}	Steady state input current (after 1 min, 3.6 V)			10	µA
I_{in}	Steady state input current for high voltage operation (after 1 min, 16 V)			260	µA
I_{in}	Steady state input current for high voltage operation (after 1 min, 32 V)			600	µA
V_{in}	Input voltage		2.5	32	V
V_{in}	Input voltage	1	1.8	32	V
Δ t/day	Time variation per day at +25 ℃			± 1.5	s/day
RTC_{res}	Real-time clock resolution		1		s

Note 1 T_{ECU} = -40. . . +85 ℃

5 Internal Structure

5.1 Safety Concept

If HY-TTC 500 is used in safety-critical applications, you must follow the requirements specified in the Safety Manual [36].

5.1.1 Overview Safety Concept

The following picture descibes all possible (application-controlled) enable/disable paths for the HY-TTC 500 powerstages.

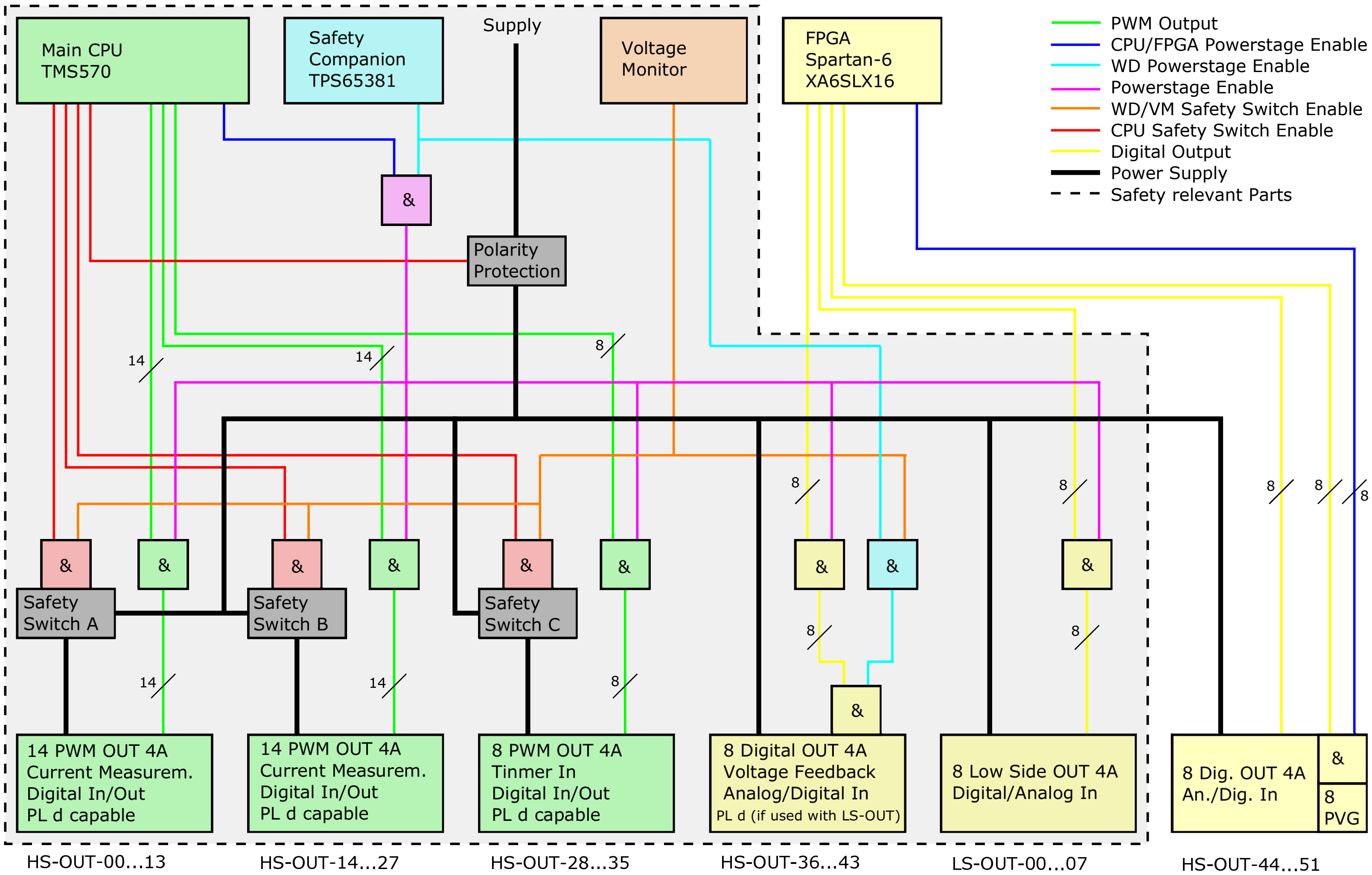

Figure 5.1: Overview safety concept HY-TTC 500

5.2 Thermal Management

5.2.1 Board Temperature Sensor

5.2.1.1 Pinout

Pin No.	Function	SW-define
No connector pin	Internal board temperature sensor	IO_ADC_BOARD_TEMP

5.2.1.2 Functional Description

To measure the temperature T_{ECU} within the housing, there is a temperature sensor located on the printed circuit board. This sensor allows monitoring of the internal board temperature for diagnostic purposes and monitoring of safety features.

Dependent on the T_{ECU} board temperature, the maximum current limit for High- and Low-side output stages is adjusted, see e.g. Section 4.12.4.1 on page 117. This is a strategy to allow higher current consumption at lower temperatures and to bring the ECU immediately to a safe state and switch off loads if over temperature is detected. See Section 5.2.2 on the facing page for temperature limits triggering the safe state.

All input/output tolerance characteristics stated in this System Manual are worst case tolerances and are respected to the internal worst ECU temperature T_{ECU}. At lower $T_{ECU}/T_{ambient}$ temperature and/or lower loads, the tolerances are better.

The (internal) ECU temperature T_{ECU} is close to ambient temperature, when there is no load driven by the power stages at all. The ECU temperature may rise by 40 K above ambient temperature, when there is significant output load (many outputs activated with high load current at the same time) and no airflow supports cooling of the housing. Many applications tend to be somewhere in the middle. Reading out the ECU temperature during system development is a useful feature to analyze the application-specific thermal load and mounting situation. For more information please ask your local sales representative.

5.2.2 Characteristics

T_{ECU} = -40. . . +125 °C

Symbol	Parameter	Note	Min.	Max.	Unit
T_m	Temperature measurement range		-40	+125	°C
$T_{tol\text{-}m}$	Temperature tolerance at 120 °C		-6	+6	K
$T_{safe\ state}$	Temperature limit	1	-40	+125	°C

Note 1 A temperature (including measurement tolerance) below or above the specified limits immediately triggers the safe state.

6 Application Notes

6.1 Wiring Harness

The following general layout rules for the wiring harness must obeyed to enable safe operation of HY-TTC 500.

The ECU is limited to a total load current for the power stages connected to the BAT+ power pins. When all loads are tied toward ground, the load current will be carried by these supply pins as well. As each contact pin is thermally limited to 10 A, multiple supply pins work in parallel for the power stages supply. So, the system designer must be careful with the cable harness design to ensure an even supply current distribution on all pins.

Example: Do not use a cable with a length of two meters and a large diameter for a connection between a fuse box and the ECU and do not crimp it to some piggy tails with a small diameter in the connector area. Small differences between the contact pressure can lead to a big imbalance. In the worst case, one contact carries most of the current load and is overloaded at maximum current. It is necessary to use six wires with the same total cross sectional area as one thick cable. All wires must have the same length and diameter. In this case, an even current distribution can be achieved, even with slightly different contact resistances.

6.2 Handling of High-Load Current

6.2.1 Load Distribution

The permanent input current of the HY-TTC 500 $I_{in\text{-}max}$ is limited due to thermal limits and contact current limits. As the power stages do not have negligible power dissipation, each load current leads to a rise in temperature within the device. To ensure proper operation of the HY-TTC 500 in its temperature range (-40 ℃ to +85 ℃) the total current $I_{in\text{-}total}$ driven by the power stages must be limited and the load must be evenly distributed. If two output stages are operated in mutually exclusive states (e.g. output channel 00 is only activated in state 1 and output channel 01 is only activated in state 2; never activated at the same time), then, as a rule of thumb, these outputs should be driven by a double-channel high-side power stage due to only one active channel at a time.

6.2.2 Total Load Current

Operating all power stages ON with maximum rated current (4 A) that would result in a load current $I_{in\text{-}total}$ in excess of 200 A, which is far beyond any allowed limit to ensure no violation of the allowed contact current limit as well as overall thermal limits.

Therefore, the maximum allowed load current, which can be controlled simultaneously with different power stages, is separately given as the maximum total load current $I_{in\text{-}total}$. This value can only be applied if an equal load distribution over different power stages is ensured - which implicit a different $I_{in\text{-}total}$ limit set by the different number of power stages in each HY-TTC 500 variant, e.g. the same current distributed over 52 output stages cannot be driven with 38 output stages due to the square rise of the power dissipation of the output stages and the respective thermal limit.

Maximum total load current $I_{in\text{-}total}$ for each variant at different T_{ECU}:

Variant	$I_{in\text{-}total}$ [A], T_{ECU} = -40...+85 ℃	$I_{in\text{-}total}$ [A], T_{ECU} = -40...+125 ℃
HY-TTC 580	60	45
HY-TTC 540	50	40
HY-TTC 520	40	30
HY-TTC 510	40	30

Table 6.1: Total Load Current $I_{in\text{-}total}$

6.2.2.1 Calculation of the battery supply current

For the PWM- and digital output stage supply pins BAT+ Power, up to 6 pins can be used in parallel to increase the overall current capability.

For the maximum battery supply current of 60 A, all 6 pins must be used in parallel with the maximum possible wire size (FLRY type) to reduce voltage drop and prevent overheating of the crimp contact. To define a proper cabling, it is important to calculate the maximum average battery supply current first.

For a single digital output power stage it is simply calculated as:

$$I_{\text{bat}} = I_{\text{power-stage}} \tag{6.1}$$

If for example one power stage is loaded with 2 A, it will also load the battery supply with 2 A.

For PWM output stages with inductive load it is calculated as:

$$I_{\text{bat}} = \frac{I_{\text{power-stage}} * (duty\ cycle)}{100} \tag{6.2}$$

Example: A load current of 2 A with a duty cycle of 25 % results in an average battery current of 0.5 A. More precisely explained: a single power stage with 100 Hz PWM frequency will draw 2.5 s in duration 2 A out of the battery followed by 7.5 ms with 0 A. The average current is 0.5 A, the rms current is higher.

However, with a couple of used PWM power stages, there is no significant difference between average and rms current, due to different phase operation of the individual power stages.

Once, the maximum battery supply current for the individual application is calculated, the required minimum number of battery supply wires and/or cabling diameter can be defined.

6.3 Inductive Loads

Inductive loads in PWM operation generate current through the freewheeling diodes, but these diodes have, at the same current, a power dissipation that is several times greater than the high-side switches themselves.

Therefore the duty cycle has a great influence on the power dissipation of the output devices. The duty cycle results from the relationship between coil resistance and supply voltage. A low resistance at a high supply voltage is the worst combination, because it results in a low duty cycle and, thus, in a long conduction time of the diodes.

6.4 Ground Connection of Housing

The HY-TTC 500 housing is not directly connected to ground. In case of chassis mounting this prevents ground loops or excessive current flow through the Negative Power Supply (BAT-) pins and cables in case of partly power connection loss in the negative supply rail of the vehicle.

Instead of direct connection, the housing is capacitively connected to BAT-. In order to discharge static electricity, a 1 MΩ resistor is equipped between housing and BAT-.

Note: It is allowed to mount the housing without any additional isolation directly to the vehicle chassis.

6.5 Motor Control

This application note describes control options for directly driving a DC-motor with outputs of the HY-TTC 500.

6.5.1 Motor Types supported

- Standard brushed DC-motors with defined power rating
- Motors with end position switches and diodes for rotation direction dependent automatic stop at full excursion.
- Motors with integrated electronic control (including brushless DC-motors) up to a well-defined current limit and a maximum input capacitance.

Motors running at different speed by using different windings are not supported.

6.5.2 Direct Control Options

The following control options are described into details in the following chapters:

6.5.2.1 Depending on Direction

- Unidirectional drive
- Bidirectional drive

6.5.2.2 Depending on Single Motor / Motor Group

- Single motor drive
- Motor cluster

6.5.2.3 Depending on Control

- PWM operation
- Steady state operation

6.5.2.4 Depending on stalled Motor Current

- 0 to 2 A max. motor current
- 2 to 4 A max. motor current
- 4 to 16 A max. motor current
- 16 A and higher

6.5.3 Motor Details

6.5.3.1 Start Current

A typical standard (brushed) DC-motor is driven by a current through the rotor winding and permanent magnets in the stator. The rotor winding is normally made out of copper wire with a well-defined DC-resistance. The DC-resistance defines the maximum possible current, when the motor does not rotate (stalled motor). The same current shows as peak current in the moment when supply voltage is switched on the motor. By taking up speed the motor builds up an electro motoric force (EMF) that works against the voltage applied on the motor terminals thus reducing the virtual voltage on the winding -> the current drops. When the motor stabilizes at nominal speed (depending on motor voltage) and the motor runs idle (no mechanical load) the current drops to a value 10 to 100 times smaller than the start current, depending on motor friction in the bearings and other constructive details.

This stall current (= max. motor current with blocked motor) is required to classify the motor power. This has a main impact on possible motor control options and even number of paralleled power stages, if required.

6.5.3.2 Start Current at low Temperatures

The motor windings are made out of copper. The winding resistance goes down towards lower temperature, by approx. 0.4 % / K. This means that at -40 ℃ the winding resistance is lower than the nominal value at +25 ℃ by 26 %. Depending on the lowest operating temperature this effect needs to be considered. In our example that turns a 2 A motor (@+25 ℃) into a 2.7 A motor (@-40 ℃).

6.5.3.3 Battery Voltage Impact

The start current increases proportionally with higher battery voltage. In our assumption the battery voltage of a 12 V system stays below 14 V and for a 24 V system below 28 V. Often the DC resistance of the motor is not specified but the maximum current at nominal voltage. Any maximum current rating of the motor for a nominal voltage needs to be modified for the maximum voltage that the system actually can have.

6.5.3.4 Motors with Limit Switch

Some motor assemblies are fitted with an limit switch turning off the motor power when a maximum position is reached. To allow reversing the motor to move out of this position a diode is placed in parallel to the switch. Operating these kinds of motors is no problem. Care should be taken that diagnostics may detect an open load in the end position which is intended with this kind of operation and no failure.

6.5.3.5 Motors with Electronics Inside

There is no general do or don't with that kind of motors. Essential is only the maximum start current in excess of 50 ms (to be found in chapter Motor classification) and the maximum capacitive load of the motor electronics:

- Max 5 µF when used with H-bridge
- Max 200 µF when used with a single high side power stage (unidirectional supply)

6.5.3.6 Unidirectional Motor Drive

This uses in the simplest case just one high side power stage.

6.5.3.7 Bidirectional Motor Drive

By configuring an H-bridge out of 4 power stages a reverse operation of the motor is possible.

6.5.3.8 Active Motor Brake

With bidirectional motor drive active braking by shorting the motor terminals is supported.

6.5.3.9 Motors Reversing with / without Braking

Motor reversing out of full speed can lead to unexpected high motor current. The cause is that for example a 24 V motor running at nominal speed has a back EMF of almost 24 V with the same sign as the voltage applied. By sudden revering the EMF-voltage adds to the terminal voltage, which means the motor current is as high as with starting at 48 V battery voltage. In other words, the peak motor current is doubled by sudden reverse. For small motors this is neither a problem to the motor nor to the driving power stages. For bigger motors a braking phase needs to be inserted first followed by the reversing command.

6.5.3.10 PWM Operation / steady State Control

In general PWM operation is not recommended as the PWM frequency for high efficient motors often require frequencies in excess of 10 kHz. A PWM frequency so high would cause lots of unacceptable electromagnetic noise (EMI) or dramatic switching losses or both. For small motors we can use a 200 Hz PWM instead. This causes a significant torque ripple which might be desirable for slow positioning of the motor with the need to prevent static friction.

6.5.3.11 Maximum Average Current (PWM and Digital Power Stages)

The maximum steady state motor current is 4 A (up to +85 °C internal ECU temperature) or 3 A (over +85 °C). This comes from the limits of the power stage used. Normally the start current limits the choice of suitable motors and not the supply current, when the motor is running. The motor current is measured in the ECU and can be used for control / diagnostic purposes.

6.5.3.12 Peak Current Bidirectional Drive (Digital Power Stages only)

The digital (non-PWM) power stages tolerate inrush current of up to 6 A for 100 ms. Fast accelerating motors show first an inrush current peak followed by a reduction in current to less than 50 % of the initial current after 20 ms. In other words, most motors that start with 6 A drop to less than 4 A in 100 ms. Motors with that kind of fast acceleration will work properly even with a winding resistance that is lower by a factor of 1.5 (stall current 6 A instead of 4 A).

6.5.3.13 Peak Current Unidirectional Drive (Digital Power Stages only)

Single high side power stages tolerate an inrush current that linearly declines from 16 to 6 A in 50 ms. Fast accelerating motors show after an initial current peak a reduction in current to less than 50 % after 20 ms. Motors with that kind of fast acceleration may have a winding resistance that is lower by a factor of 4 (stall current 16 A instead of 4 A).

6.5.4 Motor Classification

6.5.4.1 Independent from Speed Class

The most conservative limits apply for slow accelerating motors, found in the following section:

6.5.4.1.1 Tiny Motors (0 to 2 A)

This class gives the widest choice of control options.

	12 V System	24 V System
Max. battery voltage [1]	14 V	28 V
Motor blocked current	0 to 2 A	
DC-resistance	>=7.0 Ω	>=14.0 Ω
Unidirectional drive	yes	yes
Bidirectional drive	yes	yes
Active motor braking	yes	yes
Reverse without brake	yes	yes
PWM operation	yes	yes
Suitable high side power stages	PWM power stages, digital HS power stages	

[1] For other max. battery voltage please recalculate winding resistance (R = U / I).

6.5.4.1.2 Small Motors (2 to 4 A)

For this class some restrictions apply.

	12 V System	24 V System
Max. battery voltage [1]	14 V	28 V
Motor blocked current	2 to 4 A	
DC-resistance	>=3.5 Ω	>=7.0 Ω
Unidirectional drive	yes	yes
Bidirectional drive	yes	yes
Active motor braking	yes	yes
Reverse without brake	no	no
PWM operation	no	no
Suitable high side power stages	PWM power stages, digital HS power stages	
Maximum periodic DC peak current [2]	4 A (T_{ECU} < +85 °C)	
Maximum periodic DC peak current [2]	3 A (T_{ECU} > +85 °C)	

[1] For other max. battery voltage please recalculate winding resistance (R = U / I).
[2] Some motors draw a significant ripple current. The highest periodic peaks shall be below that limit.

6.5.4.1.3 Medium Power Motors (4 to 14 A)

This class needs paralleled power stages. In the example below each switch utilizes 4 power stages in parallel for each switch. This in theory quadruples the power rating. In practice depending on close-to-perfect cabling a de-rating of maximum current (14 A instead of 16 A) is foreseen.

	12 V System	24 V System
Max. battery voltage [1]	14 V	28 V
Motor blocked current	4 to 14 A	
DC-resistance	>=1.0 Ω	>=2.0 Ω
Unidirectional drive	yes	yes
Bidirectional drive	yes	yes
Active motor braking	yes	yes
Reverse without brake	no	no
PWM operation	no	no
Suitable high side power stages	digital HS power stages only	
Maximum periodic DC peak current [2]	12 A (T_{ECU} < +85 °C)	
Maximum periodic DC peak current [2]	10 A (T_{ECU} > +85 °C)	

[1] For other max. battery voltage please recalculate winding resistance (R = U/I).
[2] Some motors draw a significant ripple current. The highest periodic peaks shall be below that limit.

6.5.4.2 Fast Accelerating Motors only

Higher inrush current,is allowed for fast accelerating motors. Most actuator motors fall into this category. Depending on operation mode different limits apply:

6.5.4.2.1 Bidirectional (max. 6 A)

For this class some restrictions apply.

	12 V System	24 V System
Max. battery voltage [1]	14 V	28 V
Motor blocked current	<6 A	
DC-resistance	>=2.35 Ω	>=4.7 Ω
Unidirectional drive	yes	yes
Bidirectional drive	yes	yes
Active motor braking	yes	yes
Reverse without brake	no	no
PWM operation	no	no
Suitable high side power stages	digital HS power stages only	
Maximum periodic DC peak current [2]	4 A (T_{ECU} < +85 °C)	
Maximum periodic DC peak current [2]	3 A (T_{ECU} > +85 °C)	
Inrush current drops after 50ms to	<6 A	
Inrush current drops after 50ms to	<4 / 3 A (depending on ECU temperature)	

[1] For other max. battery voltage please recalculate winding resistance (R = U/I).
[2] Some motors draw a significant ripple current. The highest periodic peaks shall be below that limit.

6.5.4.2.2 Bidirectional (max. 16 A)

For this class each low side switch is built out of 4 switches working in parallel to increase current handling. The high side switches are still single switches.

	12 V System	24 V System
Max. battery voltage [1]	14 V	28 V
Motor blocked current	<16 A	
DC-resistance	>=0.88 Ω	>=1.75 Ω
Unidirectional drive	yes	yes
Bidirectional drive	yes	yes
Active motor braking	yes	yes
Reverse without brake	no	no
PWM operation	no	no
Suitable high side power stages	digital HS power stages only	
Maximum periodic DC peak current [2]	4 A (T_{ECU} < +85 ℃)	
Maximum periodic DC peak current [2]	3 A (T_{ECU} > +85 ℃)	
Inrush current drops after 50ms to	<6 A	
Inrush current drops after 50ms to	<4 / 3 A (depending on ECU temperature)	

[1] For other max. battery voltage please recalculate winding resistance (R = U/I).
[2] Some motors draw a significant ripple current. The highest periodic peaks shall be below that limit.

6.5.4.2.3 Bidirectional (max. 20 A)

For this class each high and low side switch is built out of 4 switches working in parallel to increase current handling. This in theory quadruples the power rating. In practice depending on close-to-perfect cabling a de-rating of maximum current (20 A instead of 24 A) is foreseen.

	12 V System	24 V System
Max. battery voltage [1]	14 V	28 V
Motor blocked current	<20 A	
DC-resistance	>=0.7 Ω	>=1.4 Ω
Unidirectional drive	yes	yes
Bidirectional drive	yes	yes
Active motor braking	yes	yes
Reverse without brake	no	no
PWM operation	no	no
Suitable high side power stages	digital HS power stages only	
Maximum periodic DC peak current [2]	12 A (T_{ECU} < +85 ℃)	
Maximum periodic DC peak current [2]	10 A (T_{ECU} > +85 ℃)	
Inrush current drops after 50ms to	<12 / 10 A (depending on ECU temperature)	

[1] For other max. battery voltage please recalculate winding resistance (R = U/I).
[2] Some motors draw a significant ripple current. The highest periodic peaks shall be below that limit.

6.5.4.2.4 Unidirectional (max. 16 A)

	12 V System	24 V System
Max. battery voltage [1]	14 V	28 V
Motor blocked current	<16 A	
DC-resistance	>=0.88 Ω	>=1.75 Ω
Unidirectional drive	yes	yes
Bidirectional drive	no	no
Active motor braking	no	no
Reverse without brake	no	no
PWM operation	no	no
Suitable high side power stages	digital HS power stages only	
Maximum periodic DC peak current [2]	4 A (T_{ECU} < +85 °C)	
Maximum periodic DC peak current [2]	3 A (T_{ECU} > +85 °C)	
Inrush current drops after 50ms to	<6 A	
Inrush current drops after 50ms to	<4 / 3 A (depending on ECU temperature)	

[1] For other max. battery voltage please recalculate winding resistance (R = U/I).

[2] Some motors draw a significant ripple current. The highest periodic peaks shall be below that limit.

6.5.4.2.5 Unidirectional (max. 32 A)

For this class two digital high side power stages are operated in parallel.

	12 V System	24 V System
Max. battery voltage [1]	14 V	28 V
Motor blocked current	<32 A	
DC-resistance	>=0.4 Ω	>=0.8 Ω
Unidirectional drive	yes	yes
Bidirectional drive	no	no
Active motor braking	no	no
Reverse without brake	no	no
PWM operation	no	no
Suitable high side power stages	digital HS power stages only	
Maximum periodic DC peak current [2]	8 A (T_{ECU} < +85 °C)	
Maximum periodic DC peak current [2]	6 A (T_{ECU} > +85 °C)	
Inrush current drops after 50ms to	<12 A	
Inrush current drops after 50ms to	<8 / 6 A (depending on ECU temperature)	

[1] For other max. battery voltage please recalculate winding resistance (R = U/I).
[2] Some motors draw a significant ripple current. The highest periodic peaks shall be below that limit.

6.5.5 Connection

6.5.5.1 Unidirectional Single Power Stage

Figure 6.1 on the current page shows the battery wiring for maximum total load current. This kind of wiring is used to increase output current capability. It is strongly recommended to support equal distribution of current between the power supply pins. This implies to use cables of same diameter and same wire length in parallel. In this example all power supply pins (BAT+ and GND) are connected to cable of maximum diameter supported by the appropriate connector pin. The cables in parallel are going to a distribution point (for example in the fuse box) and from there with a bigger diameter all the way to the battery.

The return pin of the motor is connected not direct to the ECU ground but to somewhere else, perhaps to the chassis. In this case significant voltage drops may occur between ECU and motor ground connection. This can lead to unexpected fluctuations in motor voltage and motor current.

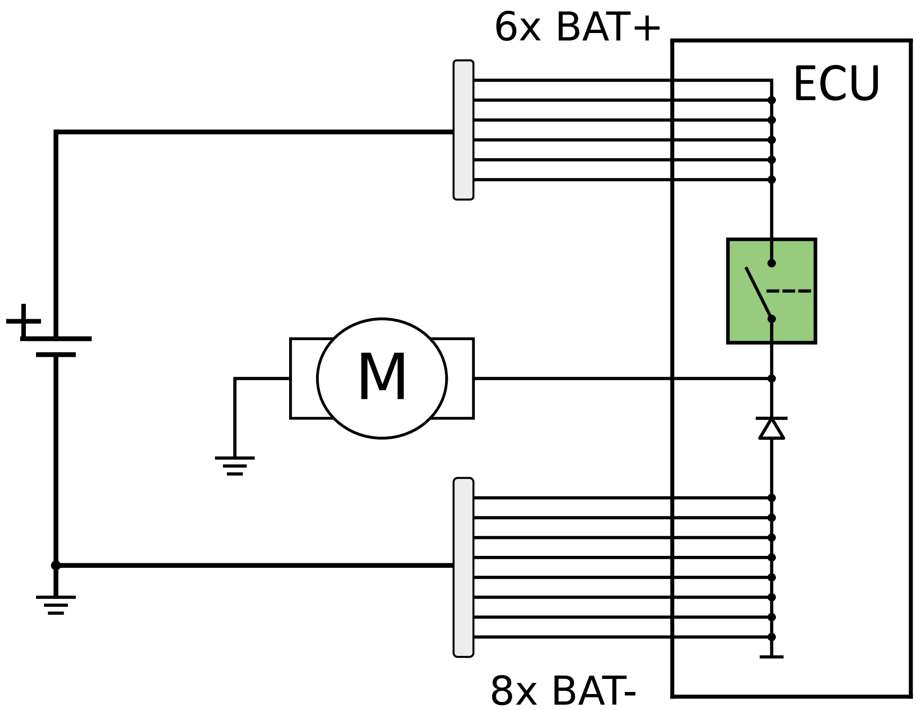

Figure 6.1: Unidirectional Single Power Stage

A better solution for achieving less voltage drop in the return path shows the Figure 6.2 on this page.

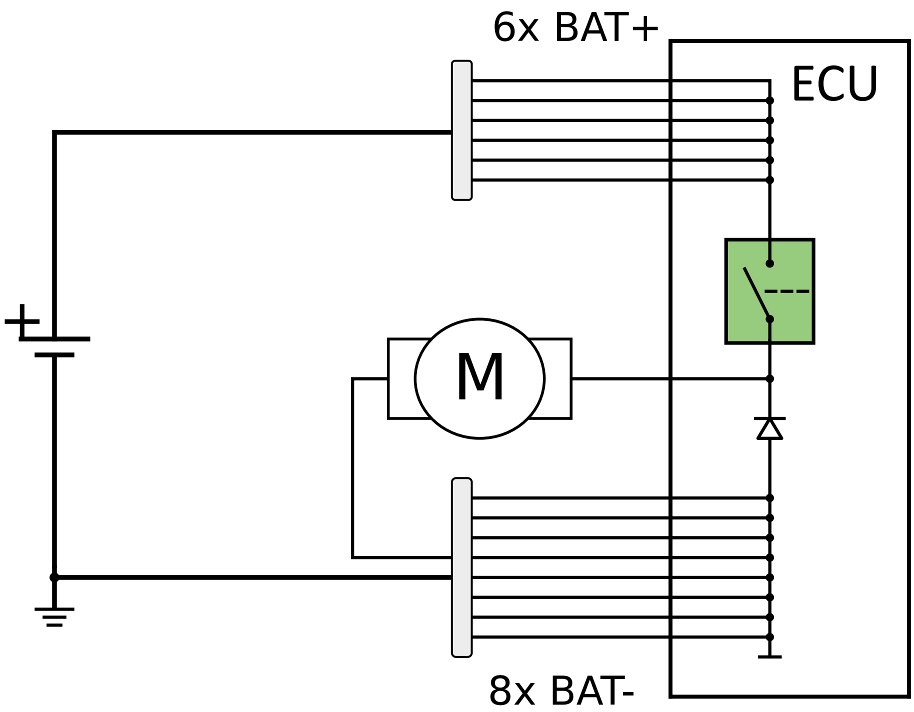

Figure 6.2: Unidirectional Single Power Stage

6.5.5.2 Unidirectional Double Power Stage

Higher current can be achieved by paralleling output stages, see Figure 6.3 on the current page. Please observe the use of cables of same diameter and same wire length in parallel on the power stages as well.

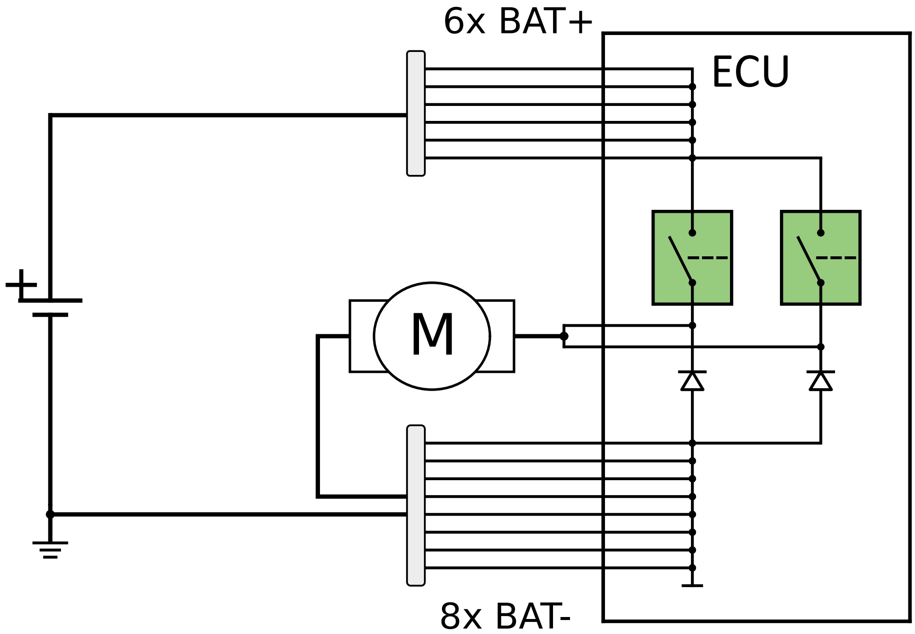

Figure 6.3: Unidirectional Double Power Stage

6.5.5.3 Bidirectional H-bridge (Single Power Stages)

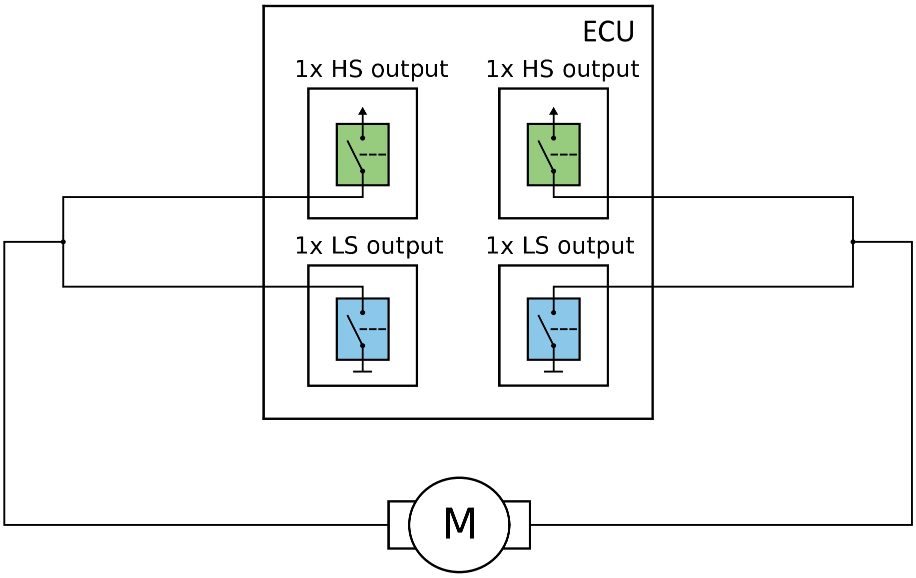

Figure 6.4: Bidirectional H-bridge (Single Power Stages)

6.5.5.4 Bidirectional H-bridge (Multiple Low Side Power Stages)

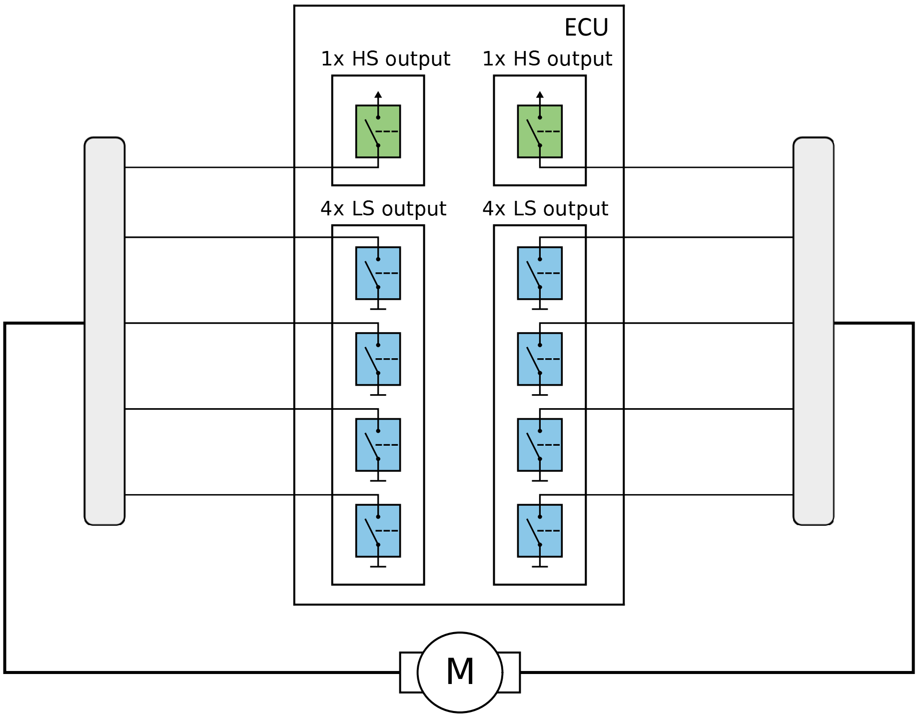

Figure 6.5: Bidirectional H-bridge (Multiple Low Side Power Stages)

6.5.5.5 Bidirectional H-bridge (Multiple High and Low Side Power Stages)

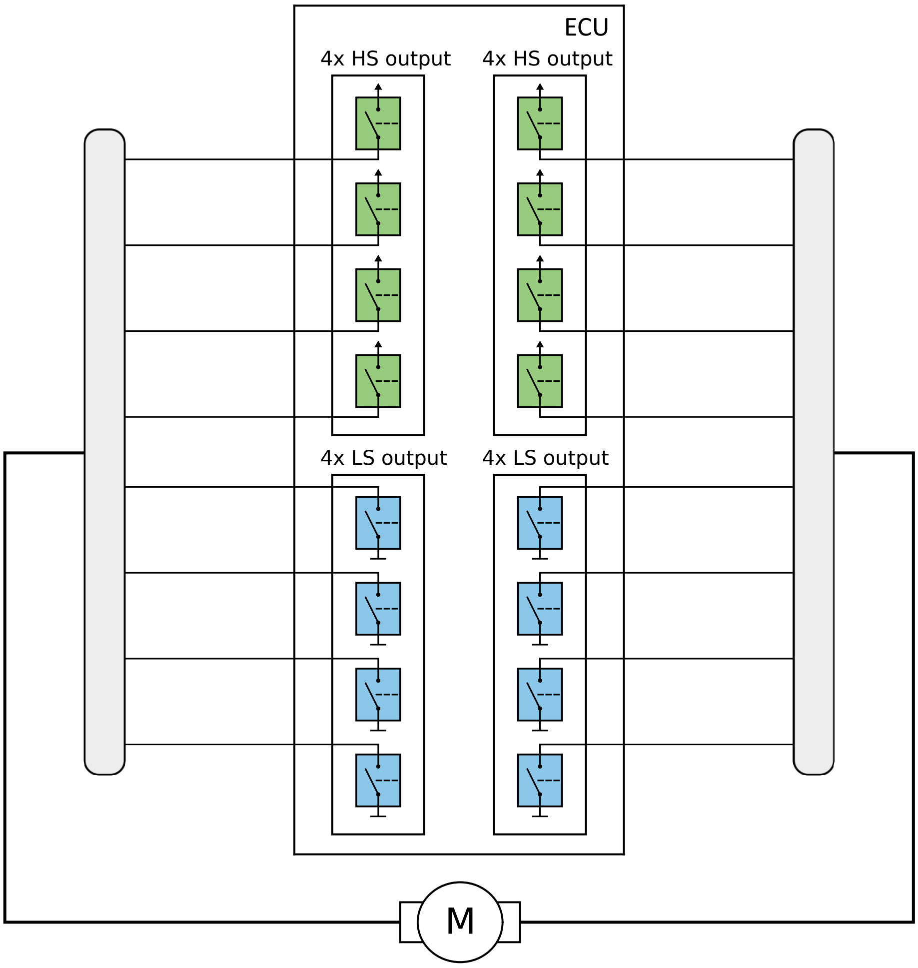

Figure 6.6: Bidirectional H-bridge (Multiple High and Low Side Power Stages)

6.5.5.6 Motor Cluster

This is a usual configuration to control 2 motors, that are never operated at the same time, with 3 half bridges (3 x high side + 3 x low side power stage).

Example: Outside mirror control

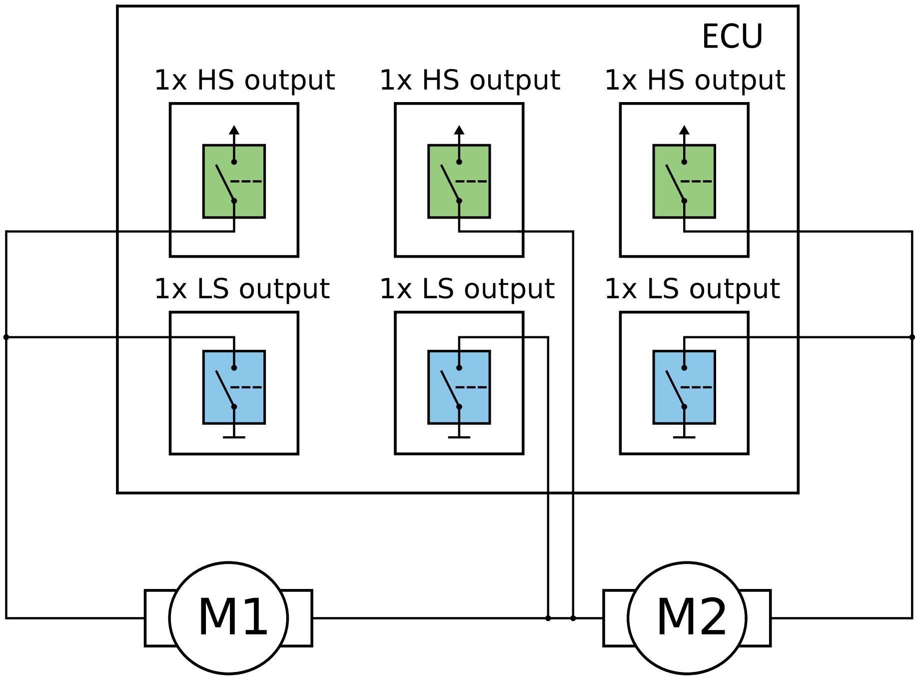

Figure 6.7: Motor Cluster (Example: Outside Mirror Control)

6.5.6 Power Stages for Motor Control

6.5.6.1 High Side Power Stages for PWM Operation

Depending on the HY-TTC 500 variant, up to 36 high side PWM power stages are provided:

- HY-TTC 580: HS channel 00 .. 35
- HY-TTC 540: HS channel 00 .. 27
- HY-TTC 520: HS channel 00 .. 09, 14 .. 21
- HY-TTC 510: HS channel 00 .. 07, 14 .. 21

6.5.6.2 High Side Power Stages for Static Operation

Depending on the HY-TTC 500 variant, up to 16 digital high side power stages are provided:

- HY-TTC 580: HS channel 36 .. 51
- HY-TTC 540: HS channel 36 .. 43
- HY-TTC 520: HS channel 36 .. 43
- HY-TTC 510: HS channel 36 .. 51

6.5.6.3 Low Side Power Stages

8 digital low side power stages are provided independent from the HY-TTC 500 variant:

- HY-TTC 580: HS channel 00 .. 07
- HY-TTC 540: HS channel 00 .. 07
- HY-TTC 520: HS channel 00 .. 07
- HY-TTC 510: HS channel 00 .. 07

6.5.7 Parallel Operation of Power Stages

This kind of wiring is used to increase output current capability. It is strongly recommended to support equal distribution of current between those power stages. This implies to use cables of same diameter and same wire length in parallel as well as to use matched power stages inside the ECU. Matched power stages are always grouped in pairs of even and following odd signal number. For example a good pair is HS-OUT-36 and 37. See Section 4.15.2.1 on page 143 for more information about Power Stage Pairing.

6.5.8 Cabling

The cable resistance adds to the rotor resistance of the motor. While the impact of a small voltage drop is negligible for normal operation it can be helpful to reduce inrush current when switching on the motor. If there is significant cable length used for motor cabling, please add the wire resistance to the motor current classification to relax need of paralleling power stages.

For power stages operated in parallel please observe that this cable resistance also helps to equally distribute the load current over the power stages. On the other hand, if there is significant difference in wire length of paralleled output stages this enhances non-symmetric load, which means one power stage might carry significant more load current than the others. This ends up in reduced power capability.

4 power stages of the HY-TTC 500 would theoretically deliver a total current of 16 A if switched on concurrently.

So the system designer must be careful with the cable harness design to guarantee evenly distribution of supply current on all 4 pins. For example, it is not OK to use 1 cable with large diameter to connect from a fuse box to the ECU for, let's say 2 meters and crimp it to 4 piggy tails with small diameter in the connector area. Small differences in the contact pressure can lead to a big imbalance. In worst case condition 2 contacts carry almost the full current load and are overloaded at maximum current. It is better to use 4 wires with the same total cross sectional area than this one thick cable. All wires must have exactly the same length and diameter. In this case an evenly distribution of current will be the case even with slightly different contact resistance.

6.5.9 Control Sequence for Bidirectional Drive

The complexity of SW control for bidirectional motor operation depends more or less on the motor type. Up to 2 A peak current a sudden reverse is still limited to 4 A maximum current, which is below the over load threshold of the power stages.

Above 2 A an active braking sequence close to speed 0 must be inserted prior to reversing the motor.

6.5.9.1 Motor Running

Forward or backward rotation is managed by diagonally switch on a high side and a low side power stage simultaneously. Always switch on low side first followed by high side. Never switch on high side first and then switch on low side.

6.5.9.2 Motor Breaking

Motor braking is managed by switching on both high side power stages simultaneously.

Always switch off low side first before switching on high side. Never switch on high side while low side is still on.

Please observe that the high side power stages can only measure current in positive direction (out of the terminal towards ground)

While the high side power stages are actively braking, one power stage (this one that was operated before entering brake mode) shows a high current (close to start current) declining to 0 A, while the other is reverse operated and does not show any current at all.

The brake sequence is considered to be finished when the brake current falls below 1 A. By using multiple power stages the limit is 1 A per power stage.

A reverse motor run now can be started.

6.5.9.3 Paralleled High Side Power Stage Control

This configuration can be needed for to increasing inrush current capability. However, in case of a short circuit this also dramatically increases the peak current that might be drawn out of the battery terminals. It is strongly recommended to first switch on only one single power stage and measure the output voltage. If the output voltage is > 0.75xVBAT this indicates that no short circuit is on the motor terminals and the other 1 to 3 high side power stages working in parallel can be switched on. The time that only one single power stage is working shall be limited to <= 20 ms. Important: this sequencing applies only on the high side power stages. The low side outputs needs to be switched on first.

6.6 Power Stage Alternative Functions

This application note describes the DO and DON'TS when using the alternative functions of HY-TTC 500 high- and low-side power stages.

6.6.1 High-Side Output Stages

High-side power stages can alternatively used as analog, digital or frequency inputs.

BAT+Power +0,5 V
In all high-side power stages there is a parasitic diode that conducts if the output voltage or, in case of alternative input function, the input voltage is externally driven higher than the voltage on the BAT+Power supply pins. The input voltage on all high-side stages, including the alternative input functions, must never exceed the power stage supply BAT+ Power +0,5 V.

To counteract such fault scenarios, maximum ratings and specified wiring examples must be followed and are essential for safe operation.

6.6.1.1 Wiring examples

Attention!
In many applications external switches (open collector, open drain (NPN) or a push-pull type), push-buttons or analog sensors have to be monitored by an alternative input of the ECU. Switches which are directly switching to battery voltage must not be used with alternative inputs.

For safety critical applicactions, however, additional restrictions apply, see the following sections.

Workaround for safety critical applications:

- **1.**: Usage of external switches connected to GND.
  Short circuits to battery supply needs to be excluded in the system architecture.
- **2**: Battery supplied switches and sensors need to be supplied via a digital output of the HY-TTC 500.

6.6.1.1.1 Valid Wiring Examples

The following wiring examples for external switches and analog sensors connected to battery supply (PWM/digital high-side output) and to GND are allowed and can help avoiding system fault scenarios:

Switch connected to GND:

Usage of external switches connected to GND.
Short circuits to battery supply needs to be excluded in the system architecture.

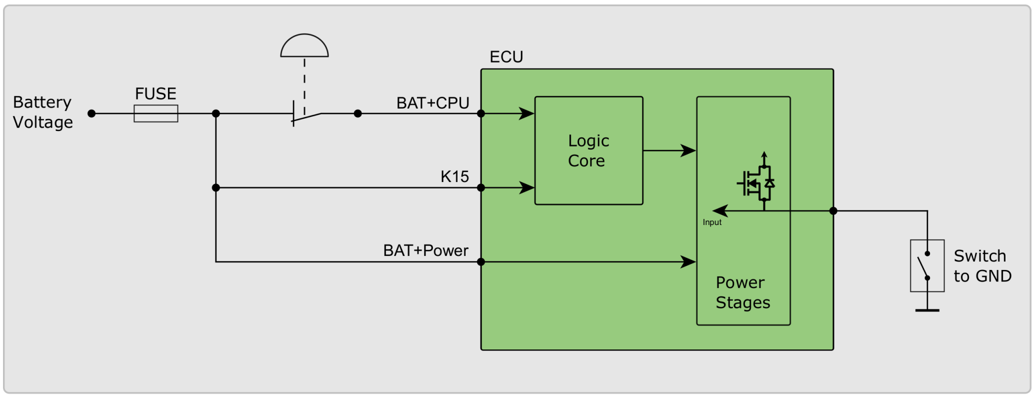

Figure 6.8: Switch connected to GND

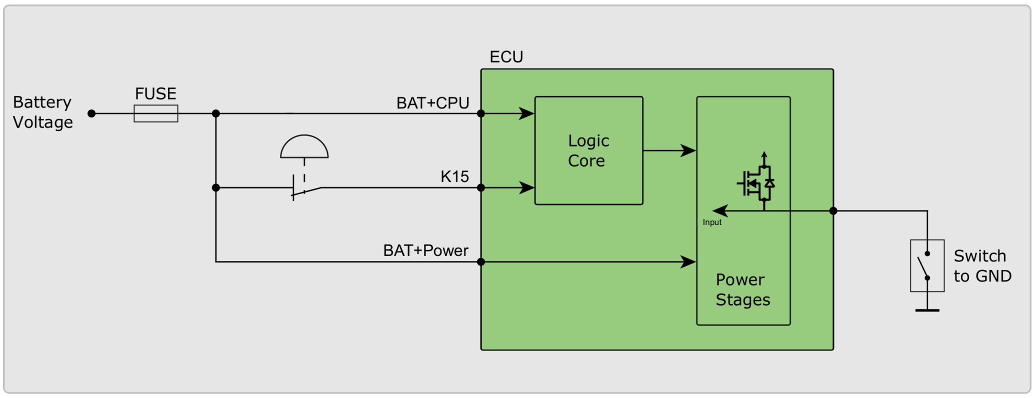

Figure 6.9: Switch connected to GND

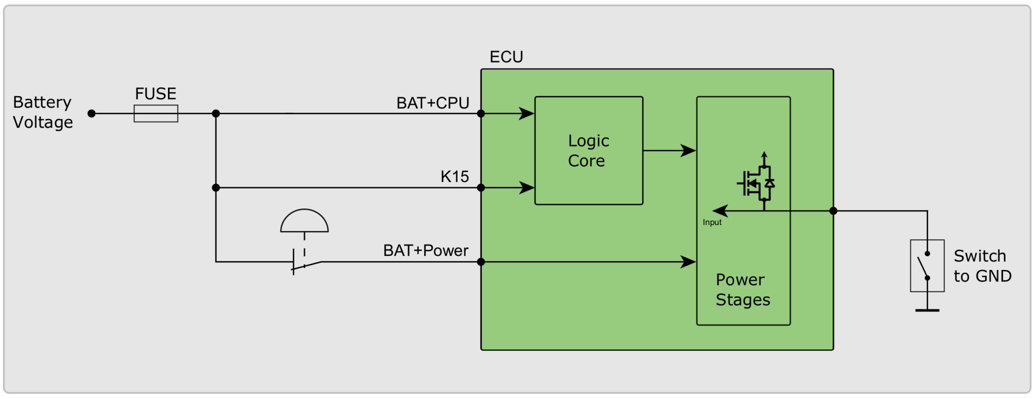

Figure 6.10: Switch connected to GND

Switches connected to PWM high-side output stage:

Digital switches and analog sensors are supplied via an HY-TTC 500 PWM high-side output pin, the switch/sensor output is monitored by an alternative (PWM high-side) input.

Caution! - The sourcing PWM high-side output stage (IO_PWM_00 - IO_PWM_35) must be out of the same secondary shut-off path (A, B or C) as the alternative input pin. For example IO_PWM_00 (output/source) supplies the digital sensor and the sensor output is monitored by IO_PWM_13 (input), both IO's are out of secondary shut-off path A. See Table 4.3 on page 113 for secondary shut-off paths and their relation to the alternative inputs.

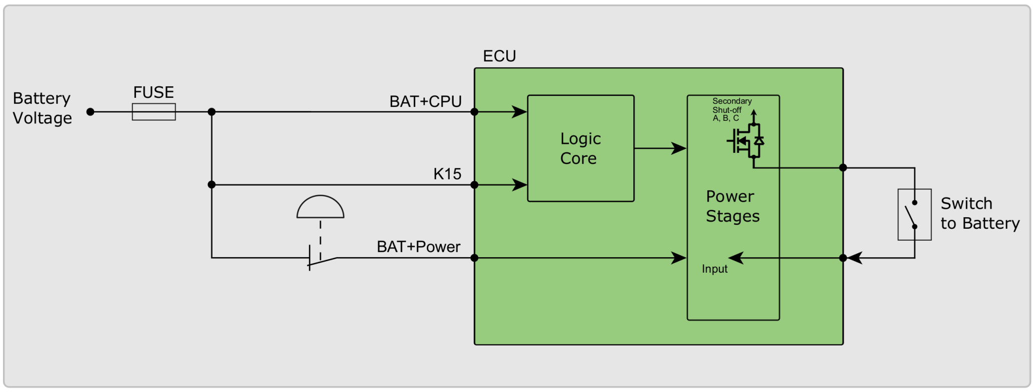

Figure 6.11: Switch connected to PWM high-side output stage

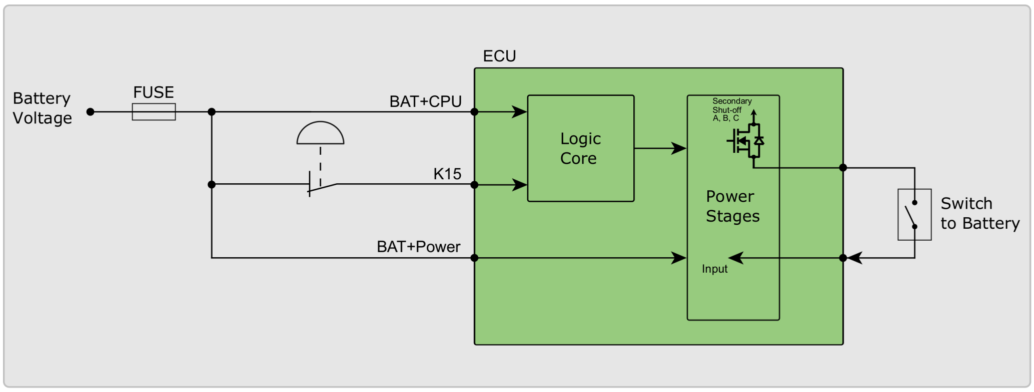

Figure 6.12: Switch connected to PWM high-side output stage

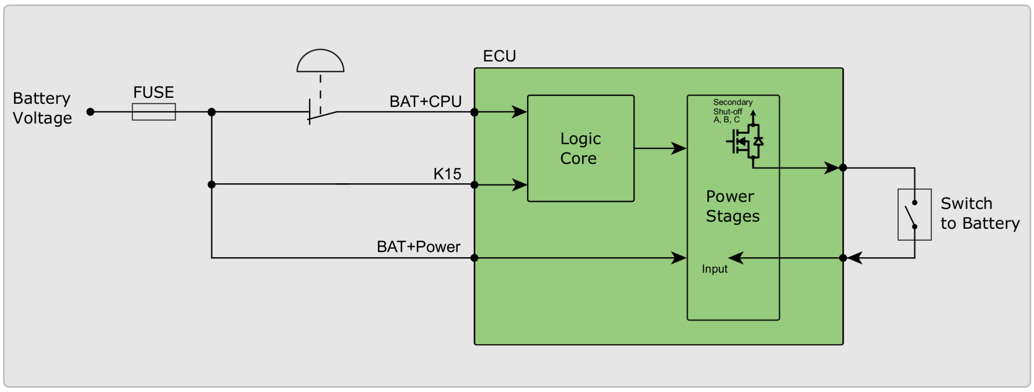

Figure 6.13: Switch connected to PWM high-side output stage

Switches connected to digital high-side output stage:

Digital switches and analog sensors are supplied via an HY-TTC 500 digital high-side output pin, the switch/sensor output is monitored by an alternative (digital high-side) input.

IO_DO_00 - IO_DO_07 and IO_PVG_00 - IO_PVG_07 are not equipped with an secondary shut-off paths. These high-side output stages can by themselves not be used for safety critical applications. If the alternative input function of IO_DO_00 - IO_DO_07 and IO_PVG_00 - IO_PVG_07 shall be used, the connected sensor must be supplied by an digital output stage out of these outputs.

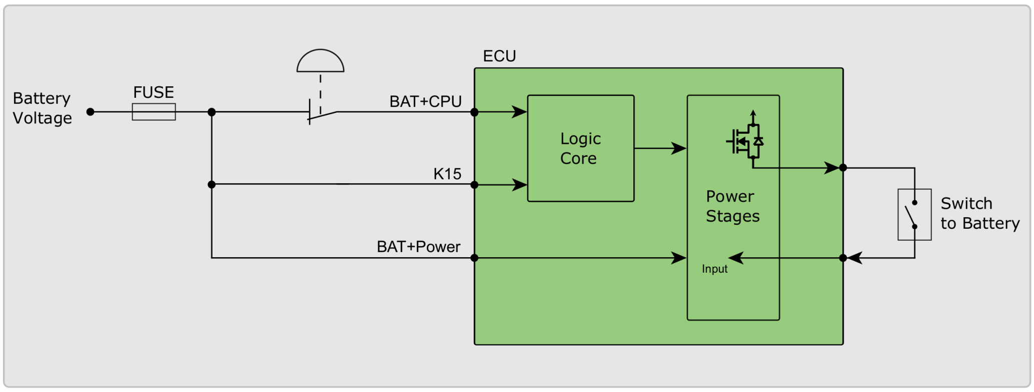

Figure 6.14: Switch connected to digital high-side output stage

6.6.1.1.2 Invalid Wiring Examples

The following wiring examples of external switches or analog sensors connected to battery supply are not allowed and must be avoided for safety critical systems.

Nonconforming wiring can lead to destruction of the HY-TTC 500.

Stop switch / blown fuse / K15 wiring

Digital switches and analog sensors are supplied directly from the battery.

If for example the fuse is blown, BAT+Power is disconnected (loose connection) or the stop switch is pressed, digital switches or analog sensors are still supplied. The current flows over the closed switch and the parasitic body diode of the output stage used as input. All the load current of all other outputs now flows via the body diode of this single output stage. This may overload and even destroy this output stage.

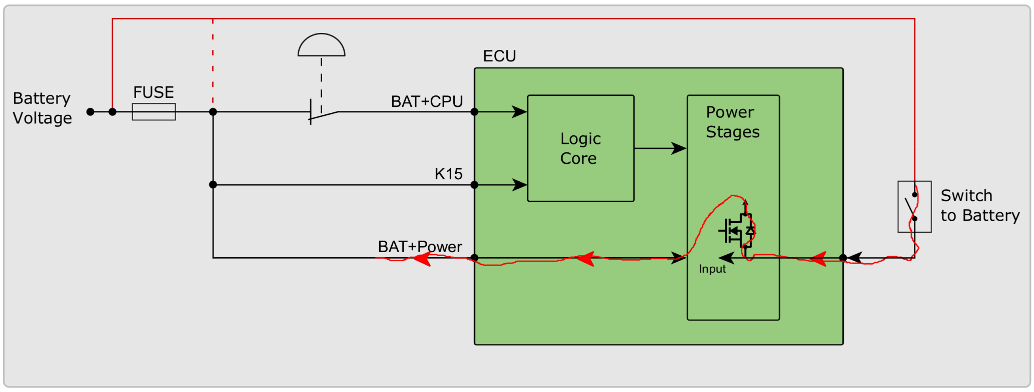

Figure 6.15: Switch connected to battery supply

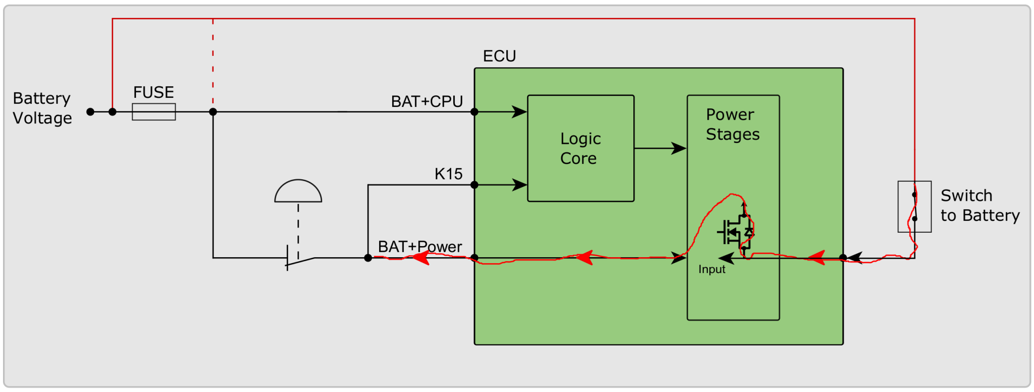

Figure 6.16: Switch connected to battery supply

6.6.2 Low-Side Output Stages

Low-side power stages can be alternatively used as analog or digital inputs.

BAT+Power +0,5 V
In all low-side power stages there is a clamping diode that conducts if the input voltage is externally driven higher than the voltage on the BAT+Power supply pins. The input voltage on all low-side stages, including the alternative input functions, must never exceed the power stage supply BAT+ Power +0,5 V.

To counteract such fault scenarios, maximum ratings and specified wiring examples must be followed and are essential for safe operation.

6.6.2.1 Wiring examples

Attention
In many applications external switches (open collector, open drain (NPN) or a push-pull type), push-buttons or analog sensors have to be monitored by an alternative input of the ECU. Switches which are directly switching to battery voltage must not be used with alternative inputs.

Workaround for safety critical applications:

- **1.**: Usage of external switches connected to GND.
  Short circuits to battery supply needs to be excluded in the system architecture.
- **2**: Battery supplied switches and sensors need to be supplied via a digital output of the HY-TTC 500.

See Section 6.6.1.1 on page 194 for valid and invalid wiring examples.

7 Debugging

7.1 Functional Description

After connecting the HY-TTC 500 and the Lauterbach Debugging Device with the TTC JTAG-Adapter Board (for the connector pinning, see Figure 7.1 on the current page), open the Trace32 Software for ARM CPUs. Double-click the batch file `flash.cmm`, which is located in the corresponding template directory. When prompted by the dialog whether to flash the application or to load only the symbols for debugging, click **Yes** to start the flashing procedure.

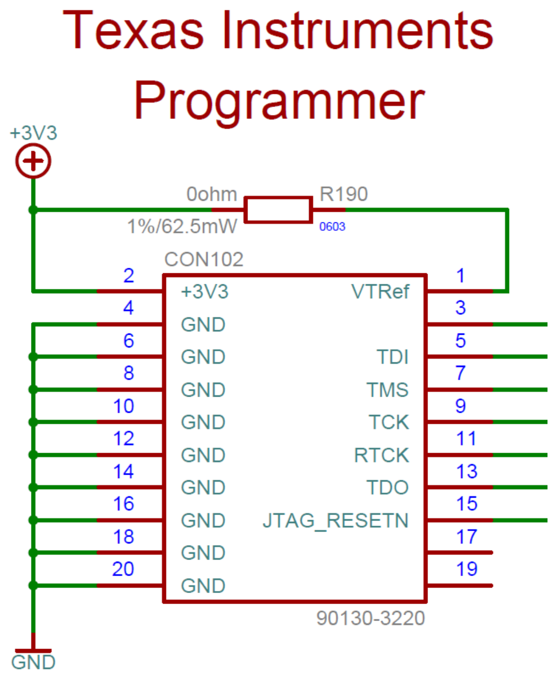

Figure 7.1: Pinning of JTAG Connector on the TTC JTAG-Adapter Board

Manufacturer:	TE CONNECTIVITY / AMP
Manufacturer Article Code:	2-1634689-0
Farnell Order Number:	8396027

Table 7.1: Order code of JTAG connector on the TTC JTAG-Adapter Board

For debugging details, see Part III on page 206.

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Part II

Software Description

8 API Documentation

Please refer to [38] for the API documentation of the HY-TTC 500 I/O driver.

Part III

Quick Start Guide

9 Overview

The purpose of this document is to give a short overview of how to setup HY-TTC 500 development kit for C- and CODESYS-programming.

In case of errors or bugs in documents or workshop examples, please send a feedback to our support team (support@ttcontrol.com).

10 Information and latest version of software

Get the latest version of the Quick Start Guide and information about new product features, improvements and bug fixes from our Service Area at http://www.ttcontrol.com/service-area/.

11 Getting Started

TTControl GmbH recommends using the HY-TTC 500 development kit, because it includes all components required for smooth development. There are two variants of the HY-TTC 500 development kit available: one for C-programming with JTAG-Adapter and open housing (see Figure 11.2 on the next page) and one for CODESYS programming (see Figure 11.3 on page 210). Note that the Ethernet USB adapter is only part of the development kit if the corresponding HY-TTC 500 ECU is equipped with an Ethernet interface.

		C programming	CODESYS programming
1	HY-TTC 500 ECU open housing	✓	
2	HY-TTC 500 ECU CODESYS		✓
3	JTAG adapter	✓	
4	JTAG adapter cable	✓	
5	Interface board	✓	✓
6	Cable harness 1.5m	✓	✓
7	Installation CD C-Programming	✓	
8	Installation CD CODESYS		✓
9	Ethernet USB adapter with driver and software	✓	✓
10	PCAN USB adapter with driver and software	✓	✓
11	Datacable 2m	✓	✓
12	Quick Start Guide	✓	✓

Figure 11.1: HY-TTC 500 Development Kit

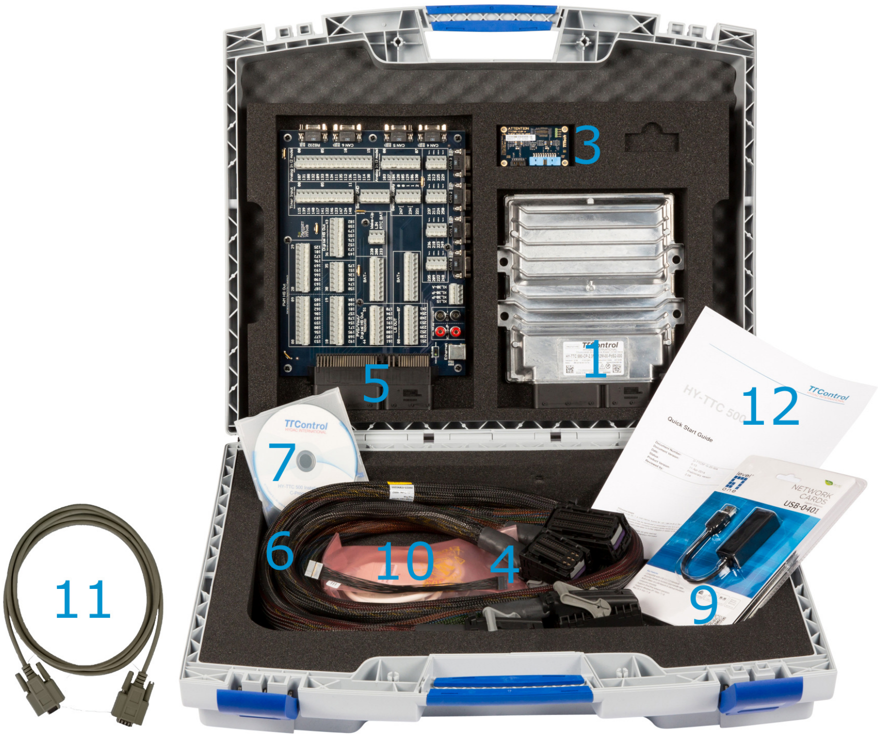


Figure 11.2: HY-TTC 500 Development Kit for C programming

Figure 11.3: HY-TTC 500 Development Kit for CODESYS programming

12 Connections

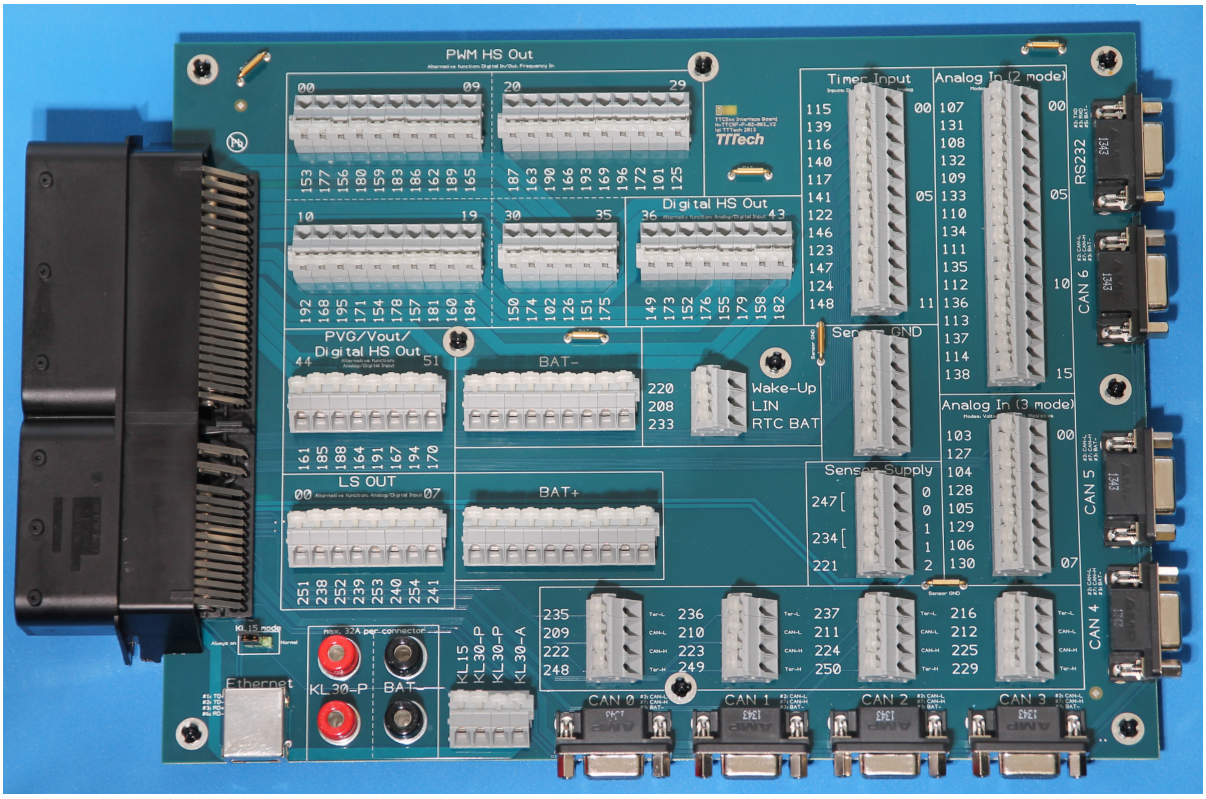


Figure 12.1: HY-TTC 500 Interface Board

12.1 HY-TTC 500 Cable Harness

Connect the HY-TTC 500 to the connector interface board with the provided cable harness.

12.2 HY-TTC 500 CAN Termination

Connect **Ter-L** with **CAN-L** and **Ter-H** with **CAN-H** and further to the terminating CAN interface if termination (2 x 60 Ωsplit termination) is required.

The CAN termination networks can be used for any CAN interface.

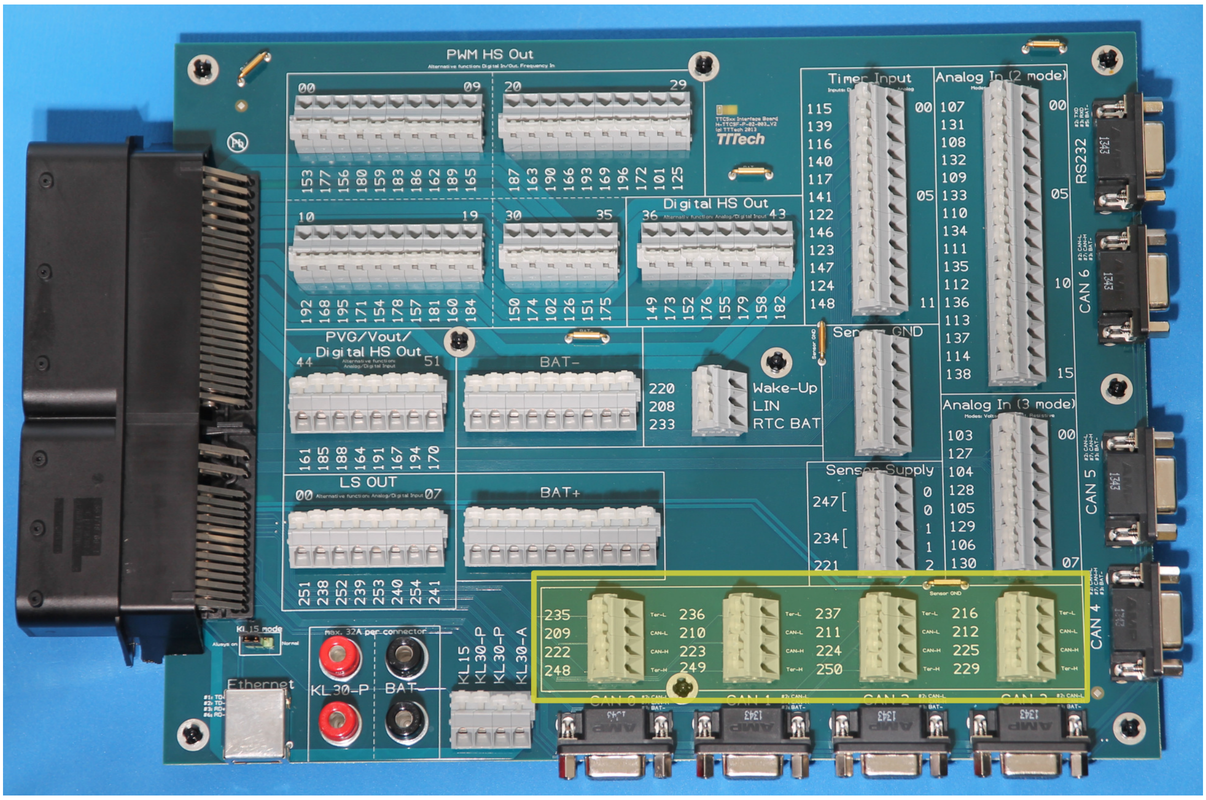


Figure 12.2: HY-TTC 500 Interface Board CAN Termination

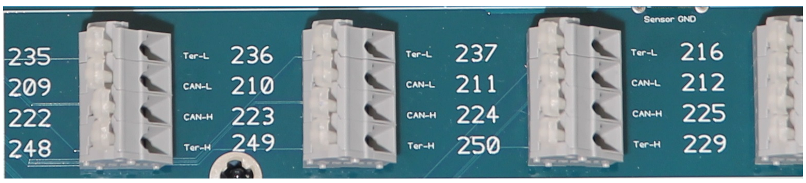


Figure 12.3: HY-TTC 500 Interface Board CAN Termination

12.3 HY-TTC 500 Power Supply

- Connect the power supply GND to the black socket named **BAT-**.
- Connect the positive power supply to the red socket named **KL30-P** and to **KL30-A** on the connector terminal block, e.g. by short-circuit of **KL30-A** to **KL30-P**.

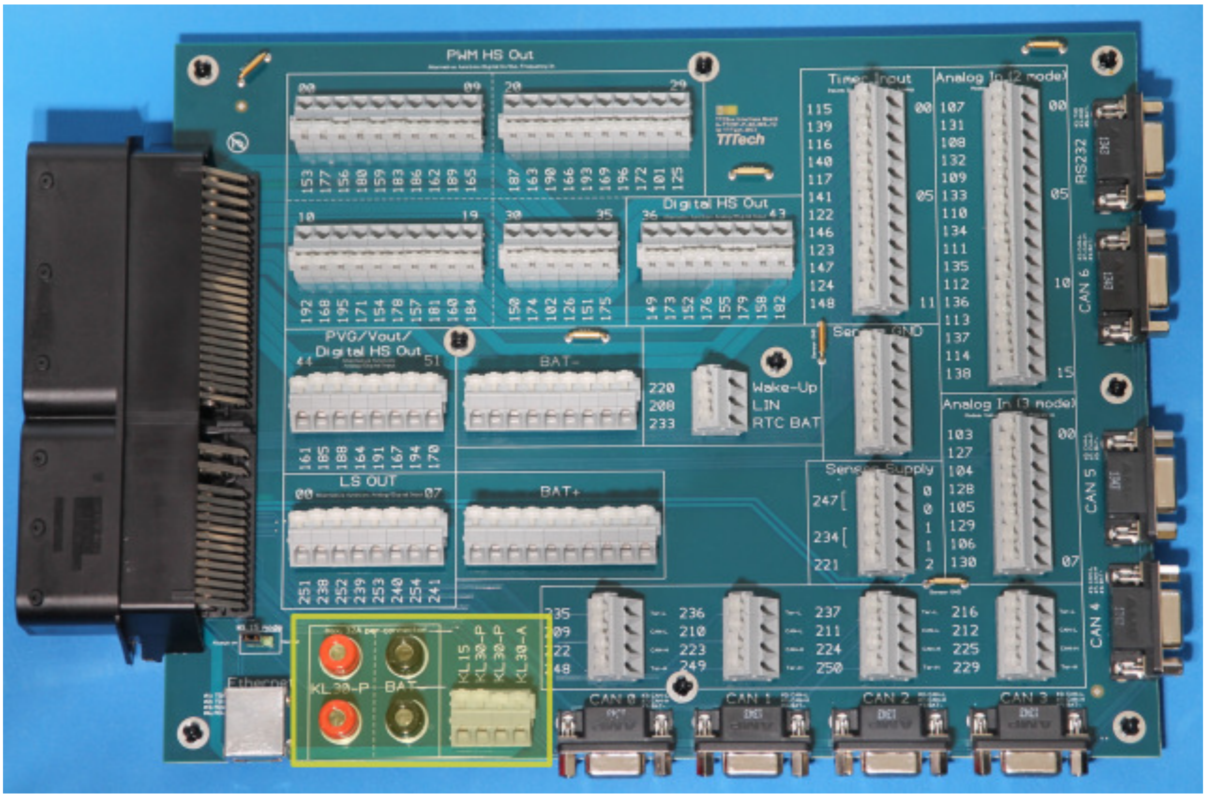


Figure 12.4: HY-TTC 500 Interface Board Power Supply

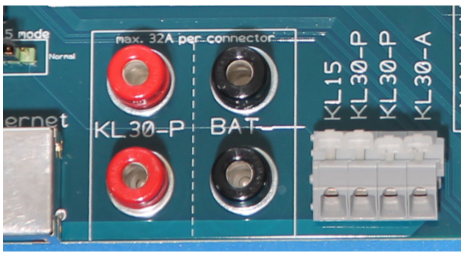


Figure 12.5: HY-TTC 500 Interface Board Power Supply

12.4 HY-TTC 500 Terminal 15 (KL15) Modes

- **Always on Mode**:
  Put the jumber to **Always on** mode if the HY-TTC 500 shall be in active state independent of **KL15**. **Always on** is connected to **KL30-P** on the interface board, **KL30-P** need to be supplied in that case.
- **Normal Mode**:
  Put the jumber to **Normal** mode if the HY-TTC 500 power-up shall be dependent of **KL15**.

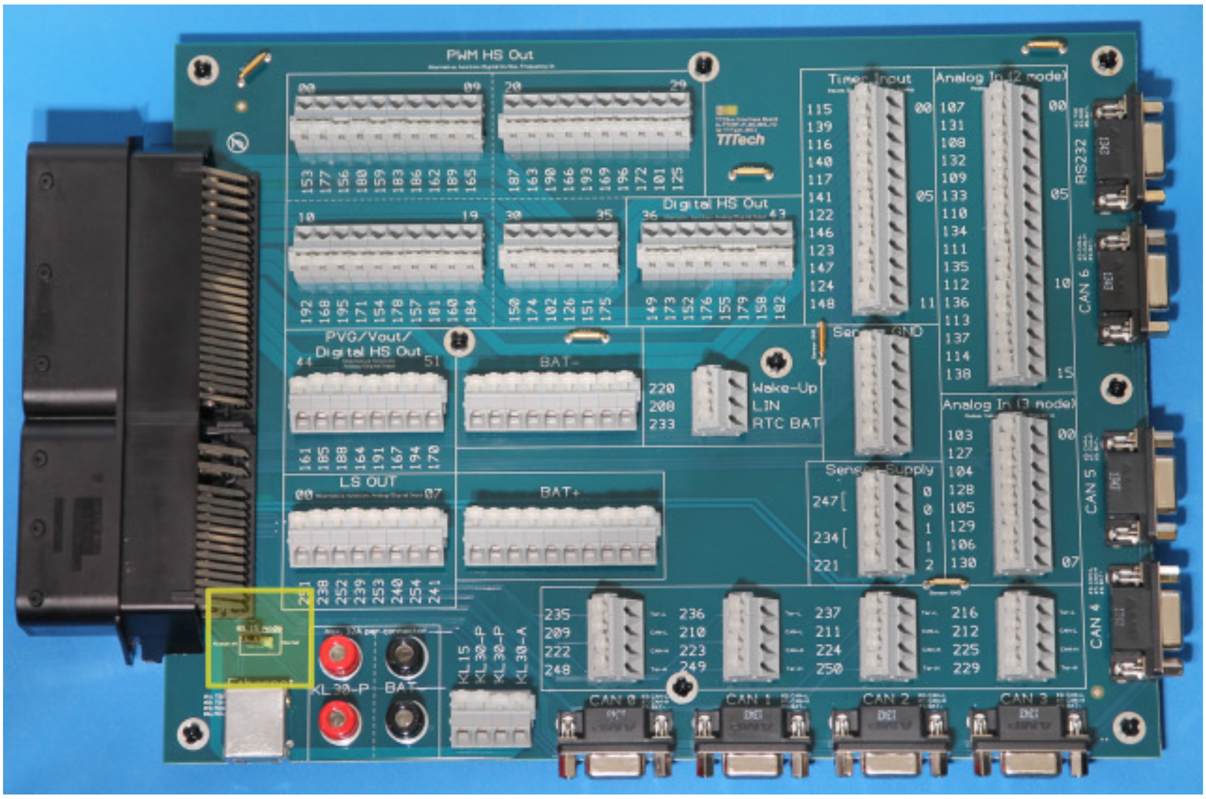


Figure 12.6: HY-TTC 500 Interface Board Terminal 15 (KL15)

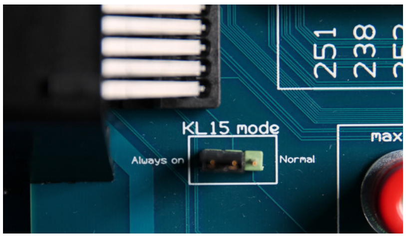


Figure 12.7: HY-TTC 500 Interface Board Terminal 15 (KL15)

12.5 HY-TTC 500 Download and Debugging

For application download to and debugging of HY-TTC 500 there are three options:

- For application download and debugging via **JTAG** it is possible to use the **Lauterbach Power Debug Module** with the **TTControl JTAG Adapter Board** and Lauterbach Batch files (`*.cmm`).
- For application download via **CAN** it is possible to use the **TTControl Downloader (TTC-Downloader)** and the **Peak PCAN-USB Adapter**.
- For application download via **Ethernet** it is possible to use the **TTControl Downloader (TTC-Downloader)** and an **Ethernet Interface**.

13 C Programming Howto for HY-TTC 500

13.1 Overview

The following C programming Howto gives customers who purchase a product of the HY-TTC 500 family a quick overview of how to program these devices in C. This Howto also provides a step-by-step guide on how to flash the software with a Lauterbach Debugging Device. Finally, this Howto lists basic C code to demonstrate how to implement main functions and safety-relevant functions of the device.

13.2 Endianess

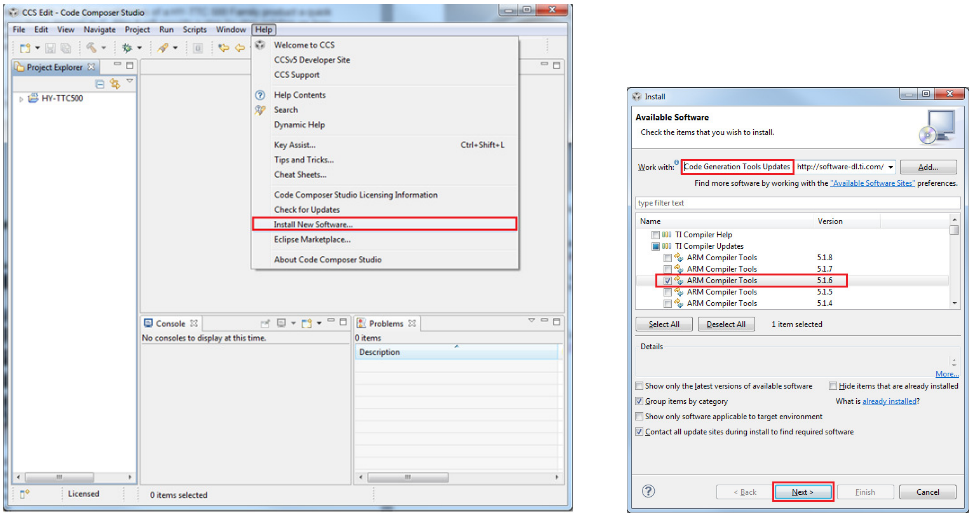
The endianess of the ARM®Cortex™-R4F core of the TI TMS570 CPU is configured to BE32. Big-endian systems store the most-significant byte of a multi-byte data field in the lowest memory address. Also, the address of the multi-byte data field is the lowest address. The endianess of the HY-TTC 500 controllers can not be changed.

13.3 Tool Chain

The required tool chain is **TI ARM Code Generation Tools**, version 5.1.6. It conforms to the ISO/IEC 9899:1990 C standard, which was previously ratified as ANSI X3.159-1989 (ANSI C89).

An integrated development environment (IDE) named **TI Code Composer Studio (TI CSS)**, is available online from `http://processors.wiki.ti.com/index.php/Download_CCS#` or from your local distributor.

To upgrade or downgrade the tool chain of your TI CCS installation to a specific version click **Help – Install new software** (see Figure 13.1(a) on the next page) to open the **Install** dialog (see Figure 13.1(b) on the facing page, **Code Generation Tools Updates**).



(a) CCS Edit – Code Composer Studio

(b) Installing a new compiler version in the TI CCS 6

Figure 13.1: TI Code Composer Studio 6

13.4 File Structure

The developer package is extracted into two directories:

- **Environment** and
- **User Manuals**

Figure 13.2 on the next page shows the contents of the **Environment** directory.

Figure 13.3(a) on the following page shows the contents of the **examples** directory. The **examples** directory has some prebuilt projects to explain the usage of certain functions. Figure 13.3(b) on the next page shows the contents of the **template** directory, which contains a template for starting a new project.

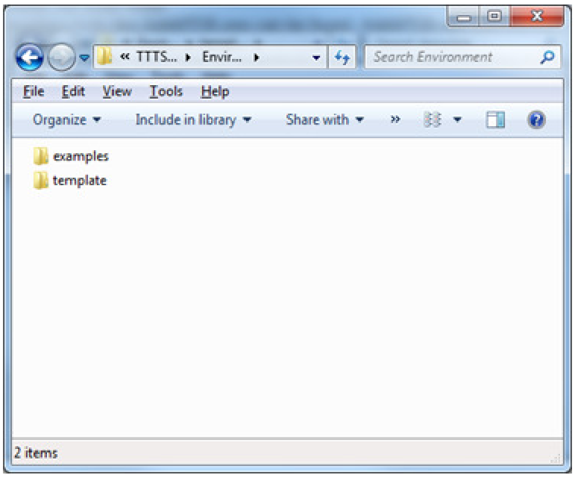
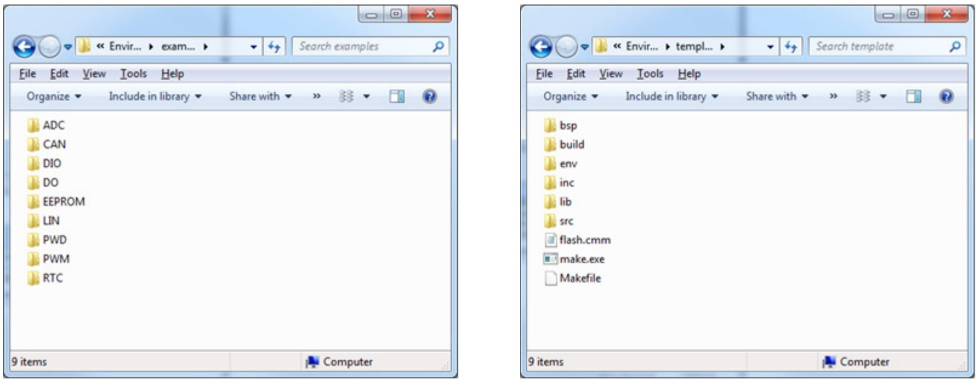


Figure 13.2: Contents of the *Environment* directory



(a) Contents of the *examples* directory (b) Contents of the *template* directory

Figure 13.3: The *examples* and *template* directories

The template should help you getting started. Every new project should start with this folder. The **template** directory contains several subdirectories that are necessary for a project:

- **bsp**: Subdirectory **bsp** (Board Support Package) contains files for the linker and board specific objects.
- **build**: Subdirectory **build** is an empty folder in which the generated files are put during the compile process.
- **env**: Subdirectory **env** (Environment) contains auxiliary tools like `AddAPDB.exe` and `nowECC.exe`.
  - File `AddAPDB.exe` patches the `.hex` files with the CRC checksum, application size and build date.
  - File `nowECC.exe` generates a second file which contains the corresponding ECC data if the Lauterbach Device is used for flashing the device.
- **inc**: Subdirectory **inc** (Include) contains all the header files that are necessary to create a project with the C programming language.
- **lib**: Subdirectory **lib** (Libraries) contains the libraries for the hardware driver and the startup code.
- **src**: Subdirectory **src** (Source) contains the source code files to create a project with the C programming language.

13.5 Development Environment

TTControl GmbH recommends the **TI Code Composer Studio (TI CSS)** as your primary development environment.

13.6 Settings.mk

The `Settings.mk` is needed to tell the `Makefile` what settings to use. Some basic settings can be modified in this file.

```
#############################################################################
#                                                                           #
#  TTC500                                                                   #
#                                                                           #
#  Settings for the make file                                               #
#                                                                           #
#############################################################################

# TTC-Downloader hardware type
DOWNLOADER_HW_TYPE = 0x00400807

# Bootloader version
BOOTLOADER_VERSION = 2.1

# path with C compiler
ifndef C_COMP_PATH
    C_COMP_PATH = C:\TI\ccsv6\tools\compiler\arm_5.1.6\bin
endif

#---------------------------------------------------------------------------
# optimization / debugging features
#---------------------------------------------------------------------------
# compile code for profiling or debugging
#   0 ... profiling
#   1 ... debugging
#   2 ... release
OPT_TYPE = 1
USE_FPU = 1

ifeq ($(OPT_TYPE), 0)
    # optimize for profiling
    OPT_STR := PROFILING
endif
ifeq ($(OPT_TYPE), 1)
    # optimize for debugging
    OPT_STR := DEBUGGING
endif
ifeq ($(OPT_TYPE), 2)
    # optimize for release
    OPT_STR := RELEASE
endif
```

Figure 13.4: Settings.mk

- `DOWNLOADER_HW_TYPE` tells `AddAPDB.exe` for which HW type to compile.
  - `0x00400807` compiles for a **HY-TTC 580**.
  - `0x00600807` compiles for a **HY-TTC 540**.
  - `0x00A00807` compiles for a **HY-TTC 520** [1].
  - `0x00C00807` compiles for a **HY-TTC 510**.
- `C_COMP_PATH` tells Make what compiler to use. **TI ARM Code Generation** tools are the only ones recommended by TTControl GmbH.
- `OPT_TYPE` configures the compiler options to predefined values, for code optimizations and debugging symbols. Select **1** for **debugging** and **2** for a **release build**. If any other compiler or linker options are needed, please refer to the *TI ARM Code Generation Tools Manual*.

[1] customer-specific variant only

Additionally all the paths for the different folders are specified in `Settings.mk`. They must be adjusted if any of the folders are moved.

13.7 Flashing the HY-TTC 500

There are two ways how to flash a HY-TTC 500 device: Using the Lauterbach Debugging Device or the TTC-Downloader Tool via CAN or Ethernet. Both ways will be described step by step. For further information on each product, please refer to the respective manual.

13.7.1 The Lauterbach Debugging Device

After connecting the HY-TTC 500 and the Lauterbach Debugging Device using the TTC JTAG-Adapter, open the **Trace32 Software** for ARM CPUs. Open the batch file `flash.cmm` located in the corresponding template folder. A Popup will ask whether to flash the application or whether to load only the symbols for debugging. Select **Yes** to start the flashing procedure.

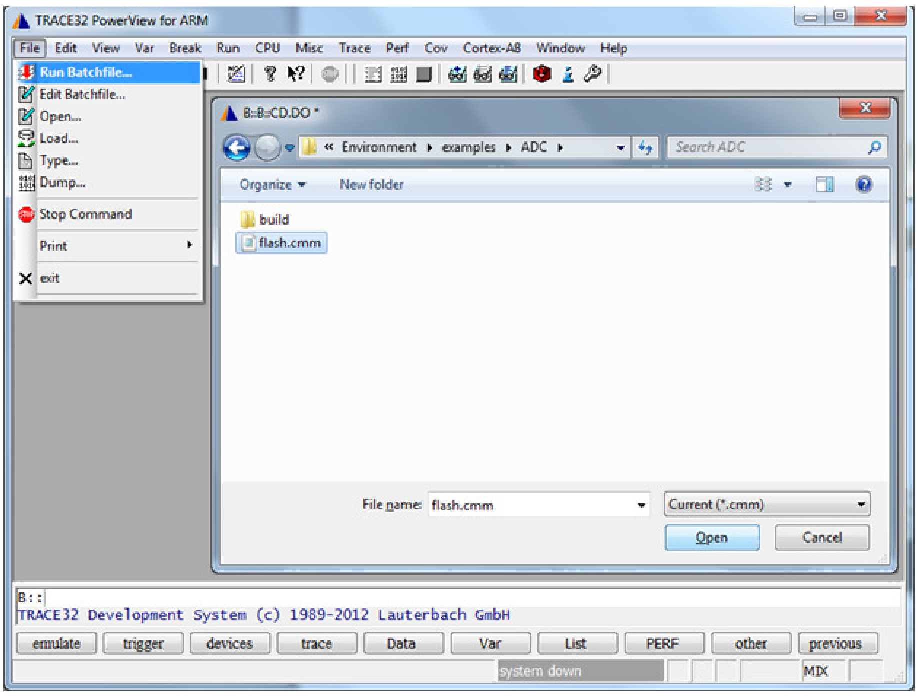


Figure 13.5: Trace32 batch file selection

13.7.2 The TTC-Downloader Tool

13.7.2.1 Connecting via CAN

To connect to the device via CAN, use a CAN connector (e.g., **Peak PCAN-USB**) and connect it to **CAN0** of the HY-TTC 500.

Then power off the device and open the TTC-Downloader.

Make sure to use the correct CAN settings if the HY-TTC 500 already holds a valid application, e.g., C application or CODESYS runtime system (press **Ctrl + P** to open the preferences dialog and change to the CAN tab). However, if no (valid) application is present, the HY-TTC 500 performs an automatic baud rate detection for the following baud rates: **125 kbit/s**, **250 kbit/s** and **500 kbit/s**. If baud rate detection fails, **500 kbit/s** is used as the **default** baud rate.

Note:

- The automatic baud rate detection and the default settings use the standard CAN identifier 1 for the download direction and the standard CAN identifier 2 for the upload direction.
- The automatic baud rate detection can be enforced by connecting pin sensor supply 0 and sensor supply 1 of the HY-TTC 500 to ground if the correct CAN settings are not known.

To start connecting press **F2** and quickly power on the device, while the Downloader progress bar appears (see Figure 13.6(a) on the facing page).

After successful connection, the downloader will identify the device and download the flash routines to the device Figure 13.6(b) on the next page.

Now you can open your `.hex`-file and flash the device. For further information about how to update the bootloader/FPGA IP, read/write from/to EEPROM, and other features of the TTC-Downloader, please refer to the online help of the tool.

13.7.2.2 Connecting via Ethernet

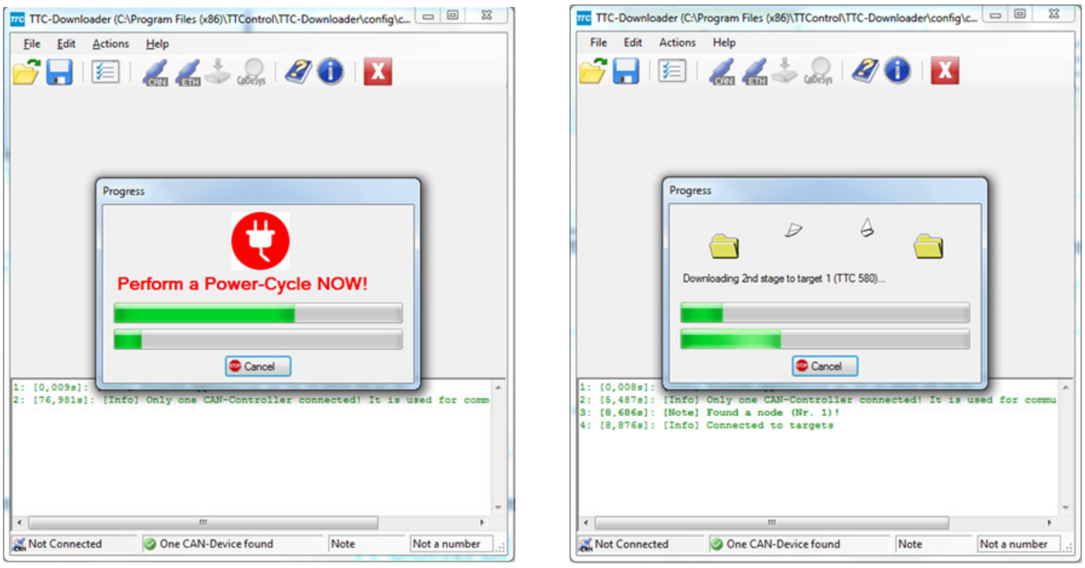
To connect to the device via Ethernet, use the Ethernet port of your computer and connect it to the Ethernet port of the HY-TTC 500. Notice: a point-to-point ethernet connection is mandatory.

Open the TTC-Downloader. Make sure to use the correct Ethernet settings if the HY-TTC 500 already holds a valid application, e.g., C application or CODESYS runtime system (press **Ctrl + P** to open the preferences dialog and change to the Ethernet tab).

Then power on the device and wait until the Ethernet link has sucessfully been established. Note that the establishment of the Ethernet link may take up to three seconds after powering up the device. Start the connection procedure by pressing **F5**. After successful connection, the downloader will identify the device and download the flash routines to the device Figure 13.6(b) on the facing page.

Note: In order to successfully connect to a device via Ethernet, one of the following conditions must be true:

- There is no valid application flashed to the device.
- There is a valid application flashed to the device that is listening for download requests on the Ethernet interface.



(a) Connecting via the TTC-Downloader

(b) Connection progress of the TTC-Downloader

Figure 13.6: TTC-Downloader

13.8 Help for C Driver Functions

The I/O driver API documentation is available in the compiled HTML format (`HY-TTC 500 Series C API Manual.chm`) and can be found in the *User Manual* folder. It provides a short overview of the available I/O driver functions as well as detailed descriptions of those functions, including parameters, return values and examples.

13.9 Application Examples

A basic example is demonstrated.

13.9.1 General

Every program must include at least the header files `IO_Driver.h` and `APDB.h`. The Application Database (APDB) is a 128-byte structure that is used by download tools and the bootloader to update the application and do CRC checks. The structure `Apdb_t` must be defined (see Figure 13.8 on page 225).

```c
/***************************************************************************
 * Includes
 ***************************************************************************/
#include "APDB.h"
#include "IO_Driver.h"
#include "IO_DIO.h"
#include "IO_POWER.h"
#include "IO_RTC.h"
```

Main include files for I/O driver and APDB

Other includes

Figure 13.7: Includes

The first step is to initialize the `IO_Driver` in the `main()`. After that every channel, function or interface can be initialized with individual parameters (Figure 13.9 on page 226).

```
/*******************************************************************************
 * Application Descriptor Block,
 * needed for TTC-Downloader
 ******************************************************************************/
#pragma SET_DATA_SECTION(".APDB_SEC")                                  } APDB used by bootloader, placed at
volatile const BL_APDB Apdb_t =                                          memory address 0x000A 0000
{
    0,                          /* ubyte4 APDBVersion;            */
    { 0 },                      /* BL_T_DATE FlashDate;           */
    {
        (ubyte4) (
            ((((ubyte4) RTS_TTC_FLASH_DATE_YEAR   ) & 0x0FFF) <<  0) |
            ((((ubyte4) RTS_TTC_FLASH_DATE_MONTH  ) & 0x000F) << 12) |
            ((((ubyte4) RTS_TTC_FLASH_DATE_DAY    ) & 0x001F) << 16) |
            ((((ubyte4) RTS_TTC_FLASH_DATE_HOUR   ) & 0x001F) << 21) |
            ((((ubyte4) RTS_TTC_FLASH_DATE_MINUTE) & 0x003F) << 26)
        )
    },                          /* BL_T_DATE BuildDate;           */
    0,                          /* ubyte4 NodeType;               */
    0,                          /* ubyte4 CRCStartAddress;        */
    0,                          /* ubyte4 CodeSize;               */
    0,                          /* ubyte4 LegacyApplicationCRC;   */
    0,                          /* ubyte4 ApplicationCRC;         */
    1,                          /* ubyte4 NodeNumber;             */
    0,                          /* ubyte4 CRCSeed;                */
    0,                          /* ubyte4 Flags;                  */
    0,                          /* ubyte4 Hook1;                  */
    0,                          /* ubyte4 Hook2;                  */
    0,                          /* ubyte4 Hook3;                  */
    APPL_START,                 /* ubyte4 MainAddress;            */
    { 0, 1 },                   /* BL_T_CAN_ID CANDownloadID;     */ } Default settings for CAN
    { 0, 2 },                   /* BL_T_CAN_ID CANUploadID;       */
    0,                          /* ubyte4 LegacyHeaderCRC;        */
    0,                          /* ubyte4 ApplicationVersion;     */
    500,                        /* ubyte4 CANBaudrate;            */ } Default settings for CAN
    0,                          /* ubyte4 CANChannel;             */
    0,                          /* ubyte4 Password;               */
    0,                          /* ubyte4 MagicSeed;              */
    {  10, 120,  30, 200 },     /* ubyte1 TargetIPAddress[4];     */
    { 255, 255,   0,   0 },     /* ubyte1 SubnetMask[4];          */ } Default settings for Ethernet
    { 239,   0,   0,   1 },     /* ubyte1 DLMulticastIPAddress[4]; */
    0,                          /* ubyte4 DebugKey;               */ } Debug key for debugging with Lauterbach
    0,                          /* ubyte4 ABRDTimeout;            */
    0xFF,                       /* ubyte1 ManufacturerID;         */
    0,                          /* ubyte1 ApplicationID;          */
    0,                          /* ubyte2 Reserved;               */
    0                           /* ubyte4 HeaderCRC;              */
};
#pragma SET_DATA_SECTION()
```

Figure 13.8: Application Database (APDB)

```c
/**************************************************************************
 * Main Task
 **************************************************************************/
#pragma TASK( main );
void main(void)
{
    volatile IO_ErrorType io_error = IO_E_OK;
    ubyte4 timestamp = 0;
    ubyte2 pwm0_current = 0;
    bool pwm0_fresh = FALSE;

    /* safety configuration */
    IO_DRIVER_SAFETY_CONF safety_conf;

    /* 10ms cycle period with 25% window size,
     * no resets,
     * 30ms glitch filter,
     * no callbacks
     */
    safety_conf.command_period     = 10000;
    safety_conf.windows_size       = SAFETY_CONF_WINDOW_SIZE_25_PERCENT;
    safety_conf.reset_behavior     = SAFETY_CONF_RESETS_DISABLED;
    safety_conf.glitch_filter_time = 30;
    safety_conf.error_callback     = NULL;
    safety_conf.notify_callback    = NULL;

    /* initialize I/O driver with safety configuration */
    io_error = IO_Driver_Init(&safety_conf);
```

1) Initialize I/O driver

```c
    /* setup a PWM output with current measurement */
    io_error = IO_PWM_Init( IO_PWM_00 /* PWM channel 0         */
                          , 100       /* frequency is 100Hz    */
                          , TRUE      /* positive polarity     */
                          , FALSE     /* no diagnostic margin */
                          , NULL      /* not safety critical  */
                          );
```

2) Initialize I/Os

```c
    /* turn on power stage */
    io_error = IO_POWER_Set( IO_INT_POWERSTAGE_ENABLE
                           , IO_POWER_ON
                           );

    /* turn on safety switch 0 */
    io_error = IO_POWER_Set( IO_INT_SAFETY_SW_0
                           , IO_POWER_ON
                           );
```

3) Enable outputs

```c
    /* start the RTC */
    IO_RTC_StartTime(&timestamp);
```

4) Start timestamp

Figure 13.9: Application initialization

After initializiation, typically a while loop (**while (1)**) is executed. This is the main loop of the applicatio, and it will be executed each cycle time ms. See Figure 13.10 on the current page.

```
    /* main loop */
    while (1)
    {
        /* task begin */
        io_error = IO_Driver_TaskBegin();

        /* set PWM duty cycle */
        io_error = IO_PWM_SetDuty( IO_PWM_00 /* PWM channel 0                        */
                                 , 0x8000    /* set duty cycle to 50%                */
                                 , NULL      /* duty cycle feedback measurement ignored */
                                 , NULL      /* period feedback measurement ignored     */
                                 );

        io_error = IO_PWM_GetCur( IO_PWM_00     /* PWM channel 0                     */
                                , &pwm0_current /* variable to store the current value */
                                , &pwm0_fresh   /* variable to store the freshness bit */
                                );

        if (pwm0_fresh != FALSE)
        {
            /* pwm0_current contains the latest current information for PWM channel 0 */
        }

        /* task end */
        io_error = IO_Driver_TaskEnd();

        /* wait until the configured cycle time is over */
        while (IO_RTC_GetTimeUS(timestamp) < safety_conf.command_period);
        timestamp += safety_conf.command_period;
    }
} /* END OF main */
```

1) Call TaskBegin

2) Execute application

3) Call TaskEnd

4) Wait until application cycle time has passed and update the timestamp

Figure 13.10: Application loop

Be aware that the duration of each cycle is 10000 µs in this case.

13.9.2 Linking Constant Data

Constant data is linked to the application region by default. However, it can also be linked to the external flash memory (if available) or to the application configuration data region. The major difference is the integrity protection of their respective contents. While constant data linked to the external flash memory is protected by the application CRC, the application configuration data region's integrity must be ensured by the application itself. Figure 13.11 depicts an example of how to link constant data to the external flash memory as well as to the application configuration data region.

13.9.3 Safety Configuration

For safety-critical applications, every safety-critical IO can be initialized with a safety configuration. Please refer to the *C API manual* for further details.

To enable any outputs, the IO Driver has to be configured with a valid safety configuration. The IO Driver safety configuration consists of several elements:

- `command_period [µs]`: Sets the period at which the application cycle is executed. **Important:** The `command_period` is checked by the watchdog and has to meet the window configuration.

```
/**************************************************************************
 * Global data
 **************************************************************************/

#pragma SET_DATA_SECTION(".EXT_FLASH")
/*
 * After setting the active data section to ".EXT_FLASH", the linker places
 * ALL of the subsequently defined variables to the external flash memory.
 *
 * Attention: Be aware that the constant data linked to the external flash
 *            memory is protected by the application CRC! Thus, any modification
 *            will lead to an invalid application CRC.
 */
const ubyte1 foo = 'f';     /* foo is linked to external flash      */
const ubyte4 bar = 123456; /* bar is also linked to external flash */

#pragma SET_DATA_SECTION(".CFG_FLASH")
/*
 * After setting the active data section to ".CFG_FLASH", the linker places
 * ALL of the subsequently defined variables to the configuration flash memory.
 *
 * Attention: Be aware that the content of the configuration flash memory is
 *            NOT protected by the application CRC! This means that the data
 *            integrity of the configuration flash memory must be ensured by
 *            the application itself.
 */
const ubyte2 wayne = 256; /* wayne is linked to configuration flash */

#pragma SET_DATA_SECTION()
/*
 * After placing the respective global variables to the external and
 * configuration flash memory, the active data section needs to be reset
 * to its default value.
 */
```

Figure 13.11: Linking constant data to the external flash and/or application configuration data

- `window_size [%]`: Configures the window size for the watchdog which checks the `command_period`. The application cycle has to fulfill those timing requirements.
- `reset_behavior`: Configures how often the watchdog can restart the CPU in case of an error. If a safety-critical error occurs, the watchdog will reset the CPU and start again. After the maximum number of resets, a permanent safe state will be entered.
- `glitch_filter_time [ms]`: Sets the time a temporary error condition must persist to cause an error reaction by the safety function.
- `error_callback`: Configures the error callback for the application. It can be set to `NULL` to disable the error callback.
- `notify_callback`: Configures the notification callback for the application. It can be set to `NULL` to disable the notification callback.

13.9.4 Debugging of a safety-critical application

If an IO Driver is configured with a valid safety configuration and it is intended to debug, it is necessary to enable debugging with Lauterbach. Therefore, the debug key of the APDB can be set to `0xC0FFEE` to enable debugging of a safety-critical IO driver with Lauterbach. This can be done directly in the definition of the APDB (Figure 13.8 on page 225) or with the TTC-Downloader (Figure 13.12 on the current page).

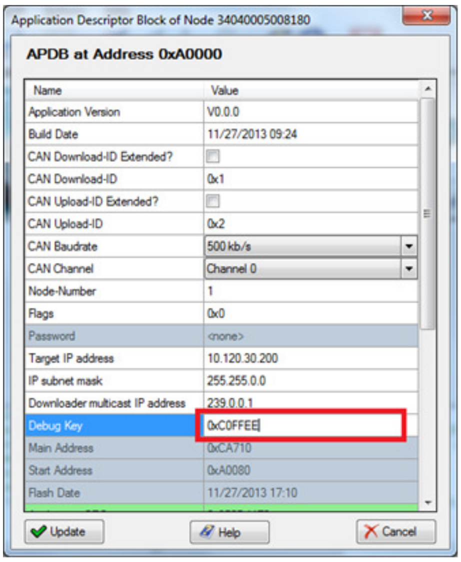


Figure 13.12: Enable debugging for the Lauterbach

After power up the device will directly enter the safe state if the debug key is set. When the device will be started again with the Lauterbach (Reset & Go), the device runs normally until the first breakpoint is hit. After the first breakpoint was hit, the device enters the permanent safe state and remains in this state until the next power cycle is performed.

The debug key has to be set to **0** if no Lauterbach is connected to the device or for any production release. Otherwise, the device does not start up normally and enters the safe state directly.

13.10 Memory Map for Internal Flash and RAM

The following table specifies the start address and size of the **bootloader**, **FPGA IP**, **APDB**, **application** and **application configuration data** for the HY-TTC 500 controllers.

Note: The application configuration data region is **not** protected by the application CRC and thus, its integrity must be ensured by the application itself. Refer to the *Modifying Flash/EEPROM Memory* page of the TTC-Downloader online help for detailed information about modifying the flash memory.

HY-TTC 500	Start Address	Size
Bootloader	0x00000000	131072 Bytes
FPGA IP	0x00020000	524288 Bytes
APDB + CRC	0x000A0000	256 Bytes
Application	0x000A0100	2490112 Bytes
Application Cfg. Data	0xF0200000	65536 Bytes

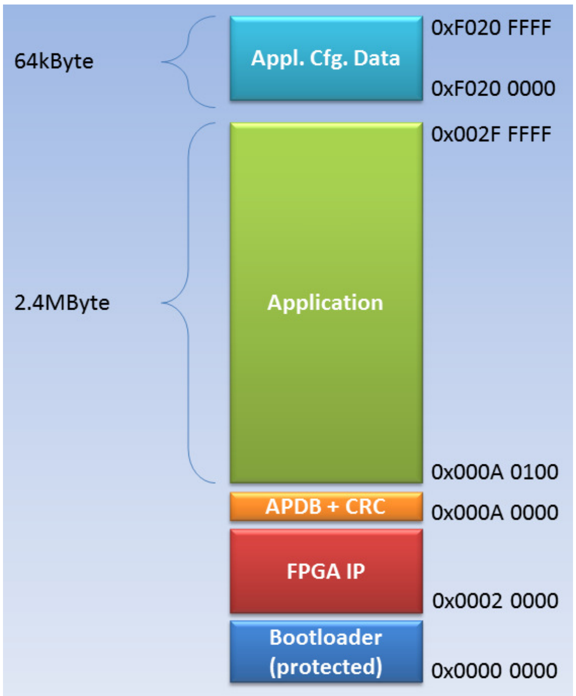


Figure 13.13: Memory map for HY-TTC 500 internal Flash

The following table specifies the start address and size of the **internal RAM** for the HY-TTC 500 controllers.

HY-TTC 500	Start Address	Size
Int. RAM	0x08003000	217088 Bytes

13.11 Memory Map for External Flash and RAM

The following tables specify the start address of the **external RAM** and **Flash** for the HY-TTC 500 controllers.

HY-TTC 580	Start Address	Size
Ext. RAM	0x60000000	2097152 Bytes
Ext. Flash	0x64000000	8388608 Bytes

HY-TTC 540	Start Address	Size
Ext. RAM	0x60000000	2097152 Bytes

HY-TTC 520	Start Address	Size
Ext. RAM	0x60000000	2097152 Bytes

HY-TTC 510	Start Address	Size
Ext. RAM	0x60000000	2097152 Bytes

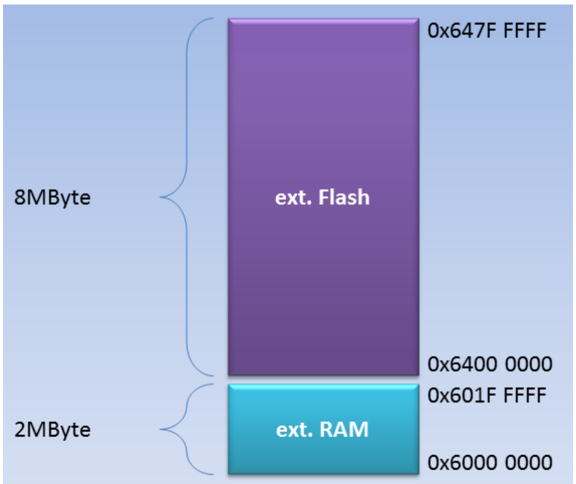


Figure 13.14: Memory map for HY-TTC 500 external RAM and Flash

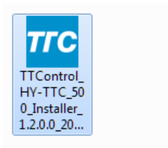
14 CODESYS Howto for HY-TTC 500

14.1 Installation

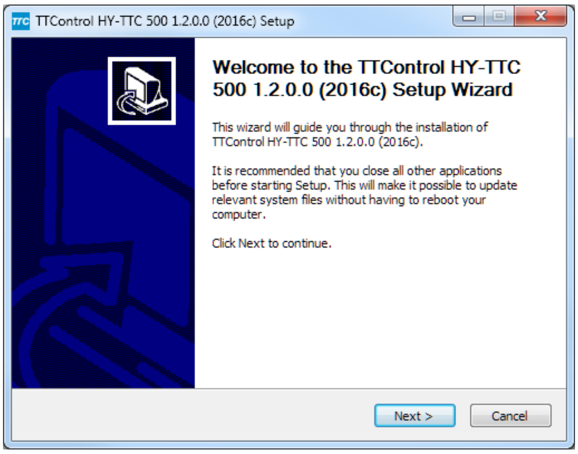
This section provides step-by-step instructions on how to install, configure and use the CODESYS environment for the HY-TTC 500 Series. You will then be able to control an I/O pin of the HY-TTC 580 SIL2 with the CODESYS runtime system.

14.1.1 CODESYS Installation

When obtaining the CODESYS release package, you get a setup file with the name **TTControl_HY-TTC_500_Installer_x.x.x.x_yyyyy.exe**, where `x.x.x.x` indicates the current version and `yyyyy` the release name (e.g. 2016c). Make sure you have administrative privileges and double-click this setup file to start the installation process and open the **TTControl HY-TTC 500 Installer Setup** wizard (see Figure 14.1(b) on this page.)



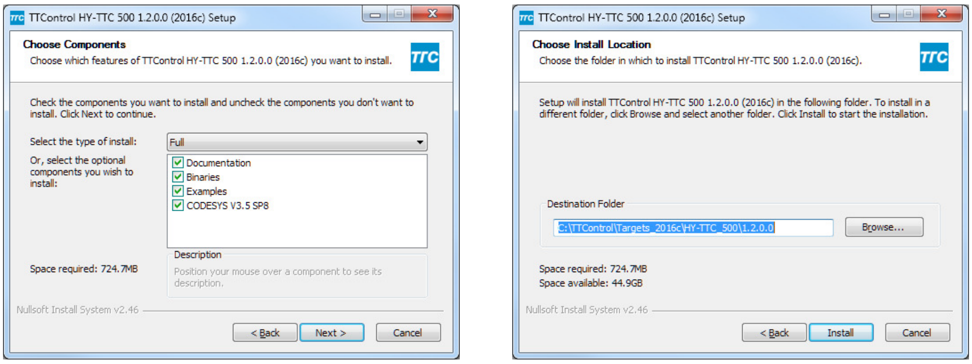
(a) The CODESYS installation file



(b) TTControl HY-TTC 500 Installer setup wizard

Figure 14.1: CODESYS installation 1

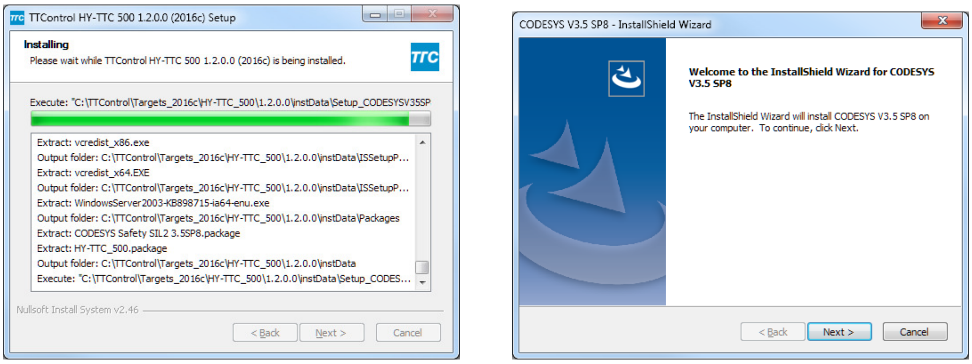
In the setup wizard, click **Next** to proceed to the **TTControl HY-TTC 500 Installer Setup** dialog, which prompts you to select the components to be installed. In the **Select the type of install** list, select **Full**. Select **Documentation**, **Binaries**, **Examples** and the respective **CODESYS IDE** and click **Next** to confirm the settings made and proceed to the next step, where you are prompted to select the destination folder for your installation (Figure 14.2(a) on the next page). Click the **Browse** button to select the destination directory of your choice and click **Next** to confirm the settings for your installation directory.



(a) Selecting the components to be installed (b) Selecting installation directory

Figure 14.2: CODESYS installation 2

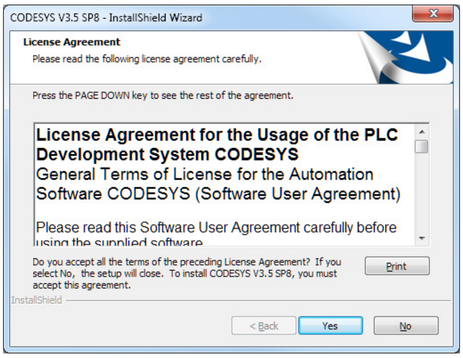
At this point, the installer starts extracting all the components, which can take a few seconds (see Figure 14.3(a) on this page). Once the extraction of the components is finished, the setup of CODESYS is started, and the **CODESYS IDE - InstallShield Wizard** dialog will appear. Click **Next** to confirm installation.



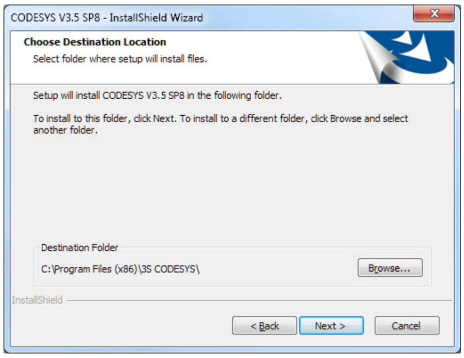
(a) Installation in progress (b) InstallShield Wizard for CODESYS IDE

Figure 14.3: CODESYS installation 3

Click **Yes** to accept the license agreement and select the desired destination directory for your installation. Click **Next** to confirm the settings made (Figure 14.4(b) on the next page).



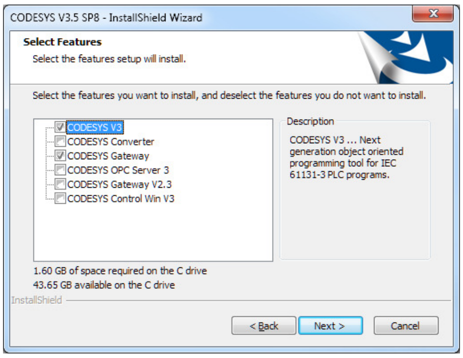
(a) Confirming license agreement



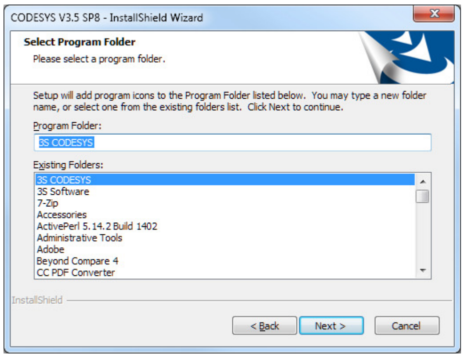
(b) Selecting destination directory

Figure 14.4: CODESYS installation 4

In the **Select Features** dialog, select **CODESYS V3** and **CODESYS Gateway** and click **Next** to confirm the settings made. In the **Select Program Folder** dialog select the default program folder which is to appear in the start menu. Click **Next** to confirm your settings and proceed to the **Start Copying Files** dialog. In the **Start Copying Files** dialog, which shows a summary of the CODESYS installa-



(a) Selecting the features to be installed

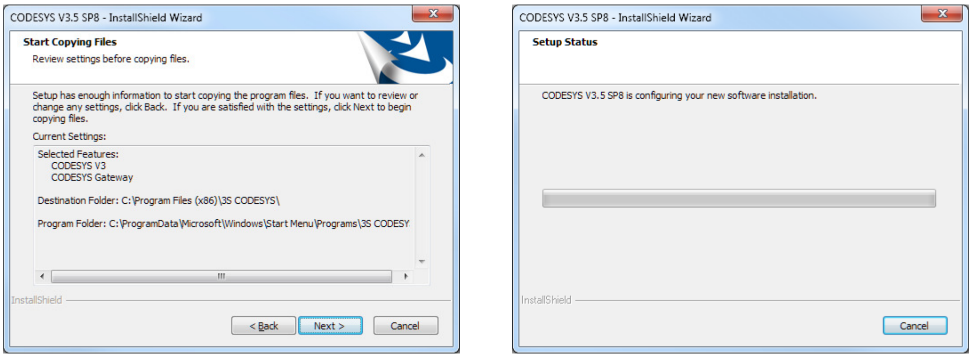


(b) Selecting a program folder

Figure 14.5: CODESYS installation 5

tion settings, click **Next** to start copying the program files (Figure 14.6(a) on the facing page). Depending on the performance of your PC, this step can take around 10 minutes, until it is completed. When the InstallShield Wizard finishes copying the CODESYS core files, the **Select Packages** dialog appears (Figure 14.7(a) on the next page), in which you can select additional components to be installed.

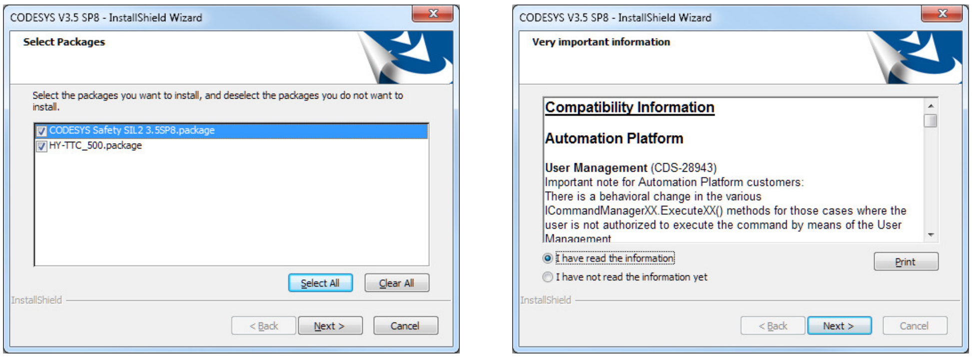
Note: Make sure to select **all** the packages listed so that the additional components from TTControl will be installed.



(a) Copying files (b) Setup status

Figure 14.6: CODESYS installation 6

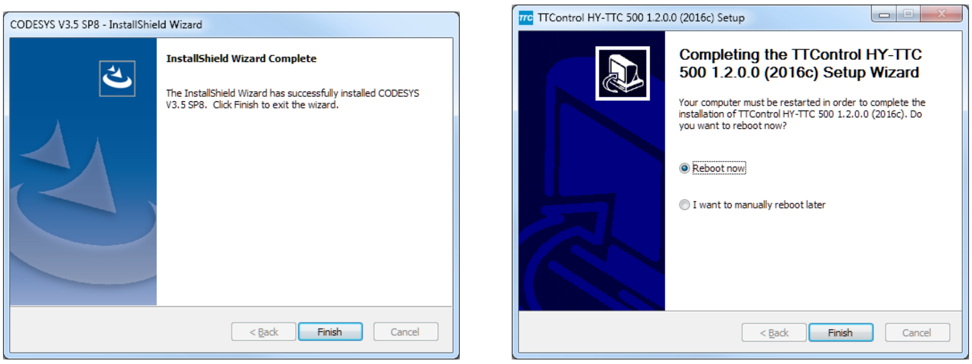
Click **Next** to continue and open the **Very important information** dialog, which provides information about whether compatibility issues affect your project or not. Select **I have read the information** and click **Next** to confirm you have read this information and to continue.



(a) Selecting packages to be installed (b) Compatibility information

Figure 14.7: CODESYS installation 7

The **InstallShield Wizard complete** dialog shows that the installation of CODESYS and its additional components has completed successfully. Click **Finish** to complete the installation of the **3S CoDeSys 3.5.SP8** and finally exit the setup wizard. At this point the installation of CODESYS is finished, and you are ready for starting development. Please note, that a system reboot may be necessary to finalize the software installation process.

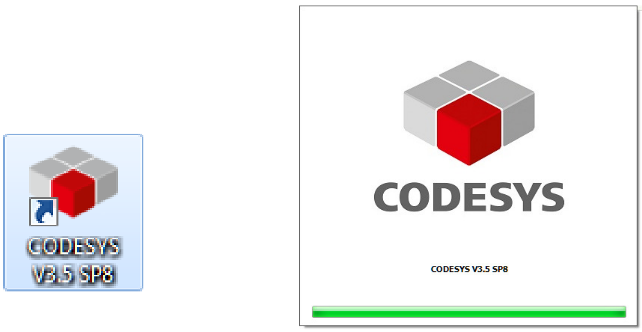


(a) InstallShield Wizard complete (b) Setup finished

Figure 14.8: CODESYS installation 8

14.1.2 First Start of CODESYS

After installation, a shortcut is available on the desktop of your PC. Double-click the **CODESYS desktop icon** (see Figure 14.9(a) on this page) to start the CODESYS environment. On first start,



(a) CODESYS desktop icon (b) CODESYS start screen

Figure 14.9: Starting CODESYS

it can take a few seconds for all the components to be fully loaded. Any future start of CODESYS will then be faster.

14.2 Configuration

After installation of CODESYS for the HY-TTC 500 series, you have to configure how to communicate with the target.

14.2.1 Communication Options

In general, the HY-TTC 500 supports two different ways to communicate with the CODESYS host machine – **CAN** and **Ethernet**. Note that each variant of the HY-TTC 500 family provides at least one CAN interface. However, the Ethernet interface is optionally equipped.

The following sections describe what to do when you use the two different communication channels and what to configure on the target and on the host so that communication can work reliably.

Note: A rescan in the **CODESYS Gateway** is necessary if you want to switch between these two different communication types. This has to do with the internal addressing algorithm of the CODESYS Gateway. When both communication channels have been setup correctly, the Gateway will prefer Ethernet over CAN.

14.2.1.1 CAN

When you select **CAN** as the communication channel between the HY-TTC 500 and the host system, note the following. Currently, the CODESYS Gateway only supports devices from **Peak Systems** to work with their Gateway. TTControl tested the **standard PeakCAN USB adapter** and the **PeakCAN Pro USB adapter**. Both should be working without any limitations.

Note: If you attach more than one of these adapters to the host system, the internal enumeration of the driver can mix up the devices. It is therefore highly recommended to have **only one** PeakCAN adapter attached to the system.

Make sure that the cabling setup is properly done with termination resistors on both ends of the bus. The CODESYS CAN communication works via a CANopen-based protocol. With respect to communication, the target acts as a server, and the host system acts as a client. Note that the target's node ID must be unique in the entire net. The baud rates must be configured to the same values on every node of the bus. For more detailed information on how to configure the CAN settings on the target and the host system, refer to the following sections.

14.2.1.2 Ethernet

When you select **Ethernet** as the communication channel between the HY-TTC 500 and the host system, note the following. Normally, the CODESYS Gateway scans all the available network interfaces. For performance and reliablity reasons, it is preferred to use a dedicated Ethernet interface for establishing communication with the HY-TTC 500. Therefore a separate USB Ethernet adapter is provided. It is not necessary to use a crossover cable. The HY-TTC 500 supports automatic crossover detection so that a normal patch cable is enough.

When you configure the IP addresses, make sure that the address range does not overlap any of your other network adapter IP ranges. The tested default IP configuration at TTControl is as follows:

- **Host IP:** `10.100.30.1`
- **Target IP:** `10.100.30.200`
- **Subnet mask:** `255.255.0.0` (on both)

When this IP range does not conflict with any of your other adapters, it is recommended to keep it this way. TTControl GmbH strongly recommends the **LevelOne USB 3.0 Ethernet adapter** for the use of Ethernet with CODESYS.

Figure 14.10: USB 3.0 Ethernet adapter

The USB 3.0 Ethernet adapter was tested to work with the described setup.

Note: TTControl GmbH does not give any guarantee that Ethernet communication will work with a changed setup. For more detailed information on how to configure the Ethernet settings on the target and the host system, refer to the following sections.

14.2.2 Target Configuration

The CODESYS runtime system needs to be flashed to any HY-TTC 500 device like a normal C application. After the installation, two files will be located in the **Binaries** directory.

- codesys_rts_sil2.hex
- codesys_rts.hex

Dependent on your needs, it is possible to choose between CODESYS Safety SIL2 and standard CODESYS. File **codesys_rts_sil2.hex** contains the Safety SIL2 CODESYS runtime system, file **codesys_rts.hex** contains the standard CODESYS runtime system.

One of these two files must be flashed to the target with the **Application Download** feature of the TTC-Downloader (see Section 13.7.2 on page 222 for details).

Note:

- Make sure to check if the CODESYS runtime system is compatible with the device's bootloader *and* FPGA IP before flashing the CODESYS runtime system. Refer to the HY-TTC 500 CODESYS Release Notes [32] for detailed information about CODESYS runtime system compatibility. For details on how to update the bootloader/FPGA IP, please refer to the online help of the TTC-Downloader.
- Before flashing the CODESYS runtime system, the CODESYS application (also denoted as CODESYS bootproject) has to be manually erased using the TTC-Downloader. Otherwise, the CODESYS runtime system may crash as a result of a version mismatch. Please refer to the online help of the TTC-Downloader for detailed information on how to erase the CODESYS application.

When the CODESYS runtime system has been flashed to the target, the communication channels can be set up. If you plan to use the default configuration, you need not change anything. If not, check the APDB configuration options inside the TTC-Downloader. The settings given in Figure 14.11 on the following page show the **default configuration**.

When using Ethernet, change the following settings according to your desired configuration:

- Target IP address
- IP subnet mask

When using CAN, change the following settings according to your desired configuration:

- CAN baud rate
- CAN channel
- Node number

The CODESYS runtime system uses these five settings, described above, to initialize the communication channels.

Note: Do not change any other settings.

The type of the CODESYS runtime can be seen at the "Application ID" field:

- **0x09** indicates the Standard CODESYS RTS
- **0x0A** indicates the CODESYS Safety SIL2 RTS

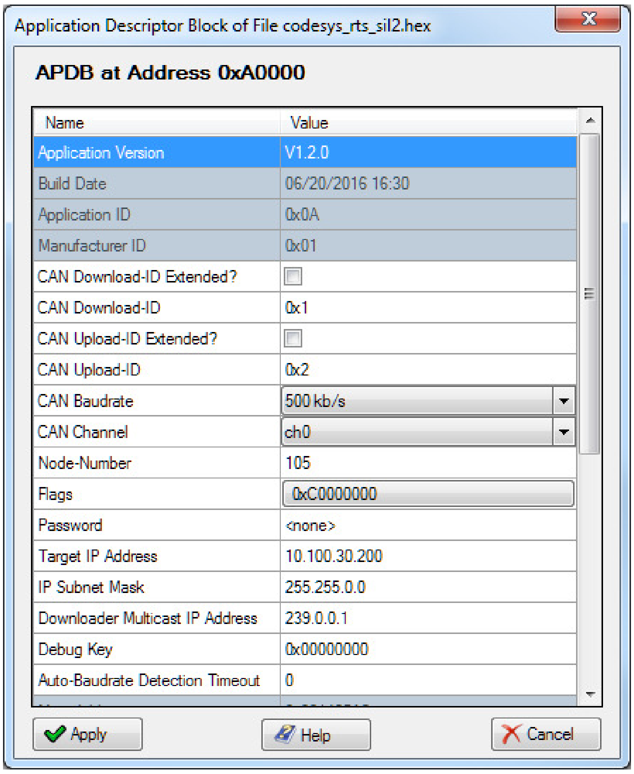


Figure 14.11: APDB configuration settings

14.2.3 Host Configuration

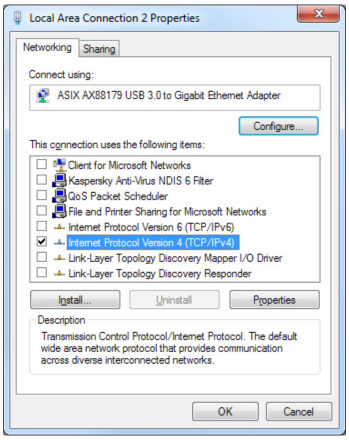
Depending on whether you use CAN or Ethernet, you must make sure that the corresponding device drivers have been installed successfully. All CAN related host settings can be configured directly out of the CODESYS IDE, with no further steps being necessary. If Ethernet is used, you must configure the provided USB Ethernet Adapter in the Windows control panel.

14.2.3.1 Ethernet Adapter Configuration

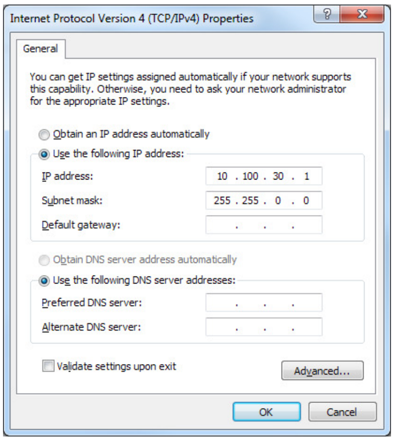
This step is necessary, if you want to use Ethernet for communication with the HY-TTC 500. To configure the USB Ethernet adapter, open the Windows **control panel** and select **Network and Sharing Center**. In the **Network and Sharing Center** window, click **Change adapter settings** on the left to list the available network adapters. The USB Ethernet adapter must appear as **Local Area Connection X** or similar. As an additional label, **ASIX AX88179 USB 3.0 to Gigabit Ethernet Adapter** must appear at the right network adapter.

Right-click the network adapter to open the **Local Area Connections X Properties** dialog. Select only **Internet Protocol Version 4 (TCP/IPv4)**, as shown in Figure 14.12(a) on the next page, and click the **Properties** button to open the **Internet Protocol Version 4 (TCP/IPv4) Properties** dialog.

Select **Use the following IP address** and enter the respective IP configuration. Figure 14.12(b) on this page shows the **default settings**.



(a) The *Local Area Connections X Properties* dialog



(b) The *Internet Protocol Version 4 (TCP/IPv4) Properties* dialog

Figure 14.12: Starting CODESYS

Click the **Advanced** button in the **Internet Protocol Version 4 (TCP/IPv4) Properties** dialog to open the **Advanced TCP/IP Settings** dialog. Click the **WINS** tab and select **Disable NetBIOS over TCP/IP**, as shown in Figure 14.13 on the current page.

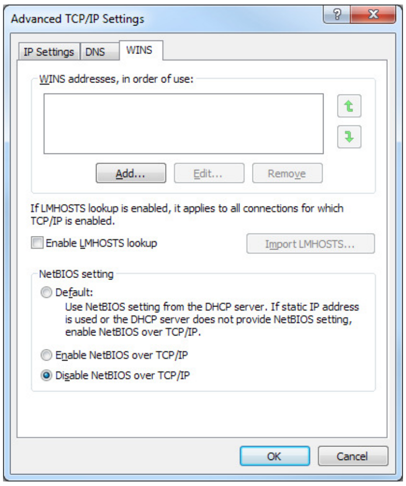


Figure 14.13: The *Advanced TCP/IP Settings* dialog

The configuration of the Ethernet adapter now is complete. Close all the open dialogs by clicking **OK** to apply all your settings made.

14.2.3.2 CODESYS Gateway Configuration

The Gateway can be configured directly out of the CODESYS IDE. To do so, double click onto the main device and ensure that the tab shows **Communication Settings**. A click to **Gateway** and **Configure the local Gateway...** will open the Gateway configuration dialog.

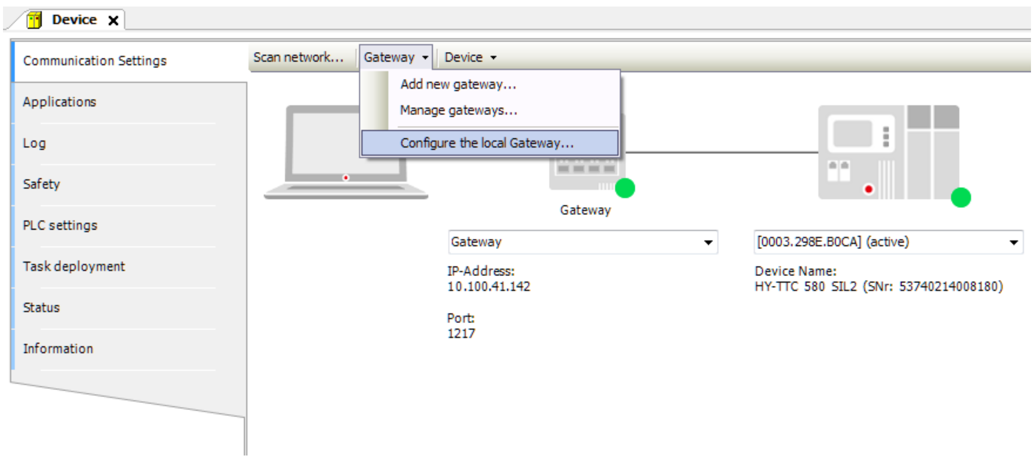


Figure 14.14: The *Configure the local Gateway...* menu

The **Gateway Configuration** dialog will appear. Here, all the CAN specific settings can be altered directly. If CAN communication is not desired at all, the node **CAN Client** can be removed. This will free the CAN adapter to make it exclusively available to other applications.

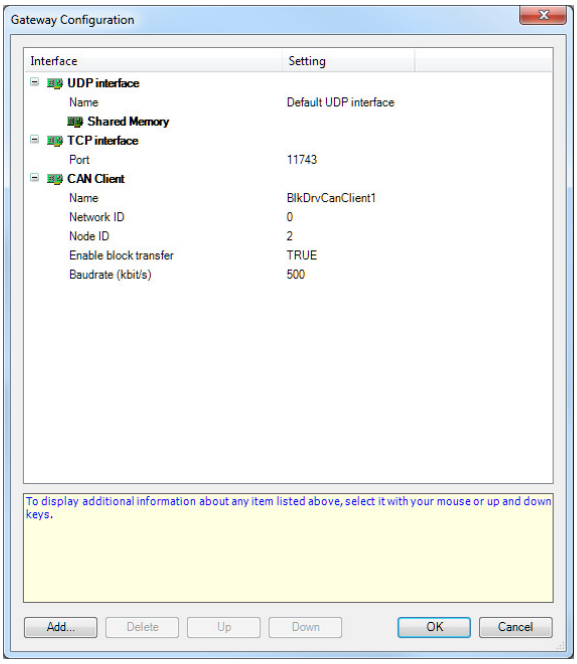


Figure 14.15: The *Gateway Configuration* dialog

At the node **CAN Client**, the following configuration options shall be configured:

- **Network ID:** Index of the CAN adapter (usually 0)
- **Node ID:** Node ID of the CODESYS Gateway (usually 2)
- **Baudrate (kbit/s):** CAN baud rate (usually 500)

Please leave all other settings in this dialog untouched! After performing the changes, the settings are applied automatically. No manual restart of the Gateway is needed.

For Ethernet nothing special needs to be considered at this step.

Note: It can be necessary that you need to restart the Gateway if you remove and plug in again the hardware (PeakCAN or USB Ethernet adapter). This step is necessary to make the gateway detect the changed hardware environment.

After successful installation, the **CODESYS Gateway SysTray** tray icon will appear in the right-hand bottom corner of the desktop. Right-click this tray icon and select **Stop Gateway**. After that, right-click the tray icon again and select **Start Gateway**.

At this point, the configuration of CAN and Ethernet communication settings is finished. When your target is powered on and connected properly, you can locate the target when doing a target scan in CODESYS.

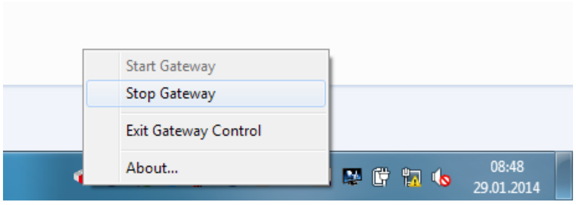


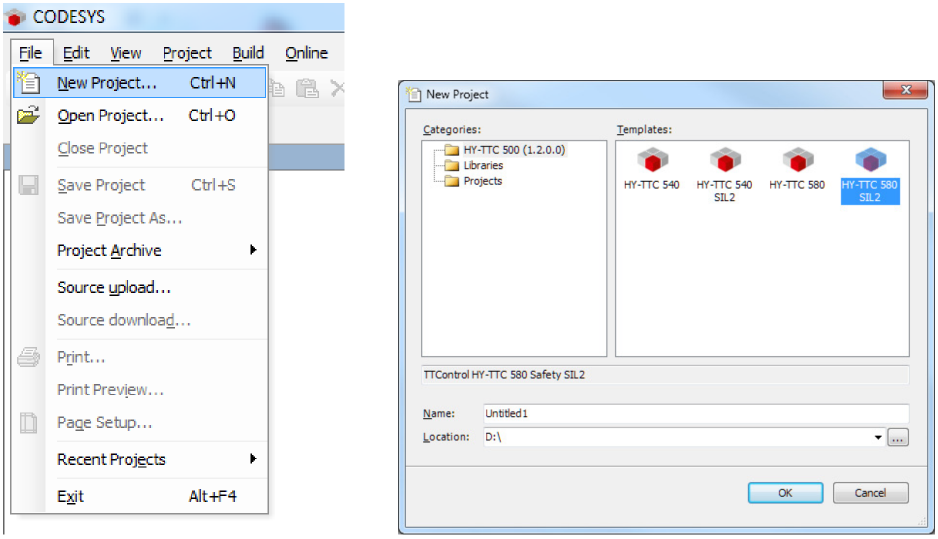
Figure 14.16: Starting the CODESYS gateway

14.3 Creating an Application

This section describes how to create a new application for the HY-TTC 580 SIL2.

14.3.1 Creating a New Project

To create a new project click the **File** menu and select **New Project** to open the **New Project** dialog (Figure 14.17(b) on this page). In the **Name** field, enter the name of your project. In the **Location**



(a) Creating a new project (b) Selecting the *HY-TTC 580 SIL2* device template

Figure 14.17: Starting CODESYS

field, select the path where your project should be saved.

In **Templates**, select **HY-TTC 580 SIL2**, because the application is intended for this device. Click **OK** to create the project. All the common libraries, device drivers and objects will be added automatically.

14.3.2 Adding I/Os and Changing Configuration Properties

To add an I/O to the project created click **Device (HY-TTC 580 SIL2)** (Figure 14.18 on the current page) on the left and select **IODriver**, **I/O group**, **Pin number**, and then **Plug Device** to open the **Plug Device** dialog (Figure 14.19 on the following page).

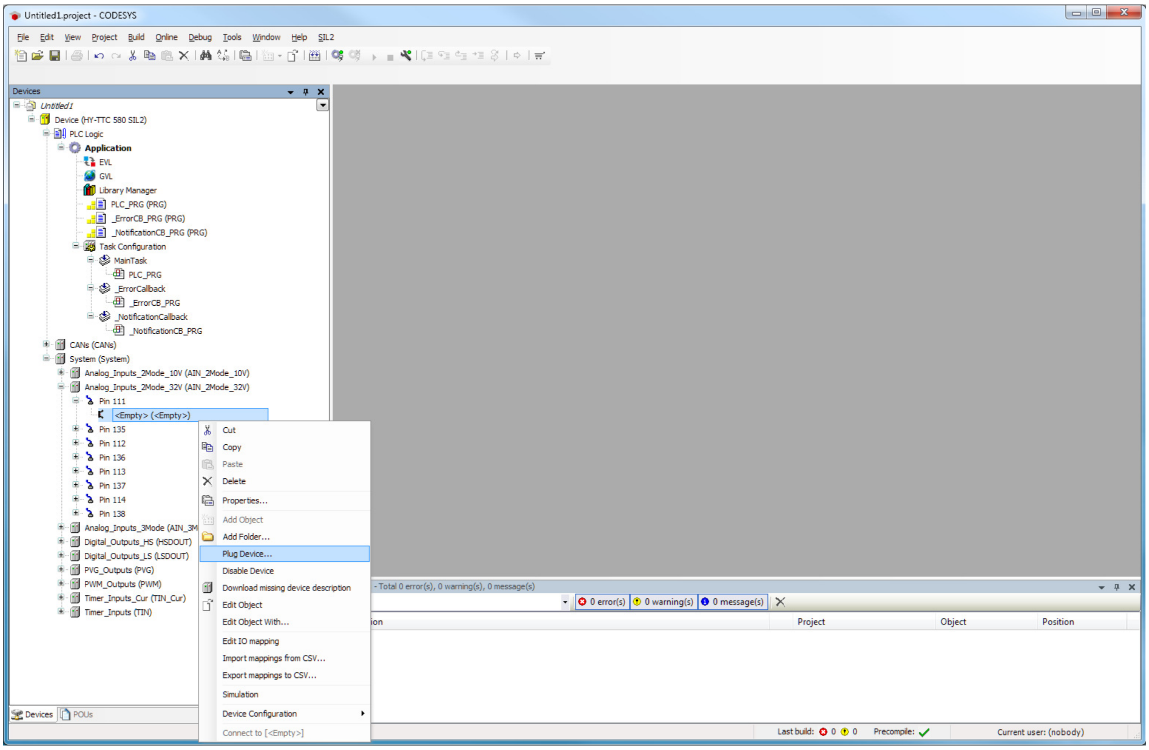


Figure 14.18: Adding an I/O to the project

In the **Plug Device** dialog, select the intended functionality for the selected I/O and click **Plug Device** to set up the I/O with the functionality selected. **In needs to be ensured that the selected I/O driver's device version matches with the main device's version!**

To change the configuration of the I/O double-click the previously plugged device on the left to open a new tab where the properties of the selected pin can be changed. To change a configuration item, click the **I/O Configuration** tab and select a value from the drop down list or enter a value, depending on the respective configuration item (see Figure 14.20 on the next page).

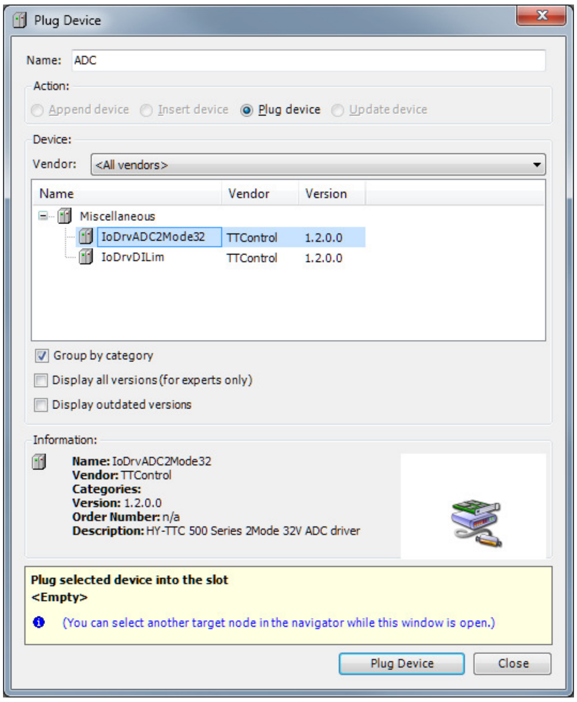


Figure 14.19: Selecting the pin functionality

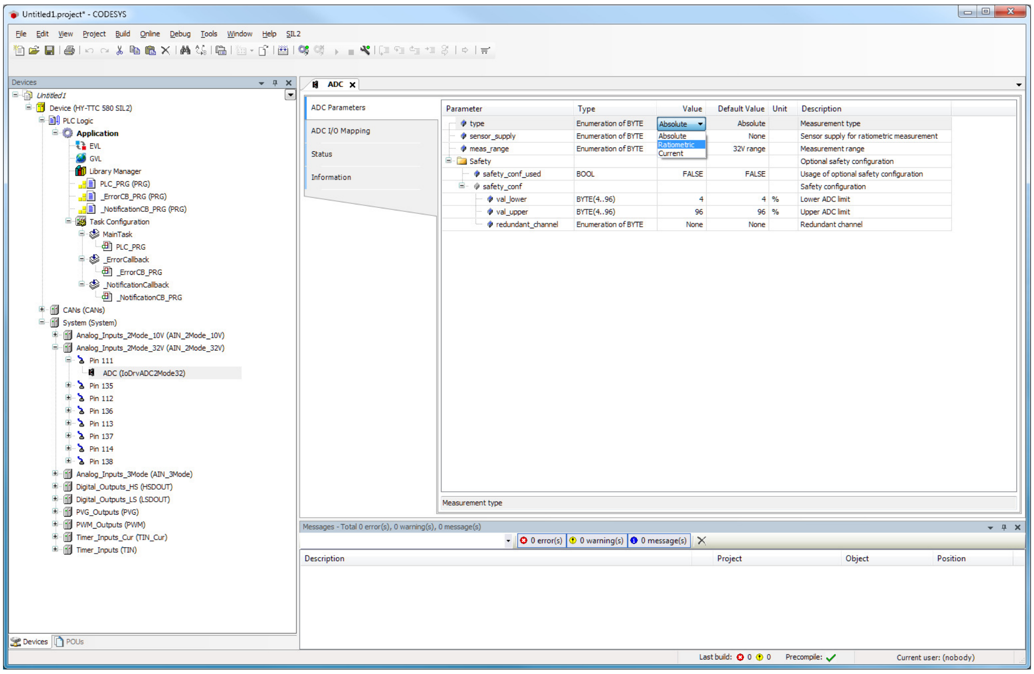


Figure 14.20: Changing the pin configuration

14.3.3 Defining the Main Program

As a part of the template, there is already the definition of a task and a program attached to that task. The default cycle of the task time is 10 ms. So, the default program **PLC_PRG** will be executed every 10 ms. Any inputs will be read at the beginning of the tasks. Any outputs will be written at the end of the task.

In **Devices**, double-click **PLC_PRG (PRG)**, a sub-item of the node **Application**, to open a kind of program editor (see Figure 14.21 on this page), which is divided into two sections. The top section is used for defining variables. The bottom section is used for implementing the program logic.

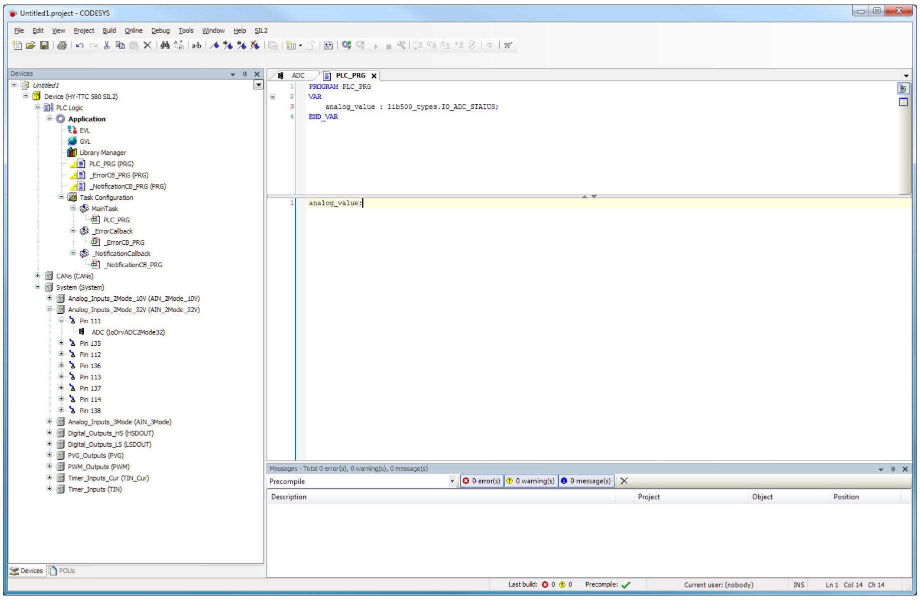


Figure 14.21: CODESYS editor

The example given in Figure 14.21 on the current page shows an instance of a structure type defined as **lib500_types.IO_ADC_STATUS**, which contains, for example, all information about an ADC channel. The bottom section of the editor shows the variable just referenced without any logic implementation. This prevents the optimizer to keep the variable, although it is somehow unused.

14.3.4 Mapping I/Os to Variables

At this point, one I/O driver has been plugged and configured, and an application has been defined. The next step is to establish a link between I/Os and application variables. Click the **ADC I/O Mapping** tab of the I/O driver you want to configure. Click a variable or structure that can be mapped to open a button with the label **. . .** .

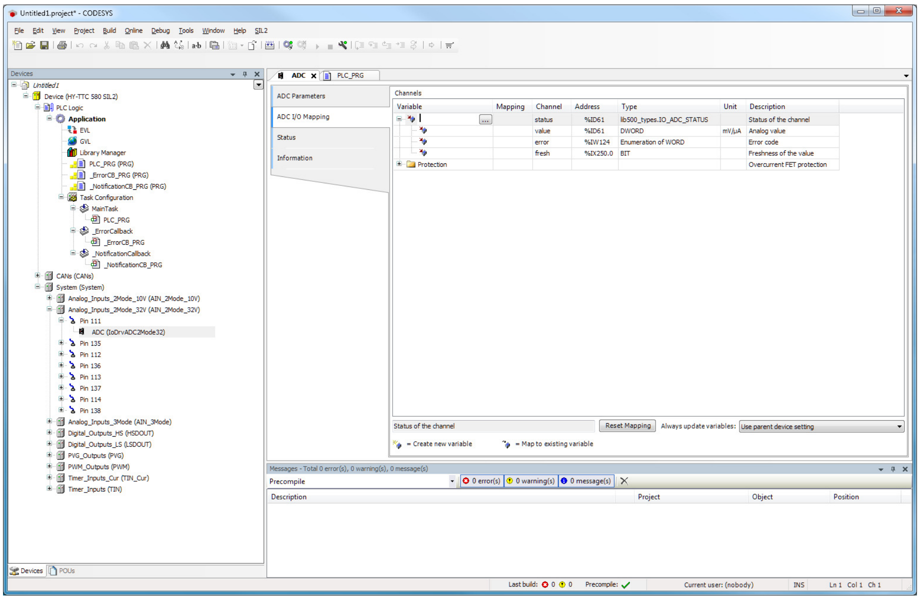
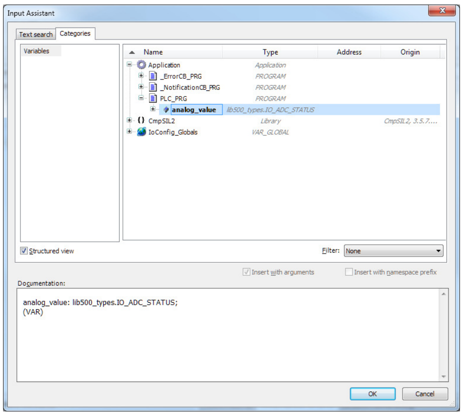
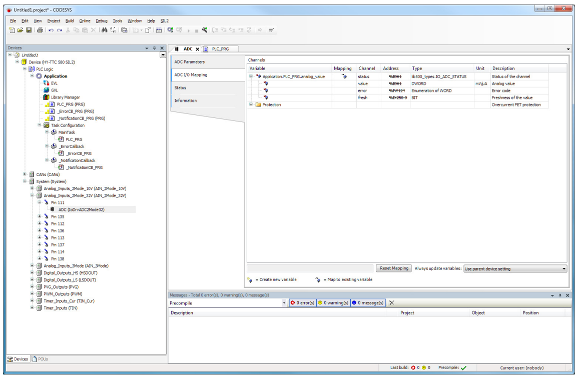


Figure 14.22: Mapping I/Os to variables

Click this button to open the **Input Assistant** dialog and map I/O parts to application variables. After finishing mapping, click **OK** to confirm the settings made and make the mapping appear in the **ADC I/O Mapping** tab.



(a) Mapping I/O parts to application variables



(b) Mapping appears in *ADC I/O Mapping* tab

Figure 14.23: *Input Assistant* dialog

14.3.5 Downloading an Application

To download the application to your HY-TTC 580, you must first scan the network for a connected device. In the CODESYS project window, click **Device (HY-TTC 580 SIL2)** on the left and then the **Communication Settings** tab. Click the button **Scan network** to scan the network for a connected device (Figure 14.24 on this page).

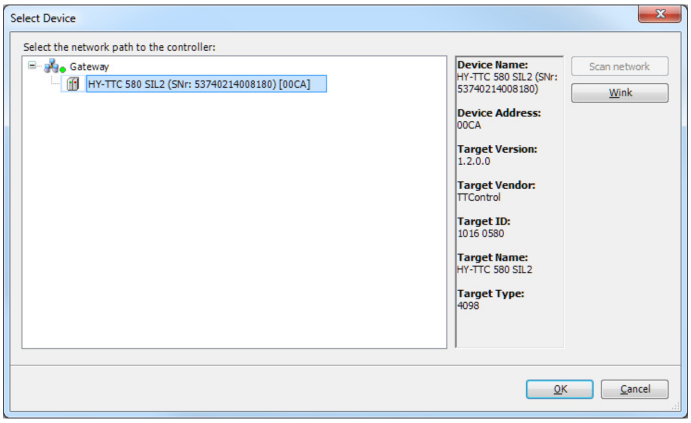


Figure 14.24: Scanning the network

In the **Select Device** dialog (Figure 14.25 on the current page), select the device found and click **OK**.

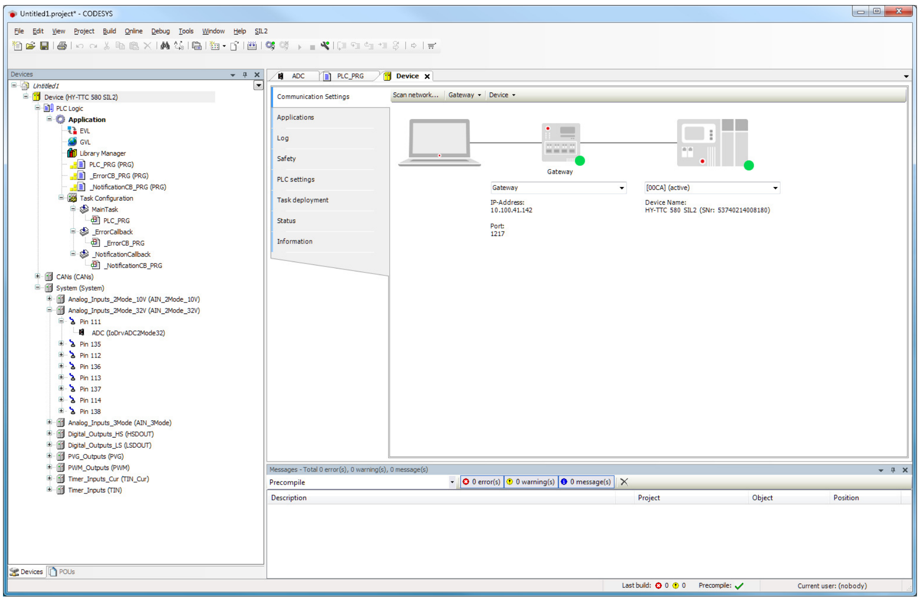


Figure 14.25: Selecting the device

Before you can download an application to a Safety SIL2 control unit, the device must be in **Debug mode**. With the standard CODESYS runtime system, this step can be skipped.

In the CODESYS project window, click the **SIL2** menu and click **Enter debug mode** (Figure 14.26 on this page).

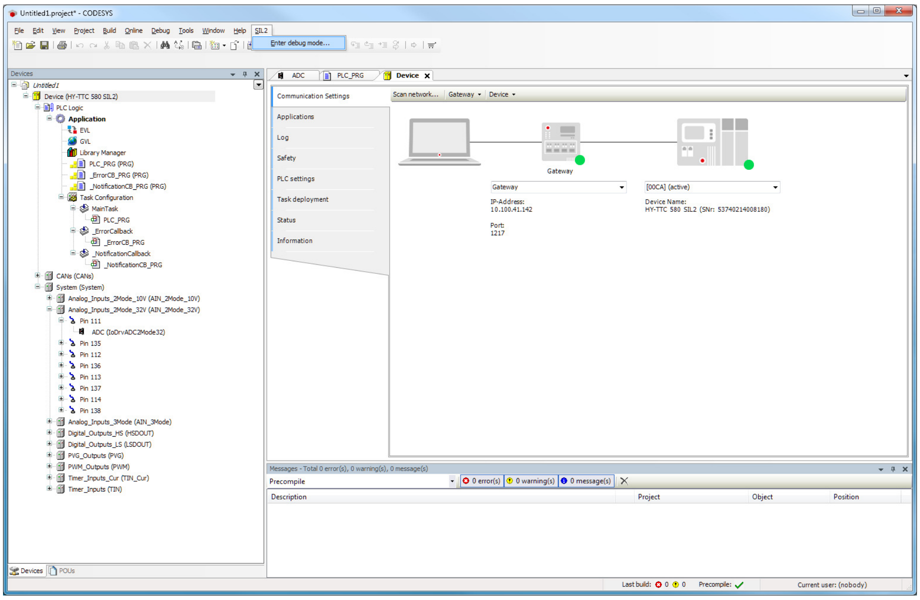


Figure 14.26: Entering the debug mode

In the **CODESYS** message (Figure 14.27 on the current page), click **OK** to confirm the debug mode.

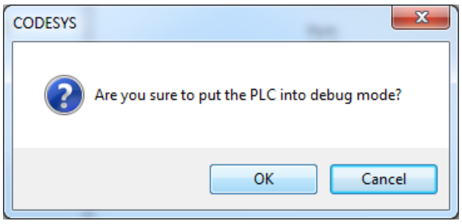


Figure 14.27: Confirming the debug mode

To download the application click the **Online** menu in the CODESYS project window and select **Login** (Figure 14.28 on the facing page).

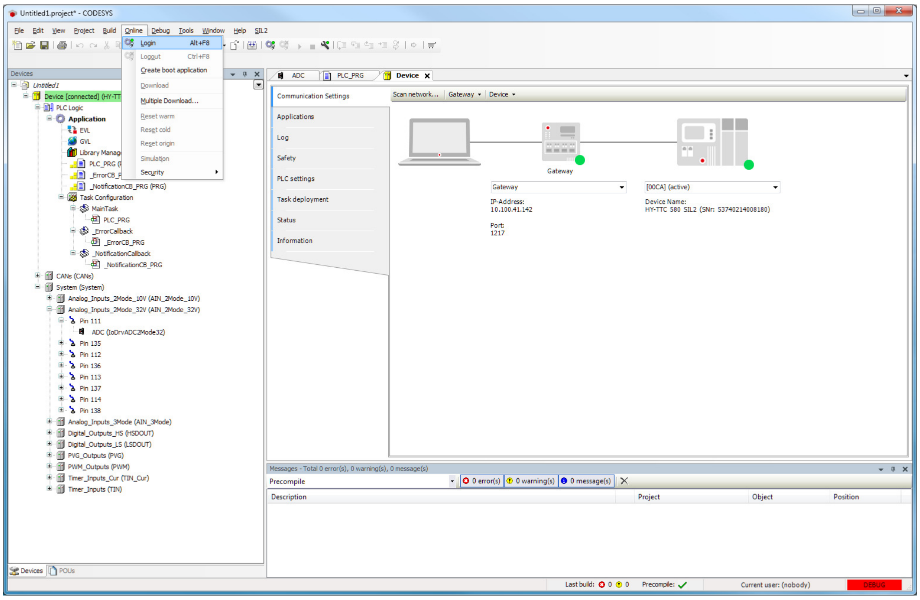


Figure 14.28: Logging on to the device

In the **CODESYS** message (Figure 14.29 on the current page), click **OK** to confirm the download of the application to the device.

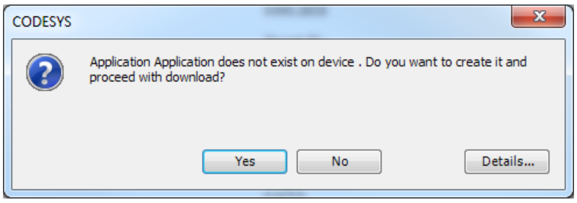


Figure 14.29: Confirming the download

To start the application after successful download, press **F5** or click the **Debug** menu in the CODESYS project window and select **Start**. If the application runs correctly, all the configured I/Os should appear in green and no errors should appear in the log window. I/O mappings will be updated continuously if they were mapped to variables. The variables to which those mappings apply will be refreshed automatically as well.

14.3.6 Available Memory

Please refer to the *Available Memory* page of the HY-TTC 500 CODESYS online help for detailed information about the available memory for CODESYS applications.

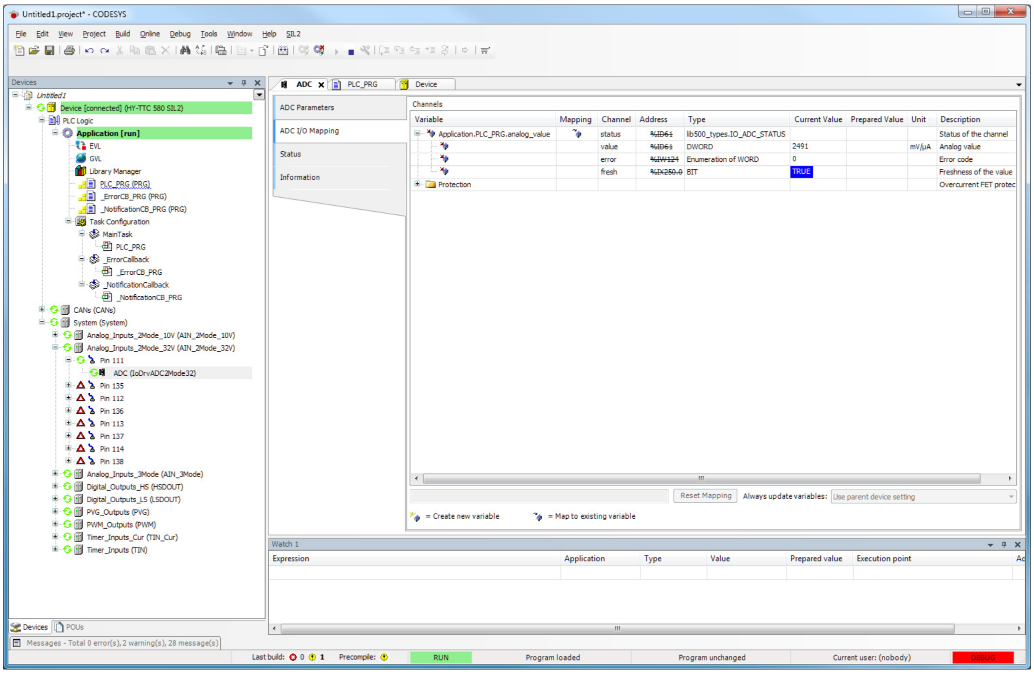


Figure 14.30: Running the application

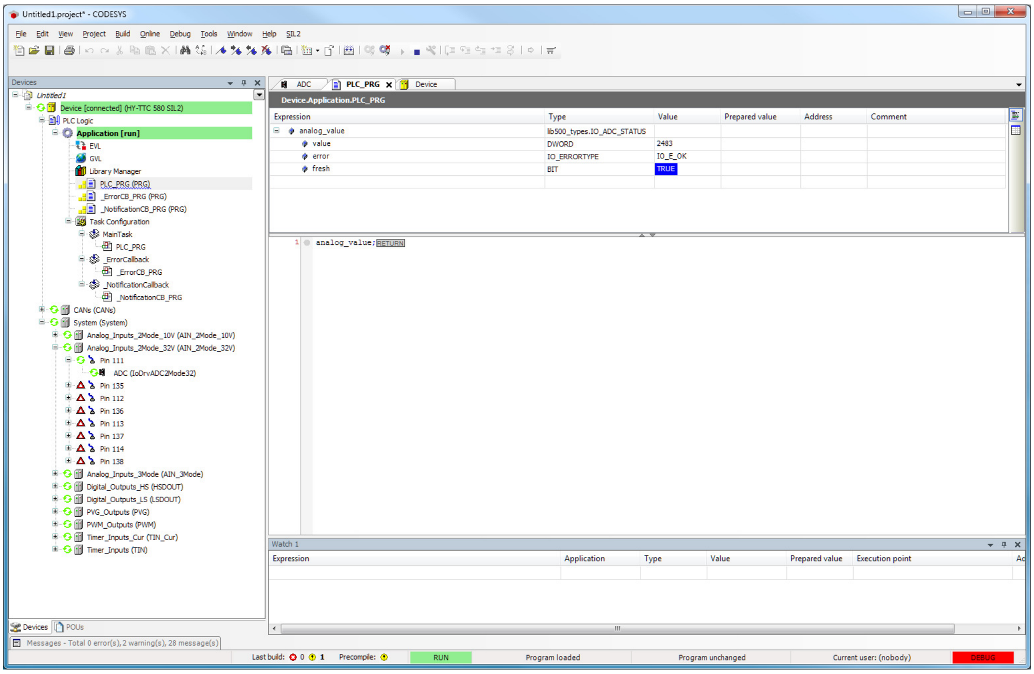


Figure 14.31: Logged in to running application

Glossary

Entry	Description
2M	2 Mode
3M	3 Mode
A/D	Analog and Digital
ADC	Analog-to-Digital Converter
API	Application Programming Interface
CAN	Controller Area Network
CPU	Central Processing Unit
CTS	Clear to Send
DC	Direct Current
DI	Digital Input
DOUT	Digital Output
DO	Digital Output
ECU	Electronic Control Unit
EEPROM	Electrically Erasable Programmable Read-Only Memory
EMC	Electromagnetic Compatibility
Flash	Nonvolatile computer storage
GND	Ground
HS	High Side
HW	Hardware
I/O	Input and Output
IN	Input
LIN	Local Interconnect Network
LSB	Least Significant Bit
LS	Low Side
MRD	Mounting Requirements Document
Max.	Maximal
Min.	Minimal
OUT	Output
PC	Personal Computer
PD	Pull Down
PID	Proportional Integral Differential
PL	Performance Level
PU	Pull Up
PVG	Proportional Valve Group
PWD	Pulse Width Demodulation
PWM	Pulse With Modulation
RTC	Real-Time Clock
RTS	Request to Send

Entry	Description
SIL	Safety Integrity Level
SRC	Signal Range Check
SSUP	Sensor Supply
SW	Software
TBD	To Be Defined
$\mathbf{T_{ambient}}$	Ambient Temperature
Terminal 15	Battery+ from ignition switch
UNECE	United Nations Economic Commission for Europe
$\mathbf{U_{bat}}$	Battery Voltage
VOUT	Voltage Output

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Disposal

NOTICE

If disposal has to be undertaken after the life span of the device, the respective applicable country-specific regulations are to be observed.

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